

MODERN BIOLOGY[®]



**John H. Postlethwait
Janet L. Hopson**



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ABOUT THE AUTHORS

The legacy of *Modern Biology* extends from 1921, when the first edition was published under the title *Biology for Beginners*. The original author was Truman Moon, a high school teacher in Middletown, New York. Moon believed that a strong grounding in the basic concepts and vocabulary of biology was the best way to prepare students for college biology. Paul Mann, a science teacher in New York City, shared Moon's passion for teaching biology and later became a coauthor.

In the mid-1940s, James Otto, a biology teacher from George Washington High School in Indianapolis, took over authorship of the book, which by then was titled *Modern Biology*. Otto added another hallmark to the program—an emphasis on the latest developments in biology at a level appropriate for young learners. In 1958, Dr. Albert Towle, a professor of biology at San Francisco State University, was added to the program as a coauthor. After the death of Otto in 1972, Towle remained the primary author until he completed the 2002 edition after which he passed away.

Now, to continue the legacy of more than eight decades of exceptional authorship, *Modern Biology* adds two more distinguished authors: John Postlethwait and Janet Hopson.



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John Postlethwait received his Ph.D. in developmental genetics from Case Western Reserve University and completed his postdoctoral fellowship in genetics at Harvard University. During a teaching career that has spanned more than 30 years, he has received several research awards and the Ersted Distinguished Teaching Award. Dr. Postlethwait now teaches general biology, embryology, and genetics at the University of Oregon and, is the coauthor of several college-level textbooks.



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On the cover: The white-faced owl is found in most regions of Africa south of the Sahara. Recent DNA research suggests that the owl differs significantly from members of the genus *Otus*, to which it was once assigned. As a result, taxonomists have reclassified this owl into two species: the Southern white-faced owl—*Ptilopsis granti* and the Northern white-faced owl—*Ptilopsis leucotis*.

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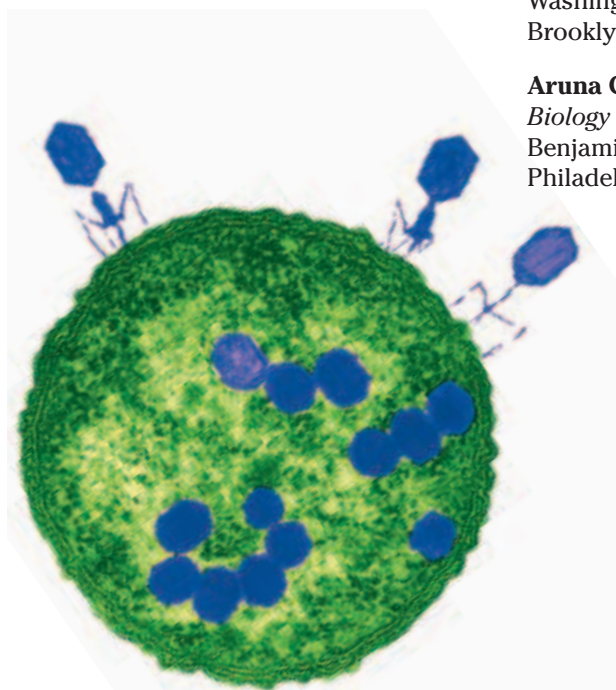
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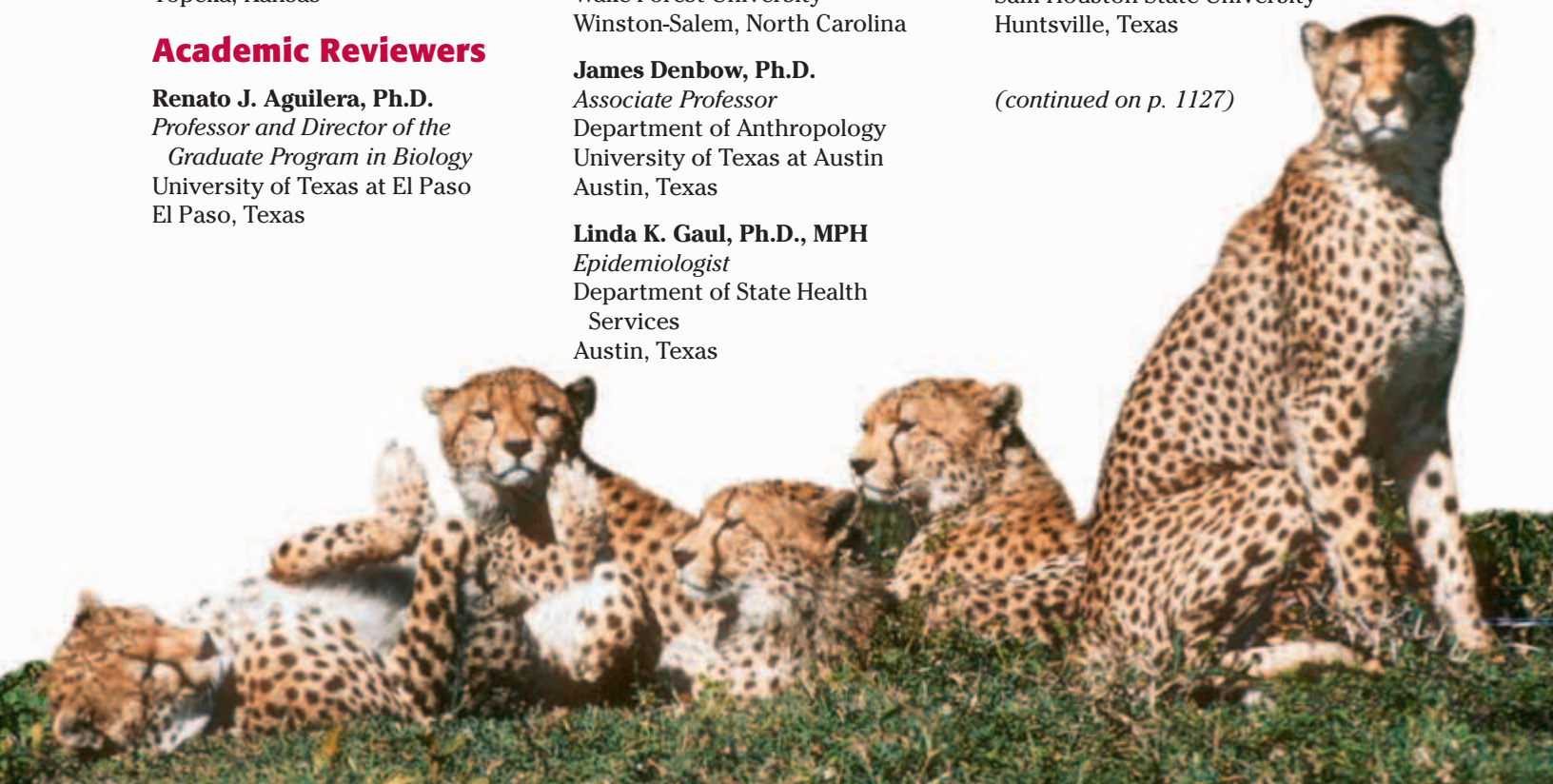
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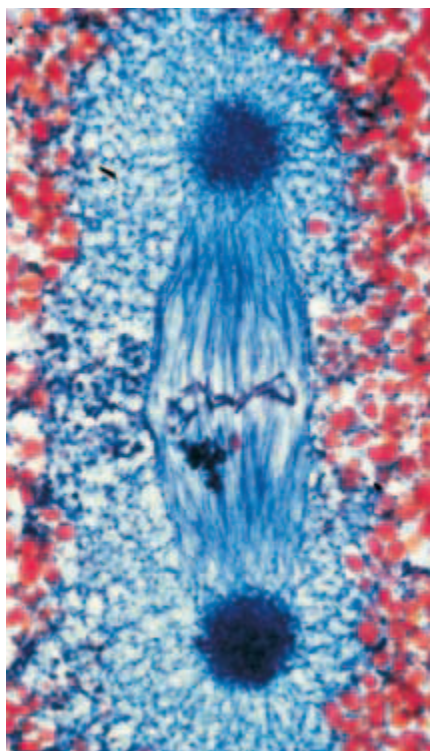


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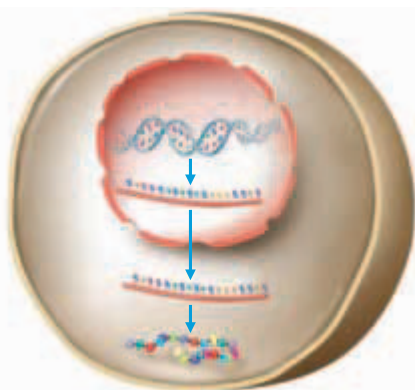
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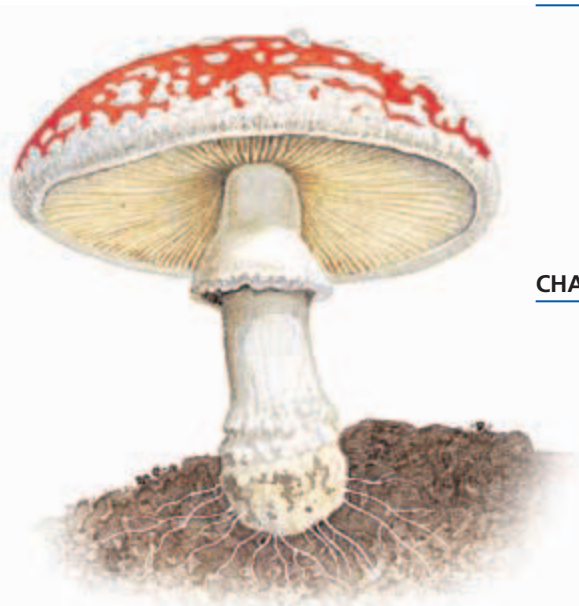
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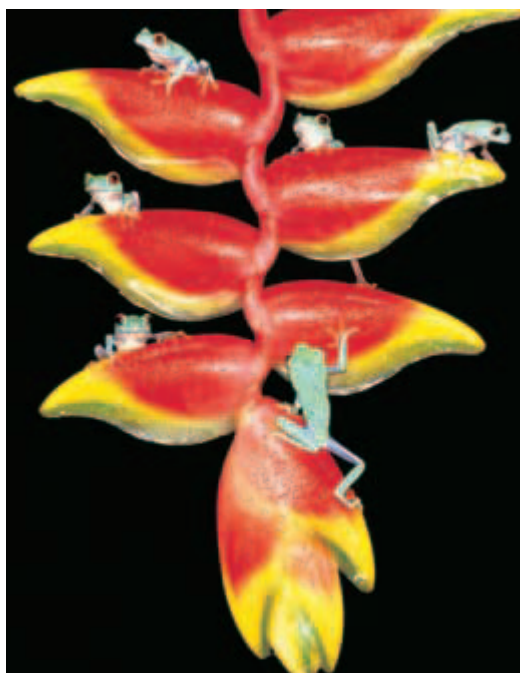
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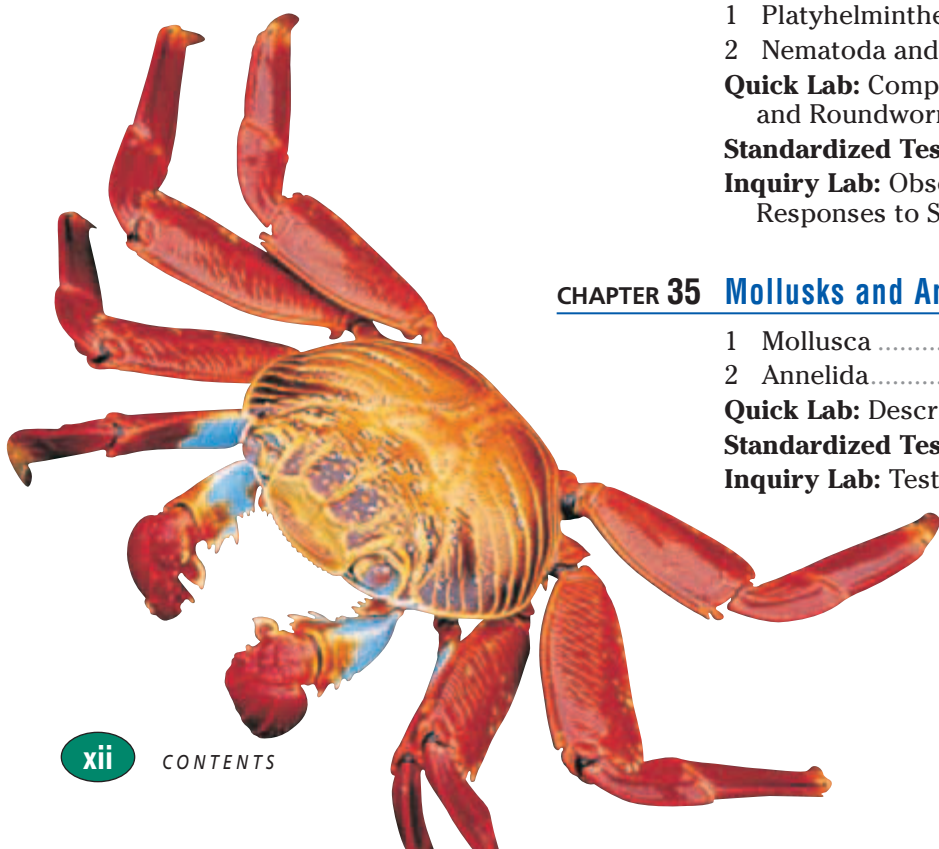
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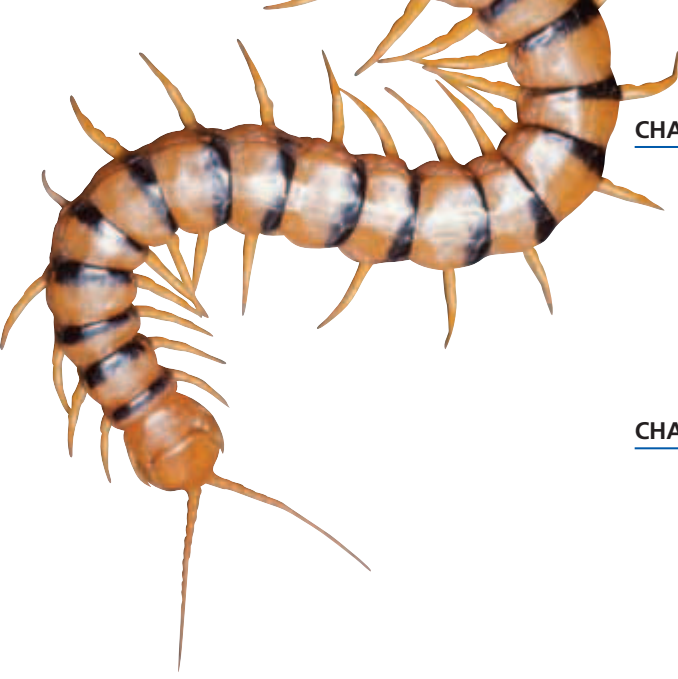
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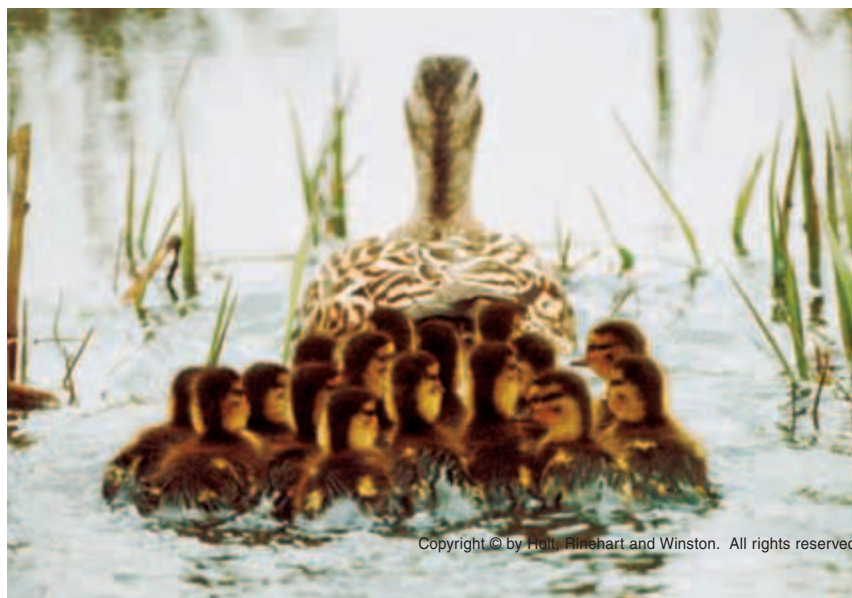
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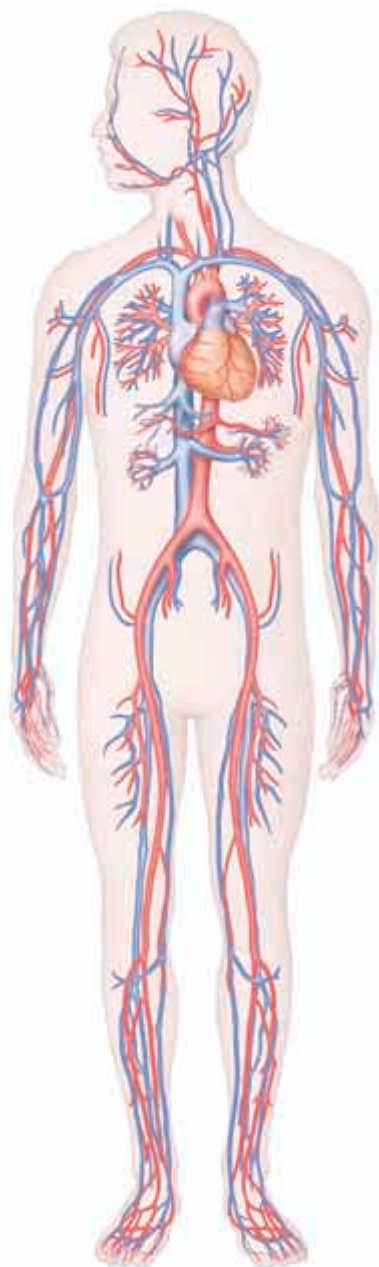
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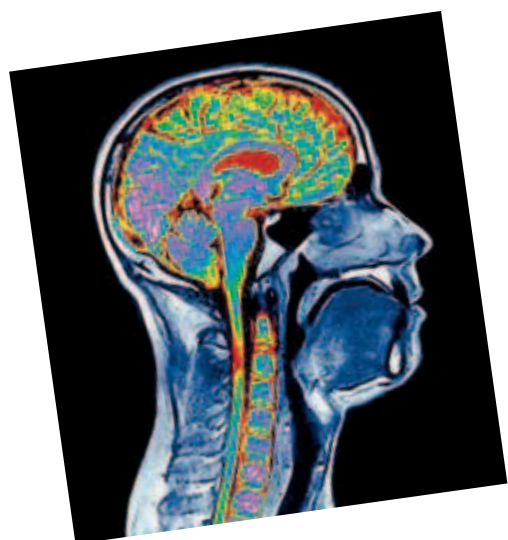
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CHAPTER LABS

Inquiry Labs allow you to form hypotheses and draw conclusions based on your work.

Exploration Labs allow you to model a phenomenon.

Skills Practice Labs teach you lab skills used by biologists.

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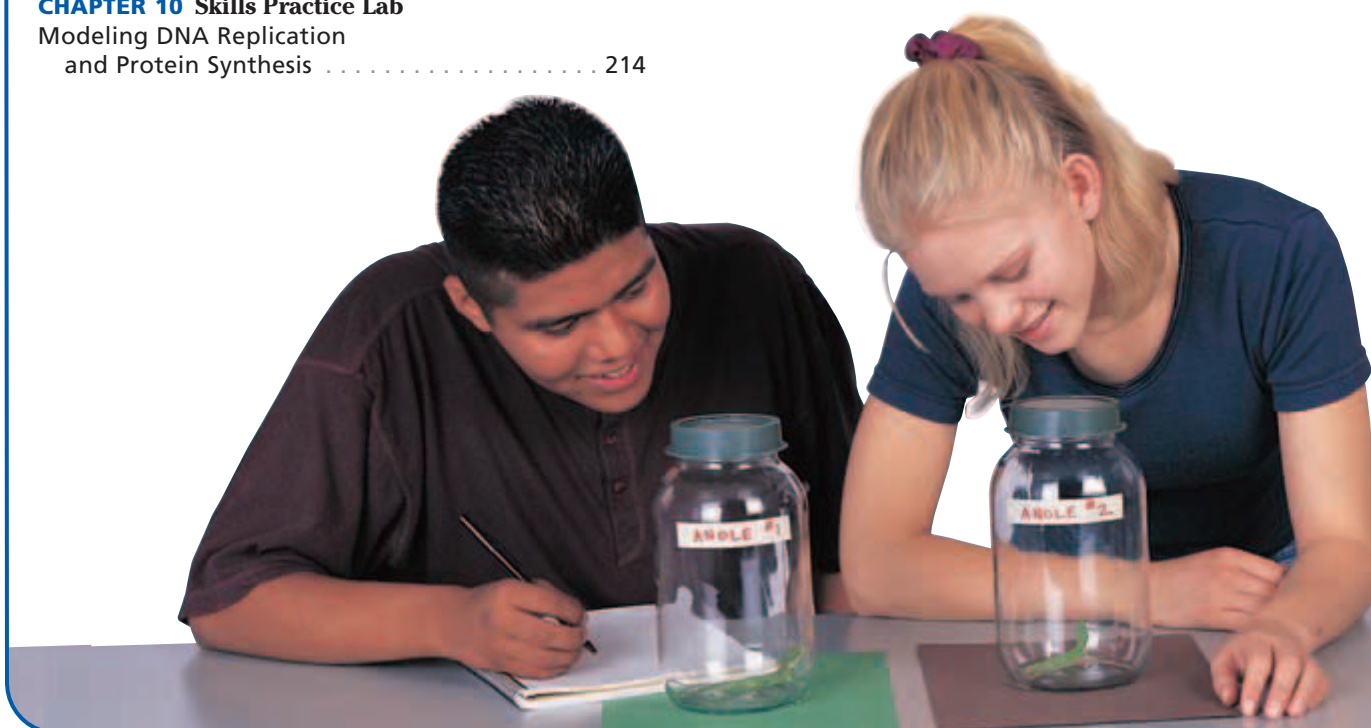
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The *SciLinks* topics below take you to more resources on the topic.

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How to Use Your Textbook

Your Roadmap for Success with *Modern Biology*

Read for Meaning

Read the **Objectives** at the beginning of each section because they will tell you what you'll need to learn. **Vocabulary** terms are also listed for each section. Each Vocabulary term is highlighted in the text and defined in text and in the **Glossary** in the **Appendix**.

After reading each chapter, turn to the **Chapter Highlights** page and review the **key concepts**, which are brief summaries of the chapter's main ideas. You may want to do this even before you read the chapter.

STUDY TIP If you don't understand a definition, reread the page on which the term was introduced. The surrounding text should help make the definition easier to understand.

SECTION 4

OBJECTIVES

- Outline the flow of genetic information in cells from DNA to protein.
- Compare the structure of RNA with that of DNA.
- Summarize the process of transcription.
- Describe the importance of the genetic code.
- Compare the role of mRNA, rRNA, and tRNA in translation.
- Identify the importance of learning about the human genome.

VOCABULARY

ribonucleic acid (RNA)
transcription
translation
protein synthesis
ribose
messenger RNA (mRNA)
ribosomal RNA (rRNA)
transfer RNA (tRNA)
RNA polymerase
promoter
termination signal
genetic code
codon
anticodon
genome

PROTEIN SYNTHESIS

Characteristics such as hair color are largely determined by genetic factors. But how does inheriting a particular form of a gene result in the appearance of a specific hair color? The structure of DNA helps explain how genes function in making proteins that determine traits in organisms.

FLOW OF GENETIC INFORMATION

A gene is a segment of DNA that is located on a chromosome and that codes for a hereditary character. For example, a gene in hair follicle cells determines a person's hair color. The gene directs the making of the protein called *melanin* (a pigment) through an intermediate—the nucleic acid called **ribonucleic acid**, or **RNA**.

Figure 10-12 summarizes the flow of genetic information in a eukaryotic cell. During **transcription**, DNA acts as a template for the synthesis of RNA. In **translation**, RNA directs the assembly of proteins. Forming proteins based on information in DNA and carried out by RNA is called **protein synthesis**, or gene expression. This central concept can be symbolized as $DNA \rightarrow RNA \rightarrow protein$. Proteins do important work in cells, such as protecting the body against infections and carrying oxygen in red blood cells.

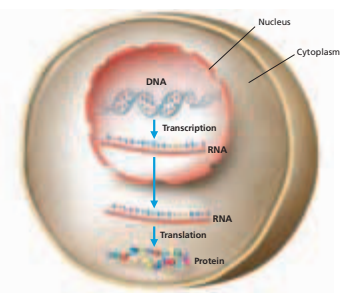


FIGURE 10-12
DNA contains the instructions for building a protein. DNA transfers the instructions to an RNA molecule in a process called transcription. The RNA moves out into the cytoplasm, where its instructions are read and the protein is assembled in a process called translation.

204 CHAPTER 10

Be Resourceful—Use the Web



SciLinks boxes in your textbook take you to resources that you can use for science projects, reports, and research papers. Go to scilinks.org, and type in the **SciLinks code** to get information on a topic.



Visit go.hrw.com

Find resources and reference materials that go with your textbook at go.hrw.com. Enter the keyword **HM6 Home** to access the home page for your textbook.

Word Roots and Origins

transcription

from the Latin *scribere*, meaning "to write," and *trans*, meaning "across"

TRANSCRIPTION

Transcription is the process by which the genetic instructions in a specific gene are transcribed or "rewritten" into an RNA molecule. Transcription takes place in the nucleus of eukaryotic cells and in the DNA-containing region in the cytoplasm of prokaryotic cells.

Steps of Transcription

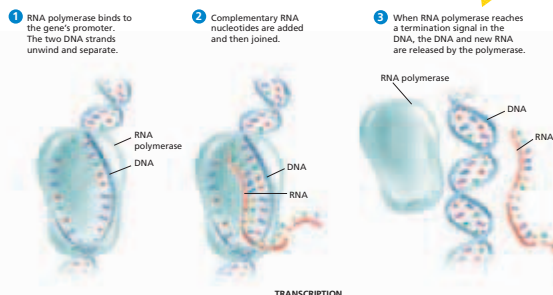
Transcription occurs in three steps, as shown in Figure 10-15. In step 1, **RNA polymerase**, an enzyme that catalyzes the formation of RNA on a DNA template, binds to a promoter. A **promoter** is a specific nucleotide sequence of DNA where RNA polymerase binds and initiates transcription. After RNA polymerase binds to the promoter, the DNA strands unwind and separate.

In step 2, RNA polymerase adds free RNA nucleotides that are complementary to the nucleotides on one of the DNA strands. The resulting chain is an RNA molecule. As in DNA replication, complementary base pairing determines the nucleotide sequence in the newly made RNA. For example, if the bases on the DNA strand was ATCGAC, the bases on the RNA strand would be UAGCUG. Unlike DNA replication, transcription uses only a specific region (a gene) on one of the two DNA strands to serve as the template. As RNA polymerase moves past, the separated DNA strands rewind.

During step 3, RNA polymerase reaches a **termination signal**, a specific sequence of nucleotides that marks the end of a gene. Upon reaching this "stop" signal, RNA polymerase releases both the DNA and the newly formed RNA. The RNA made during transcription can be one of many types including mRNA, tRNA, or rRNA. The newly made RNA can now perform its job in the cell, and the RNA polymerase can transcribe another gene.

FIGURE 10-15

During transcription, the enzyme RNA polymerase "reads" one of the chains, the template strand. RNA polymerase adds and then joins complementary RNA nucleotides, resulting in an RNA strand.



206 CHAPTER 10

To learn where and when human cells use each of the proteins coded for in the approximately 30,000 genes in the human genome will take much more analysis. This information is important because learning which gene sequences control particular biological functions may help diagnose, treat, and prevent genetic disorders, cancer, and infectious diseases in the future.

SECTION 4 REVIEW

1. Summarize the flow of genetic information.
2. List the four ways in which the structure of RNA differs from that of DNA.
3. Describe the structure and function of each of the three types of RNA.
4. Sequence the main steps of transcription.
5. What is the genetic code?
6. Compare the roles of the three different types of RNA during translation.
7. Describe the significance of identifying the entire sequence of the human genome.

CRITICAL THINKING

8. **Making Comparisons** How does the role of RNA polymerase in transcription differ from that of DNA polymerase in DNA replication?
9. **Applying Information** What amino acids would translation of the mRNA with the sequence UAACAAGGAGCAUCC produce?
10. **Analyzing Processes** Discuss why it is important which of the two DNA strands serves as a template during transcription.
11. **Drawing Conclusions** How does the structure of tRNA relate to its function in translation?

210 CHAPTER 10

Use the Illustrations and Photos

Art shows complex ideas and processes. Learn to analyze the art so that you better understand the material in the text.

Tables and graphs display important information in an organized way to help you see relationships.

A picture is worth a thousand words. Look at the photographs to see relevant examples of science concepts that you are reading about.

Prepare for Tests

Section Reviews and **Chapter Reviews** test your knowledge of the main points of the chapter. **Critical Thinking** items challenge you to think about the material in different ways and in greater depth. The **Standardized Test Preparation** that is located after each Chapter Review helps you sharpen your test-taking abilities.

STUDY TIP Reread the Objectives and Chapter Highlights when studying for a test to be sure you know the material.

Use the Appendix

The **Appendix** contains a variety of resources designed to enhance your learning experience. These resources include **Using a Compound Light Microscope**, **Using SI Units for Measurement**, and **Analyzing Word Parts**. The Appendix also contains reference materials that will be useful in your exploration of biology, such as **Biologist's Guide to the Periodic Table**, **Classification**, and **Geologic Time Scale**.



Visit Holt Online Learning

If your teacher gives you a special password to log onto the **Holt Online Learning** site, you'll find your complete textbook on the Web. In addition, you'll find some great learning tools and practice quizzes. You'll be able to see how well you know the material from your textbook.



For advanced-level project ideas from *Scientific American*, visit go.hrw.com and type in the keyword **HM6SAX**.

UNIT 1

CHAPTERS

- 1 *The Science of Life*
- 2 *Chemistry of Life*
- 3 *Biochemistry*

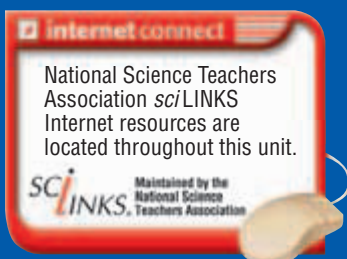
FOUNDATIONS OF BIOLOGY

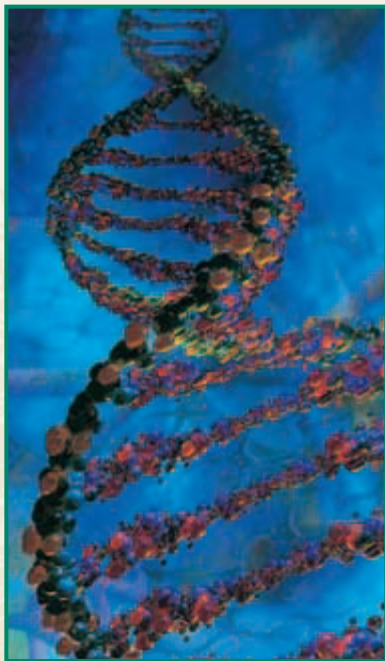
“Our ideas are only instruments which we use to break into phenomena; we must change them when they have served their purpose, as we change a blunt lancet that we have used long enough.”

Claude Bernard



Organisms living in this taiga ecosystem are adapted to dry, cold weather and to reduced availability of food in winter.





DNA is responsible for transmitting genetic information to offspring.



This giant panda gets its energy by eating bamboo leaves. The Amazon water lily, above left, lives in shallow eutrophic ponds.



Red-eyed tree frog

THE SCIENCE OF LIFE

Snowy owls are keen hunters that live in Arctic forests and on open tundra. These juveniles of the species *Nyctea scandiaca* hatched on a flower-filled alpine meadow.



SECTION 1 *The World of Biology*

SECTION 2 *Themes in Biology*

SECTION 3 *The Study of Biology*

SECTION 4 *Tools and Techniques*

THE WORLD OF BIOLOGY

Owls, such as the young snowy owls on the previous page, have for centuries been symbols of both wisdom and mystery. To many cultures their piercing eyes have conveyed a look of intelligence. Their silent flight through darkened landscapes in search of prey has projected an air of power or wonder. For this chapter and this book, owls are an engaging example of a living organism from the world of biology—the study of life.

BIOLOGY AND YOU

Living in a small town, in the country, or at the edge of the suburbs, one may be lucky enough to hear an owl's hooting. This experience can lead to questions about where the bird lives, what it hunts, and how it finds its prey on dark, moonless nights. **Biology**, or the study of life, offers an organized and scientific framework for posing and answering such questions about the natural world. Biologists study questions about how living things work, how they interact with the environment, and how they change over time. Biologists study many different kinds of living things ranging from tiny organisms, such as bacteria, to very large organisms, such as elephants.

Each day, biologists investigate subjects that affect you and the way you live. For example, biologists determine which foods are healthy. As shown in Figure 1-1, everyone is affected by this important topic. Biologists also study how much a person should exercise and how one can avoid getting sick. Biologists also study what your air, land, and food supply will be like in the near future.



FIGURE 1-1

Biology, the study of life, directly applies to your health, life, and future in ways as simple as daily food choices.

OBJECTIVES

- **Relate** the relevance of biology to a person's daily life.
- **Describe** the importance of biology in human society.
- **List** the characteristics of living things.
- **Summarize** the hierarchy of organization within complex multicellular organisms.
- **Distinguish** between homeostasis and metabolism and between growth, development, and reproduction.

VOCABULARY

biology
organization
cell
unicellular
multicellular
organ
tissue
organelle
biological molecule
homeostasis
metabolism
cell division
development
reproduction
gene

Biology and Society

By studying biology you can make informed decisions on issues that impact you and our society. Every day newspapers, television, and the Internet contain issues that relate to biology. For example, you may read that your local water or air supply is polluted. How will that pollution affect your health and the health of other living things? You may hear about new technologies or tools that biologists have invented. How will we control how those technologies and tools are used? Biologists actively work to solve these and other real-world issues and problems, including improving our food supply, curing diseases and preserving our environment.

CHARACTERISTICS OF LIFE

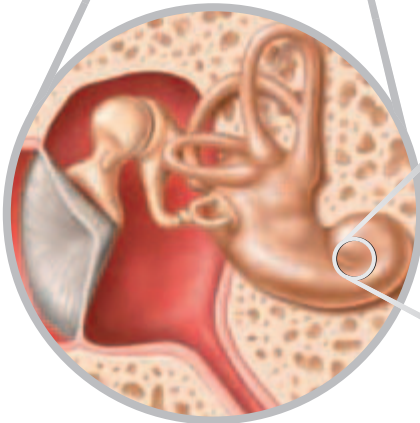
FIGURE 1-2

Every living organism has a level of organization. The different levels of organization for a complex multicellular organism, such as an owl, are shown in the figure below.

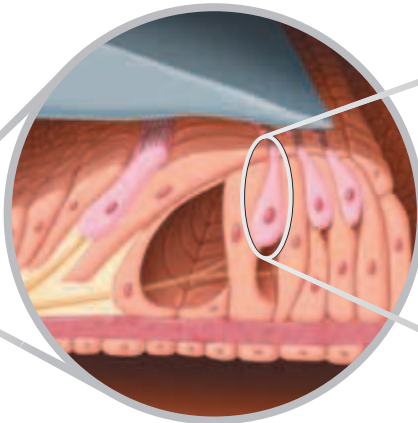
ORGANISM
(Barn Owl)



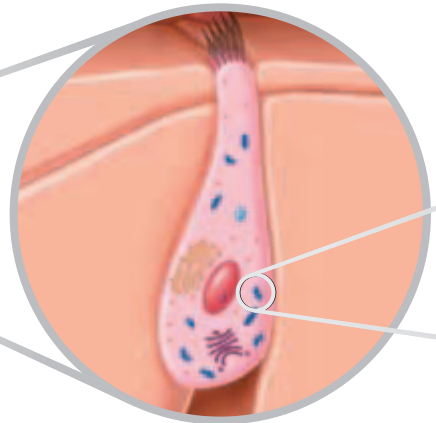
ORGAN
(Owl's Ear)



TISSUE
(Nervous Tissue Within the Ear)



CELL
(Nerve Cell)



The world is filled with familiar objects, such as tables, rocks, plants, pets, and automobiles. Which of these objects are living or were once living? What are the criteria for assigning something to the living world or the nonliving world? Biologists have established that living things share seven characteristics of life. These characteristics are organization and the presence of one or more cells, response to a stimulus (plural, *stimuli*), homeostasis, metabolism, growth and development, reproduction, and change through time.

Organization and Cells

Organization is the high degree of order within an organism's internal and external parts and in its interactions with the living world. For example, compare an owl to a rock. The rock has a specific shape, but that shape is usually irregular. Furthermore, different rocks, even rocks of the same type, are likely to have different shapes and sizes. In contrast, the owl is an amazingly organized individual, as shown in Figure 1-2. Owls of the same species have the same body parts arranged in nearly the same way and interact with the environment in the same way.

All living organisms, whether made up of one cell or many cells, have some degree of organization. A **cell** is the smallest unit that can perform all life's processes. Some organisms, such as bacteria, are made up of one cell and are called **unicellular** (YOON-uh-SEL-yoo-luhr) organisms. Other organisms, such as humans or trees, are made up of multiple cells and are called **multicellular** (MUHL-ti-SEL-yoo-luhr) organisms. Complex multicellular organisms have the level of organization shown in Figure 1-2. In the highest level, the organism is made up of *organ systems*, or groups of specialized parts that carry out a certain function in the organism. For example, an owl's nervous system is made up of a brain, sense organs, nerve cells, and other parts that sense and respond to the owl's surroundings.

Organ systems are made up of organs. **Organs** are structures that carry out specialized jobs within an organ system. An owl's ear is an organ that allows the owl to hear. All organs are made up of tissues. **Tissues** are groups of cells that have similar abilities and that allow the organ to function. For example, nervous tissue in the ear allows the ear to detect sound. Tissues are made up of cells. A cell must be covered by a membrane, contain all genetic information necessary for replication, and be able to carry out all cell functions.

Within each cell are organelles. **Organelles** are tiny structures that carry out functions necessary for the cell to stay alive. Organelles contain **biological molecules**, the chemical compounds that provide physical structure and that bring about movement, energy use, and other cellular functions. All biological molecules are made up of *atoms*. Atoms are the simplest particle of an element that retains all the properties of a certain element.

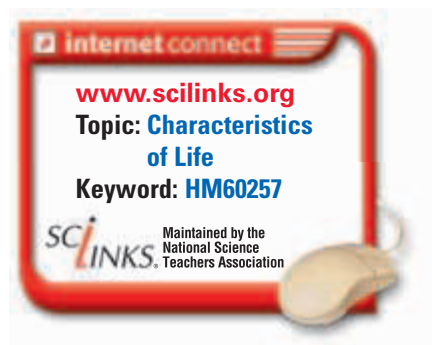
Response to Stimuli

Another characteristic of life is that an organism can respond to a *stimulus*—a physical or chemical change in the internal or external environment. For example, an owl dilates its pupils to keep the level of light entering the eye constant. Organisms must be able to respond and react to changes in their environment to stay alive.

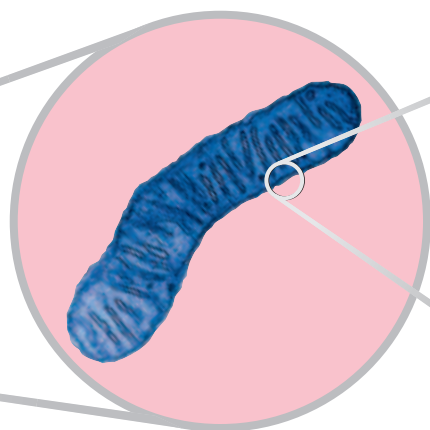
Word Roots and Origins

cell

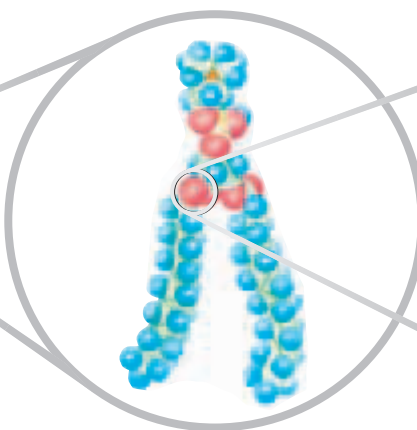
from the Latin, *cella*
meaning "small room," or "hut"



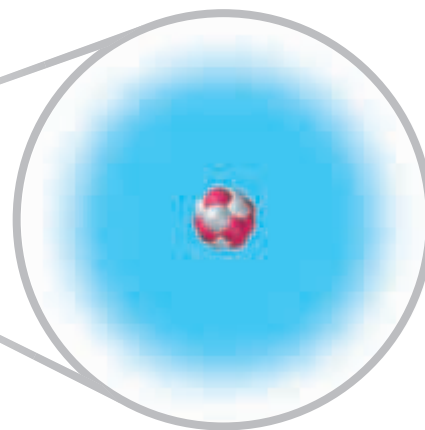
ORGANELLE
(Mitochondrion)



BIOLOGICAL MOLECULE
(Phospholipid)



ATOM
(Oxygen)





Quick Lab

Observing Homeostasis

Materials 500 mL beakers (3), wax pen, tap water, thermometer, ice, hot water, goldfish, small dip net, watch or clock with a second hand

Procedure



1. Use a wax pen to label three 500 mL beakers as follows: 27°C (80°F), 20°C (68°F), 10°C (50°F). Put 250 mL of tap water in each beaker. Use hot water or ice to adjust the temperature of the water in each beaker to match the temperature on the label.
2. Put the goldfish in the beaker of 27°C water. Record the number of times the gills move in 1 minute.
3. Move the goldfish to the beaker of 20°C water. Repeat observations. Move the goldfish to the beaker of 10°C. Repeat observations.

Analysis What happens to the rate at which gills move when the temperature changes? Why? How do gills help fish maintain homeostasis?

FIGURE 1-3

This unicellular organism, *Escherichia coli*, inhabits the human intestines. *E. coli* reproduces by means of cell division, during which the original cell splits into two identical offspring cells.



Homeostasis

All living things, from single cells to entire organisms, have mechanisms that allow them to maintain stable internal conditions. Without these mechanisms, organisms can die. For example, a cell's water content is closely controlled by the taking in or releasing of water. A cell that takes in too much water will rupture and die. A cell that doesn't get enough water will also shrivel and die.

Homeostasis (HOH-mee-OH-STAY-sis) is the maintenance of a stable level of internal conditions even though environmental conditions are constantly changing. Organisms have regulatory systems that maintain internal conditions, such as temperature, water content, and uptake of nutrients by the cell. In fact, multicellular organisms usually have more than one way of maintaining important aspects of their internal environment. For example, an owl's temperature is maintained at about 40°C (104°F). To keep a constant temperature, an owl's cells burn fuel to produce body heat. In addition, an owl's feathers can fluff up in cold weather. In this way, they trap an insulating layer of air next to the bird's body to maintain its body temperature.

Metabolism

Living organisms use energy to power all the life processes, such as repair, movement, and growth. This energy use depends on metabolism (muh-TAB-uh-LIZ-uhm). **Metabolism** is the sum of all the chemical reactions that take in and transform energy and materials from the environment. For example, plants, algae, and some bacteria use the sun's energy to generate sugar molecules during a process called *photosynthesis*. Some organisms depend on obtaining food energy from other organisms. For instance, an owl's metabolism allows the owl to extract and modify the chemicals trapped in its nightly prey and use them as energy to fuel activities and growth.

Growth and Development

All living things grow and increase in size. Some nonliving things, such as crystals or icicles, grow by accumulating more of the same material of which they are made. In contrast, the growth of living things results from the division and enlargement of cells. **Cell division** is the formation of two new cells from an existing cell, as shown in Figure 1-3. In unicellular organisms, the primary change that occurs following cell division is cell enlargement. In multicellular life, however, organisms mature through cell division, cell enlargement, and development.

Development is the process by which an organism becomes a mature adult. Development involves cell division and cell differentiation, or specialization. As a result of development, an adult organism is composed of many cells specialized for different functions, such as carrying oxygen in the blood or hearing. In fact, the human body is composed of trillions of specialized cells, all of which originated from a single cell, the fertilized egg.

Reproduction

All organisms produce new organisms like themselves in a process called **reproduction**. Reproduction, unlike other characteristics, is not essential to the survival of an individual organism. However, because no organism lives forever, reproduction is essential for the continuation of a species. Glass frogs, as shown in Figure 1-4, lay many eggs in their lifetime. However, only a few of the frogs' offspring reach adulthood and successfully reproduce.

During reproduction, organisms transmit hereditary information to their offspring. Hereditary information is encoded in a large molecule called *deoxyribonucleic acid*, or *DNA*. A short segment of DNA that contains the instructions for a single trait of an organism is called a **gene**. DNA is like a large library. It contains all the books—genes—that the cell will ever need for making all the structures and chemicals necessary for life.

Hereditary information is transferred to offspring during two kinds of reproduction. In *sexual reproduction*, hereditary information recombines from two organisms of the same species. The resulting offspring are similar but not identical to their parents. For example, a male frog's sperm can fertilize a female's egg and form a single fertilized egg cell. The fertilized egg then develops into a new frog.

In *asexual reproduction*, hereditary information from different organisms is not combined; thus the original organism and the new organism are genetically the same. A bacterium, for example, reproduces asexually when it splits into two identical cells.

Change Through Time

Although individual organisms experience many changes during their lifetime, their basic genetic characteristics do not change. However, populations of living organisms *evolve* or change through time. The ability of populations of organisms to change over time is important for survival in a changing world. This factor is also important in explaining the diversity of life-forms we see on Earth today.



FIGURE 1-4

Like many animal species, this glass frog, *Centrolenella* sp., produces and lays a large number of eggs. However, a high percentage of these eggs die. In contrast, the offspring of animals that give birth to just a few live offspring typically have a high rate of survival.

SECTION 1 REVIEW

1. How does biology affect a person's daily life?
2. How does biology affect society?
3. Name the characteristics shared by living things.
4. Summarize the hierarchy of organization found in complex multicellular organisms.
5. What are the different functions of homeostasis and metabolism in living organisms?
6. How does the growth among living and nonliving things differ?
7. Why is reproduction an important characteristic of life?

CRITICAL THINKING

8. **Applying Information** Crystals of salt grow and are highly organized. Why don't biologists consider them to be alive?
9. **Analyzing Models** When a scientist designs a space probe to detect life on a distant planet, what kinds of things should it measure?
10. **Making Comparisons** Both cells and organisms share the characteristics of life. How are cells and organisms different?

SECTION 2

OBJECTIVES

- **Identify** three important themes that help explain the living world.
- **Explain** how life can be diverse, yet unified.
- **Describe** how living organisms are interdependent.
- **Summarize** why evolution is an important theme in biology.

VOCABULARY

domain
kingdom
ecology
ecosystem
evolution
natural selection
adaptation

THEMES IN BIOLOGY

A snowy owl, with all its beauty and complexity, is just one species among the millions of species on Earth. How can one understand so many different living organisms? Important unifying themes help explain the living world and are part of biology. These themes are found within this book and include the diversity and unity of life, the interdependence of living organisms, and the evolution of life.

DIVERSITY AND UNITY OF LIFE

The *diversity*, or variety, of life, is amazing. For example, there are single-celled organisms that thrive inside thick Antarctic ice that never thaws. There are whales that contain about 1,000 trillion cells that can easily cruise the Pacific and migrate each year from Alaska to Mexico. There are even plants that can capture and eat insects. Biologists have identified more than 1.5 million species on Earth. And there may be many more species that remain to be identified.

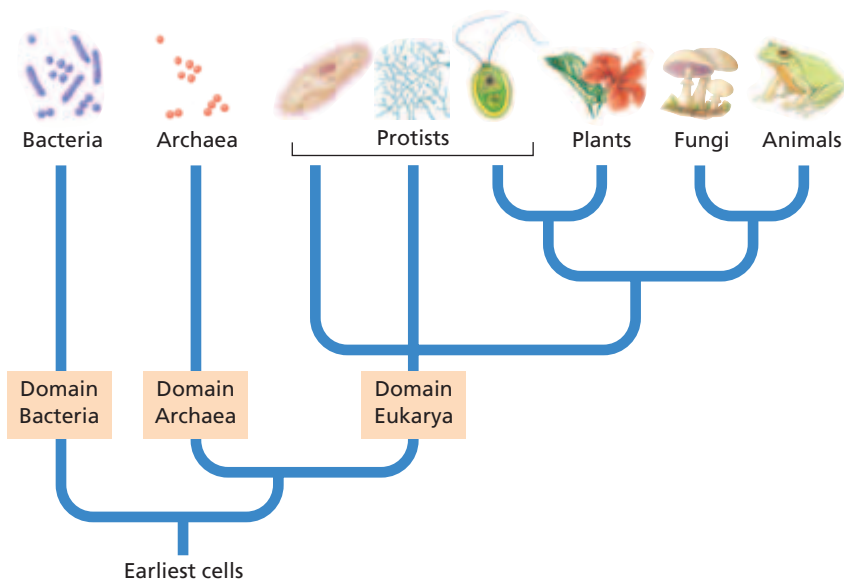
Unity in the Diversity of Life

Life is so diverse, yet life is also characterized by *unity*, or features that all living things have in common. One feature is the *genetic code*, the rules that govern how cells use the hereditary information in DNA. Another unifying feature is the presence of organelles that carry out all cellular activities.

The “tree of life” shown in Figure 1-5 is a model of the relationships by ancestry among organisms. All living things share certain genes, yet no two types of organisms have the same full sets of genes. One way biologists build a “tree of life” is to place organisms that have more similar sets of genes on closer branches, or *lineages*, of the “tree.” They place the more distantly related organisms on more distant branches. The placement of all kinds of organisms produces a “tree” that relates and unites life’s diversity.

FIGURE 1-5

This “tree of life,” is a model of the relationships by ancestry among all major groups of organisms. The model is based on comparisons of organisms’ characteristics, including body structures and genetic information. For updates on such “trees,” visit go.hrw.com and enter the keyword **HM6 Phylo**.



So, how does the “tree of life” represent the unity in the diversity of life? Scientists think that all living things have descended with modification from a single common ancestor. Thus, all of life is connected. Yet, there are many different lineages representing different species. This diversity stems from the fact that genetic changes accumulate over the years. Also, organisms change as they become suited to their own special environments.

Three Domains of Life

Notice in Figure 1-5 that the “tree” has three main branches. Biologists call these major subdivisions of all organisms **domains**. The three domains are Bacteria, Archaea, and Eukarya. Bacteria and Archaea have less complex cells than those of Eukarya. Later chapters describe these domains in more detail. Notice that the largest and most familiar organisms, the animals, plants, and fungi, occupy parts of the Eukarya branch of the “tree.”

Another system of grouping organisms divides all life into six major categories called **kingdoms**. The six kingdoms consist of four kingdoms within the domain Eukarya (the Kingdoms Animalia, Plantae, Fungi, and Protista), one kingdom in the domain Archaea (Kingdom Archaea) and one kingdom in the domain Bacteria (Kingdom Bacteria). Many biologists recognize these six kingdoms and three domains, but some biologists use other systems of grouping.

Word Roots and Origins

ecosystem

from the Greek word, *eco*, meaning “house,” and *system*, meaning “ordered parts in a whole”

INTERDEPENDENCE OF ORGANISMS

Organisms interact with each other throughout the living world. **Ecology** is the branch of biology that studies organisms interacting with each other and with the environment. Ecologists study single species as well as ecosystems (EK-oh-SIS-tuhmz).

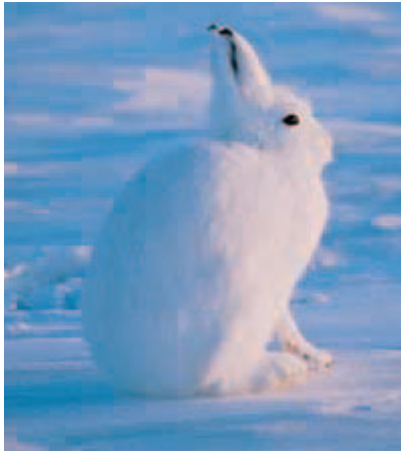
Ecosystems are communities of living species and their physical environments. Such studies reveal that organisms depend on each other as well as on minerals, nutrients, water, gases, heat, and other elements of their physical surroundings. For example, a panther eats a bird, which eats nuts from trees. The tree needs carbon dioxide and water. Carbon dioxide is a main byproduct of all animals.

Scientists now recognize the huge effect that humans have had on the world’s environment. For millions of years, tropical rain forests, as shown in Figure 1-6, have existed as stable—but fragile—environments. These forests play a vital role in the global environment. Humans have cleared vast areas of these forests in recent years. The destruction of these forests, in addition to other ecological changes in other regions, could impact all life on Earth.

FIGURE 1-6

Tropical rain forests, such as this one in the Amazon River basin in Ecuador, support an extraordinary variety and number of plants and animals, which are all on top of a very thin layer of fertile topsoil.





(a)



(b)

FIGURE 1-7

(a) This short-eared arctic hare, *Lepus arcticus*, is hidden from predators and protected from frostbite in a snowy environment. (b) The mottled brown coats of desert rabbits blend in with the dirt and dry grasses, and their long ears help them radiate excess heat and thus avoid overheating.

EVOLUTION OF LIFE

Individual organisms change during their lifetime, but their basic genetic characteristics do not change. However, populations of living organisms do change through time, or evolve. **Evolution**, or descent with modification, is the process in which the inherited characteristics within populations change over generations, such that genetically distinct populations and new species can develop.

Evolution as a theme in biology helps us understand how the various branches of the “tree of life” came into existence and have changed over time. It also explains how organisms alive today are related to those that lived in the past. Finally, it helps us understand the mechanisms that underlie the way organisms look and behave.

Natural Selection

The ability of populations of organisms to change over time is important for survival in a changing world. According to the theory of evolution by **natural selection**, organisms that have certain favorable traits are better able to survive and reproduce successfully than organisms that lack these traits.

One product of natural selection is the adaptation of organisms to their environment. **Adaptations** are traits that improve an individual’s ability to survive and reproduce. For example, rabbits with white fur and short ears in a snowy place, such as the one in Figure 1-7a, may avoid predators and frostbitten ears more often than those with dark fur and long ears. Thus, the next generation of rabbits will have a greater percentage of animals carrying the genes for white fur and short ears. In contrast, the brown, long-eared rabbit, as shown in Figure 1-7b, would survive and reproduce more successfully in a hot desert environment.

The survival and reproductive success of organisms with favorable traits cause a change in populations of organisms over generations. This descent with modification is an important factor in explaining the diversity of organisms we see on Earth today.

SECTION 2 REVIEW

1. Name three unifying themes found in biology.
2. How is the unity and diversity in the living world represented?
3. Identify the three domains and the kingdoms found in each domain.
4. How are organisms interdependent?
5. Describe why evolution is important in explaining the diversity of life.
6. Distinguish between evolution and natural selection.

CRITICAL THINKING

7. **Applying Information** Assign the various toppings you put on pizza to the appropriate domains and kingdoms of life.
8. **Analyzing Graphics** According to the “tree” in Figure 1-5, which of these pairs are more closely related: Archaea:Bacteria or Archaea:Eukarya?
9. **Making Hypotheses** Fossil evidence shows that bats descended from shrewlike organisms that could not fly. Write a hypothesis for how natural selection might have led to flying bats.

THE STUDY OF BIOLOGY

Curiosity leads us to ask questions about life. Science provides a way of answering such questions about the natural world. Science is a systematic method that involves forming and testing hypotheses. More importantly, science relies on evidence, not beliefs, for drawing conclusions.

SCIENCE AS A PROCESS

Science is characterized by an organized approach, called the **scientific method**, to learn how the natural world works. The methods of science are based on two important principles. The first principle is that events in the natural world have natural causes. For example, the ancient Greeks believed that lightning and thunder occurred because a supernatural god Zeus hurled thunderbolts from the heavens. By contrast, a scientist considers lightning and thunder to result from electric charges in the atmosphere. When trying to solve a puzzle from nature, all scientists, such as the one in Figure 1-8, accept that there is a natural cause to solve that puzzle.

A second principle of science is uniformity. *Uniformity* is the idea that the fundamental laws of nature operate the same way at all places and at all times. For example, scientists assume that the law of gravity works the same way on Mars as it does on Earth.

Steps of the Scientific Method

Although there is no single method for doing science, scientific studies involve a series of common steps.

1. The process of science begins with an observation. An **observation** is the act of perceiving a natural occurrence that causes someone to pose a question.
2. One tries to answer the question by forming hypotheses (singular, *hypothesis*). A **hypothesis** is a proposed explanation for the way a particular aspect of the natural world functions.
3. A **prediction** is a statement that forecasts what would happen in a test situation if the hypothesis were true. A prediction is recorded for each hypothesis.
4. An **experiment** is used to test a hypothesis and its predictions.
5. Once the experiment has been concluded, the data are analyzed and used to draw conclusions.
6. After the data have been analyzed, the data and conclusions are communicated to scientific peers and to the public. This way others can verify, reject, or modify the researcher's conclusions.

OBJECTIVES

- **Outline** the main steps in the scientific method.
- **Summarize** how observations are used to form hypotheses.
- **List** the elements of a controlled experiment.
- **Describe** how scientists use data to draw conclusions.
- **Compare** a scientific hypothesis and a scientific theory.
- **State** how communication in science helps prevent dishonesty and bias.

VOCABULARY

scientific method
observation
hypothesis
prediction
experiment
control group
experimental group
independent variable
dependent variable
theory
peer review



FIGURE 1-8

All researchers, such as the one releasing an owl above, use the scientific method to answer the questions they have about nature.

OBSERVING AND ASKING QUESTIONS

The scientific method generally begins with an unexplained observation about nature. For example, people have noticed for thousands of years that owls can catch prey in near total darkness. As shown in steps ① and ② of Figure 1-9, an observation may then raise questions. The owl observation raises the question: How does an owl detect prey in the dark?

FORMING A HYPOTHESIS

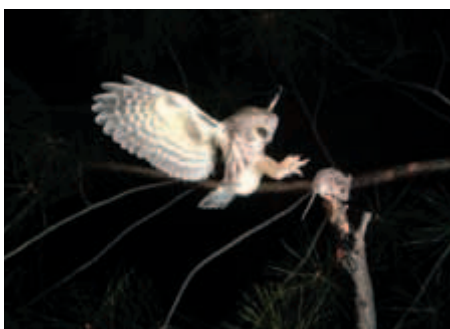
After stating a question, a biologist lists possible answers to a scientific question—hypotheses. Good hypotheses answer a question and are testable in the natural world. For example, as shown in step ③ Figure 1-9, there are several possible hypotheses for the question of how owls hunt at night: (a) owls hunt by keen vision in the dark; (b) owls hunt by superb hearing; or (c) owls hunt by detecting the prey's body heat.

Predicting

To test a hypothesis, scientists make a prediction that logically follows from the hypothesis. A prediction is what is expected to happen if each hypothesis were true. For example, if hypothesis (a) is true, (owls hunt by keen night vision) then one can predict that the owl will pounce only on the mouse in either a light or a dark room. If hypothesis (b) is true (owls hunt by hearing), then one can predict that in a lighted room, the owl will pounce closer to the mouse's head. But, in a dark room, the owl should pounce closer to a rustling leaf attached to the mouse. Finally, if hypothesis (c) is true (owls hunt by sensing body heat), then an owl would strike only the prey no matter the room conditions, because owls hunt by detecting the prey's body heat.

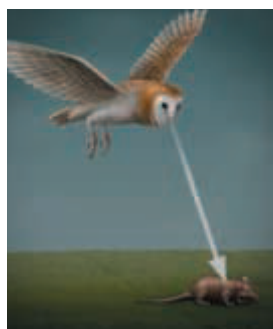
FIGURE 1-9

A scientific study includes observations, questions, hypotheses, predictions, experiments, data analysis, and conclusions. A biologist can use the scientific method to set up an experiment to learn how an owl captures prey at night.

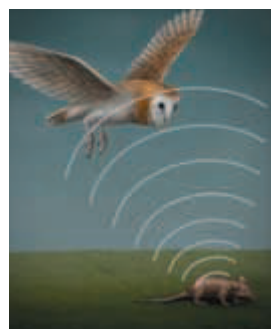


1 OBSERVATION
Owls capture prey on dark nights.

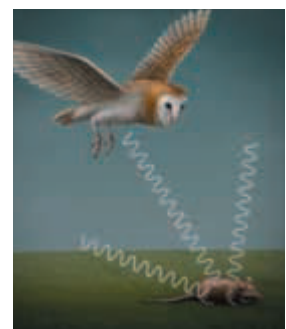
2 QUESTION
How do owls detect prey on dark nights?



a) Owls hunt in the dark by vision.



b) Owls hunt in the dark by hearing.



c) Owls hunt in the dark by sensing body heat.

Notice that these predictions make it difficult to distinguish between the vision and body heat hypotheses. The reason is that both hypotheses predict that the owl could grab the mouse in a dark room. Also, these three hypotheses do not eliminate all other factors that could influence how the owl finds its prey. However, testing predictions can allow one to begin rejecting hypotheses and thus to get closer to determining the answer(s) to a question.

DESIGNING AN EXPERIMENT

Biologists often test hypotheses by setting up an experiment. Step 4 in Figure 1-9 outlines an experiment to test the hypotheses about how an owl hunts at night. First, experimenters set up a room with an owl perch high on one side and a small trap door on the other side for releasing mice. Then, they tied a leaf to each mouse's tail with a string and released each mouse into the room. Next, each mouse ran silently across the room, but the leaf trailed behind, making a rustling noise. During half of the trials, the lights were on. During the other half, the room was dark. Technicians videotaped all the action in the chamber with an infrared light, which owls cannot see. The researchers then viewed the videos and measured the position of the owl's strike relative to each mouse's head.

Performing the Experiment

Many scientists use a controlled experiment to test their hypotheses. A *controlled experiment* compares an experimental group and a control group and only has one variable. The **control group** provides a normal standard against which the biologist can compare results of the experimental group. The **experimental group** is identical to the control group except for one factor, the **independent variable**. The experimenter manipulates the independent variable, sometimes called the *manipulated variable*.



Quick Lab

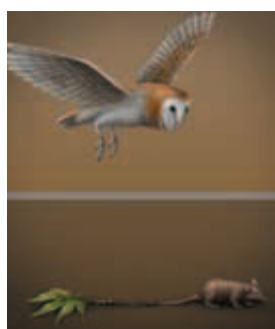
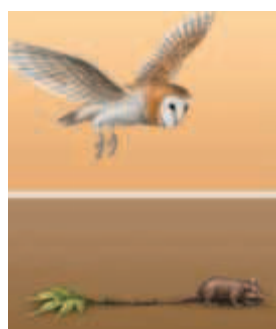
Predicting Results

Materials 2 Petri dishes with agar, cellophane tape, wax pen

Procedure

1. Open one of the Petri dishes, and streak your finger across the surface of the agar.
2. Replace the lid, and seal it with the tape. Label this Petri dish with your name and a number 1.
3. Seal the second Petri dish without removing the lid. Label this Petri dish with your name and the number 2.
4. Write a prediction about what will happen in each dish. Store your dishes as your teacher directs. Record your observations.

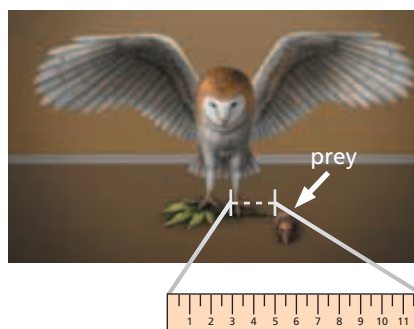
Analysis Was your prediction accurate? What evidence can you cite to support your prediction? If you did not obtain the results you predicted, would you change your testing method or your prediction? Explain. Evaluate the importance of obtaining a result that does not support your prediction.



4 EXPERIMENT

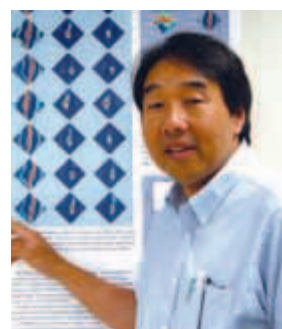
Test predictions of the three hypotheses.

Control: In the light **Experimental:** In the dark



5 DATA COLLECTION AND ANALYSIS

Measure and compare the distance from the owl's strike to the mouse and to the leaf in light and dark.



6 CONCLUSION

Data supported the hearing hypothesis: Owls hunt in the dark by hearing.

The independent variable in the owl experiment is the presence or absence of light. In the owl experiment, the control group hunts in the light, and the experimental group hunts in the dark. In addition to varying the independent variable, a scientist observes or measures another factor called the **dependent variable**, or *responding variable*, because it is affected by the independent variable. In the owl experiment, the dependent variable is distance from the owl's strike to the mouse's head.

Testing the Experiment

Some controlled experiments are conducted “blind.” In other words, the biologist who scores the results is unaware of whether a given subject is part of the experimental or control group. This factor helps eliminate experimenter bias. Experiments should also be repeated, because living systems are variable. Moreover, scientists must collect enough data to find meaningful results.

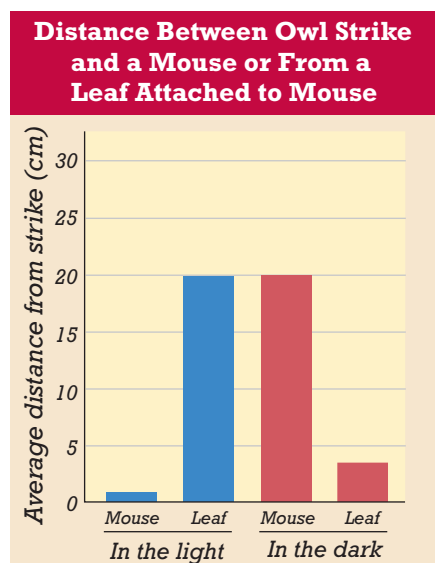
COLLECTING AND ANALYZING DATA

Most experiments measure a variable—the dependent variable. This measurement provides *quantitative data*, data measured in numbers. For example, in the experiment above, scientists measured the distance of an owl's strike from the prey's head in centimeters, as shown in step 5 of Figure 1-9. An event's duration in milliseconds is also an example of quantitative data.

Biologists usually score the results of an experiment by using one of their senses. They might see or hear the results of an experiment. Scientists also extend their senses with a microscope for tiny objects or a microphone for soft sounds. In the owl experiment, biologists extended their vision with infrared cameras.

FIGURE 1-10

The data below are hypothetical results that might occur from the described owl experiment. The independent variable is the darkness of the room, and the dependent variable is how far the owl struck from the mouse's head. The data show that the owl strikes more accurately at the mouse in the light but strikes more accurately at the leaf in the dark.



Analyzing and Comparing Data

After collecting data from a field study or an experiment and then organizing it, biologists then analyze the data. In analyzing data, the goal is to determine whether the data are reliable, and whether they support or fail to support the predictions of the hypothesis. To do so, scientists may use statistics to help determine relationships between the variables involved.

They can then compare their data with other data that were obtained in other similar studies. It is also important at this time to determine possible sources of error in the experiment just performed. Scientists usually display their data in tables or graphs when analyzing it. For the owl study, biologists could have made a bar graph such as the one in Figure 1-10, which shows the average distance from the owl's strike relative to the mouse's head or the leaf in the light and in the dark.

DRAWING CONCLUSIONS

Biologists analyze their tables, graphs, and charts to draw conclusions about whether or not a hypothesis is supported, as shown in step 6 of Figure 1-9. The hypothetical owl data show that in the light, owls struck with greater accuracy at the mouse than at the leaf, but in the dark, owls struck with greater accuracy at the leaf than the mouse. Thus, the findings support the hearing hypothesis, but not the vision hypothesis.

An experiment can only disprove, not prove, a hypothesis. For example, one cannot conclude from the results that the hearing hypothesis is *proven* to be true. Perhaps the owl uses an unknown smell to strike at the mouse. One can only reject the vision hypothesis because it did not predict the results of the experiment correctly.

Acceptance of a hypothesis is always tentative in science. The scientific community revises its understanding of phenomena, based on new data. Having ruled out one hypothesis, a biologist will devise more tests to try to rule out any remaining hypotheses.

Making Inferences

Scientists often draw inferences from data gathered during a field study or experiment. An *inference* (IN-fuhr-uhns) is a conclusion made on the basis of facts and previous knowledge rather than on direct observations. Unlike a hypothesis, an inference is not directly testable. In the owl study, it is inferred that the owl detects prey from a distance rather than by direct touch.

Applying Results and Building Models

As shown in Figure 1-11, scientists often apply their findings to solve practical problems. They also build models to represent or describe things. For example in 1953, James Watson and Francis Crick used cardboard balls and wire bars to build physical models of atoms in an attempt to understand the structure of DNA. Mathematical models are sets of equations that describe how different measurable items interact in a system. The experimenter can adjust variables to better model the real-world data.



FIGURE 1-11

Biologists often apply their knowledge of the natural world to practical problems. Studies on the owl's keen ability to locate sounds in space despite background noise are helping biotechnologists and bioengineers develop better solutions for people with impaired hearing, such as the people shown in this picture.

CONSTRUCTING A THEORY

When a set of related hypotheses is confirmed to be true many times, and it can explain a great amount of data, scientists often reclassify it as a **theory**. Some examples include the quantum theory, the cell theory, or the theory of evolution. People commonly use the word “theory” in a different way than scientists use the word. People may say “It’s just a theory” suggesting that an idea is untested, but scientists view a theory as a highly tested, generally accepted principle that explains a vast number of observations and experimental data.

COMMUNICATING IDEAS



An essential aspect of scientific research is scientists working together. Scientists often work together in research teams or simply share research results with other scientists. This is done by publishing findings in scientific journals or presenting them at scientific meetings, as shown in Figure 1-12. Sharing information allows others working independently to verify findings or to continue work on established results. For example, Roger Payne published the results of his owl experiments in a journal in 1971. Then, other biologists could repeat it for verification or use it to study the mechanisms introduced by the paper. With the growing importance of science in solving societal issues, it is becoming increasingly vital for scientists to be able to communicate with the public at large.

Publishing a Paper

Scientists submit research papers to scientific journals for publication. A typical research paper has four sections. First, the *Introduction* poses the problem and hypotheses to be investigated. Next, the *Materials and Methods* describe how researchers proceeded with the experiment. Third, the *Results* state the findings the experiment presented, and finally, the *Discussion* gives the significance of the experiment and future directions the scientists will take.

Careers in BIOLOGY

Forensic Biologist

Job Description Forensic biologists are scientists who study biological materials to investigate potential crimes and other legal issues against humans and animals. Forensic scientists have knowledge in areas of biology, such as DNA and blood pattern analysis, and work in private sector and public laboratories.

Focus On a Forensic Biologist

As a law enforcement forensic specialist for the Texas Parks and Wildlife Department, Beverly Villarreal assists the game warden in investigations of fish and wildlife violations, such as illegal hunting and fishing. Villarreal analyzes

blood and tissue samples to identify species of animals such as fish, birds, and reptiles. Her work helps game wardens as they enforce state laws regarding hunting and fishing. Most people think of forensic scientists as the glamorous crime investigators on TV, but according to Villarreal real forensic scientists "spend a great deal of time at a lab bench running analysis after analysis." Many of the methods used in animal forensics, such as DNA sequencing, are also used in human forensics.

Education and Skills

- **High school**—three years of science courses and four years of math courses.



- **College**—bachelor of science in biology, including course work in zoology and genetics, plus experience in performing DNA analyses.
- **Skills**—patience, attention to detail, and ability to use fine tools.



For more about careers, visit go.hrw.com and type in the keyword **HM6 Careers**.

After scientists submit their papers to a scientific journal, the editors of that journal will send the paper out for peer review. In a **peer review**, scientists who are experts in the field anonymously read and critique that research paper. They determine if a paper provides enough information so that the experiment can be duplicated and if the author used good experimental controls and reached an accurate conclusion. They also check if the paper is written clearly enough for broad understanding. Careful analysis of each other's research by fellow scientists is essential to making scientific progress and preventing scientific dishonesty.



FIGURE 1-12

Scientists present their experiments in various forms. The scientists above are presenting their work in the form of a poster at a scientific meeting.

HONESTY AND BIAS

The scientific community depends on both honesty and good science. While designing new studies, experimenters must be very careful to prevent previous ideas and biases from tainting both the experimental process and the conclusions. Scientists have to keep in mind that they are always trying to disprove their favorite ideas. Scientists repeat experiments to verify previous findings. This allows for science to have a method for self-correction and it also keeps researchers honest and credible to their peers in the field.

Conflict of Interest

For most scientists, maintaining a good reputation for collecting and presenting valid data is more important than temporary prestige or income. So, scientists try to avoid any potential conflicts of interest. For example, a scientist who owns a biotechnology company and manufactures a drug would not be the best researcher to critically test that drug's safety and effectiveness. To avoid this potential conflict of interest, the scientist allows an unaffected party, such as a research group, to test the drug's effectiveness. The threat of a potential scandal based on misleading data or conclusions is a powerful force in science that helps keep scientists honest and fair.

SECTION 3 REVIEW

1. What two principles make the scientific method a unique process?
2. Define the roles of observations and hypotheses in science.
3. Summarize the parts of a controlled experiment.
4. Summarize how we make conclusions about the results of an experiment.
5. Why is the phrase, "it's just a theory" misleading?
6. Give another example of a conflict of interest.

CRITICAL THINKING

7. **Making Hypotheses** On a nocturnal owl's skull, one ear points up, and the other ear points down. Suggest a hypothesis for this observation.
8. **Designing Experiments** Design an experiment to establish if owls hunt by keen sight or hunt by heat seeking.
9. **Calculating Information** What was the average distance between the owl's strike and the mouse if the recorded differences in this experiment were 25, 22, 19, 19, and 15?

SCIENCE ON THE INTERNET: A New Information Age

In the past, students researching a science topic would typically begin their research by visiting a library to use printed reference materials, such as encyclopedias. Today, most students research topics by using a computer and searching for information on the Internet.

The Internet can provide students with a wealth of information. But which Web sites have accurate information, and which Web sites do not?

Checking Web Addresses

Students should use the Web address, or *URL*, to establish the Web site's credibility. Usually, the domain name can suggest who has published the Web site. Web sites can be published by governmental agencies (ends in "dot gov" or .gov), by educational institutions (ends in "dot edu" or .edu), by organizations (ends in "dot

org" or .org), or by commercial businesses (ends in "dot com" or .com).

Government Web sites are usually reliable. Examples of credible governmental Web sites are the National Institutes of Health (NIH) and the Food and Drug Administration (FDA). University and medical school sites are also reliable sources of information. Many organizations that research and teach the public about specific diseases and conditions can also provide reliable information. Examples of such organizations are the American Cancer Society and the American Heart Association.

Evaluating Web Sites

The credibility of the author of the Web site should also be checked. Make sure the author is not trying to sell anything and is established in his or her field. For example, a health Web

site's author should be a medical professional.

It is also important to check the date that the information was posted on the Web to ensure that the information is current. Also, the Web site should provide references from valid sources, such as scientific journals or government publications.

Finally, the student should always double-check information between several reliable Web sites. If two or three reliable sites provide the same information, the student can feel confident in using that information.

Web Sites for Students

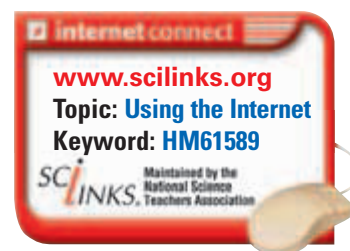
The Internet Connect boxes in this textbook have all been reviewed by professionals at the National Science Teachers Association (NSTA). Students can trust that these sites are reliable sources for science- or health-related topics.



The Internet can provide a wealth of scientific information for a report, but the information may not always be credible or accurate. You can use the methods above to check the accuracy and credibility of your sources.

REVIEW

1. Which types of Web addresses are the most reliable?
2. List four important features to evaluate when using a Web site for research.
3. **Supporting Reasoned Opinions**
Why do you think a Web site that is advertising a product may not offer accurate information?



TOOLS AND TECHNIQUES

With proper equipment and good methods, biologists can see, manipulate, and understand the natural world in new ways. Microscopes are one of many useful tools used to unlock nature's biological secrets.

MICROSCOPES AS TOOLS

Tools are objects used to improve the performance of a task. *Microscopes* are tools that extend human vision by making enlarged images of objects. Biologists use microscopes to study organisms, cells, cell parts, and molecules. Microscopes reveal details that otherwise might be difficult or impossible to see.

Light Microscopes

To see small organisms and cells, biologists typically use a light microscope, such as the one shown in Figure 1-13. A **compound light microscope** is a microscope that shines light through a specimen and has two lenses to magnify an image. To use this microscope, one first mounts the specimen to be viewed on a glass slide. The specimen must be thin enough for light to pass through it. For tiny pond organisms, such as the single-celled paramecium, light passing through the organism is not a problem. For thick objects, such as plant stems, biologists must cut thin slices for viewing. There are four major parts of a compound light microscope. For further description of the parts of a microscope, see the Appendix.

- 1. Eyepiece** The **eyepiece (ocular (AHK-yoo-luhr) lens)** magnifies the image, usually 10 times.
- 2. Objective Lens** Light passes through the specimen and then through the objective lens, which is located directly above the specimen. The **objective lens** enlarges the image of the specimen. Scientists sometimes use stains to make the image easier to see.
- 3. Stage** The **stage** is a platform that supports a slide holding the specimen. The slide is placed over the opening in the stage of the microscope.
- 4. Light Source** The **light source** is a light bulb that provides light for viewing the image. It can be either light reflected with a mirror or an incandescent light from a small lamp.

OBJECTIVES

- **List** the function of each of the major parts of a compound light microscope.
- **Compare** two kinds of electron microscopes.
- **Describe** the importance of having the SI system of measurement.
- **State** some examples of good laboratory practice.

VOCABULARY

compound light microscope
 eyepiece (ocular lens)
 objective lens
 stage
 light source
 magnification
 nosepiece
 resolution
 scanning electron microscope
 transmission electron microscope
 metric system
 base unit

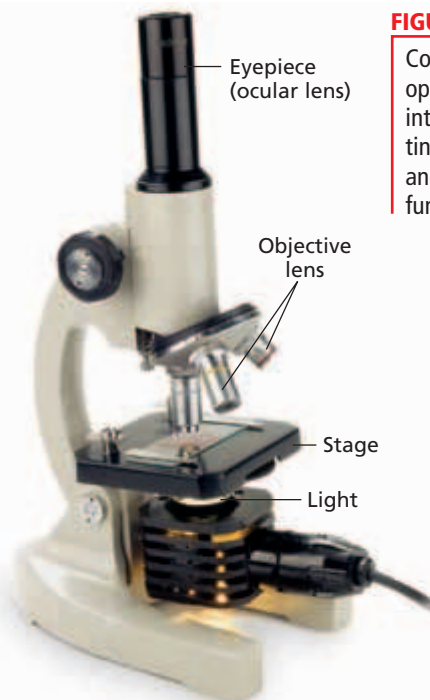
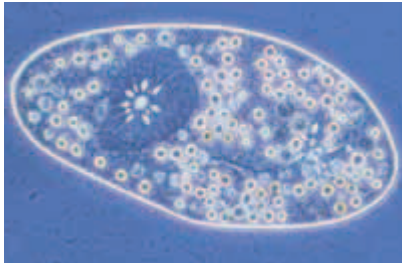
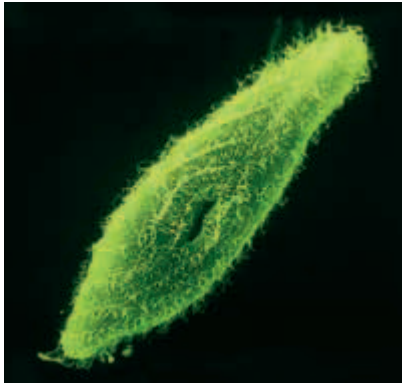


FIGURE 1-13

Compound light microscopes open the human eye to an interesting world including tiny pond organisms, healthy and diseased cells, and the functioning of cell parts.



(a) *Paramecium* (light microscope)



(b) *Paramecium* (scanning electron microscope)



(c) *Paramecium* (transmission electron microscope)

FIGURE 1-14

The images above show a *Paramecium* as viewed under three different types of microscopes. (a) Light microscopes can produce an image that is up to 1,000 times larger than the actual specimen. (b) Scanning electron microscopes produce images up to 100,000 times larger than the specimen. SEMs provide a view of surface features. (c) Transmission electron microscopes produce images up to 200,000 times larger than the actual specimen.

Magnification and Resolution

Microscopes vary in powers of magnification and resolution. **Magnification** is the increase of an object's apparent size. Revolving the **nosepiece**, the structure that holds the set of objective lens, rotates these lenses into place above the specimen. In a typical compound light microscope, the most powerful objective lens produces an image up to 100 times ($100\times$) the specimen's actual size. The degree of enlargement is called the *power of magnification* of the lens. The standard ocular lens magnifies a specimen 10 times ($10\times$). To compute the power of magnification of a microscope, the power of magnification of the strongest objective lens (in this case, $100\times$) is multiplied by the power of magnification of the ocular lens ($10\times$). The result is a total power of magnification of $1000\times$.

Resolution (REZ-uh-LOO-shuhn) is the power to show details clearly in an image. The physical properties of light limit the ability of light microscopes to resolve images, as shown in Figure 1-14a. At powers of magnification beyond about $2,000\times$, the image of the specimen becomes fuzzy. For this reason, scientists use other microscopes to view very small cells and cell parts.

Electron Microscopes

To examine cells in more detail or to view cell parts or viruses, scientists can use other microscopes, such as an electron microscope. In an *electron microscope*, a beam of electrons produces an enlarged image of the specimen. Electron microscopes are more powerful in magnification and resolution than light microscopes. Some electron microscopes can even show the contours of individual atoms in a specimen. Electron microscope images are always in black and white. However, scientists can use computers to add artificial colors to help identify structures in the image. Also, the specimen must be placed in a vacuum chamber. Because cells cannot survive in a vacuum, electron microscopes cannot be used to view living specimens.

There are two main types of electron microscopes. The first type of electron microscope is the **scanning electron microscope (SEM)**. The SEM passes a beam of electrons over the specimen's surface. SEMs provide three-dimensional images of the specimen's surface, as shown in Figure 1-14b. First, the specimen is sprayed with a fine metal coating. Then, a beam of electrons is aimed at the specimen, which causes the metal coating to emit a shower of electrons. These electrons project onto a fluorescent screen or photographic plate. The result is an image of the object's surface. SEMs can magnify objects up to 100,000 times.

The second type is the **transmission electron microscope (TEM)**. The TEM transmits a beam of electrons through a very thinly sliced specimen. Magnetic lenses enlarge the image and focus it on a screen or photographic plate. The result is an image such as the one shown in Figure 1-14c. Note the great resolution of the paramecium's internal structure. Transmission electron microscopes can magnify objects up to 200,000 times.

TABLE 1-1 SI Base Units

Base quantity	Name	Abbreviation
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Amount of substance	mole	mol
Luminous (light) intensity	candela	cd

TABLE 1-2 Some SI prefixes

Prefix	Abbreviation	Factor of base unit
giga	G	1,000,000,000 (10^9)
mega	M	1,000,000 (10^6)
kilo	k	1,000 (10^3)
hecto	h	100 (10^2)
deka	da	10 (10^1)
base unit		1
deci	d	0.1 (10^{-1})
centi	c	0.01 (10^{-2})
milli	m	0.001 (10^{-3})
micro	μ	0.000001 (10^{-6})
nano	n	0.000000001 (10^{-9})
pico	p	0.000000000001 (10^{-12})

UNITS OF MEASUREMENT

Scientists use a common measurement system so that they can compare their results. Scientists use a single, standard system of measurement, called the **metric system**. The metric system is a decimal system and thus is based on powers of 10. The official name of this measurement system is *Système International d'Unités*. The English translation of this French title is the International System of Units, or simply SI. Biology students use SI while making measurements in the laboratory.

Base and Other Units

The SI has seven fundamental **base units** that describe length, mass, time, and other quantities, as shown in Table 1-1. Multiples of a base unit (in powers of 10) are designated by prefixes, as shown in Table 1-2. For example, the base unit for length is the meter. One kilometer (km) is equal in length to 1,000 meters (m).

Although the base units in Table 1-1 are extremely useful, they can't be applied to certain measurements, such as surface area or velocity. Scientists use other important units called derived units for these types of quantities. *Derived units* are produced by the mathematical relationship between two base units or between two derived units. Table 1-3 shows some common derived units. There are some additional units of measurement that are not part of the SI but that can be used with SI units such as units of time, volume, and mass, as shown in Table 1-3.

TABLE 1-3 Some Derived and Other Units

Quantity	Name	Abbreviation
Area	square meter	m ²
Volume	cubic meter	m ³
Density	kilogram per cubic meter	kg/m ³
Specific volume	cubic meter per kilogram	m ³ /kg
Celsius temperature	degree Celsius	°C
Time	minute	1 min = 60 s
Time	hour	1 h = 60 min
Time	day	1 d = 24 h
Volume	liter	1 L = 1,000 cm ³
Mass	kilogram	1,000 g = 1 kg
	metric ton	1 t = 1,000 kg



FIGURE 1-15

Good laboratory practice involves protecting yourself and others by being safe.

SAFETY

Studying living things is interesting, fun, and rewarding, but it can be hazardous. The hazards can be chemical, physical, radiological, or biological and can vary between the the lab and the field. For example, getting splashed in the eye with a blinding chemical is more likely to occur in the laboratory, but falling down a cliff or getting bitten by a poisonous spider is more likely to occur in the field.

Good Laboratory Practice

Lab safety involves good laboratory practice, which means establishing safe, common-sense habits, as shown in Figure 1-15. Never work alone in the lab or without proper supervision by the teacher, and always ask your teacher before using any equipment. The diagram below shows the safety symbols used in this book. More information on lab safety and the safety symbols can be found in the Appendix.



Eye Safety



Hand Safety



Safety with Gases



Sharp-Object Safety



Clothing Protection



Animal Care and Safety



Heating Safety



Hygienic Care



Glassware Safety



Proper Waste Disposal



Electrical Safety



Plant Safety



Chemical Safety

SECTION 4 REVIEW

1. List the four major parts of a compound light microscope.
2. What is the difference between the magnification and resolution of an image under a microscope?
3. Compare the function of a transmission electron microscope with that of a scanning electron microscope.
4. What is the importance of scientists using a common SI system of measurement?
5. How would you convert kilometers to millimeters?
6. Name the safety symbols used in this textbook.

CRITICAL THINKING

7. **Applying Information** A biologist thinks a virus, which is much smaller than a cell, is likely to cause a disease. Which type of microscope is most likely to be used to view the internal structure of a virus?
8. **Calculating Information** How would you convert cubic meters to cubic centimeters?
9. **Calculating Information** On a light microscope, an objective lens magnifies the view of some pond water 25 times, and the ocular lens magnifies it 10 times further. What is the final magnification of the image?

CHAPTER HIGHLIGHTS

SECTION 1

The World of Biology

- Biology is the study of life and can be used to both solve societal problems and explain aspects of our daily lives.
- Living things share the same 7 characteristics: organization and cells, response to stimuli, homeostasis, metabolism, growth and development, reproduction, and evolution.
- Multicellular organisms show a hierarchy of organization going from the organism to the atom.
- To stay alive, living things must maintain homeostasis, obtain and use energy, and pass on hereditary information from parents to offspring, also called reproduction.

Vocabulary

biology (p. 5)
organization (p. 6)
cell (p. 7)
unicellular (p. 7)

multicellular (p. 7)
organ (p. 7)
tissue (p. 7)
organelle (p. 7)

biological molecule (p. 7)
homeostasis (p. 8)
metabolism (p. 8)
cell division (p. 8)

development (p. 8)
reproduction (p. 9)
gene (p. 9)

SECTION 2

Themes in Biology

- Three themes in biology are the unity of life's diversity, the interdependence of organisms, and evolution of life.
- Living organisms show diversity and can be classified into domains and kingdoms.
- Organisms live in interdependent communities and interact with both organisms and the environment.
- Evolution helps to explain how species came to exist, have changed over time, and adapt to their environment.

Vocabulary

domain (p. 11)
kingdom (p. 11)

ecology (p. 11)
ecosystem (p. 11)

evolution (p. 12)
natural selection (p. 12)

adaptation (p. 12)

SECTION 3

The Study of Biology

- The scientific method involves making observations, asking questions, forming hypotheses, designing experiments, analyzing data, and drawing conclusions.
- Trying to answer questions about observations helps scientists form hypotheses.
- A controlled experiment has a control and experimental group, and tests independent and dependent variables.
- Scientists analyze data to draw conclusions about the experiment performed.
- A theory is a set of related hypotheses confirmed to be true many times.
- Communication between scientists about their methods and results helps prevent dishonesty and bias in science.

Vocabulary

scientific method (p. 13)
observation (p. 13)
hypothesis (p. 13)

prediction (p. 13)
experiment (p. 13)
control group (p. 15)

experimental group (p. 15)
independent variable (p. 15)

dependent variable (p. 16)
theory (p. 17)
peer review (p. 19)

SECTION 4

Tools and Techniques

- Four major parts of a compound light microscope are the ocular lens, objective lens, stage, and light source.
- Transmission and scanning electron microscopes provide greater magnification than light microscopes.
- Scientists use the metric system to take scientific measurements.
- Lab safety is a good laboratory practice.

Vocabulary

compound light microscope (p. 21)
eyepiece (ocular lens) (p. 21)

objective lens (p. 21)
stage (p. 21)
light source (p. 21)
magnification (p. 22)

nosepiece (p. 22)
resolution (p. 22)
scanning electron microscope (SEM) (p. 22)

transmission electron microscope (TEM) (p. 22)
metric system (p. 23)
base unit (p. 23)

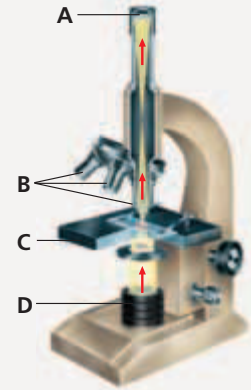
CHAPTER REVIEW

USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
 - unicellular* and *multicellular*
 - homeostasis* and *metabolism*
 - natural selection* and *adaptation*
 - hypothesis* and *theory*
 - magnification* and *resolution*
- Explain the relationship between an independent variable and a dependent variable.
- Use the following terms in the same sentence: *observation*, *hypothesis*, *prediction*, and *experiment*.
- Word Roots and Origins** The word *magnification* is derived from the Latin *magnificus* or *magnus*, which means “large” or “great.” Using this information, explain why the term *magnification* is a good name for the function it describes.

UNDERSTANDING KEY CONCEPTS

- Describe** why learning about biology is relevant to a person’s life.
- Describe** one way in which biology affects our society.
- Summarize** the characteristics of living things.
- List** the hierarchy of organization in a snowy owl.
- Explain** how homeostasis and metabolism are interrelated.
- Compare** the processes of growth, development, and reproduction.
- State** three major themes found in biology.
- Identify** how the “tree of life” can help explain both the unity and diversity of life.
- Describe** the interdependence of living organisms.
- Summarize** how evolution helps explain the diversity of life.
- Sequence** the main steps of the scientific method.
- Explain** how observations are used to form hypotheses.
- Summarize** how biologists set up controlled experiments.
- State** the purpose of analyzing data that are collected during an experiment.
- Summarize** how a hypothesis becomes a theory.
- Describe** two types of scientific models.
- Identify** how a peer review keeps scientists honest.
- Name** the part of the compound light microscope denoted by each letter in the figure below.
- Differentiate** between the scanning electron microscope and the transmission electron microscope.
- Describe** the relationship between a kilometer, meter, and micrometer.
- Explain** why scientists throughout the world use the SI system.
- List** three safety symbols used in this textbook.
- CONCEPT MAPPING** Use the following terms to create a concept map that outlines the steps of the scientific method: *observations*, *experiments*, *conclusions*, *questions*, *hypotheses*, *data analyses*, *predictions*, *theories*, and *communication*.



CRITICAL THINKING

- Forming Hypotheses** Go to a window or outside, and observe a bird’s behavior for a few minutes. Record your observations, and write down one question about bird behavior and one hypothesis that answers the question.
- Analyzing Concepts** One of the most important parts of any scientific publication is the part called Methods and Materials, in which the scientist describes the procedure used in the experiment. Why do you think such details are so important?
- Making Calculations** Determine the number of liters that are in 150 kiloliters.
- Making Comparisons** Look at the photographs below. The TEM (left) is a photo of a paramecium. The SEM (right) is also a photo of a paramecium. Compare and contrast what each electron micrograph reveals to you about this organism.



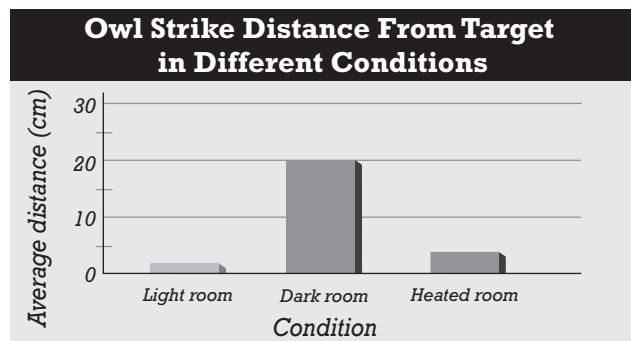


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- Which of the following does evolution help explain?
 - how organisms reproduce
 - how organisms grow and develop
 - how organisms are related to each other
 - how organisms obtain and metabolize energy
- Which of the following is the hereditary material in most living things?
 - DNA
 - lipids
 - oxygen
 - carbon dioxide
- Which of the following does the hierarchy of organization within an organism describe?
 - metabolism
 - homeostasis
 - internal structures
 - relationship to the physical environment
- To which of the following does the resolution of a microscope refer?
 - its ability to show detail clearly
 - its power to scan the surface of an object
 - its series of interchangeable objective lenses
 - its power to increase an object's apparent size

INTERPRETING GRAPHICS: The graph below shows the distance it takes an owl to strike a mouse under different conditions. Use the graph to answer the question that follows.



- An owl strikes a mouse more closely and on target in which of the following rooms?
 - dark room
 - light room
 - heated room
 - dark and lighted rooms

DIRECTIONS: Complete the following analogy.

- compound light microscope : light :: TEM:
 - tissues
 - electrons
 - organelles
 - organ systems

INTERPRETING GRAPHICS: The figure below shows a newspaper clipping. Use the figure to answer the question that follows.



- Which of the following terms most accurately reflects the use of the term theory in the newspaper headline above?
 - law
 - fact
 - hypothesis
 - experiment

SHORT RESPONSE

Dolly was cloned from mammary cells from an adult female sheep. She was an exact genetic copy of her mother.

Explain whether Dolly represents a product of sexual reproduction or asexual reproduction.

EXTENDED RESPONSE

Life is so diverse, yet it is characterized by a unity. The tree of life can relate life's unity and diversity.

Part A Describe the relationship between animals, plants, fungi, protists, bacteria, and archaea in the "tree of life."

Part B Explain how the "tree of life" represents and relates both the unity and diversity of life.

Test TIP

When faced with similar answers, define the answer choices and then use that definition to narrow down the choices on a multiple-choice question.

Using SI Units

OBJECTIVES

- Express measurements in SI units.
- Read a thermometer.
- Measure liquid volume using a graduated cylinder.
- Measure mass using a balance.
- Determine the density (mass-to-volume ratio) of two different liquids.

PROCESS SKILLS

- measuring
- calculating

MATERIALS

- 75 mL light-colored sand
- 75 mL dark-colored sand
- 1 100 mL graduated cylinder
- Celsius thermometers, alcohol filled (2)
- 5 oz plastic cups (4)
- graph paper
- heat-protective gloves
- light source
- stopwatch or clock
- ring stand or lamp support
- 25 mL corn oil
- 25 mL water
- clear-plastic cup
- balance

Background

1. What does the abbreviation *S*/stand for?
2. List the seven SI base units.

PART A Measuring Temperature

1. In your lab report, prepare a data table similar to Table A, above right.
2. Using a graduated cylinder, measure 75 mL of light-colored sand and pour it into one of the small plastic cups. Repeat this procedure with the dark-colored sand and another plastic cup.
3. Level the sand by placing the cup on your desk and sliding the cup back and forth.

TABLE A SAND TEMPERATURE

Time (min)	Temperature (°C)	
	Dark-colored sand	Light-colored sand
Start		
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

4. Insert one thermometer into each cup. The zero line on the thermometer should be level with the sand, as shown in the figure below. Re-level the sand if necessary.



5.  **CAUTION** Wear heat-protective gloves when handling the lamp. The lamp will become very hot and may burn you. Using a ring stand or lamp support, position the lamp approximately 9 cm from the top of the sand, as shown in the figure on p. 28. Make sure the lamp is evenly positioned between the two cups.
6. Before turning on the lamp, record the initial temperature of each cup of sand in your data table.
7. Note the time or start the stopwatch when you turn on the lamp. The lamp will become hot and warm the sand. Check the temperature of the sand in each container at one-minute intervals for 10 minutes. Record the temperature of the sand after each minute in your data table.

PART B Comparing the Density of Oil and Water

8. In your lab report, prepare a data table similar to Table B below.

TABLE B DENSITY OF TWO LIQUIDS

a. Mass of empty oil cup	_____ g
b. Mass of empty water cup	_____ g
c. Mass of cup and oil	_____ g
d. Mass of cup and water	_____ g
e. Volume of oil	25 mL
f. Volume of water	25 mL
Calculating Actual Mass	
Oil Item c – Item a =	_____ g
Water Item d – Item b =	_____ g
g. Density of oil	_____ g/ml
h. Density of water	_____ g/ml

9. Label one clean plastic cup "oil," and label another "water." Using a balance, measure the mass of each plastic cup, and record the value in your data table.
10. Using a clean graduated cylinder, measure 25 mL of corn oil and pour it into the plastic cup labeled "oil." Using a balance, measure the mass of the plastic cup containing the corn oil, and record the mass in your data table.

11. Using a clean graduated cylinder, measure 25 mL of water and pour it into the plastic cup labeled "water." Using a balance, measure the mass of the plastic cup containing the water, and record the mass in your data table.
12. To find the mass of the oil, subtract the mass of the empty cup from the mass of the cup and the oil together.
13. To find the density of the oil, divide the mass of the oil by the volume of the oil, as shown in the equation below:

$$\text{Density of oil} = \frac{\text{mass of oil}}{\text{volume of oil}} = \text{_____ g/mL}$$

14. To find the mass of water, subtract the mass of the empty cup from the mass of the cup and the water together.
15. To find the density of the water, divide the mass of the water by the volume of the water, as shown in the equation below:

$$\text{Density of water} = \frac{\text{mass of water}}{\text{volume of water}} = \text{_____ g/mL}$$

16. Combine the oil and water in the clear cup, and record your observations in your lab report.
17.   Clean up your materials, and wash your hands before leaving the lab.

Analysis and Conclusions

1. Graph the data you collected in Part A. Plot time on the *x*-axis and temperature on the *y*-axis.
2. Based on your data from Part A, what is the relationship between color and heat absorption?
3. How might the color of the clothes you wear affect how warm you are on a sunny 90° day?
4. In Part B, what did you observe when you combined the oil and water in the clear cup? Relate your observation to the densities that you calculated.
5. What could you infer about the value for the density of ice if you observe it to float in water?
6. How would your calculated values for density be affected if you misread the volume measurement on the graduated cylinder?

Further Inquiry

Pumice is a volcanic rock that has a density less than 1.00 g/cm³. How would you prove this if you did not have a balance to weigh the pumice? (Hint: The density of water is 1.00 g/cm³.)

CHEMISTRY OF LIFE



All living things are made of the same basic materials: carbon, hydrogen, oxygen, and nitrogen. Living things, such as this jellyfish, *Pseudorhiza haeckeli*, are made of cells that are composed primarily of water. The chemical reactions of life occur in the aqueous environment of the cell.

SECTION 1 *Composition of Matter*

SECTION 2 *Energy*

SECTION 3 *Water and Solutions*

COMPOSITION OF MATTER

Earth supports an enormous variety of organisms. The structure and function of all living things are governed by the laws of chemistry. An understanding of the basic principles of chemistry will give you a better understanding of living things and how they function.

MATTER

Everything in the universe is made of matter. **Matter** is anything that occupies space and has mass. **Mass** is the quantity of matter an object has. Mass and weight are not the same; *weight* is defined as the force produced by gravity acting on mass. The same mass would have less weight on the moon than it would on Earth because the moon exerts less force on the object than the Earth does.

Chemical changes in matter are essential to all life processes. Biologists study chemistry because all living things are made of the same kinds of matter that make up nonliving things. By learning how changes in matter occur, you will gain an understanding of the life processes of the organisms you will study.

ELEMENTS AND ATOMS

Elements are substances that cannot be broken down chemically into simpler kinds of matter. More than 100 elements have been identified, though fewer than 30 are important to living things. In fact, more than 90 percent of the mass of all kinds of living things is composed of combinations of just four elements: oxygen, carbon, hydrogen, and nitrogen.

Information about the elements is summarized on a chart known as the *periodic table*, which appears in the Appendix. Each element has a different chemical symbol. A chemical symbol consists of one, two, or three letters, as shown in Figure 2-1. In most cases, the symbol derives from the first letter or other letters in the name of the element, such as Cl for chlorine. Most of the other symbols are derived from the Latin names of elements. One example is sodium's symbol, Na, from the Latin word *natrium*.

OBJECTIVES

- Define the term *matter*.
- Explain the relationship between elements and atoms.
- Draw and label a model of the structure of an atom.
- Explain how compounds affect an atom's stability.
- Contrast covalent and ionic bonds.

VOCABULARY

matter
mass
element
atom
nucleus
proton
neutron
atomic number
mass number
electron
orbital
isotope
compound
chemical bond
covalent bond
molecule
ion
ionic bond

FIGURE 2-1

The periodic table lists information about the elements, including the atomic number, the chemical symbol, and the atomic mass for each element. A complete periodic table can be found in the Appendix.

Atomic number	2
Chemical symbol	He
Atomic mass	4
	Helium
9	10
F	Ne
Fluorine	Neon
19.00	20.18

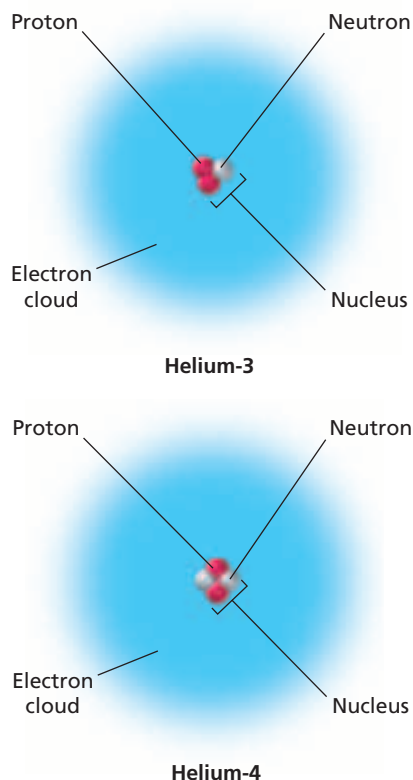


FIGURE 2-2

Most scientists consider the electron cloud model, shown above, to be the most accurate model of an atom. Here the models show the difference between elements and isotopes. Helium-3 has two protons and one neutron, while Helium-4 has two protons and two neutrons.

The simplest particle of an element that retains all of the properties of that element is an **atom**. The properties of different kinds of atoms determine the structure and properties of the matter they compose. Atoms are so small that their structure cannot be directly observed. However, scientists have developed models that describe the structure of the atom. One model is shown in Figure 2-2.

The Nucleus

The central region, or **nucleus**, makes up the bulk of the mass of the atom and consists of two kinds of subatomic particles, a **proton** and a neutron. The proton is positively charged and the neutron has no charge. The number of protons in an atom is called the **atomic number** of the element. In the periodic table of elements, the atomic number generally appears directly above the chemical symbol, as shown in Figure 2-1. The atomic number of fluorine is 9, which indicates that each atom of the element fluorine has nine protons. The **mass number** of an atom is equal to the total number of protons and neutrons of the atom. The mass number of fluorine is 19, which indicates that each atom of fluorine has 10 neutrons.

Electrons

In an atom, the number of positively charged protons is balanced by an equal number of small, negatively charged particles called **electrons**. The net electrical charge of an atom is zero. Electrons are high-energy particles that have very little mass. They move about the nucleus at very high speeds and are located in orbitals. An **orbital** is a three-dimensional region around a nucleus that indicates the probable location of an electron. Electrons in orbitals that are farther away from the nucleus have greater energy than electrons that are in orbitals closer to the nucleus. When all orbitals are combined, there is a cloud of electrons surrounding the nucleus, as shown in Figure 2-2.

Orbitals correspond to specific energy levels. Each energy level corresponds to a group of orbitals that can hold only a certain, total number of electrons. For example, the orbital that corresponds to the first energy level can hold only two electrons. The first energy level is the highest energy level for the elements hydrogen and helium. There are four orbitals in the second energy level, and that energy level can hold up to eight total electrons, with a maximum of two electrons in each orbital.

Isotopes

All atoms of an element have the same number of protons. However, all atoms of an element do not necessarily have the same number of neutrons. Atoms of the same element that have a different number of neutrons are called **isotopes**. Additional neutrons change the mass of the element. Most elements are made up of a mixture of isotopes, as shown in Figure 2-2. The *average atomic mass* of an element takes into account the relative amounts of each isotope in the element, and this average is the mass found in the periodic table.



COMPOUNDS

Under natural conditions, most elements do not exist alone; atoms of most elements can readily combine with the same or different atoms or elements to make compounds. **Compounds** are made up of atoms of two or more elements in fixed proportions. A chemical formula shows the kinds and proportions of atoms of each element that forms a particular compound. For example, water's chemical formula, H_2O , shows that the atoms always combine in a proportion of two hydrogen (H) atoms to one oxygen (O) atom.

The physical and chemical properties differ between the compounds and elements that compose them. In nature, the elements oxygen and hydrogen are usually found as gases with the formulas O_2 and H_2 . However, when oxygen gas and hydrogen gas combine at room temperature, they form liquid H_2O . How elements combine and form compounds depends on the number and arrangement of electrons in their orbitals. An atom is chemically stable when the orbitals that correspond to its highest energy level are filled with the maximum number of electrons. Some elements, such as helium and neon, consist of atoms that have the maximum number of electrons in the orbitals of their highest energy levels. These elements, also called *noble* or *inert elements*, do not react with other elements under normal conditions.

Most atoms are not stable in their natural state, so they tend to react with other atoms in different ways to become more stable. Carbon, nitrogen, and oxygen atoms have unfilled orbitals that correspond to their highest energy levels. Similar to these elements, most elements tend to interact with other atoms to form chemical bonds. **Chemical bonds** are the attractive forces that hold atoms together.

Covalent Bonds

A **covalent bond** forms when two atoms share one or more pairs of electrons. For example, water is made up of one oxygen atom and two hydrogen atoms held together by covalent bonds. In Figure 2-3, step ①, an atom of hydrogen needs a second electron to achieve stability. Having two electrons in the orbital that corresponds to hydrogen's highest energy level allows the atom to be more stable. The oxygen atom needs two more electrons to give it a stable arrangement of eight electrons, which fill oxygen's orbitals to its highest energy level. Thus, hydrogen atoms and oxygen atoms share pairs of electrons in a ratio of two atoms of hydrogen to one atom of oxygen. The resulting stable compound, H_2O (water), is shown in step ②.

A **molecule** is the simplest part of a substance that retains all of the properties of that substance and can exist in a free state. For example, one molecule of the compound water is H_2O , and one molecule of oxygen gas is O_2 . Some molecules that biologists study are large and complex.

Word Roots and Origins

compound

from the Latin *componere*, meaning "to put together"

FIGURE 2-3

Two atoms of hydrogen and one atom of oxygen share electrons in covalent bonds and thus become stable. Covalent bonding results in the formation of molecules.

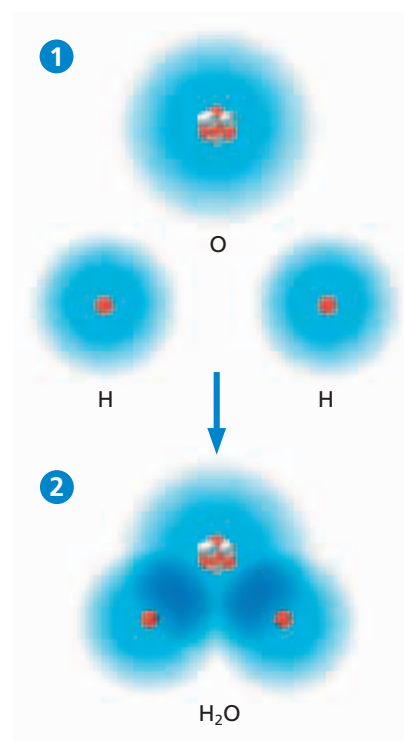
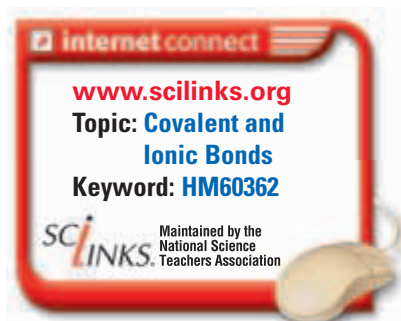
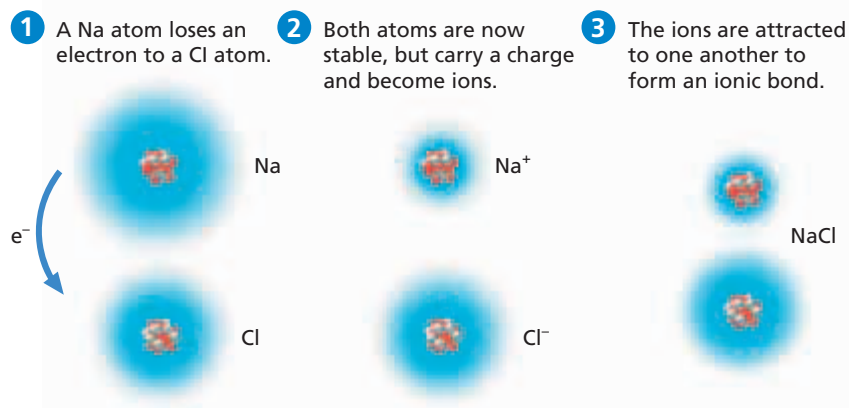


FIGURE 2-4

By losing its outermost electron (e^-), a sodium atom becomes a Na^+ ion. By gaining one electron, a chlorine atom becomes a Cl^- ion. Because of their opposite charges, the Na^+ and Cl^- ions are attracted to each other and form an ionic bond.



Ionic Bonds

In Figure 2-4, step **1**, both sodium and chlorine atoms have unfilled outermost energy levels and are therefore reactive. Both atoms achieve stability in the presence of one another. The one outer electron (e^-) of a sodium atom is transferred to a chlorine atom. This transfer makes both atoms more stable than they were before. The orbitals that correspond to the sodium atom's new outermost energy level are filled with eight electrons. But it also results in a sodium atom with a net positive electrical charge. The sodium atom has 11 protons (11 positive charges) balanced by only 10 electrons (10 negative charges). An atom or molecule with an electrical charge is called an **ion**. The sodium ion is written as Na^+ .

Chlorine, in step **2**, gained an electron from a sodium atom. The chlorine atom now has eight electrons in its orbitals that correspond to its outermost energy level. This makes the chlorine atom more stable. With this additional electron, chlorine becomes a negatively charged ion which is abbreviated as Cl^- .

Because positive and negative electrical charges attract each other, the sodium ion and the chloride ion attract each other. This attraction is called an **ionic bond**. The resulting compound, sodium chloride, NaCl , shown in step **3**, is an ionic compound and is familiar to you as common table salt.

SECTION 1 REVIEW

1. What is matter?
2. What is the relationship between elements and atoms?
3. Describe the arrangement within energy levels of the six electrons of an atom of carbon.
4. How are isotopes of the same element alike?
5. How can we predict which elements are reactive under normal conditions and which are unreactive?
6. Distinguish between covalent and ionic bonds.

CRITICAL THINKING

7. **Distinguishing Differences** Explain why the terms *mass* and *weight* should not be used interchangeably.
8. **Applying Information** Classify each of the following as an element or a compound: HCl , CO_2 , Cl , Li , and H_2O .
9. **Applying Information** Given that elements are pure substances, how many types of atoms make up the structure of a single element? Explain your answer.

ENERGY

All living things use energy. The amount of energy in the universe remains the same over time, but energy can change from one form to another. It is the transfer of energy—from the sun to and through almost every organism on Earth—that biologists seek to understand when they study the chemistry of living things.

ENERGY AND MATTER

Scientists define **energy** as the ability to do work. Energy can occur in various forms, and one form of energy can be converted to another form. In a light bulb's filament, electrical energy is converted to radiant energy (light) and thermal energy (heat). Some forms of energy important to biological systems include chemical energy, thermal energy, electrical energy, and mechanical energy. Inside any single organism, energy may be converted from one form to another. For example, after you eat a meal, your body changes the chemical energy found in food into thermal and mechanical energy, among other things.

States of Matter

Although it is not apparent when we observe matter, all the atoms and molecules in any substance are in constant motion. The motion of and spacing between atoms or molecules of a substance determine the substance's state: solid, liquid, or gas, as shown in Figure 2-5. In general, the atoms or molecules of a solid are more closely linked together than in a liquid or gas. Water is an exception to this, as will be described later. Solids move less rapidly than the particles that make up a liquid or a gas. A solid maintains a fixed volume and shape. A liquid maintains a fixed volume, but its particles move more freely than those of a solid, which gives a liquid its ability to flow and to conform to the shape of any container. Particles of a gas move the most rapidly. Gas particles have little or no attraction to each other, and they fill the volume of the container they occupy. Thermal energy must be added to the substance to cause a substance to change states.

OBJECTIVES

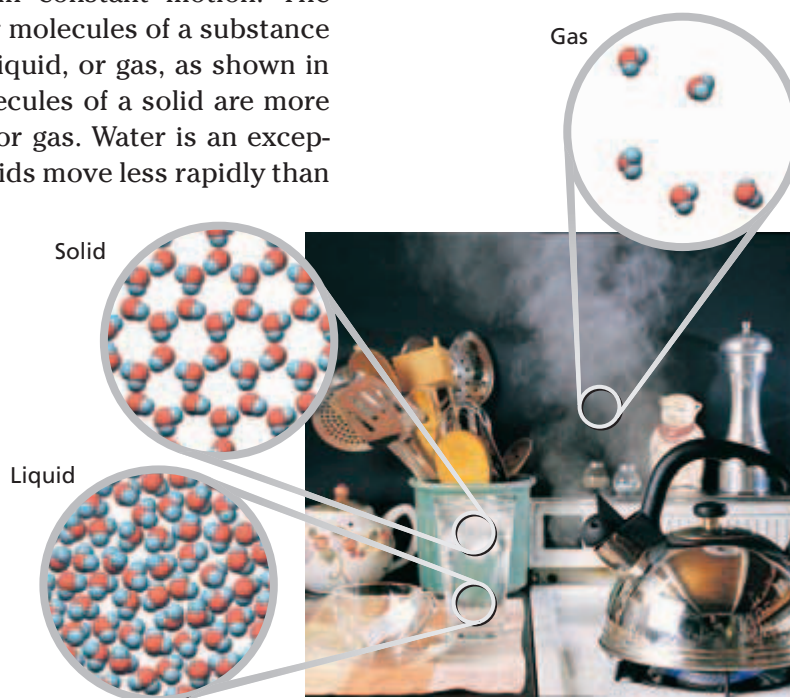
- **Describe** the physical properties of each state of matter.
- **Describe** the role of reactants and products in chemical reactions.
- **Explain** the relationship between enzymes and activation energy.
- **Explain** how oxidation and reduction reactions are linked.

VOCABULARY

energy
chemical reaction
reactant
product
metabolism
activation energy
catalyst
enzyme
redox reaction
oxidation reaction
reduction reaction

FIGURE 2-5

Matter exists as solids, liquids, and gases as shown below with water.



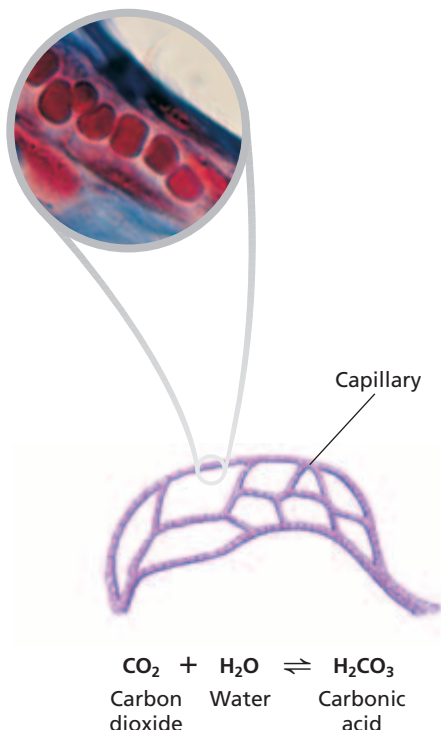


FIGURE 2-6

Chemical reactions occur in the human body. The chemical reaction shown above takes place in capillaries. Because the products of the reaction remain in the blood, the reaction is reversible and is written as: $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3$.

ENERGY AND CHEMICAL REACTIONS

In a **chemical reaction**, one or more substances change to produce one or more different substances. Energy is absorbed or released when chemical bonds are broken and new ones are formed. Living things undergo many thousands of chemical reactions every day. Reactions can vary from highly complex to very simple. The chemical reaction in Figure 2-6 takes place in your blood. Carbon dioxide is taken up from body cells and into the blood when it crosses the thin capillary walls. The carbon dioxide reacts with water in the blood to form carbonic acid. Carbon dioxide is then released into the lung's alveoli and exhaled when the carbonic acid breaks down to carbon dioxide and water.

If the reaction proceeds in only one direction, the **reactants** are shown on the left side of the equation. In the reaction in Figure 2-6, the reactants are carbon dioxide (CO_2) and water (H_2O). In a chemical reaction, bonds present in the reactants are broken, the elements are rearranged, and new compounds are formed as the products. The **products** of this reaction are shown on the right side. In this reaction, the product is carbonic acid (H_2CO_3). Notice that the number of each kind of atom must be the same on either side of the arrow. Some chemical reactions can proceed in either direction and a two-direction arrow ($\rightleftharpoons$) is used. For example, the equation in Figure 2-6 is reversible and can be written as $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3$.

The energy your body needs is provided by the sugars, proteins, and fats found in foods. Your body continuously undergoes a series of chemical reactions in which these energy-supplying substances are broken down into carbon dioxide, water, and other products. In this process, energy is released for use by your body to build and maintain body cells, tissues, and organs. **Metabolism** (MUH-TAB-uh-LIZ-uhm) is the term used to describe all of the chemical reactions that occur in an organism.

Activation Energy

For most chemical reactions to begin, energy must be added to the reactants. In many chemical reactions, the amount of energy needed to start the reaction, called **activation energy**, is large. Figure 2-7 shows the activation energy for a hypothetical chemical reaction.

Certain chemical substances, known as **catalysts** (KAT-uh-LISTS), reduce the amount of activation energy that is needed for a reaction to take place, as shown in Figure 2-7. A reaction in the presence of the correct catalyst will proceed spontaneously or with the addition of a small amount of energy. In living things enzymes act as catalysts. An **enzyme** is a protein or RNA molecule that speeds up metabolic reactions without being permanently changed or destroyed.

Activation Energy With and Without a Catalyst

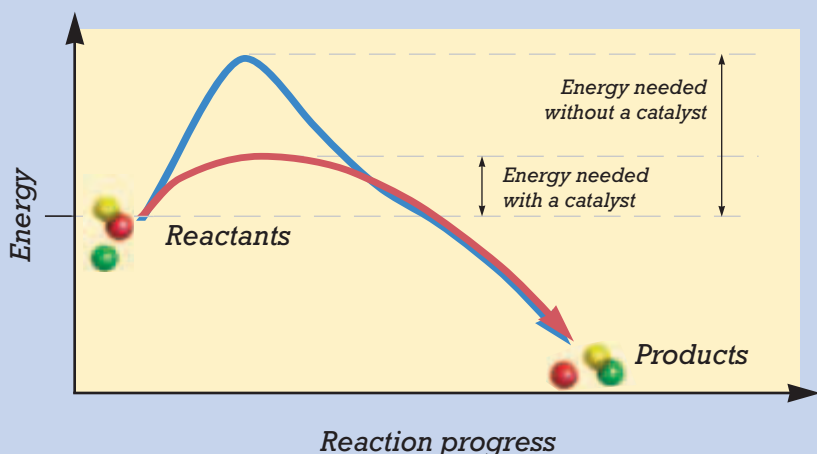


FIGURE 2-7

The blue curve shows the activation energy that must be supplied before this reaction can begin. The activation energy can be reduced, as shown by the red curve, by adding a catalyst.

Oxidation Reduction Reactions

You know that there is a constant transfer of energy into and throughout living things. Many of the chemical reactions that help transfer energy in living things involve the transfer of electrons. These reactions in which electrons are transferred between atoms are known as oxidation reduction reactions, or **redox reactions**. In an **oxidation** (AHKS-i-DAY-shuhn) **reaction**, a reactant loses one or more electrons, thus becoming more positive in charge. For example, remember that a sodium atom loses an electron to achieve stability when it forms an ionic bond, as shown in Figure 2-4. Thus, the sodium atom undergoes oxidation to form a Na^+ ion. In a **reduction reaction**, a reactant gains one or more electrons, thus becoming more negative in charge. When a chlorine atom gains an electron to form a Cl^- ion, the atom undergoes reduction. Redox reactions always occur together. An oxidation reaction occurs, and the electron given up by one substance is then accepted by another substance in a reduction reaction.



Quick Lab

Modeling Ionic Bonds

Materials toothpicks, small and large marshmallows, and raisins

Procedure



Use large marshmallows to represent chlorine, small marshmallows to represent sodium, raisins to represent electrons, and toothpicks to represent bonds or orbital place holders. Make models of Na , Cl , Na^+ , Cl^- , and NaCl (sodium chloride).

Analysis Use your models to identify each of the following: a sodium atom, a sodium ion, a chlorine atom, a chloride ion, an ionic bond, and a particle of sodium chloride.

SECTION 2 REVIEW

1. Name and describe the physical properties of the three states of matter.
2. Explain the roles of reactants and products in a chemical reaction.
3. Describe the effect of an enzyme on the activation energy in a chemical reaction.
4. Enzymes are biological catalysts. Explain what they do in living systems.
5. Why does a reduction reaction always accompany an oxidation reaction?

CRITICAL THINKING

6. **Analyzing Concepts** Living things need a constant supply of energy. Explain why.
7. **Analyzing Graphics** Carbonic anhydrase is the enzyme that catalyzes the chemical reaction illustrated in Figure 2-6. What effect might a molecule that interferes with the action of carbonic anhydrase have on your body?
8. **Analyzing Information** In a reduction reaction, the reduced atom gains one or more electrons. Why is this reaction called a reduction?

Science in Action

Is There Water on Mars?

Do other living organisms share our solar system? One way to discover a possible answer to this age-old question is to search for the presence of water on other planets. Water is essential for life on Earth, so where there is water, there might be life.



NASA's exploration rover *Opportunity*

HYPOTHESIS: Water Exists on Mars

Scientists at the National Aeronautics and Space Administration (NASA) have sent orbiters, (reusable spacecraft designed to transport people and cargo) between Earth and Mars. Recent orbiters to be sent to Mars include NASA's *Mars Global Surveyor (MGS)* and *Mars Odyssey*. These missions revealed boulders, dust, canyons, and tall volcanic peaks. Some of the most exciting images revealed gullies carved into the Martian landscape. The appearance of these gullies led geologists to hypothesize that the gullies had been carved by running water within the past few million years.

Why is the presence of water on Mars so intriguing? On our planet, where there is water, there is life. For now, scientists are assuming that life-forms on Mars would have the same dependence on water.

METHODS: Image and Analyze Martian Rock Samples

In the summer of 2003, NASA scientists launched *Spirit* and *Opportunity*, two Mars exploration rovers. The job of these two rovers was to take small-scale geologic surveys of surrounding rocks and soil and to search for ancient traces of water. These rovers landed in regions near the Martian equator that may have held water at one time.

RESULTS: Water and Minerals That Can Form in Water Are Present on Mars

The images sent back by the *Odyssey* and *MGS* orbiters reveal that water covers large areas of Mars' polar regions as well as some large areas at its equator. The water is almost certainly frozen in the form of dusty snowpacks, which may occur largely as an icy soil layer. These icepacks may resemble the permafrost of Earth's polar regions.

In addition, *Opportunity* detected the presence of hematite at its landing site, the Meridiani Planum, an area on Mars that was thought to have been a shallow lake at one time. Hematite is a mineral that often forms in pools of standing water on Earth but can also form as a result of volcanic activity.

Opportunity also found strong evidence that the rocks at Meridiani Planum were once sediments that were laid down by liquid water. This discovery also gives greater weight to the hypothesis that Mars was once a habitat for microbial life.

CONCLUSION: Mars Once Had Water

Chances are that the current mission to Mars will not determine whether life ever started on the planet, but scientists are hopeful that they will have an answer someday. Human exploration of Mars is already being planned. Astronauts would be able to carry out many experiments that robots cannot do.



This picture was taken by one of the rovers that explored Mars.

REVIEW

1. Why did geologists initially hypothesize that there might have been water on Mars?
2. What did the images sent back by the NASA orbiters and the exploration rovers in 2003 reveal?

3. Critical Thinking

Explain why the existence of hematite may not be the best indicator of water on Mars.



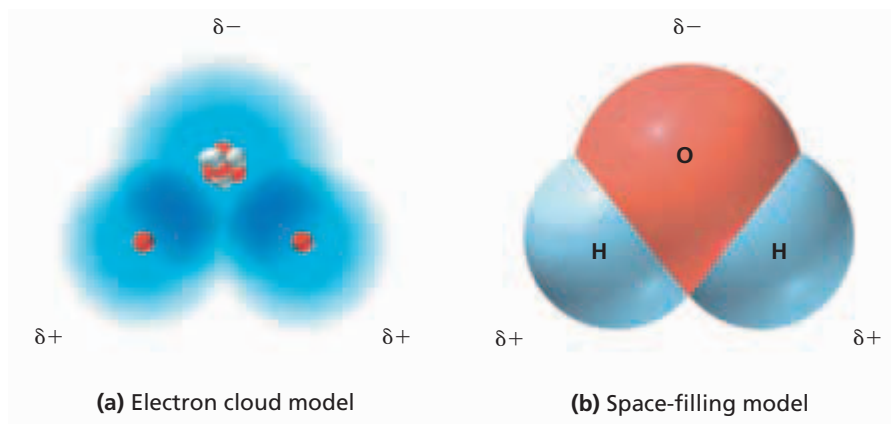
WATER AND SOLUTIONS

Compare the body of a jellyfish with your own body. A jellyfish would die if it was removed from its watery environment. Yet you can live on the driest parts of Earth. Jellyfish and humans seem unlike each other, yet the bodies of both are made of cells that consist mostly of water. The chemical reactions of all living things take place in the aqueous environment of the cell. Water has several unique properties that make it one of the most important compounds found in living things.

POLARITY

Many of water's biological functions stem from its chemical structure. Recall that in the water molecule, H_2O , the hydrogen and oxygen atoms share electrons to form covalent bonds. However, these atoms do not share the electrons equally. The oxygen atom has a greater ability to attract electrons to it because it pulls hydrogen's electrons towards its nucleus. As a result, as shown in Figure 2-8, the region of the molecule where the oxygen atom is located has a partial negative charge, denoted with a δ^- , while the regions of the molecule where each of the two hydrogen atoms are located have partial positive charges, each of which are denoted with a δ^+ . Thus, even though the total charge on a water molecule is neutral, the charge is unevenly distributed across the water molecule. Because of this uneven distribution of charge, water is called a **polar** compound.

Notice also in Figure 2-8 that the three atoms in a water molecule are not arranged in a straight line as you might expect. Rather, the two hydrogen atoms bond with the single oxygen atom at an angle.



OBJECTIVES

- **Describe** the structure of a water molecule.
- **Explain** how water's polar nature affects its ability to dissolve substances.
- **Outline** the relationship between hydrogen bonding and the different properties of water.
- **Identify** the roles of solutes and solvents in solutions.
- **Differentiate** between acids and bases.

VOCABULARY

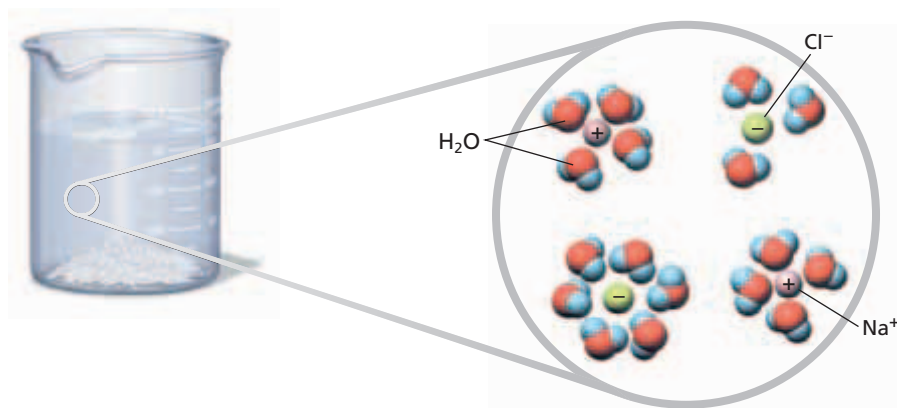
polar
hydrogen bond
cohesion
adhesion
capillarity
solution
solute
solvent
concentration
saturated solution
aqueous solution
hydroxide ion
hydronium ion
acid
base
pH scale
buffer

FIGURE 2-8

The oxygen region of the water molecule is weakly negative, and the hydrogen regions are weakly positive. Notice the different ways to represent water, H_2O . You are familiar with the electron cloud model (a). The space-filling model (b) shows the three-dimensional structure of a molecule.

FIGURE 2-9

The positive region of a water molecule attracts the negative region of an ionic compound, such as the Cl^- portion of NaCl. Similarly, the negative region of the water molecule attracts the positive region of the compound—the Na^+ portion of NaCl. As a result, NaCl breaks apart, or dissolves, in water.

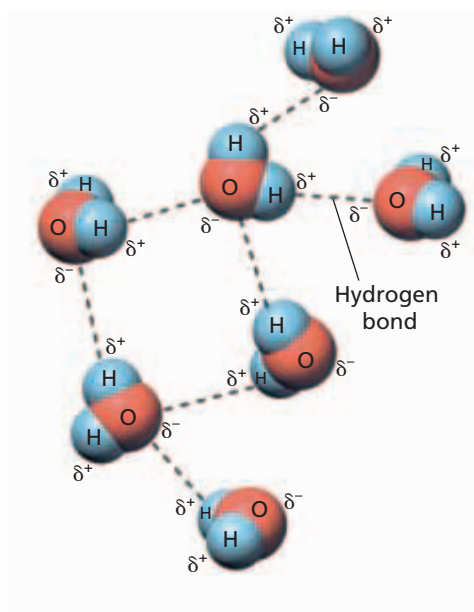


Solubility of Water

The polar nature of water allows it to dissolve polar substances, such as sugars, ionic compounds, and some proteins. Water does not dissolve nonpolar substances, such as oil because a weaker attraction exists between polar and nonpolar molecules than between two polar molecules. Figure 2-9 shows how water dissolves the ionic compound sodium chloride, NaCl. In your body, ions, such as sodium and chloride, are essential to bodily functions, such as muscle contraction and transmission of impulses in the nervous system. In fact, dissolved, or dissociated ions, are present in all of the aqueous solutions found in living things and are important in maintaining normal body functions.

FIGURE 2-10

The dotted lines in this figure represent hydrogen bonds. A hydrogen bond is a force of attraction between a hydrogen atom in one molecule and a negatively charged region or atom in a second molecule.



HYDROGEN BONDING

The polar nature of water also causes water molecules to be attracted to one another. As is shown in Figure 2-10, the positively charged region of one water molecule is attracted to the negatively charged region of another water molecule. This attraction is called a hydrogen bond. A **hydrogen bond** is the force of attraction between a hydrogen molecule with a partial positive charge and another atom or molecule with a partial or full negative charge. Hydrogen bonds in water exert an attractive force strong enough so that water “clings” to itself and some other substances.

Hydrogen bonds form, break, and reform with great frequency. However, at any one time, a great number of water molecules are bonded together. The number of hydrogen bonds that exist depends on the state that water is in. If water is in its solid state all its water molecules are hydrogen bonded and do not break. As water liquifies, more hydrogen bonds are broken than are formed, until an equal number of bonds are formed and broken. Hydrogen bonding accounts for the unique properties of water, some of which we will examine further. These properties include cohesion and adhesion, the ability of water to absorb a relatively large amount of energy as heat, the ability of water to cool surfaces through evaporation, the density of ice, and the ability of water to dissolve many substances.

Cohesion and Adhesion

Water molecules stick to each other as a result of hydrogen bonding. An attractive force that holds molecules of a single substance together is known as **cohesion**. Cohesion due to hydrogen bonding between water molecules contributes to the upward movement of water from plant roots to their leaves.

Related to cohesion is the *surface tension* of water. The cohesive forces in water resulting from hydrogen bonds cause the molecules at the surface of water to be pulled downward into the liquid. As a result, water acts as if it has a thin “skin” on its surface. You can observe water’s surface tension by slightly overfilling a drinking glass with water. The water will appear to bulge above the rim of the glass. Surface tension also enables small creatures such as spiders and water-striders to run on water without breaking the surface.

Adhesion is the attractive force between two particles of different substances, such as water molecules and glass molecules. A related property is **capillarity** (KAP-uh-LER-i-tee), which is the attraction between molecules that results in the rise of the surface of a liquid when in contact with a solid. Together, the forces of adhesion, cohesion, and capillarity help water rise through narrow tubes against the force of gravity. Figure 2-11 shows cohesion and adhesion in the water-conducting tubes in the stem of a flower.

Temperature Moderation

Water has a high heat capacity, which means that water can absorb or release relatively large amounts of energy in the form of heat with only a slight change in temperature. This property of water is related to hydrogen bonding. Energy must be absorbed to break hydrogen bonds, and energy is released as heat when hydrogen bonds form. The energy that water initially absorbs breaks hydrogen bonds between molecules. Only after these hydrogen bonds are broken does the energy begin to increase the motion of the water molecules, which raises the temperature of the water. When the temperature of water drops, hydrogen bonds reform, which releases a large amount of energy in the form of heat.

Therefore, during a hot summer day, water can absorb a large quantity of energy from the sun and can cool the air without a large increase in the water’s temperature. At night, the gradually cooling water warms the air. In this way, the Earth’s oceans stabilize global temperatures enough to allow life to exist. Water’s high heat capacity also allows organisms to keep cells at an even temperature despite temperature changes in the environment.

As a liquid evaporates, the surface of the liquid that remains behind cools down. A relatively large amount of energy is absorbed by water during evaporation, which significantly cools the surface of the remaining liquid. Evaporative cooling prevents organisms that live on land from overheating. For example, the evaporation of sweat from a person’s skin releases body heat and prevents overheating on a hot day or during strenuous activity.

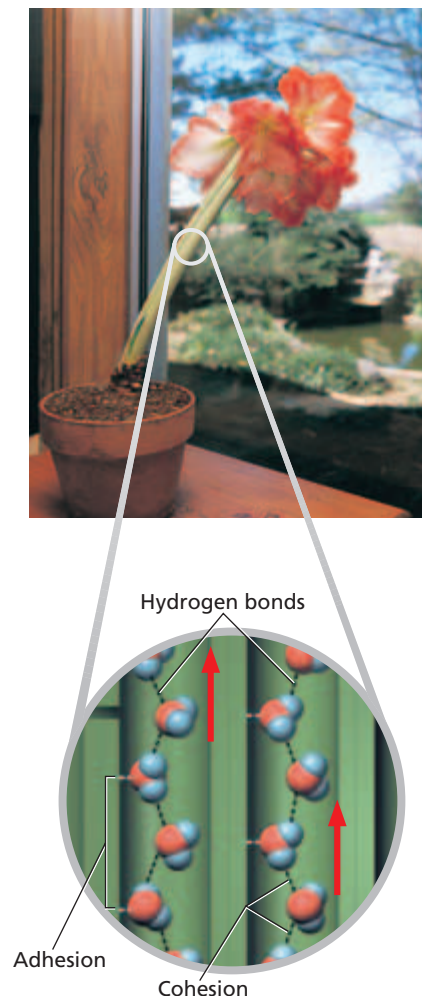
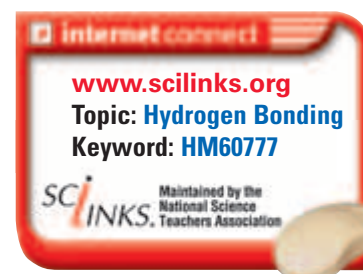


FIGURE 2-11

Cohesion, adhesion, and capillarity contribute to the upward movement of water from the roots of plants.



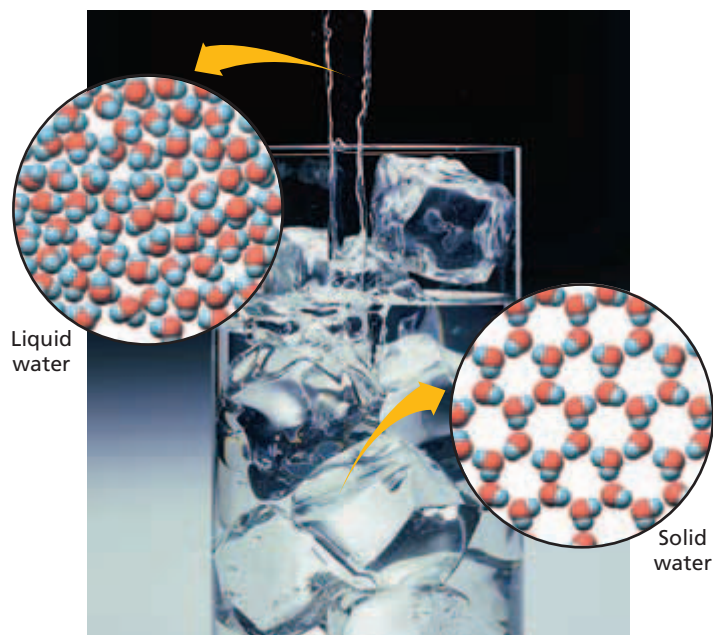


FIGURE 2-12

Ice (solid water) is less dense than liquid water because of the structure of ice crystals. The water molecules in ice are bonded to each other in a way that creates large amounts of open space between the molecules, relative to liquid water.

Density of Ice

Unlike most solids, which are denser than their liquids, solid water is less dense than liquid water. This property is due to the shape of the water molecule and hydrogen bonding. The angle between the hydrogen atoms is quite wide. So, when water forms solid ice, the angles in the molecules cause ice crystals to have large amounts of open space, as shown in Figure 2-12. This open space lattice structure causes ice to have a low density.

Because ice floats on water, bodies of water such as ponds and lakes freeze from the top down and not the bottom up. Ice insulates the water below from the cold air, which allows fish and other aquatic creatures to survive under the icy surface.

SOLUTIONS

A **solution** is a mixture in which one or more substances are uniformly distributed in another substance. Solutions can be mixtures of liquids, solids, or gases. For example, plasma, the liquid part of blood, is a very complex solution. It is composed of many types of ions and large molecules, as well as gases, that are dissolved in water. A **solute** (SAHL-YOOT) is a substance dissolved in the solvent. The particles that compose a solute may be ions, atoms, or molecules. The **solvent** is the substance in which the solute is dissolved. For example, when sugar, a solute, and water, a solvent, are mixed, a solution of sugar water results. Though the sugar dissolves in the water, neither the sugar molecules nor the water molecules are altered chemically. If the water is boiled away, the sugar molecules remain and are unchanged.

Solutions can be composed of various proportions of a given solute in a given solvent. Thus, solutions can vary in concentration. The **concentration** of a solution is the amount of solute dissolved in a fixed amount of the solution. For example, a 2 percent saltwater solution contains 2 g of salt dissolved in enough water to make 100 mL of solution. The more solute dissolved, the greater is the concentration of the solution. A **saturated solution** is one in which no more solute can dissolve.

Aqueous (AY-kwee-uhs) **solutions**—solutions in which water is the solvent—are universally important to living things. Marine microorganisms spend their lives immersed in the sea, an aqueous solution. Most nutrients that plants need are in aqueous solutions in moist soil. Body cells exist in an aqueous solution of intercellular fluid and are themselves filled with fluid; in fact, most chemical reactions that occur in the body occur in aqueous solutions.

Word Roots and Origins

solvent

from the Latin *solvere*, meaning
"to loosen"

ACIDS AND BASES

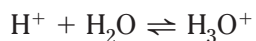
One of the most important aspects of a living system is the degree of its acidity or alkalinity. What do we mean when we use the terms *acid* and *base*?

Ionization of Water

As water molecules move about, they bump into one another. Some of these collisions are strong enough to result in a chemical change: one water molecule loses a proton (a hydrogen nucleus), and the other gains this proton. This reaction really occurs in two steps. First, one molecule of water pulls apart another water molecule, or dissociates, into two ions of opposite charge:



The OH^- ion is known as the **hydroxide ion**. The free H^+ ion can react with another water molecule, as shown in the equation below.



The H_3O^+ ion is known as the **hydronium ion**. Acidity or alkalinity is a measure of the relative amounts of hydronium ions and hydroxide ions dissolved in a solution. If the number of hydronium ions in a solution equals the number of hydroxide ions, the solution is said to be neutral. Pure water contains equal numbers of hydronium ions and hydroxide ions and is therefore a neutral solution.

Acids

If the number of hydronium ions in a solution is greater than the number of hydroxide ions, the solution is an **acid**. For example, when hydrogen chloride gas, HCl , is dissolved in water, its molecules dissociate to form hydrogen ions, H^+ , and chloride ions, Cl^- , as is shown in the equation below.



These free hydrogen ions combine with water molecules to form hydronium ions, H_3O^+ . This aqueous solution contains many more hydronium ions than it does hydroxide ions, making it an acidic solution. Acids tend to have a sour taste; however, never taste a substance to test it for acidity. In concentrated forms, they are highly corrosive to some materials, as you can see in Figure 2-13.

Bases

If sodium hydroxide, NaOH , a solid, is dissolved in water, it dissociates to form sodium ions, Na^+ , and hydroxide ions, OH^- , as shown in the equation below.



Eco Connection

Acid Precipitation

Acid precipitation, more commonly called *acid rain*, describes rain, snow, sleet, or fog that contains high levels of sulfuric and nitric acids. These acids form when sulfur dioxide gas, SO_2 , and nitrogen oxide gas, NO , react with water in the atmosphere to produce sulfuric acid, H_2SO_4 , and nitric acid, HNO_3 .

Acid precipitation makes soil and bodies of water, such as lakes, more acidic than normal. These high acid levels can harm plant and animal life directly. A high level of acid in a lake may kill mollusks, fish, and amphibians. Even in a lake that does not have a very elevated level of acid, acid precipitation may leach aluminum and magnesium from soils, poisoning water-dwelling species.

Reducing fossil-fuel consumption, such as occurs in gasoline engines and coal-burning power plants, should reduce high acid levels in precipitation.

FIGURE 2-13

Sulfur dioxide, SO_2 , which is produced when fossil fuels are burned, reacts with water in the atmosphere to produce acid precipitation. Acid precipitation, or acid rain, can make lakes and rivers too acidic to support life and can even corrode stone, such as the face of this statue.



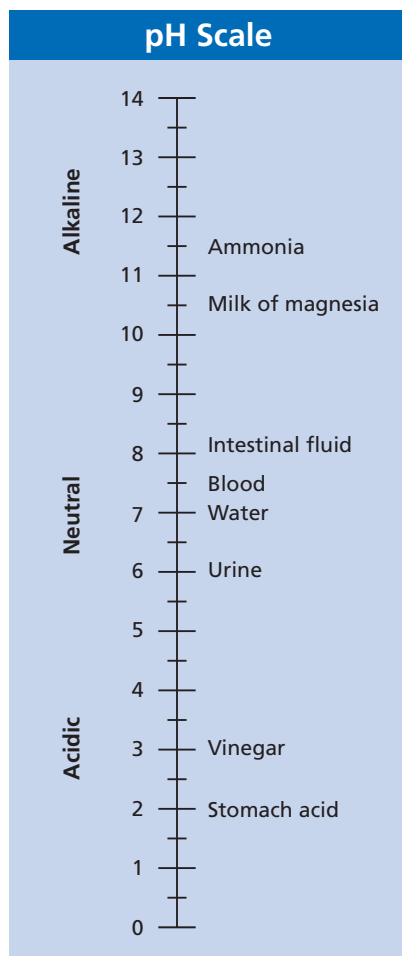


FIGURE 2-14

Some of your body fluids are acidic, and others are alkaline. A solution with a pH above 7 is alkaline, and a solution with a pH below 7 is acidic. Each unit on the pH scale reflects a 10-fold change in acidity or alkalinity.

This solution then contains more hydroxide ions than hydronium ions and is therefore defined as a **base**. The adjective *alkaline* refers to bases. Bases have a bitter taste; however, never taste a substance to test for alkalinity. They tend to feel slippery because the OH^- ions react with the oil on our skin to form a soap. In fact, commercial soap is the product of a reaction between a base and a fat.

pH

Scientists have developed a scale for comparing the relative concentrations of hydronium ions and hydroxide ions in a solution. This scale is called the **pH scale**, and it ranges from 0 to 14, as shown in Figure 2-14. A solution with a pH of 0 is very acidic, a solution with a pH of 7 is neutral, and a solution with a pH of 14 is very basic. A solution's pH is measured on a logarithmic scale. That is, the change of one pH unit reflects a 10-fold change in the acidity or alkalinity. For example, urine has 10 times the H_3O^+ ions at a pH of 6 than water does at a pH of 7. Vinegar, has 1,000 times more H_3O^+ ions at a pH of 3 than urine at a pH of 6, and 10,000 times more H_3O^+ ions than water at a pH of 7. The pH of a solution can be measured with litmus paper or with some other chemical indicator that changes color at various pH levels.

Buffers

The control of pH is important for living systems. Enzymes can function only within a very narrow pH range. The control of pH in organisms is often accomplished with buffers. **Buffers** are chemical substances that neutralize small amounts of either an acid or a base added to a solution. As Figure 2-14 shows, the composition of your internal environment—in terms of acidity and alkalinity—varies greatly. Some of your body fluids, such as stomach acid and urine, are acidic. Others, such as intestinal fluid and blood, are basic or alkaline. Complex buffering systems maintain the pH values in a normal healthy body.

SECTION 3 REVIEW

1. Illustrate the structure of a water molecule by drawing a space-filling model.
2. Why is water called a polar molecule?
3. Identify the properties of water that are important for life to be able to exist.
4. Identify the solute and solvent in a hot chocolate solution that is made of chocolate syrup and warm milk.
5. Why does pure water have a neutral pH?
6. Outline a reason why the control of pH is important in living systems.

CRITICAL THINKING

7. **Recognizing Relationships** What is the relationship among hydrogen bonds and the forces of cohesion, adhesion, and capillarity?
8. **Applying Information** The active ingredient in aspirin is acetylsalicylic acid. Why would doctors recommend buffered aspirin, especially for those with a "sensitive" stomach?
9. **Analyzing Graphics** All units on the pH scale in Figure 2-14 look equivalent, but they are not. Why is the scale drawn as though they are?

CHAPTER HIGHLIGHTS

SECTION 1

Composition of Matter

- Matter is anything that occupies space and has mass.
- Elements are made of a single kind of atom and cannot be broken down by chemical means into simpler substances.
- Atoms are composed of protons, neutrons, and electrons. Protons and neutrons make up the nucleus of the atom. Electrons move about the nucleus in orbitals.
- Compounds consist of atoms of two or more elements that are joined by chemical bonds in a fixed proportion.
- Most elements react to form chemical bonds so that their atoms become stable. An atom achieves stability when the orbitals that correspond to its highest energy level are filled with the maximum number of electrons.
- A covalent bond is formed when two atoms share electrons.
- An ionic bond is formed when one atom gives up an electron to another. The positive ion is then attracted to a negative ion to form the ionic bond.

Vocabulary

matter (p. 31)
mass (p. 31)
element (p. 35)
atom (p. 32)
nucleus (p. 32)

proton (p. 32)
neutron (p. 32)
atomic number (p. 32)
mass number (p. 32)
electron (p. 32)

orbital (p. 32)
isotope (p. 32)
compound (p. 33)
chemical bond (p. 33)
covalent bond (p. 33)

molecule (p. 33)
ion (p. 34)
ionic bond (p. 34)

SECTION 2

Energy

- Addition of energy to a substance can cause its state to change from a solid to a liquid and from a liquid to a gas.
- Reactants are substances that enter chemical reactions. Products are substances produced by chemical reactions.
- Enzymes lower the amount of activation energy necessary for a reaction to begin in living systems.
- A chemical reaction in which electrons are exchanged between atoms is called an oxidation-reduction reaction.

Vocabulary

energy (p. 35)
chemical reaction (p. 36)
reactant (p. 36)

product (p. 36)
metabolism (p. 36)
activation energy (p. 36)

catalyst (p. 36)
enzyme (p. 36)
redox reaction (p. 37)

oxidation reaction (p. 37)
reduction reaction (p. 37)

SECTION 3

Water and Solutions

- The two hydrogen atoms and one oxygen atom that make up a water molecule are arranged at an angle to one another.
- Water is a polar molecule. The electrons in the molecule are shared unevenly between hydrogen and oxygen. This polarity makes water effective at dissolving other polar substances.
- Hydrogen bonding accounts for most of the unique properties of water.
- The unique properties of water include the ability to dissolve many substances, cohesion and adhesion, the ability to absorb a relatively large amount of energy as heat, the ability to cool surfaces through evaporation, and the low density of ice.
- A solution consists of a solute dissolved in a solvent.
- Water ionizes into hydronium ions and hydroxide ions.
- Acidic solutions contain more hydronium ions than hydroxide ions. Basic solutions contain more hydroxide ions than hydronium ions.
- Buffers are chemicals that neutralize the effects of adding small amounts of either an acid or a base to a solution.

Vocabulary

polar (p. 39)
hydrogen bond (p. 40)
cohesion (p. 41)
adhesion (p. 41)
capillarity (p. 41)

solution (p. 42)
solute (p. 42)
solvent (p. 42)
concentration (p. 42)
saturated solution (p. 42)

aqueous solution (p. 42)
hydroxide ion (p. 43)
hydronium ion (p. 43)
acid (p. 43)
base (p. 44)

pH scale (p. 44)
buffer (p. 44)

CHAPTER REVIEW

USING VOCABULARY

- For each pair of terms, explain how the meanings of the terms differ.
 - oxidation* and *reduction*
 - reactants* and *products*
 - acid* and *base*
- Explain the relationship between electrons, neutrons, and protons.
- Choose the term that does not belong in the following group, and explain why it does not belong: *element*, *compound*, *chemical bonds*, and *adhesion*.
- Word Roots and Origins** The term *catalyst* comes from the Greek *katalysis*, meaning “dissolution.” Give reasons that this term is appropriate in describing the function of enzymes.

UNDERSTANDING KEY CONCEPTS

- Differentiate** between the mass and the weight of an object.
- Name** the subatomic particles that are found in the nucleus of an atom.
- Describe** the arrangement within energy levels of the seven electrons of an atom of nitrogen.
- Describe** how compounds affect an atom’s stability.
- Differentiate** between covalent and ionic bonds.
- Compare** the motion and spacing of the molecules in a solid to the motion and spacing of the molecules in a gas.
- Identify** the reactants and the products in the following chemical reaction.
$$A + B \rightarrow C + D$$
- Explain** the relationship between enzymes and activation energy.
- Explain** oxidation and reduction in terms of electron transfer and charge.
- Describe** the structure of a water molecule.
- Outline** what happens when water ionizes.
- Describe** how water behaves at its surface and the role hydrogen bonding plays in this behavior.
- Identify** the solute(s) and solvent(s) in a cup of instant coffee with sugar.
- Name** two ions that are the products of the dissociation of water.

- Compare** an acid to a base in terms of the hydroxide ion concentration.

- CONCEPT MAPPING** Use the following terms to create a concept map that shows how the properties of water help it rise in plants: *water*, *hydrogen bonds*, *cohesion*, *adhesion*, *capillarity*, and *plants*.

CRITICAL THINKING

- Analyzing Concepts** In nature, the elements oxygen and hydrogen are usually found as gases with the formulas O_2 and H_2 . Why? Are they compounds? Are they molecules?
- Analyzing Data** The table below shows melting and boiling points at normal pressure for five different elements or compounds. Above the boiling point, a compound or element exists as a gas. Between the melting point and the boiling point, a compound or element exists as a liquid. Below the melting point, a compound or element exists as a solid. Use the table to answer the following questions:
 - At 20°C and under normal pressure conditions, which substances exist as solids? as liquids? as gases?
 - Which substance exists as a liquid over the broadest range of temperature?
 - Which substance exists as a liquid over the narrowest range of temperature?
 - Which one of the substances are you least likely to encounter as a gas?

Melting and Boiling Points at Normal Pressure

Substance	Melting point ($^\circ\text{C}$)	Boiling point ($^\circ\text{C}$)
Aluminum	658	2,330
Argon	-190	-186
Chlorine	-104	-34
Mercury	-39	357
Water	0	100

- Recognizing Relationships** Cells contain mostly water. What would happen to the stability of an organism’s internal temperature with respect to environmental temperature changes if cells contained mostly oil, which does not have extensive hydrogen bonding?

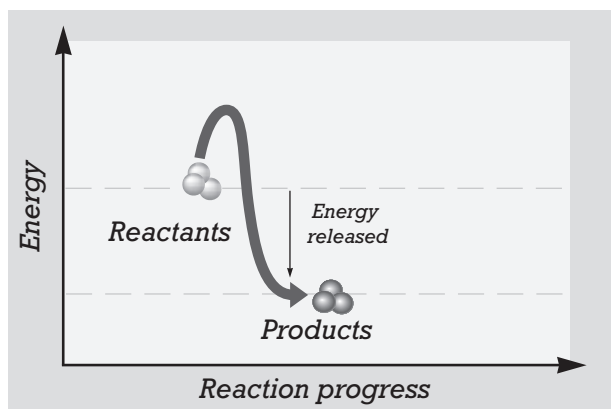


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- The way in which elements bond to form compounds depends on which of the following?
A. the model of the atom
B. the structural formula of the compound
C. the dissociation of the ions in the compound
D. the number and arrangement of electrons in the atoms of the elements
- If an atom is made up of 6 protons, 7 neutrons, and 6 electrons, what is its atomic number?
F. 6
G. 7
H. 13
J. 19

INTERPRETING GRAPHICS: The graph below shows the energy in a chemical reaction as the reaction progresses. Use the graph to answer the questions that follow.

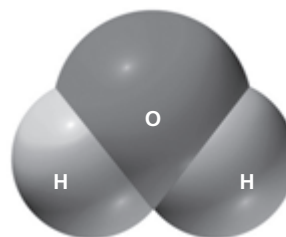


- The amount of energy needed for this chemical reaction to begin is shown by the line rising from the reactants. What is this energy called?
A. chemical energy
B. electrical energy
C. activation energy
D. mechanical energy
- Suppose that this reaction needs a catalyst to proceed. In the absence of a catalyst, the activation energy would be which of the following?
F. larger than what is shown
G. the same as what is shown
H. smaller than what is shown
J. not much different from what is shown
- What is an aqueous solution that contains more hydroxide ions than hydronium ions called?
A. a gas
B. a base
C. a solid
D. an acid

DIRECTIONS: Complete the following analogy.

6. Oxidation : loss :: reduction :
F. win
G. gain
H. take
J. forfeit

INTERPRETING GRAPHICS: The illustration below is a space-filling model of water. Use the model to answer the following question.



- The water molecule above has partial positive charges on the hydrogen atoms and a partial negative charge on the oxygen atom. What can you conclude from this information and the diagram of the water molecule?
A. Water is an ion.
B. Water is a polar molecule.
C. Water needs a proton and two electrons to be stable.
D. Oxygen atoms and hydrogen atoms have opposite charges.

SHORT RESPONSE

Covalent bonding is a sharing of electrons between atoms. Why do some atoms share electrons?

EXTENDED RESPONSE

Pure water contains equal numbers of hydronium ions and hydroxide ions and is therefore a neutral solution.

Part A What is the initial cause of the dissociation of water molecules into hydrogen and hydroxide ions? Explain the process.

Part B After water dissociates, hydronium ions are formed. Explain this process.

Test TIP

When possible use the text in the test to answer other questions. For example use a multiple choice answer to "jump start" your thinking about another question.

Measuring the Activity of Enzymes in Detergents

OBJECTIVES

- Recognize the function of enzymes in laundry detergents.
- Relate the factors of temperature and pH to the activity of enzymes.

PROCESS SKILLS

- designing an experiment
- making observations
- measuring volume, mass, and pH
- graphing
- analyzing data
- making conclusions

MATERIALS

- safety goggles
- lab apron
- balance
- graduated cylinder
- glass stirring rod
- 150 mL beaker
- 18 g regular instant gelatin or 1.8 g sugar-free instant gelatin
- 0.7 g Na_2CO_3 , sodium carbonate
- tongs or a hot mitt
- 50 mL boiling water
- thermometer
- pH paper
- 6 test tubes
- test-tube rack
- pipet with bulb
- plastic wrap
- tape
- 50 mL beakers (6)
- 50 mL distilled water
- 1 g each of 5 brands of laundry detergent
- wax pencil
- metric ruler

Background

1. Write a definition for the term *enzyme*.
2. From what you know about enzymes, why might enzymes be added to detergents?

Procedure

PART A Making a Protein Substrate

1.   **CAUTION** Always wear safety goggles and a lab apron to protect your eyes and clothing. Put on safety goggles and a lab apron.
2.   **CAUTION** Use tongs or a hot mitt to handle heated glassware. Put 18 g of regular (1.8 g of sugar-free) instant gelatin in a 150 mL beaker. Slowly add 50 mL of boiling water to the beaker, and stir the mixture with a stirring rod. Test and record the pH of this solution.
3.  **CAUTION** Do not touch or taste any chemicals. Very slowly add 0.7 g of Na_2CO_3 to the hot gelatin while stirring. Note any reaction. Test and record the pH of this solution.
4.   **CAUTION** Glassware is fragile. Notify the teacher of broken glass or cuts. Do not clean up broken glass or spills with broken glass unless the teacher tells you to do so. Remember to use tongs or a hot mitt to handle heated glassware. Place 6 test tubes in a test-tube rack. Pour 5 mL of the gelatin- Na_2CO_3 mixture into each tube. Use a pipet to remove any bubbles from the surface of the mixture in each tube. Cover the tubes tightly with plastic wrap and tape. Cool the tubes, and store them at room temperature until you begin Part C. Complete step 11 in Part C.



PART B Designing Your Experiment

- Based on the objectives for this lab, write a question you would like to explore about enzymes in detergents. To explore the question, design an experiment that uses the materials listed for this lab.
- Write a procedure for your experiment. Make a list of all the safety precautions you will take. Have your teacher approve your procedure and safety precautions before you begin the experiment.

PART C Conducting Your Experiment

-   **CAUTION** Always wear safety goggles and a lab apron to protect your eyes and clothing. Put on safety goggles and a lab apron.
- Make a 10 percent solution of each laundry detergent by dissolving 1 g of detergent in 9 mL of distilled water.
- Set up your experiment. Repeat step 11.
- Record your data after 24 hours in a data table similar to the one below.

   **CAUTION** Know the location of the emergency shower and eye-wash station and how to use them. If you get a chemical on your skin or clothing, wash it off at the sink while calling to the teacher. Notify the teacher

of a spill. Spills should be cleaned up promptly, according to your teacher's directions. Dispose of solutions, broken glass, and gelatin in the designated waste containers. Do not pour chemicals down the drain or put lab materials in the trash unless your teacher tells you to do so.

- Clean up your work area and all lab equipment. Return lab equipment to its proper place. Wash your hands thoroughly before leaving the lab and after finishing all work.

Analysis and Conclusions

- Suggest a reason for adding Na_2CO_3 to the gelatin solution.
- Make a bar graph of your data. Plot the amount of gelatin broken down (change in the depth of the gelatin) on the y -axis and the type of detergent on the x -axis. Use a separate sheet of graph paper.
- What conclusions did your group infer from the results? Explain.

Further Inquiry

Research other household products that contain enzymes, and find out their role in each of the products.

DATA TABLE

Solution	Date	Observations

BIOCHEMISTRY



All living organisms, such as those seen in this photo, are made up of molecules that contain primarily carbon atoms.

SECTION 1 *Carbon Compounds*

SECTION 2 *Molecules of Life*

CARBON COMPOUNDS

Although water is the primary medium for life on Earth, most of the molecules from which living organisms are made are based on the element carbon. Carbon's ability to form large and complex molecules has contributed to the great diversity of life.

CARBON BONDING

All compounds can be classified in two broad categories: organic compounds and inorganic compounds. **Organic compounds** are made primarily of carbon atoms. Most matter in living organisms that is not water is made of organic compounds. *Inorganic compounds*, with a few exceptions, do not contain carbon atoms.

A carbon atom has four electrons in its outermost energy level. Most atoms become stable when their outermost energy level contains eight electrons. A carbon atom therefore readily forms four covalent bonds with the atoms of other elements. Unlike other elements, however, carbon also readily bonds with other carbon atoms, forming straight chains, branched chains, or rings, as shown in Figure 3-1. This tendency of carbon to bond with itself results in an enormous variety of organic compounds.

In the symbolic shorthand of chemistry, each line shown in Figure 3-1 represents a covalent bond formed when two atoms share a pair of electrons. A bond formed when two atoms share one pair of electrons is called a *single bond*.

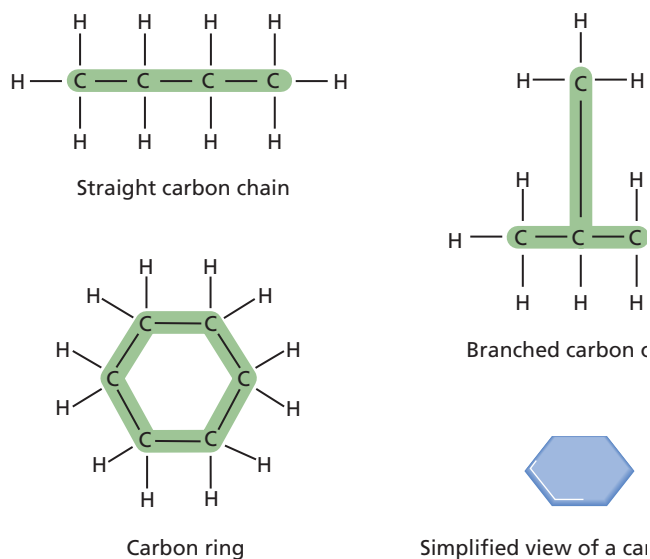


FIGURE 3-1

Carbon can bond in a number of ways to produce molecules of very different shapes, including straight chains (a), branched chains (b), and rings (c) and (d). These structures form the backbone of many different kinds of organic molecules. The carbon ring is shown with all of its atoms (c), and in a simplified version (d) commonly used in this textbook and elsewhere.

OBJECTIVES

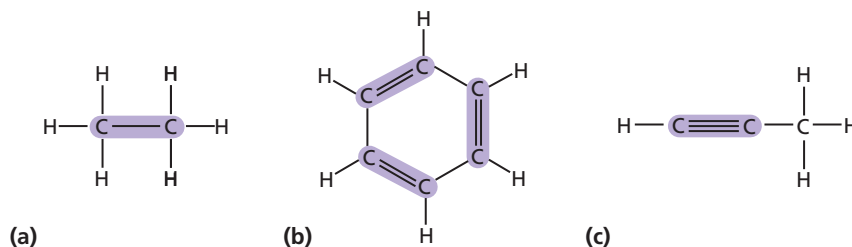
- **Distinguish** between organic and inorganic compounds.
- **Explain** the importance of carbon bonding in biological molecules.
- **Identify** functional groups in biological molecules.
- **Summarize** how large carbon molecules are synthesized and broken down.
- **Describe** how the breaking down of ATP supplies energy to drive chemical reactions.

VOCABULARY

organic compound
functional group
monomer
polymer
macromolecule
condensation reaction
hydrolysis
adenosine triphosphate (ATP)

FIGURE 3-2

Carbon atoms can form single (a), double (b), or triple (c) bonds. Organic molecules can have many different shapes and patterns of bonding.



A carbon atom can also share two or even three pairs of electrons with another atom. Figure 3-2b shows a model for an organic compound in which six carbon atoms have formed a ring. Notice that each carbon atom forms four covalent bonds: a single bond with another carbon atom, a single bond with a hydrogen atom, and a double bond with a second carbon atom. In a *double bond*—represented by two parallel lines—atoms share two pairs of electrons. A *triple bond*, the sharing of three pairs of electrons, is represented by three parallel lines in Figure 3-2c.



Quick Lab

Demonstrating Polarity

Materials disposable gloves; lab apron; safety goggles; 3 test tubes; test-tube rack; 6 mL each of cooking oil, ethanol, and water



Procedure

1. Put on disposable gloves, a lab apron, and safety goggles, and then label the test tubes "A," "B," and "C."
2. In test tube A, put 3 mL of water and 3 mL of oil.
3. In test tube B, put 3 mL of ethanol and 3 mL of oil.
4. In test tube C, put 3 mL of water and 3 mL of ethanol.
5. With your middle finger, flick each test tube to mix the contents, and allow each to sit for 10–15 minutes. Record your observations.

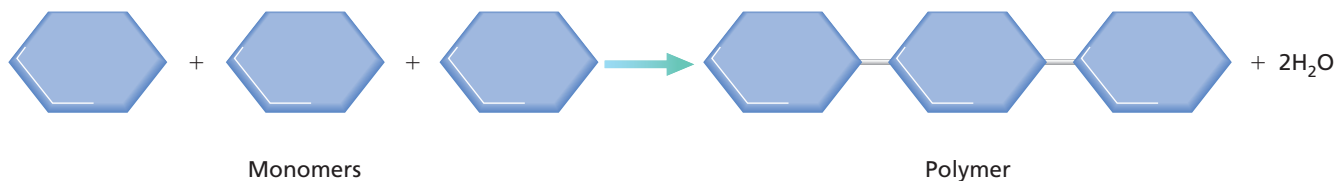
Analysis Polar molecules are soluble in water. How does this activity demonstrate polarity of molecules that contain the —OH group?

FUNCTIONAL GROUPS

In most organic compounds, clusters of atoms, called **functional groups**, influence the characteristics of the molecules they compose and the chemical reactions the molecules undergo. For example, one functional group important to living things, the hydroxyl group, —OH, can make the molecule it is attached to polar. Polar molecules are *hydrophilic* (HIE-droh-FIL-ik), or soluble in water. An *alcohol* is an organic compound with a hydroxyl group attached to one of its carbon atoms. The hydroxyl group makes an alcohol a polar molecule. The alcohol illustrated in the first row in Table 3-1 is ethanol. Other functional groups important to living things are shown in Table 3-1. These functional groups include a carboxyl group, an amino group, and a phosphate group.

TABLE 3-1 Common Functional Groups

Functional group	Structural formula	Example
Hydroxyl	—OH	
Carboxyl		
Amino		
Phosphate		



LARGE CARBON MOLECULES

Many carbon compounds are built up from smaller, simpler molecules known as **monomers** (MAH-ne-mers), such as the ones shown in Figure 3-3. As you can also see in Figure 3-3, monomers can bond to one another to form polymers (PAWL-eh-mer). A **polymer** is a molecule that consists of repeated, linked units. The units may be identical or structurally related to each other. Large polymers are called **macromolecules**. There are many types of macromolecules, such as carbohydrates, lipids, proteins and nucleic acids.

Monomers link to form polymers through a chemical reaction called a **condensation reaction**. Each time a monomer is added to a polymer, a water molecule is released. In the condensation reaction shown in Figure 3-4, two sugar molecules, glucose and fructose, combine to form the sugar sucrose, which is common table sugar. The two sugar monomers become linked by a C—O—C bridge. In the formation of that bridge, the glucose molecule releases a hydrogen ion, H^+ , and the fructose molecule releases a hydroxide ion, OH^- . The OH^- and H^+ ions that are released then combine to produce a water molecule, H_2O .

In addition to building polymers through condensation reactions, living organisms also have to break them down. The breakdown of some complex molecules, such as polymers, occurs through a process known as hydrolysis (hie-DRAHL-i-sis). In a **hydrolysis** reaction, water is used to break down a polymer. The water molecule breaks the bond linking each monomer. Hydrolysis is the reverse of a condensation reaction. The addition of water to some complex molecules, including polymers, under certain conditions can break the bonds that hold them together. For example, in Figure 3-4 reversing the reaction will result in sucrose breaking down into fructose and glucose.



FIGURE 3-3

A polymer is the result of bonding between monomers. In this example, each monomer is a six-sided carbon ring. The starch in potatoes is an example of a molecule that is a polymer.

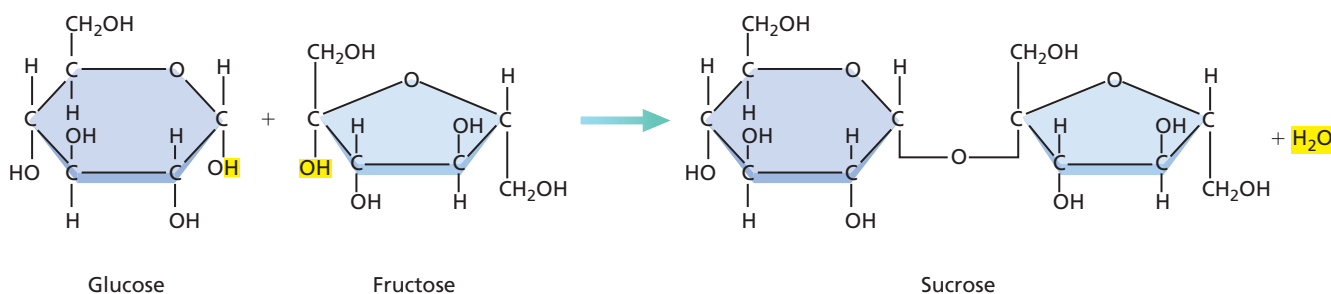
Word Roots and Origins

monomer

from the Greek *mono*, meaning "single or alone," and *meros*, meaning "a part"

FIGURE 3-4

The condensation reaction below shows how glucose links with fructose to form sucrose. One water molecule is produced each time two monomers form a covalent bond.



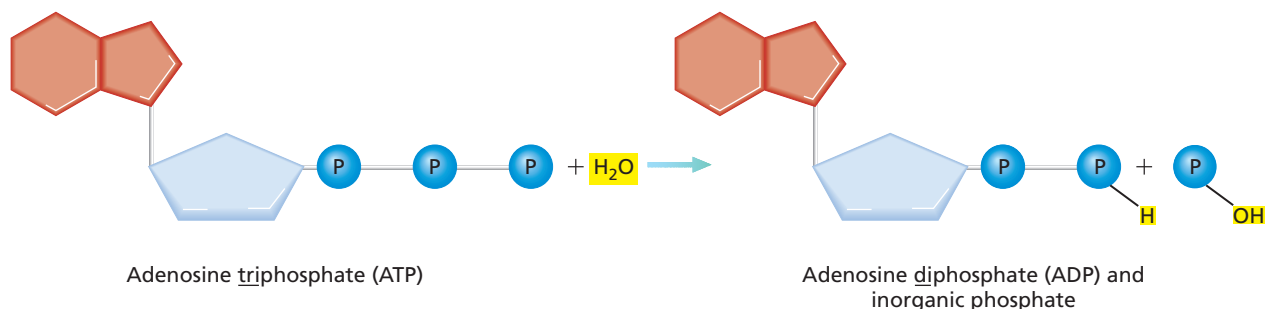


FIGURE 3-5

The hydrolysis of ATP yields adenosine diphosphate (ADP) and inorganic phosphate. In hydrolysis, a hydrogen ion from a water molecule bonds to one of the new molecules, and a hydroxide ion bonds to the other new molecule. Most hydrolysis reactions release energy.

ENERGY CURRENCY

Life processes require a constant supply of energy. This energy is available to cells in the form of certain compounds that store a large amount of energy in their overall structure. One of these compounds is **adenosine** (uh-DEN-uh-SEEN) **triphosphate**, more commonly referred to by its abbreviation, **ATP**.

The left side of Figure 3-5 shows a simplified ATP molecule structure. The 5-carbon sugar, ribose, is represented by the blue carbon ring. The nitrogen-containing compound, adenine, is represented by the 2 orange rings. The three linked phosphate groups, —PO_4^- , are represented by the blue circles with a “P.” The phosphate groups are attached to each other by covalent bonds.

The covalent bonds between the phosphate groups are more unstable than the other bonds in the ATP molecule because the phosphate groups are close together and have negative charges. Thus, the negative charges make the bonds easier to break. When a bond between the phosphate groups is broken, energy is released. This hydrolysis of ATP is used by the cell to provide the energy needed to drive the chemical reactions that enable an organism to function.

Word Roots and Origins

phosphate

from the Latin *phosphor*, meaning “morning star,” (morning stars are very bright, similar to phosphorus when it burns) and *ate*, meaning “salt”

SECTION 1 REVIEW

1. How do inorganic and organic compounds differ?
2. How do carbon’s bonding properties contribute to the existence of a wide variety of biological molecules?
3. Name four types of functional groups.
4. What role do functional groups play in the molecules in which they are found?
5. How are monomers, polymers, and macromolecules related to each other?
6. How is a polymer broken down?
7. Why is ATP referred to as the “energy currency” in living things?

CRITICAL THINKING

8. **Analyzing Concepts** Humans are about 65 percent water, and tomatoes are about 90 percent water. Yet, water is not a major building block of life. Explain.
9. **Analyzing Concepts** Carbon dioxide, CO_2 , contains carbon, yet it is considered to be inorganic. Explain.
10. **Relating Information** Condensation reactions are also referred to as dehydration synthesis. Explain how the name *dehydration synthesis* is descriptive of the process.

MOLECULES OF LIFE

Four main classes of organic compounds are essential to the life processes of all living things: carbohydrates, lipids, proteins, and nucleic acids. You will see that although these compounds are built primarily from carbon, hydrogen, and oxygen, these atoms occur in different ratios in each class of compound. Each class of compounds has different properties.

CARBOHYDRATES

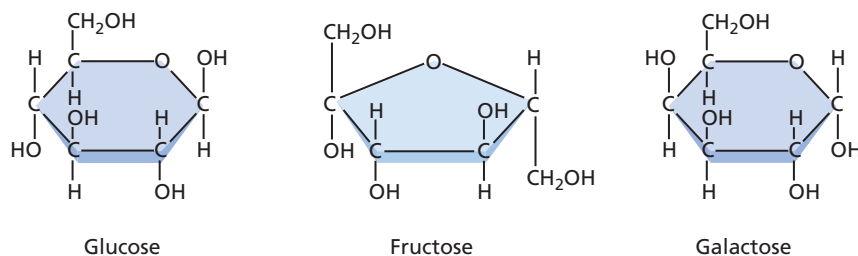
Carbohydrates are organic compounds composed of carbon, hydrogen, and oxygen in a ratio of about one carbon atom to two hydrogen atoms to one oxygen atom. The number of carbon atoms in a carbohydrate varies. Some carbohydrates serve as a source of energy. Other carbohydrates are used as structural materials. Carbohydrates can exist as monosaccharides, disaccharides, or polysaccharides.

Monosaccharides

A monomer of a carbohydrate is called a **monosaccharide** (MAHN-oh-SAK-uh-RIED). A monosaccharide—or simple sugar—contains carbon, hydrogen, and oxygen in a ratio of 1:2:1. The general formula for a monosaccharide is written as $(\text{CH}_2\text{O})_n$, where n is any whole number from 3 to 8. For example, a six-carbon monosaccharide, $(\text{CH}_2\text{O})_6$, would have the formula $\text{C}_6\text{H}_{12}\text{O}_6$.

The most common monosaccharides are glucose, fructose, and galactose, as shown in Figure 3-6. Glucose is a main source of energy for cells. Fructose is found in fruits and is the sweetest of the monosaccharides. Galactose is found in milk.

Notice in Figure 3-6 that glucose, fructose, and galactose have the same molecular formula, $\text{C}_6\text{H}_{12}\text{O}_6$, but differing structures. The different structures determine the slightly different properties of the three compounds. Compounds like these sugars, with a single chemical formula but different structural forms, are called *isomers* (IE-soh-muhrz).



OBJECTIVES

- **Distinguish** between monosaccharides, disaccharides, and polysaccharides.
- **Explain** the relationship between amino acids and protein structure.
- **Describe** the induced fit model of enzyme action.
- **Compare** the structure and function of each of the different types of lipids.
- **Compare** the nucleic acids DNA and RNA.

VOCABULARY

carbohydrate
monosaccharide
disaccharide
polysaccharide
protein
amino acid
peptide bond
polypeptide
enzyme
substrate
active site
lipid
fatty acid
phospholipid
wax
steroid
nucleic acid
deoxyribonucleic acid (DNA)
ribonucleic acid (RNA)
nucleotide

FIGURE 3-6

Glucose, fructose, and galactose have the same chemical formula, but their structural differences result in different properties among the three compounds.

Disaccharides and Polysaccharides

In living things, two monosaccharides can combine in a condensation reaction to form a double sugar, or **disaccharide** (die-SAK-e-RIED). For example in Figure 3-4, the monosaccharides fructose and glucose can combine to form the disaccharide sucrose.

A **polysaccharide** is a complex molecule composed of three or more monosaccharides. Animals store glucose in the form of the polysaccharide *glycogen*. Glycogen consists of hundreds of glucose molecules strung together in a highly branched chain. Much of the glucose that comes from food is ultimately stored in your liver and muscles as glycogen and is ready to be used for quick energy.

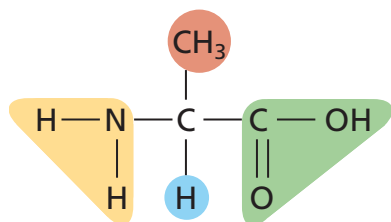
Plants store glucose molecules in the form of the polysaccharide *starch*. Starch molecules have two basic forms—highly branched chains that are similar to glycogen and long, coiled, unbranched chains. Plants also make a large polysaccharide called *cellulose*. Cellulose, which gives strength and rigidity to plant cells, makes up about 50 percent of wood. In a single cellulose molecule, thousands of glucose monomers are linked in long, straight chains. These chains tend to form hydrogen bonds with each other. The resulting structure is strong and can be broken down by hydrolysis only under certain conditions.

FIGURE 3-7

(a) Many structures, such as hair and horns are made of proteins. (b) Proteins are made up of amino acids. Amino acids differ only in the type of R group (shown in red) they carry. Polar R groups can dissolve in water, but nonpolar R groups cannot. (c) Amino acids have complex structures, so, in this and other textbooks, they are often simplified into balls.



(a)



(b) Alanine (an amino acid)



(c) Simplified version of amino acid

PROTEINS

Proteins are organic compounds composed mainly of carbon, hydrogen, oxygen, and nitrogen. Like most of the other biological macromolecules, proteins are formed from the linkage of monomers called **amino acids**. Hair and horns, as shown in Figure 3-7a, are made mostly of proteins, as are skin, muscles and many biological catalysts (enzymes).

Amino Acids

There are 20 different amino acids, and all share a basic structure. As Figure 3-7b shows, each amino acid contains a central carbon atom covalently bonded to four other atoms or functional groups. A single hydrogen atom, highlighted in blue in the illustration, bonds at one site. A carboxyl group, —COOH , highlighted in green, bonds at a second site. An amino group, —NH_2 , highlighted in yellow, bonds at a third site. A side chain called the *R group*, highlighted in red, bonds at the fourth site.

The main difference among the different amino acids is in their R groups. The R group can be complex or it can be simple, such as the CH_3 group shown in the amino acid alanine in Figure 3-7b. The differences among the amino acid R groups gives different proteins very different shapes. The different shapes allow proteins to carry out many different activities in living things. Amino acids are commonly shown in a simplified way such as balls, as shown in Figure 3-7c.

Dipeptides and Polypeptides

Figure 3-8a shows how two amino acids bond to form a *dipeptide* (die-PEP-tied). In this condensation reaction, the two amino acids form a covalent bond, called a **peptide bond** (shaded in blue in Figure 3-8a) and release a water molecule.

Amino acids often form very long chains called **polypeptides** (PAHL-i-PEP-tiedZ). Proteins are composed of one or more polypeptides. Some proteins are very large molecules, containing hundreds of amino acids. Often, these long proteins are bent and folded upon themselves as a result of interactions—such as hydrogen bonding—between individual amino acids. Protein shape can also be influenced by conditions such as temperature and the type of solvent in which a protein is dissolved. For example, cooking an egg changes the shape of proteins in the egg white. The firm, opaque result is very different from the initial clear, runny material.

Enzymes

Enzymes—RNA or protein molecules that act as biological catalysts—are essential for the functioning of any cell. Many enzymes are proteins. Figure 3-9 shows an induced fit model of enzyme action.

Enzyme reactions depend on a physical fit between the enzyme molecule and its specific **substrate**, the reactant being catalyzed. Notice that the enzyme has folds, or an **active site**, with a shape that allows the substrate to fit into the active site. An enzyme acts only on a specific substrate because only that substrate fits into its active site. The linkage of the enzyme and substrate causes a slight change in the enzyme's shape. The change in the enzyme's shape weakens some chemical bonds in the substrate, which is one way that enzymes reduce activation energy, the energy needed to start the reaction. After the reaction, the enzyme releases the products. Like any catalyst, the enzyme itself is unchanged, so it can be used many times.

An enzyme may not work if its environment is changed. For example, change in temperature or pH can cause a change in the shape of the enzyme or the substrate. If such a change happens, the reaction that the enzyme would have catalyzed cannot occur.

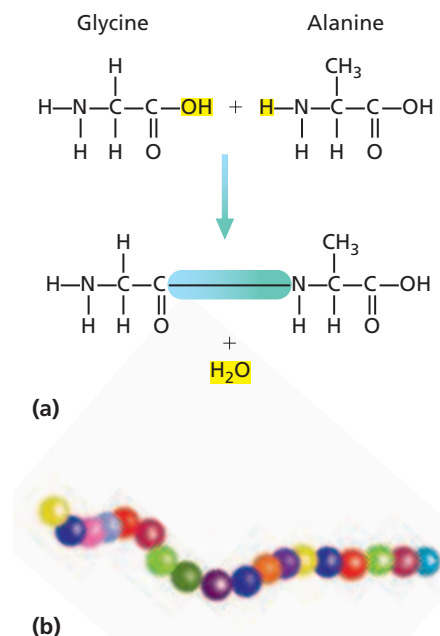


FIGURE 3-8

(a) The peptide bond (shaded blue) that binds amino acids together to form a polypeptide results from a condensation reaction that produces water. (b) Polypeptides are commonly shown as a string of balls in this textbook and elsewhere. Each ball represents an amino acid.

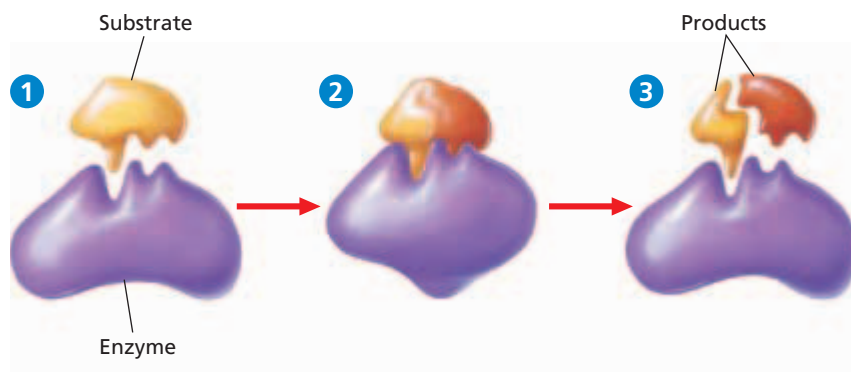


FIGURE 3-9

1 In the induced fit model of enzyme action, the enzyme can attach only to a substrate (reactant) with a specific shape. 2 The enzyme then changes and reduces the activation energy of the reaction so reactants can become products. 3 The enzyme is unchanged and is available to be used again.

TREATING AND PREVENTING DIABETES

Proteins play many important roles in living organisms. The hormone *insulin* is a protein that stimulates cells to take up glucose. More than 18 million Americans have *diabetes*, an inability of the body to make or respond to insulin.

When the body cannot make or respond to insulin, the body's cells must switch to burning mainly fat as their fuel. The resulting high levels of fat in the blood can cause cardiovascular disease. In addition, the glucose that accumulates in the blood causes other problems. For example, diabetes can have serious complications, including kidney disease, heart failure, blindness, and amputation of the lower limbs. Some symptoms of diabetes include increased thirst, frequent urination, fatigue, and weight loss.



Regular physical activity can help reduce the risk of developing type 2 diabetes.

Type 1 Diabetes

Between 5 and 10 percent of people who suffer from diabetes have type 1 diabetes, which usually starts in childhood. The body's immune system mistakenly attacks cells in the pancreas that make insulin. If untreated, type 1 diabetes is usually fatal.

Treating Type 1 Diabetes

People with type 1 diabetes require a carefully monitored diet, physical activity, home blood glucose testing several times a day, and multiple daily insulin injections. In the past, insulin was delivered by shots. Now, there are pumps that regularly deliver small amounts of insulin. The pumps can be implanted surgically and refilled periodically by injection.

Type 2 Diabetes

The majority of people who suffer from diabetes have type 2 diabetes, which can begin at any age. A diet high in sugars and fats, a sedentary lifestyle, and being overweight can each increase the chances of developing this type of diabetes. Type 2 diabetes occurs when the pancreas cannot keep up with the demand for insulin or the cells become resistant to insulin's effects.

Treating Type 2 Diabetes

For type 2 diabetes, treatment typically includes a healthy diet, regular exercise, and home blood glucose testing. Some people must also take oral medication and/or insulin.

About 40 percent of people with type 2 diabetes require insulin injections.

Preventing Diabetes

There is currently not a way to prevent type 1 diabetes. But exercise, a healthy diet, and insulin injections can allow a person to lead a normal life. Ways to prevent type 2 diabetes include exercising regularly and eating a healthy diet.

Future Treatments for Diabetes

Medical researchers are working on devices that can monitor blood sugar better. Other researchers are trying to improve the delivery of insulin by using timed-release drugs or by developing smaller implants. Some researchers are working on improving organ transplant surgery and finding genes linked to diabetes.

REVIEW

1. Distinguish between type 1 and type 2 diabetes.
2. Why is insulin important?
3. **Critical Thinking** A friend asks, "Why should I worry? By the time I'm old, they'll have diabetes cured anyway." What is your opinion? Defend your answer.



LIPIDS

Lipids are large, nonpolar organic molecules. They do not dissolve in water. Lipids include triglycerides (trie-GLIS-uh-IEDZ), phospholipids, steroids, waxes, and pigments. Lipid molecules have a higher ratio of carbon and hydrogen atoms to oxygen atoms than carbohydrates have. Because lipid molecules have larger numbers of carbon-hydrogen bonds per gram than other organic compounds do, they store more energy per gram.

Fatty Acids

Fatty acids are unbranched carbon chains that make up most lipids. Figure 3-10 shows that a fatty acid contains a long carbon chain (from 12 to 28 carbons) with a carboxyl group, —COOH , attached at one end. The two ends of the fatty-acid molecule have different properties. The carboxyl end is polar and is thus *hydrophilic* or attracted to water molecules. In contrast, the hydrocarbon end of the fatty-acid molecule is nonpolar. This end tends not to interact with water molecules and is said to be *hydrophobic* (HIE-droh-FOH-bik), or “water fearing.”

In saturated fatty acids, such as palmitic acid, which is shown in Figure 3-10, each carbon atom is covalently bonded to four atoms. The carbon atoms are in effect full, or *saturated*. In contrast, linoleic acid, also shown in Figure 3-10, has carbon atoms that are not bonded to the maximum number of atoms to which they can bond. Instead, they have formed double bonds within the carbon chain. This type of fatty acid is said to be *unsaturated*.

Triglycerides

Three classes of lipids important to living things contain fatty acids: triglycerides (fats), phospholipids, and waxes. A *triglyceride* is composed of three molecules of fatty acid joined to one molecule of the alcohol glycerol. Saturated triglycerides are composed of saturated fatty acids. They typically have high melting points and tend to be hard at room temperature. Common dietary saturated triglycerides include butter and fats in red meat. In contrast, unsaturated triglycerides are composed of unsaturated fatty acids and are usually soft or liquid at room temperature. Unsaturated triglycerides are found primarily in plant seeds where they serve as an energy and carbon source for germinating plants.

Phospholipids

Phospholipids have two, rather than three, fatty acids attached to a molecule of glycerol. They have a phosphate group attached to the third carbon of the glycerol. As shown in Figure 3-11, the cell membrane is made of two layers of phospholipids, called the *lipid bilayer*. The inability of lipids to dissolve in water allows the membrane to form a barrier between the inside and outside of the cell.

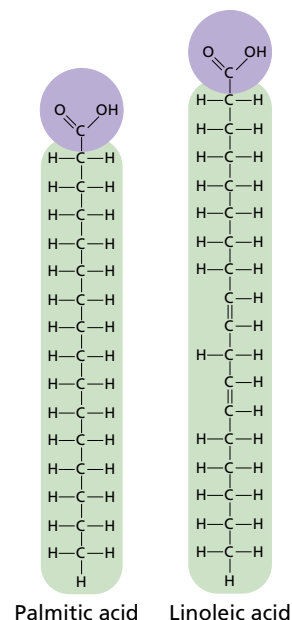
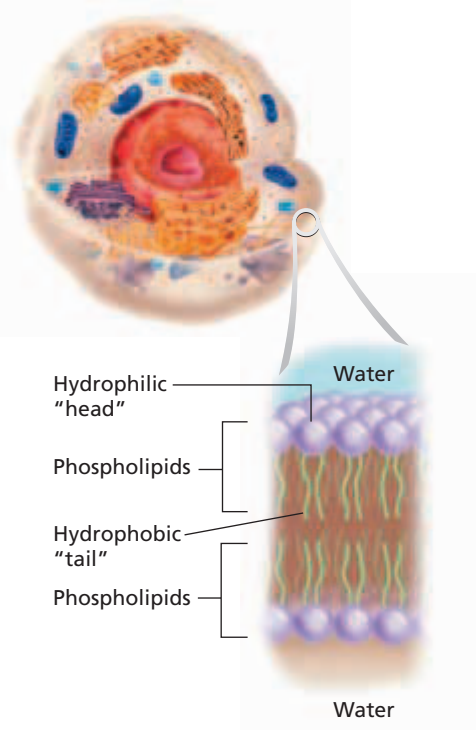


FIGURE 3-10

Fatty acids have a polar carboxyl head, highlighted in purple, and a nonpolar hydrocarbon tail, highlighted in green.

FIGURE 3-11

The lipid bilayer of a cell membrane is a double row of phospholipids. The “tails” face each other. The “head” of a phospholipid, which contains a phosphate group, is polar and hydrophilic. The two tails are two fatty acids and are nonpolar and hydrophobic.





Waxes

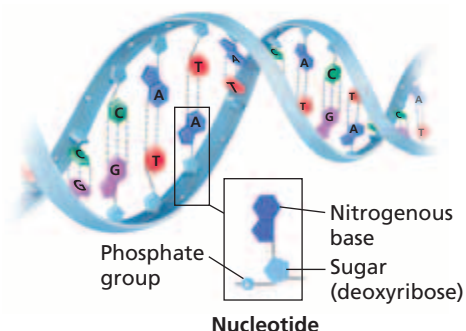
A **wax** is a type of structural lipid consisting of a long fatty-acid chain joined to a long alcohol chain. Waxes are waterproof, and in plants, form a protective coating on the outer surfaces. Waxes also form protective layers in animals. For example, earwax helps prevent microorganisms from entering the ear canal.

Steroids

Unlike most other lipids, which are composed of fatty acids, **steroid** molecules are composed of four fused carbon rings with various functional groups attached to them. Many animal hormones, such as the male hormone testosterone, are steroid compounds. One of the most familiar steroids in humans is cholesterol. Cholesterol is needed by the body for nerve and other cells to function normally. It is also a component of the cell membrane.

FIGURE 3-12

DNA as shown below, and RNA, are very large molecules formed from nucleotides linked together in a chain. A nucleotide consists of a phosphate group, a five-carbon sugar, and a ring-shaped nitrogenous base.



NUCLEIC ACIDS

Nucleic acids are very large and complex organic molecules that store and transfer important information in the cell. There are two major types of nucleic acids: deoxyribonucleic acid and ribonucleic acid.

Deoxyribonucleic acid, or **DNA**, contains information that determines the characteristics of an organism and directs its cell activities. **Ribonucleic acid** (RIE-boh-noo-KLEE-ik), or **RNA**, stores and transfers information from DNA that is essential for the manufacturing of proteins. RNA molecules can also act as enzymes. Both DNA and RNA are polymers, composed of thousands of linked monomers called *nucleotides* (NOO-klee-uh-TIEDS). As shown in Figure 3-12, each **nucleotide** is made of three main components: a phosphate group, a five-carbon sugar, and a ring-shaped nitrogenous base.

SECTION 2 REVIEW

1. Compare the structure of monosaccharides, disaccharides, and polysaccharides.
2. How are proteins constructed from amino acids?
3. How do amino acids differ from one another?
4. Describe a model of enzyme action.
5. Why do phospholipids orient in a bilayer when in a watery environment, such as a cell?
6. Describe how the three major types of lipids differ in structure from one another.
7. What are the functions of the two types of nucleic acids?

CRITICAL THINKING

8. **Applying Information** Before a long race, runners often "carbo load." This means that they eat substantial quantities of carbohydrates. How might this help their performance?
9. **Recognizing Relationships** High temperatures can weaken bonds within a protein molecule. How might this explain the effects of using a hot curling iron or rollers in one's hair?
10. **Applying Information** You want to eat more unsaturated than saturated fats. Name examples of foods you would eat more of and less of.

CHAPTER HIGHLIGHTS

SECTION 1

Carbon Compounds

- Organic compounds contain carbon atoms and are found in living things. Most inorganic compounds do not contain carbon atoms.
- Carbon atoms can readily form four covalent bonds with other atoms including other carbon atoms. The carbon bonds allow the carbon atoms to form a wide variety of simple and complex organic compounds.
- Functional groups are groups of atoms that influence the properties of molecules and the chemical reactions in which the molecules participate.
- Condensation reactions join monomers (small simple molecules) to form polymers. A condensation reaction releases water as a by-product. In a hydrolysis reaction, water is used to split polymers into monomers.
- Adenosine triphosphate (ATP) stores and releases energy during cell processes enabling organisms to function.

Vocabulary

organic compound (p. 51)
functional group (p. 52)
monomer (p. 53)

polymer (p. 53)
macromolecule (p. 53)

condensation reaction (p. 53)
hydrolysis (p. 53)

adenosine triphosphate (ATP) (p. 54)

SECTION 2

Molecules of Life

- There are four main classes of organic compounds: carbohydrates, proteins, lipids, and nucleic acids.
- Carbohydrates are made up of monomers called monosaccharides. Two monosaccharides join to form a double sugar called a disaccharide. A complex sugar, or polysaccharide, is made of three or more monosaccharides.
- Carbohydrates such as glucose, are a source of energy and are used as structural materials in organisms.
- Proteins have many functions including structural, defensive, and catalytic. Proteins are made up of monomers called amino acids. The sequence of amino acids determines a protein's shape and function. A long chain of amino acids is called a polypeptide, which is made up of amino acids joined by peptide bonds.
- Enzymes speed up chemical reactions and bind to specific substrates. The binding of a substrate with an enzyme causes a change in the enzyme's shape and reduces the activation energy of the reaction.
- Lipids are nonpolar molecules that store energy and are an important part of cell membranes. Most lipids contain fatty acids, molecules that have a hydrophilic end and a hydrophobic end.
- There are three kinds of lipids: Triglycerides consist of three fatty acids and one molecule of glycerol. Phospholipids, which make up cell membranes, consist of two fatty acids and one glycerol molecule. A wax is made of one long fatty acid chain joined to one long alcohol.
- The nucleic acid, deoxyribonucleic acid (DNA), contains all the genetic information for cell activities. Ribonucleic acid (RNA) molecules play many key roles in building of proteins and can act as enzymes.

Vocabulary

carbohydrate (p. 55)
monosaccharide (p. 55)
disaccharide (p. 56)
polysaccharide (p. 56)
protein (p. 56)
amino acid (p. 56)

peptide bond (p. 57)
polypeptide (p. 57)
enzyme (p. 57)
substrate (p. 57)
active site (p. 57)
lipid (p. 59)

fatty acid (p. 59)
triglyceride (p. 59)
phospholipid (p. 59)
wax (p. 60)
steroid (p. 60)
nucleic acid (p. 60)

deoxyribonucleic acid (DNA) (p. 60)
ribonucleic acid (RNA) (p. 60)
nucleotide (p. 60)

CHAPTER REVIEW

USING VOCABULARY

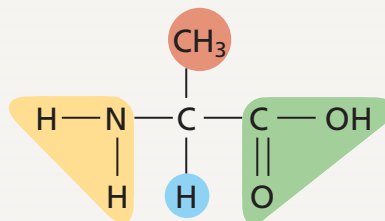
- For each pair of terms, explain how the meanings of the terms differ.
 - monomer* and *polymer*
 - functional group* and *macromolecule*
 - monosaccharide* and *disaccharide*
 - polypeptide* and *protein*
 - nucleic acid* and *nucleotide*
- For each pair of terms, explain the relationship between the terms.
 - fatty acid* and *triglyceride*
 - substrate* and *enzyme*
- Use the following terms in the same sentence: *monomer*, *polymer*, *condensation reaction*, and *hydrolysis*.
- Word Roots and Origins** The word *organic* is derived from the Greek *organikos*, which means "organ." Explain how the word *organic* is descriptive of most carbon compounds.

UNDERSTANDING KEY CONCEPTS

- Differentiate** between organic and inorganic compounds.
- Relate** the properties of carbon to the formation of organic compounds.
- Summarize** how functional groups help determine the properties of organic compounds.
- Compare** how organic compounds are built to how they are broken down.
- Explain** the role of ATP in cellular activities.
- List** the four major classes of organic compounds.
- Describe** the general structure of carbohydrates.
- Define** the term *isomer*.
- Summarize** the differences between simple sugars, double sugars, and complex sugars.
- Describe** how a protein's structure is determined by the arrangement of amino acids.
- State** the basic structure of an amino acid.
- Compare** the processes used in the formation of a dipeptide and a disaccharide.
- Summarize** the induced fit model of enzyme activity.
- Differentiate** between saturated and unsaturated triglycerides.
- Compare** the structures of triglycerides, phospholipids, and steroids.
- State** how steroids differ from other lipids.
- Identify** an important characteristic of waxes in living organisms.
- Compare** two kinds of nucleic acids.
- Name** the three parts of a nucleotide.
- CONCEPT MAPPING** Use the following terms to create a concept map that describes the different types of organic compounds: *dipeptide*, *triglycerides*, *RNA*, *phospholipids*, *carbohydrates*, *monosaccharide*, *amino acid*, *disaccharide*, *polypeptide*, *polysaccharide*, *proteins*, *DNA*, *lipids*, *nucleic acids*, *steroids*, and *waxes*.

CRITICAL THINKING

- Applying Information** What is the chemical formula for a monosaccharide that has three carbons?
- Analyzing Concepts** Starch easily dissolves in water. Cellulose does not. Both substances, however, consist of chains of glucose molecules. What structural difference between starch and cellulose might account for their different behavior in water?
- Interpreting Graphics** Identify the type of organic molecule shown below. Identify each of the functional groups shaded in yellow, red, and green.



- Making Inferences** Analysis of an unknown substance showed that it has the following characteristics: It contains carbon, hydrogen, and oxygen and it is soluble in oil but not water. Predict what kind of substance it might be. Explain your answer.
- Applying Information** Many birds store significant amounts of energy to power flight during winter migration. What type of organic molecule might be best suited for energy storage? Explain.

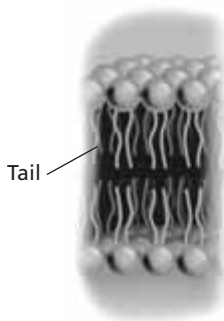


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- Which of the following is not a function of polysaccharides?
A. energy source
B. energy storage
C. structural support
D. storage of genetic information
- Which of the following statements is false?
F. A wax is a lipid.
G. Starch is a lipid.
H. Saturated fats are solid at room temperature.
J. Unsaturated fats are liquid at room temperature.
- Which of the following molecules stores hereditary information?
A. ATP
B. DNA
C. protein
D. carbohydrates
- What is the name of the molecule in plants that stores sugars?
F. starch
G. protein
H. cellulose
J. glycogen

INTERPRETING GRAPHICS: The figure below illustrates the basic structure of a cell membrane. Use the figure to answer the questions that follow.

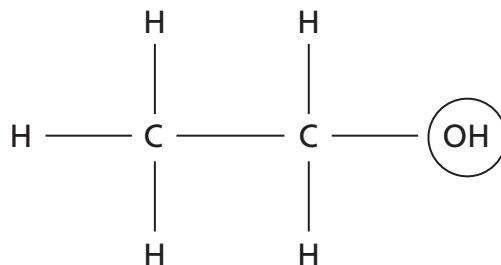


- Which of the following molecules make up the basic structure of a cell membrane?
A. waxes
B. steroids
C. fatty acids
D. phospholipids
- The "tails" of the molecules in the figure orient away from water. Which of the following describes the tail's movement away from water?
F. polar
G. adhesive
H. hydrophilic
J. hydrophobic

DIRECTIONS: Complete the following analogy.

7. simple sugars : carbohydrates :: amino acids :
A. lipids
B. proteins
C. nucleic acids
D. amino groups

INTERPRETING GRAPHICS: The figure below represents the structural formula of a molecule. Use the figure to answer the question that follows.



8. What is the name of the functional group circled in the structural formula above?
F. amino
G. hydroxyl
H. phosphate
J. carboxyl

SHORT RESPONSE

Proteins are affected by environmental conditions such as heat and pH.

Explain why the process of cooking an egg cannot be reversed.

EXTENDED RESPONSE

Enzymes are essential for the functioning of all cells.

Part A Explain what enzymes do that is essential for cell function.

Part B Explain the induced fit model of enzyme action.

Test TIP

When writing an answer to an extended response question, make an outline of what you plan to write before writing your answer. Although it may take a little more time, the answer will be easier to write.

Identifying Organic Compounds in Foods

OBJECTIVES

- Determine whether specific nutrients are present in a solution of unknown composition.
- Perform chemical tests using substances called *indicators*.

PROCESS SKILLS

- experimenting
- observing
- identifying
- measuring

MATERIALS

- lab apron
- safety goggles
- disposable gloves
- 500 mL beaker
- hot plate
- 9 test tubes
- labeling tape
- marker
- 10 mL graduated cylinder
- Benedict's solution
- 9 dropping pipets
- glucose solution
- unknown solution
- distilled water
- 9 glass stirring rods
- tongs or test-tube holder
- test-tube rack
- albumin solution
- sodium hydroxide solution
- copper sulfate solution
- vegetable oil
- Sudan III solution

Background

1. Carbohydrates, proteins, and lipids are nutrients that make up all living things. Some foods, such as table sugar, contain only one of these nutrients. Most foods, however, contain mixtures of proteins, carbohydrates, and lipids. You can confirm this fact by reading the information in the Nutrition Facts box found on any food label.
2. In this investigation, you will use chemical substances, called *indicators*, to identify the presence of specific nutrients in an unknown solution. By comparing the color change an indicator produces in the unknown food sample with the change it produces in a sample of known composition, you can determine whether specific organic compounds are present in the unknown sample.

3. Benedict's solution is used to determine the presence of monosaccharides, such as glucose. A mixture of sodium hydroxide and copper sulfate determines the presence of some proteins. This procedure is called the biuret test. Sudan III is used to determine the presence of lipids.

Procedure



CAUTION Put on a lab apron, safety goggles, and

gloves. In this lab, you will be working with chemicals that can harm your skin and eyes or stain your skin and clothing. If you get a chemical on your skin or clothing, wash it off at the sink while calling to your teacher. If you get a chemical in your eyes, immediately flush it out at the eyewash station while calling to your teacher. As you perform each test, record your data in your lab report, organized in a table such as the one on the next page.

Test 1: Monosaccharides

1.



CAUTION Do not touch the hot plate. Use tongs to move heated objects. Turn off the hot plate when not in use. Do not plug in or unplug the hot plate with wet hands. Make a water bath by filling a 500 mL beaker half full with water. Then, put the beaker on a hot plate, and bring the water to a boil.

2. While you wait for the water to boil, label one test tube "1-glucose," label the second test tube "1-unknown," and label the third test tube "1-water." Using the graduated cylinder, measure 5 mL of Benedict's solution, and add it to the "1-glucose" test tube. Repeat the procedure, adding 5 mL of Benedict's solution each to the "1-unknown" test tube and "1-water" test tube.
3. Using a dropping pipet or eyedropper, add 10 drops of glucose solution to the "1-glucose" test tube. Using a second dropping pipet, add 10 drops of the unknown solution to the "1-unknown" test tube.

IDENTIFICATION OF SPECIFIC NUTRIENTS BY CHEMICAL INDICATORS

Test	Nutrient in test solution	Nutrient category (protein, lipid, etc.)	Result for known sample	Result for unknown sample	Result for distilled water
1					
2					
3					

Using a third dropping pipet, add 10 drops of distilled water to the "1-water" test tube. Mix the contents of each test tube with a clean stirring rod. **(It is important not to contaminate test solutions by using the same dropping pipet or stirring rod in more than one solution. Use a different dropping pipet and stirring rod for each of the test solutions.)**

- When the water boils, use tongs to place the test tubes in the water bath. Boil the test tubes for 1 to 2 minutes.
-  **CAUTION** Do not touch the test tubes with your hands. They will be very hot. Use tongs to remove the test tubes from the water bath and place them in the test-tube rack. As the test tubes cool, an orange or red precipitate will form if large amounts of glucose are present. If small amounts of glucose are present, a yellow or green precipitate will form. Record your results in your data table.

Test 2: Proteins

-  Label one clean test tube "2-albumin," label a second test tube "2-unknown," and label a third test tube "2-water." Using a dropping pipet, add 40 drops of albumin solution to the "2-albumin" test tube. Using a second dropping pipet, add 40 drops of unknown solution to the "2-unknown" test tube. Using a third dropping pipet, add 40 drops of water to the "2-water" test tube.
- Add 40 drops of sodium hydroxide solution to each of the three test tubes. Mix the contents of each test tube with a clean stirring rod.
- Add a few drops of copper sulfate solution, one drop at a time, to the "2-albumin" test tube. Stir the solution with a clean stirring rod after each drop. Note the number of drops required to cause the color of the solution in the test tube to change. Then, add the same number of drops of copper sulfate solution to the "2-unknown" and "2-water" test tubes.
- Record your results in your data table.

Test 3: Lipids

- Label one clean test tube "3-vegetable oil," label a second test tube "3-unknown," and label a third test tube "3-water." Using a dropping pipet, add 5 drops of vegetable oil to the "3-vegetable oil" test tube. Using a second dropping pipet, add 5 drops of the unknown solution to the "3-unknown" test tube. Using a third dropping pipet, add 5 drops of water to the "3-water" test tube.
-  **CAUTION** Sudan III solution will stain your skin and clothing. Promptly wash off spills to minimize staining. Do not use Sudan III solution in the same room with an open flame. Using a clean dropping pipet, add 3 drops of Sudan III solution to each test tube. Mix the contents of each test tube with a clean stirring rod.
- Record your results in your data table.
-   Clean up your materials, and wash your hands before leaving the lab.

Analysis and Conclusions

- Based on the results you recorded in your data table, identify the nutrient or nutrients in the unknown solution.
- What are the experimental controls in this investigation?
- Explain how you were able to use the color changes of different indicators to determine the presence of specific nutrients in the unknown substance.
- List four potential sources of error in this investigation.

Further Inquiry

Is there a kind of macromolecule that the tests in this lab did not test for? If so, list the kinds of macromolecules not tested for, and give one reason why they were not tested for.

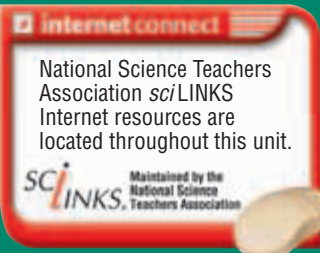
UNIT 2

CHAPTERS

- 4 **Cell Structure and Function**
- 5 **Homeostasis and Cell Transport**
- 6 **Photosynthesis**
- 7 **Cellular Respiration**
- 8 **Cell Reproduction**



References to *Scientific American* project ideas are located throughout this unit.

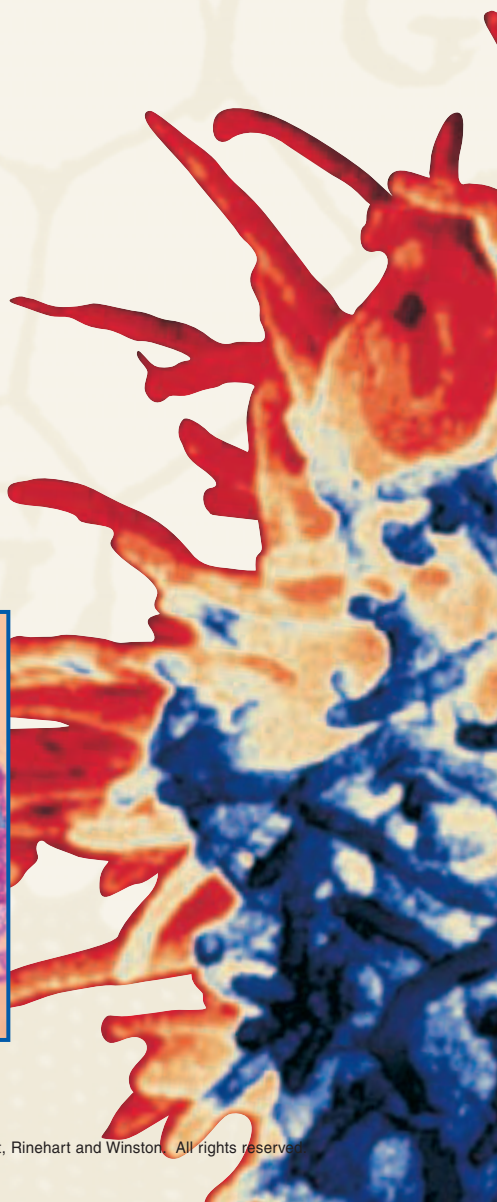
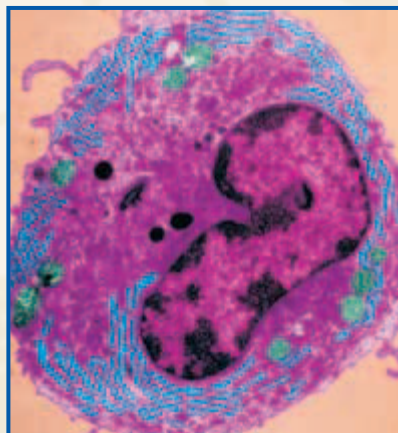


CELL BIOLOGY

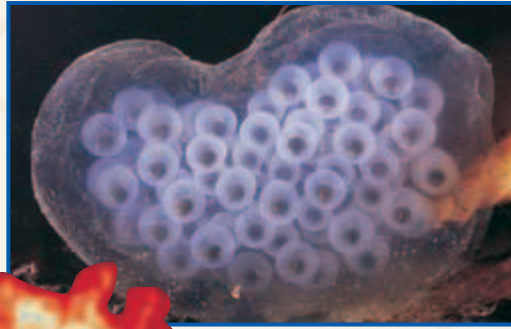
“The cell is the natural granule of life in the same way as the atom is the natural granule of simple, elemental matter. If we are to take the measure of the transit to life and determine its precise nature, we must try to understand the cell.”

From "The Advent of Life," from *The Phenomenon of Man*, by Pierre Teilhard de Chardin. Copyright © 1955 by Editions du Seuil. English translation copyright © 1959 by William Collins Sons & Co. Ltd., London and Harper & Row Publishers, Inc., New York. Reprinted by permission of HarperCollins Publishers, Inc.

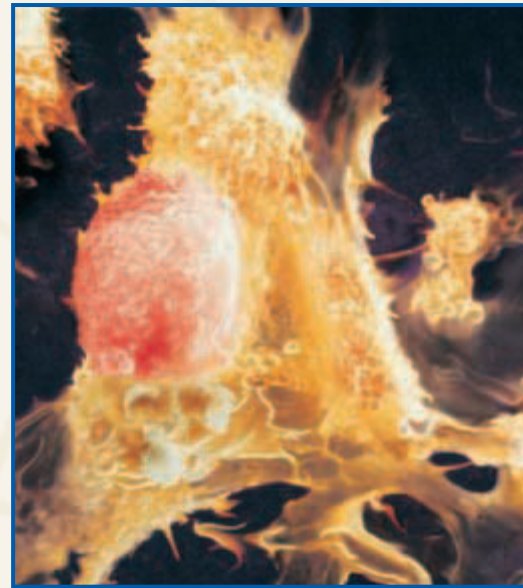
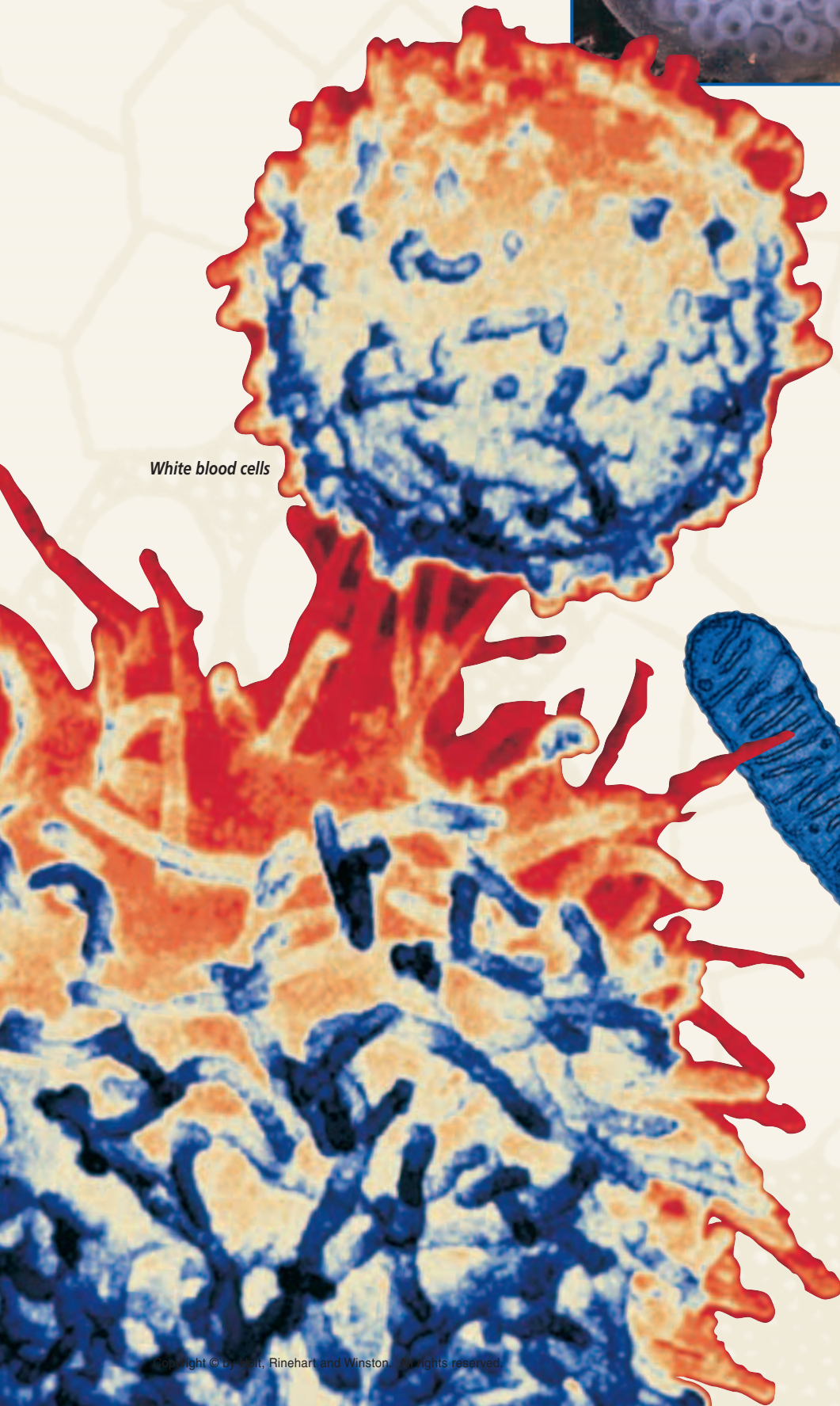
Eukaryotic cells contain a number of complex internal structures.



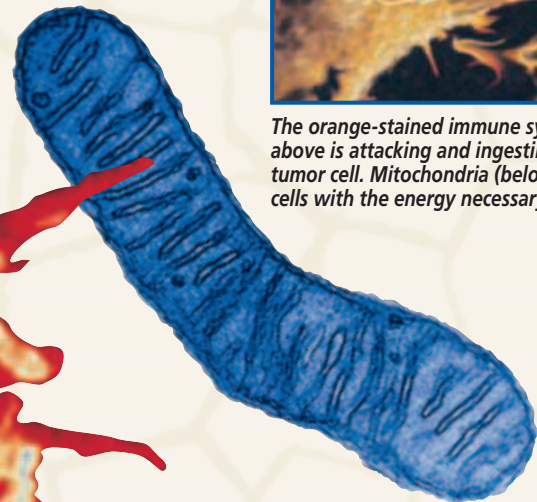
Most cells are very small, but these frog eggs can be seen with the unaided eye.



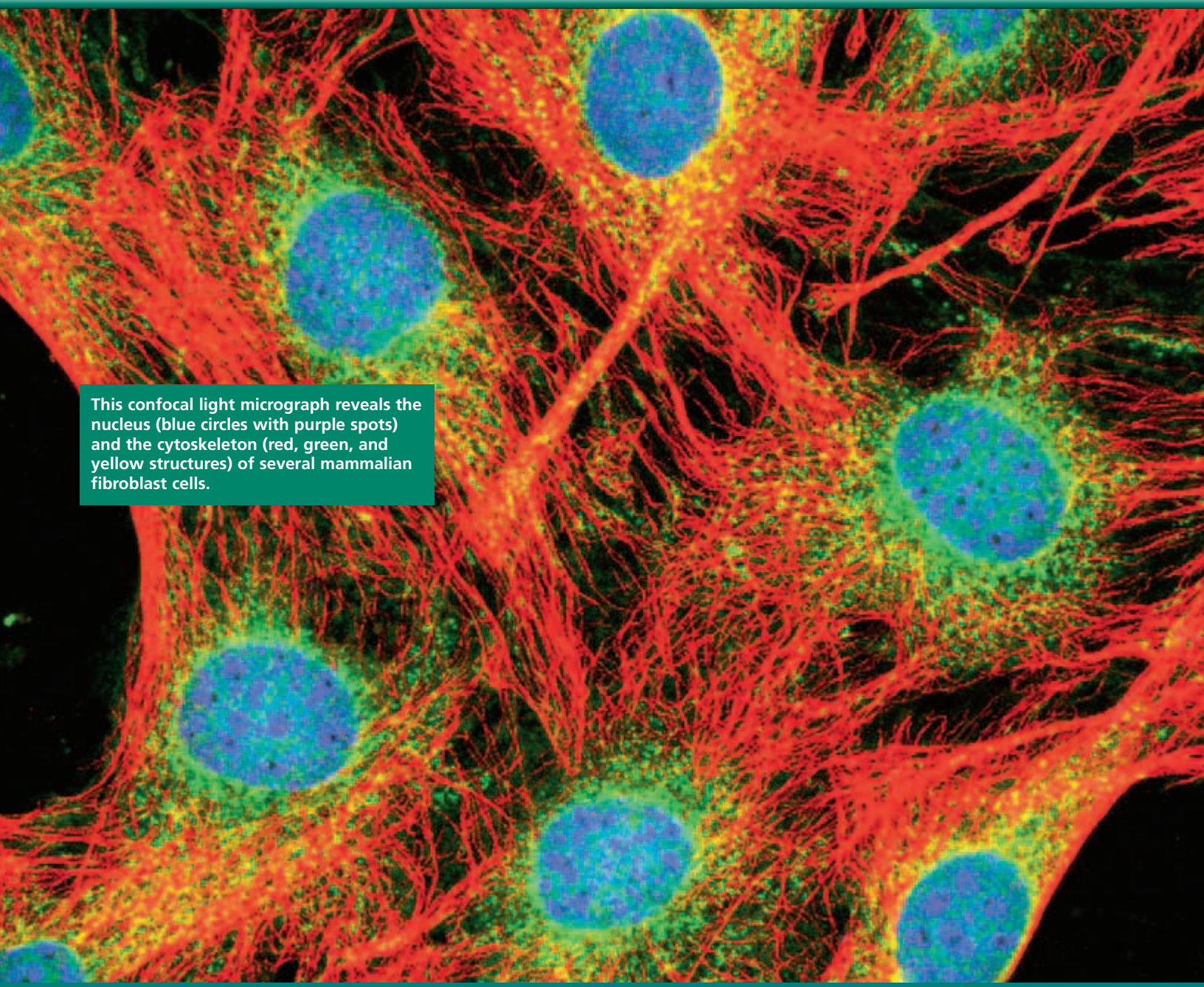
White blood cells



The orange-stained immune system cell shown above is attacking and ingesting the red-stained tumor cell. Mitochondria (below left) provide cells with the energy necessary for life.



CELL STRUCTURE AND FUNCTION



This confocal light micrograph reveals the nucleus (blue circles with purple spots) and the cytoskeleton (red, green, and yellow structures) of several mammalian fibroblast cells.

SECTION 1 *The History of Cell Biology*

SECTION 2 *Introduction to Cells*

SECTION 3 *Cell Organelles and Features*

SECTION 4 *Unique Features of Plant Cells*



Unit 1—Cell Transport and
Homeostasis
Topics 1–2

THE HISTORY OF CELL BIOLOGY

Both living and nonliving things are made of atoms, molecules, and compounds. How are living and nonliving things different? The discovery of the cell was an important step toward answering this question.

THE DISCOVERY OF CELLS

All living things are made up of one or more cells. A **cell** is the smallest unit that can carry on all of the processes of life. Beginning in the 17th century, curious naturalists were able to use microscopes to study objects too small to be seen with the unaided eye. Their studies led them to propose the cellular basis of life.

Hooke

In 1665, English scientist Robert Hooke studied nature by using an early *light microscope*, such as the one in Figure 4-1a. A light microscope is an instrument that uses optical lenses to magnify objects by bending light rays. Hooke looked at a thin slice of cork from the bark of a cork oak tree. “I could exceedingly plainly perceive it to be all perforated and porous,” Hooke wrote. He described “a great many little boxes” that reminded him of the cubicles or “cells” where monks live. When Hooke focused his microscope on the cells of tree stems, roots, and ferns, he found that each had similar little boxes. The drawings that Hooke made of the cells he saw are shown in Figure 4-1b. The “little boxes” that Hooke observed were the remains of dead plant cells, such as the cork cells shown in Figure 4-1c.

OBJECTIVES

- **Name** the scientists who first observed living and nonliving cells.
- **Summarize** the research that led to the development of the cell theory.
- **State** the three principles of the cell theory.
- **Explain** why the cell is considered to be the basic unit of life.

VOCABULARY

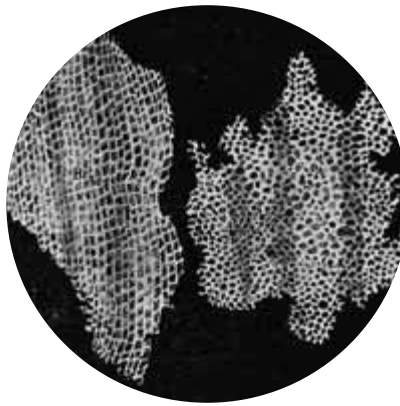
cell
cell theory

FIGURE 4-1

Robert Hooke used an early microscope (a) to see cells in thin slices of cork. His drawings of what he saw (b) indicate that he had clearly observed the remains of cork cells (300×) (c).



(a)



(b)



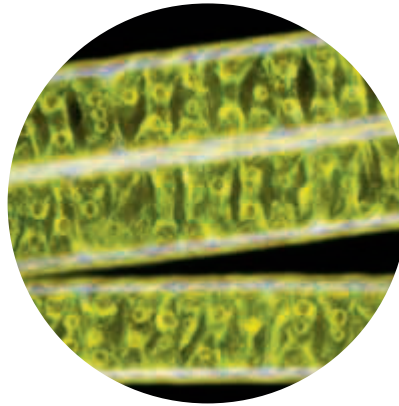
(c)



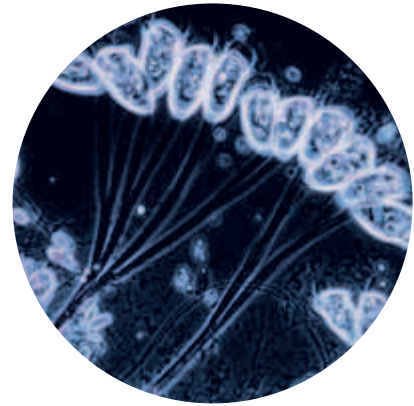
(a)

FIGURE 4-2

Anton van Leeuwenhoek (1632–1723) is shown here with one of his hand-held lenses (a). Leeuwenhoek observed an alga of the genus *Spirogyra* (b) and a protist of the genus *Vorticella* (c).



(b)



(c)

Leeuwenhoek

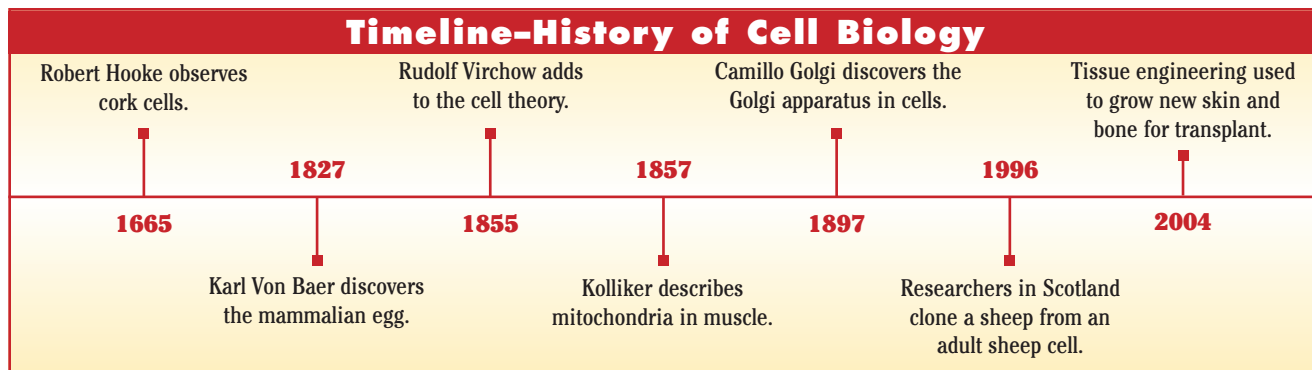
The first person to observe living cells was a Dutch trader named Anton van Leeuwenhoek. Leeuwenhoek made microscopes that were simple and tiny, but he ground lenses so precisely that the magnification was 10 times that of Hooke's instruments. In 1673, Leeuwenhoek, shown in Figure 4-2a, was able to observe a previously unseen world of microorganisms. He observed cells with green stripes from an alga of the genus *Spirogyra*, as shown in Figure 4-2b, and bell-shaped cells on stalks of a protist of the genus *Vorticella*, as shown in Figure 4-2c. Leeuwenhoek called these organisms *animalcules*. We now call them *protists*.

THE CELL THEORY

Although Hooke and Leeuwenhoek were the first to report observing cells, the importance of this observation was not realized until about 150 years later. At this time, biologists began to organize information about cells into a unified understanding. In 1838, the German botanist Matthias Schleiden concluded that all plants were composed of cells. The next year, the German zoologist Theodor Schwann concluded the same thing for animals. And finally, in his study of human diseases, the German physician Rudolf Virchow (1821–1902) noted that all cells come from other cells. These three observations were combined to form a basic theory about the cellular nature of life. The **cell theory** has three essential parts, which are summarized in Table 4-1.

TABLE 4-1 The Cell Theory

All living organisms are composed of one or more cells.
Cells are the basic units of structure and function in an organism.
Cells come only from the reproduction of existing cells.



Developments in Cell Biology

The discovery of cells and the development of the cell theory happened at the beginning of a revolutionary time in the history of science. Before the invention of the microscope, many questions about what makes up living and nonliving things could not be answered. Once cells could be observed, these questions could be explored. Scientists could then turn their attention to finding out how cells function. Figure 4-3 lists some of the major events in the history of cell biology.

FIGURE 4-3

The study of cell biology began with the discovery of the cell by Robert Hooke in 1665. Since then, constantly improving technology has allowed scientists to unlock the secrets of the cell.

The Cellular Basis of Life

Microscopes helped biologists clarify our definition of life. All living things share several basic characteristics. All living things consist of organized parts, obtain energy from their surroundings, perform chemical reactions, change with time, respond to their environments, and reproduce.

In addition, living things must be able to separate their relatively constant internal environment from the ever-changing external environment. The ability to maintain a constant internal environment, called *homeostasis*, will be discussed later. Finally, all living things share a common history. All cells share characteristics that indicate that cells are related to other living things.

SECTION 1 REVIEW

1. Describe the major contributions of Hooke and Leeuwenhoek to cell biology.
2. Identify the advance that enabled Leeuwenhoek to view the first living cells.
3. Describe the research that led to the development of the cell theory.
4. State the three fundamental parts of the cell theory.
5. List three major events in the history of cell biology.
6. Name eight characteristics that all living things share.

CRITICAL THINKING

7. **Applying Concepts** If you could go back in time, how would you explain the cell theory to someone who had never heard of cells?
8. **Making Calculations** A biologist photographs a cell in a microscope magnified at 40 times. The cell in the photo is 2 mm in diameter. What is the true diameter of the cell in micrometers (μm)?
9. **Justifying Conclusions** If organisms exist on other planets, would they consist of cells? Defend your answer.

SECTION 2

OBJECTIVES

- **Explain** the relationship between cell shape and cell function.
- **Identify** the factor that limits cell size.
- **Describe** the three basic parts of a cell.
- **Compare** prokaryotic cells and eukaryotic cells.
- **Analyze** the relationship among cells, tissues, organs, organ systems, and organisms.

VOCABULARY

plasma membrane
cytoplasm
cytosol
nucleus
prokaryote
eukaryote
organelle
tissue
organ
organ system

INTRODUCTION TO CELLS

Cells come in a variety of shapes and sizes that suit their diverse functions. There are at least 200 types of cells, ranging from flat cells to branching cells to round cells to rectangular cells.

CELL DIVERSITY

Cells of different organisms and even cells within the same organism are very diverse in terms of shape, size, and internal organization. One theme that occurs again and again throughout biology is that form follows function. In other words, a cell's function influences its physical features.

Cell Shape

The diversity in cell shapes reflects the different functions of cells. Compare the cell shapes shown in Figure 4-4. The long extensions that reach out in various directions from the nerve cell shown in Figure 4-4a allow the cell to send and receive nerve impulses. The flat, platelike shape of skin cells in Figure 4-4b suits their function of covering and protecting the surface of the body. As shown below, a cell's shape can be simple or complex depending on the function of the cell. Each cell has a shape that has evolved to allow the cell to perform its function effectively.

FIGURE 4-4

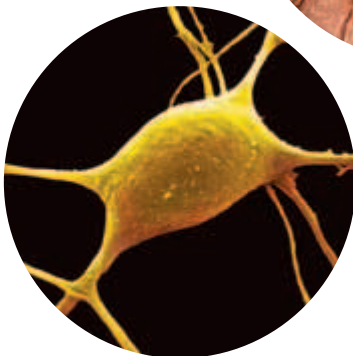
Cells have various shapes. (a) Nerve cells have long extensions. (b) Skin cells are flat and platelike. (c) Egg cells are spherical. (d) Some bacteria are rod shaped. (e) Some plant cells are rectangular.



(b) Skin cells



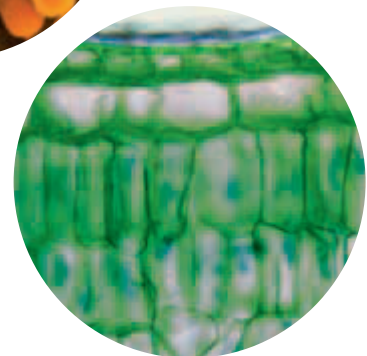
(d) Bacterial cells



(a) Nerve cell



(c) Egg cell



(e) Plant cells

Cell Size

Cells differ not only in their shape but also in their size. A few types of cells are large enough to be seen by the unaided human eye. For example, the nerve cells that extend from a giraffe's spinal cord to its foot can be 2 m (about 6 1/2 ft) long. A human egg cell is about the size of the period at the end of this sentence. Most cells, however, are only 10 to 50 μm in diameter, or about 1/500 the size of the period at the end of this sentence.

The size of a cell is limited by the relationship of the cell's outer surface area to its volume, or its *surface area-to-volume ratio*. As a cell grows, its volume increases much faster than its surface area does, as shown in Figure 4-5. This trend is important because the materials needed by a cell (such as nutrients and oxygen) and the wastes produced by a cell (such as carbon dioxide) must pass into and out of the cell through its surface. If a cell were to become very large, the volume would increase much more than the surface area. Therefore, the surface area would not allow materials to enter or leave the cell quickly enough to meet the cell's needs. As a result, most cells are microscopic in size.



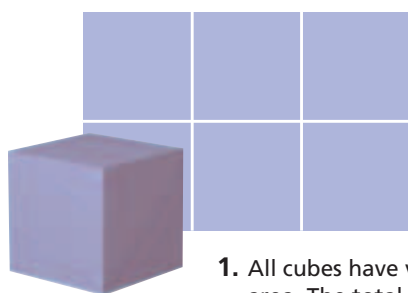
Quick Lab

Comparing Surface Cells

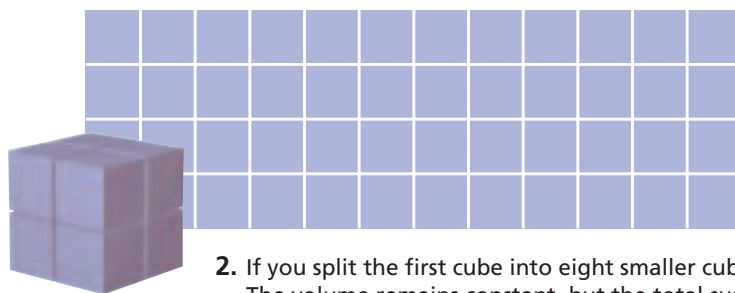
Materials microscope, prepared slides of plant (dicot) stem and animal (human) skin, pencil, paper

Procedure Examine slides by using medium magnification (100 $\times$). Observe and draw the surface cells of the plant stem and the animal skin.

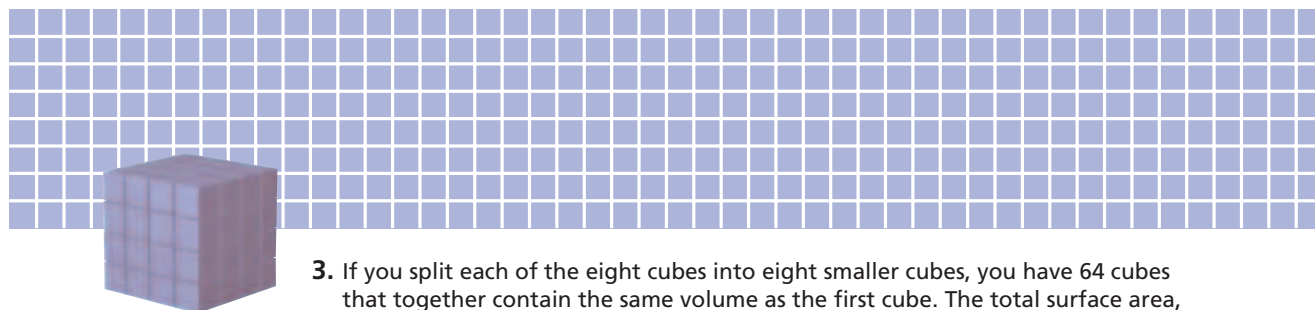
Analysis How do the surface cells of each organism differ from the cells beneath the surface cells? What is the function of the surface cells? Explain how surface cells are suited to their function based on their shape.



1. All cubes have volume and surface area. The total surface area is equal to the sum of the areas of each of the six sides (area = length $\times$ width).



2. If you split the first cube into eight smaller cubes, you get 48 sides. The volume remains constant, but the total surface area doubles.



3. If you split each of the eight cubes into eight smaller cubes, you have 64 cubes that together contain the same volume as the first cube. The total surface area, however, has doubled again.

FIGURE 4-5

Small cells can exchange substances more readily than large cells because small objects have a higher surface area-to-volume ratio.

BASIC PARTS OF A CELL

Despite the diversity among cells, three basic features are common to all cell types. All cells have an outer boundary, an interior substance, and a control region.

Plasma Membrane

The cell's outer boundary, called the **plasma membrane** (or the *cell membrane*), covers a cell's surface and acts as a barrier between the inside and the outside of a cell. All materials enter or exit through the plasma membrane. The surface of a plasma membrane is shown in Figure 4-6a.

Cytoplasm

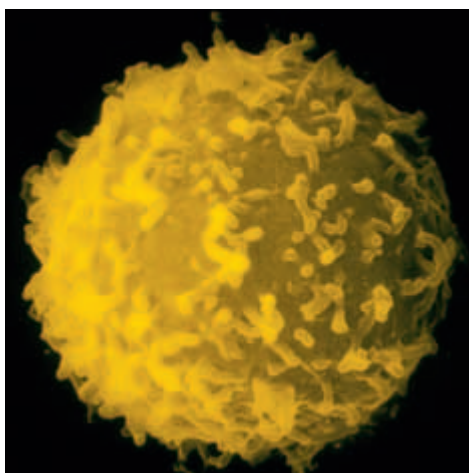
The region of the cell that is within the plasma membrane and that includes the fluid, the cytoskeleton, and all of the organelles except the nucleus is called the **cytoplasm**. The part of the cytoplasm that includes molecules and small particles, such as ribosomes, but not membrane-bound organelles is the **cytosol**. About 20 percent of the cytosol is made up of protein.

Control Center

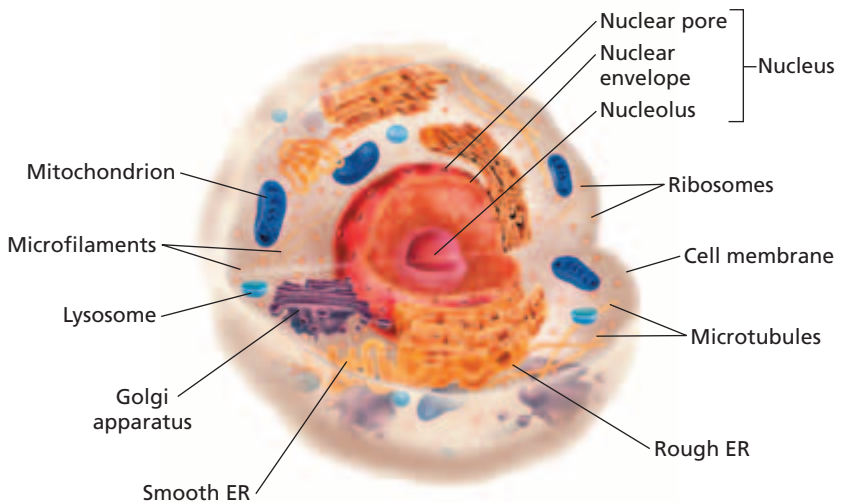
Cells carry coded information in the form of DNA for regulating their functions and reproducing themselves. The DNA in some types of cells floats freely inside the cell. Other cells have a membrane-bound organelle that contains a cell's DNA. This membrane-bound structure is called the **nucleus**. Most of the functions of a eukaryotic cell are controlled by the cell's nucleus. The nucleus is often the most prominent structure within a eukaryotic cell. It maintains its shape with the help of a protein skeleton called the *nuclear matrix*. The nucleus of a typical animal cell is shown in Figure 4-6b.

FIGURE 4-6

Most animal cells have a cell membrane, a nucleus, and a variety of other organelles embedded in a watery substance. The surface of the cell membrane can be seen in (a). The organelles inside the cell are labeled in the diagram (b).



(a)



(b)

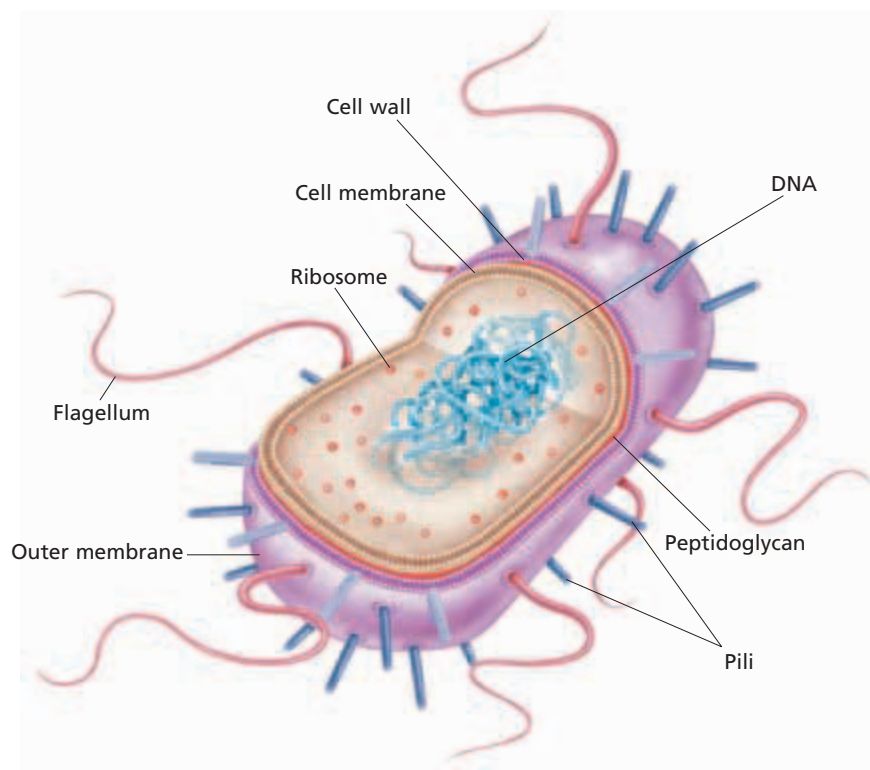


FIGURE 4-7

A prokaryotic cell lacks a membrane-bound nucleus and membrane-bound organelles. Most prokaryotic cells are much smaller than eukaryotic cells are.

TWO BASIC TYPES OF CELLS

Fossil evidence suggests that the earliest cells on Earth were simple cells similar to some present-day bacteria. As cells evolved, they differentiated into two major types: prokaryotes and eukaryotes.

Prokaryotes

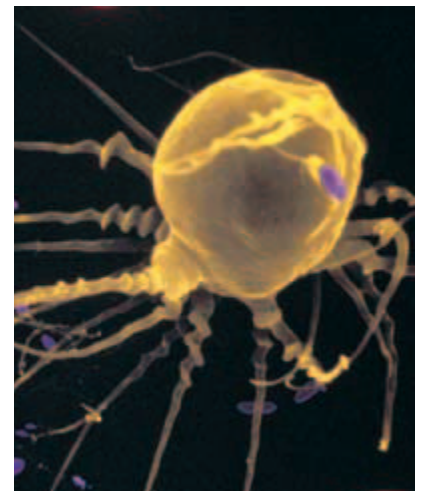
Prokaryotes (proh-KAR-ee-OHTS) are organisms that lack a membrane-bound nucleus and membrane-bound organelles. Although prokaryotic cells lack a nucleus, their genetic information—in the form of DNA—is often concentrated in a part of the cell called the *nucleoid*. Figure 4-7 shows a typical prokaryotic cell. Prokaryotes are divided into two domains: Bacteria and Archaea (ahr-KEE-uh). The domain Bacteria includes organisms that are similar to the first cellular life-forms. The domain Archaea includes organisms that are thought to be more closely related to eukaryotic cells found in all other kingdoms of life.

Eukaryotes

Organisms made up of one or more cells that have a nucleus and membrane-bound organelles are called **eukaryotes** (yoo-KAR-ee-OHTS). Eukaryotic cells also have a variety of subcellular structures called **organelles**, well-defined, intracellular bodies that perform specific functions for the cell. Many organelles are surrounded by a membrane. The organelles carry out cellular processes just as a person's pancreas, heart, and other organs carry out a person's life processes. Eukaryotic cells are generally much larger than prokaryotic cells, as seen in Figure 4-8, which shows a white blood cell (eukaryote) destroying tiny bacterial cells (prokaryotes).

FIGURE 4-8

A white blood cell (eukaryotic) changes shape as it attacks purple-stained bacterial cells that are much smaller (prokaryotic).



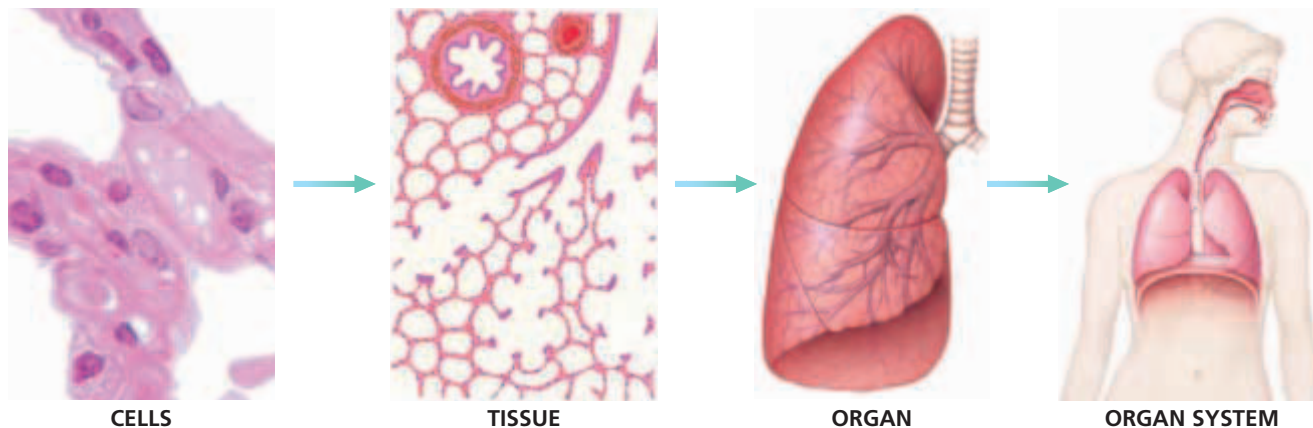


FIGURE 4-9

In a multicellular eukaryotic organisms, cells organize into tissues. Tissues organize into organs. Organs are part of organ systems, in which organs work together to perform body functions.

CELLULAR ORGANIZATION

Over time, cells began to form groups that functioned together. Some cells retained the ability to live outside a group. Others became dependent on each other for survival.

Colonies

A *colonial organism* is a collection of genetically identical cells that live together in a connected group. Colonial organisms are not truly multicellular because few cell activities are coordinated.

True Multicellularity

As organisms evolved, their cells became more specialized and eventually were unable to survive independently. Groups of cells took on specific roles within the organism. A group of similar cells and their products that carry out a specific function is called a **tissue**. Groups of tissues that perform a particular job in an organism are called **organs**. An **organ system** is a group of organs that accomplish related tasks. The stomach and liver are organs that are part of the digestive system. Finally, several organ systems combine to make up an organism. This hierarchical organization found in multicellular organisms is shown in Figure 4-9.

SECTION 2 REVIEW

1. Describe the relationship between a cell's shape and its function.
2. Explain the factor that limits cell size.
3. Identify and describe three basic parts of a cell.
4. Summarize the differences between prokaryotic cells and eukaryotic cells.
5. List four levels of organization that combine to form an organism.

CRITICAL THINKING

6. **Making Calculations** If a cube-shaped cell grew from 1 cm per side to 3 cm per side, how much would its volume change?
7. **Forming Reasoned Opinions** Why do you think there are three basic structures common to all cell types? Support your answer.
8. **Analyzing Processes** How are the functions of prokaryotic cells controlled without a nucleus?

CELL ORGANELLES AND FEATURES

Eukaryotic cells have many membrane systems. These membranes divide cells into compartments that function together to keep a cell alive.

PLASMA MEMBRANE

The plasma membrane (also called the *cell membrane*) has several functions. For example, it allows only certain molecules to enter or leave the cell. It separates internal metabolic reactions from the external environment. In addition, the plasma membrane allows the cell to excrete wastes and to interact with its environment.

Membrane Lipids

The plasma membrane, as well as the membranes of cell organelles, is made primarily of phospholipids. Phospholipids have a polar, hydrophilic (“water-loving”) phosphate head and two nonpolar, hydrophobic (“water-fearing”) fatty acid tails. Water molecules surround the plasma membrane. The phospholipids line up so that their heads point outward toward the water and their tails point inward, away from water. The result is a double layer called a **phospholipid bilayer**, as shown in Figure 4-10. The cell membranes of eukaryotes also contain lipids, called *sterols*, between the tails of the phospholipids. The major membrane sterol in animal cells is cholesterol. Sterols in the plasma membrane make the membrane more firm and prevent the membrane from freezing at low temperatures.

OBJECTIVES

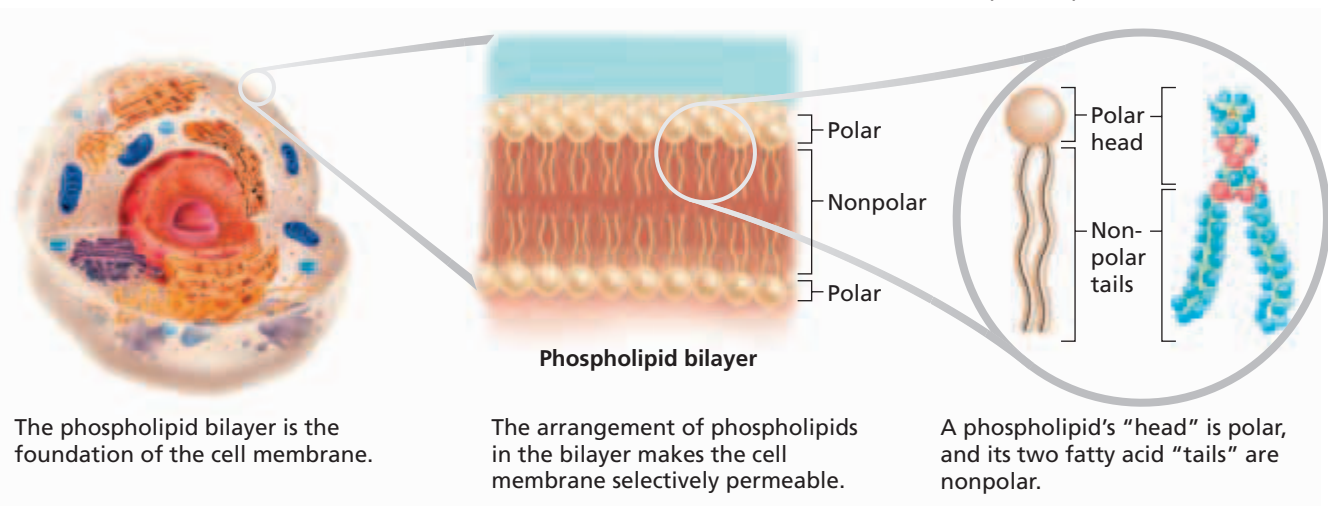
- **Describe** the structure and function of a cell’s plasma membrane.
- **Summarize** the role of the nucleus.
- **List** the major organelles found in the cytosol, and describe their roles.
- **Identify** the characteristics of mitochondria.
- **Describe** the structure and function of the cytoskeleton.

VOCABULARY

phospholipid bilayer
chromosome
nuclear envelope
nucleolus
ribosome
mitochondrion
endoplasmic reticulum
Golgi apparatus
lysosome
cytoskeleton
microtubule
microfilament
cilium
flagellum
centriole

FIGURE 4-10

Cell membranes are made of a phospholipid bilayer. Each phospholipid molecule has a polar “head” and a two-part nonpolar “tail.”



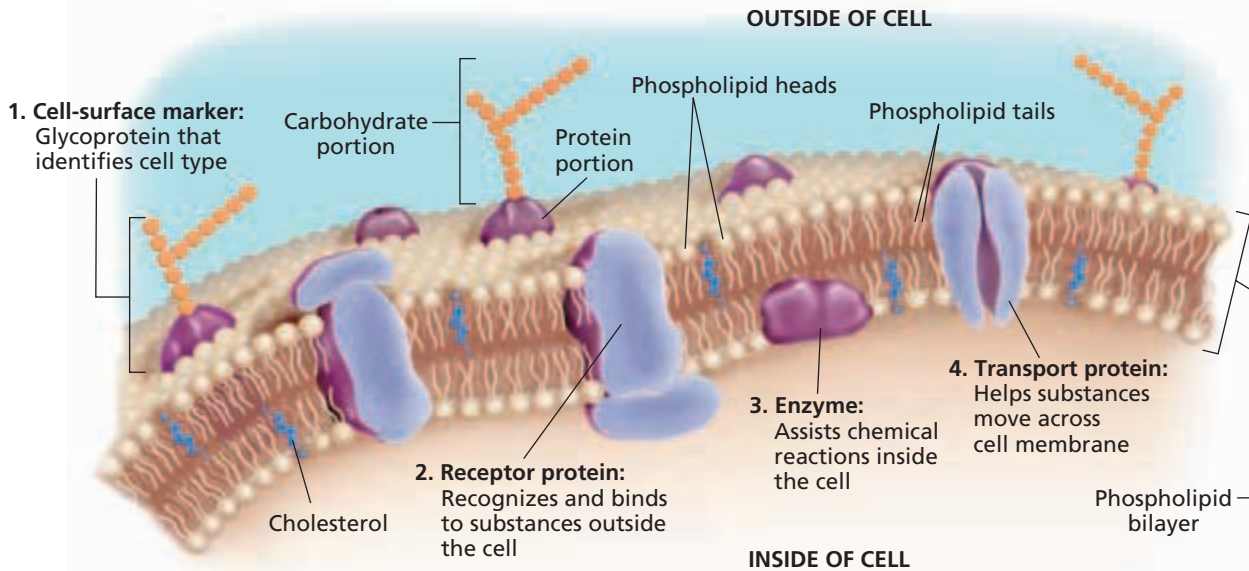


FIGURE 4-11

Cell membranes often contain proteins. Integral proteins include cell-surface markers, receptor proteins, and transport proteins. Enzymes are examples of peripheral proteins.

Membrane Proteins

Plasma membranes often contain specific proteins embedded within the lipid bilayer. These proteins are called *integral proteins*. Figure 4-11 shows that some integral proteins, such as cell surface markers, emerge from only one side of the membrane. Others, such as receptor proteins and transport proteins, extend across the plasma membrane and are exposed to both the cell's interior and exterior environments. Proteins that extend across the plasma membrane are able to detect environmental signals and transmit them to the inside of the cell. *Peripheral proteins*, such as the enzyme shown in Figure 4-11, lie on only one side of the membrane and are not embedded in it.

As Figure 4-11 shows, integral proteins exposed to the cell's external environment often have carbohydrates attached. These carbohydrates can act as labels on cell surfaces. Some labels help cells recognize each other and stick together. Viruses can use these labels as docks for entering and infecting cells.

Integral proteins play important roles in actively transporting molecules into the cell. Some act as channels or pores that allow certain substances to pass. Other integral proteins bind to a molecule on the outside of the cell and then transport it through the membrane. Still others act as sites where chemical messengers such as hormones can attach.

Fluid Mosaic Model

A cell's plasma membrane is surprisingly dynamic. Scientists describe the cell membrane as a fluid mosaic. The *fluid mosaic model* states that the phospholipid bilayer behaves like a fluid more than it behaves like a solid. The membrane's lipids and proteins can move laterally within the bilayer, like a boat on the ocean. As a result of such lateral movement, the pattern, or "mosaic," of lipids and proteins in the cell membrane constantly changes.

NUCLEUS

Most of the functions of a eukaryotic cell are controlled by the nucleus, shown in Figure 4-12. The nucleus is filled with a jellylike liquid called the *nucleoplasm*, which holds the contents of the nucleus and is similar in function to a cell's cytoplasm.

The nucleus houses and protects the cell's genetic information. The hereditary information that contains the instructions for the structure and function of the organism is coded in the organism's DNA, which is contained in the nucleus. When a cell is not dividing, the DNA is in the form of a threadlike material called *chromatin*. When a cell is about to divide, the chromatin condenses to form **chromosomes**. Chromosomes are structures in the nucleus made of DNA and protein.

The nucleus is the site where DNA is transcribed into ribonucleic acid (RNA). RNA moves through nuclear pores to the cytoplasm, where, depending on the type of RNA, it carries out its function.

Nuclear Envelope

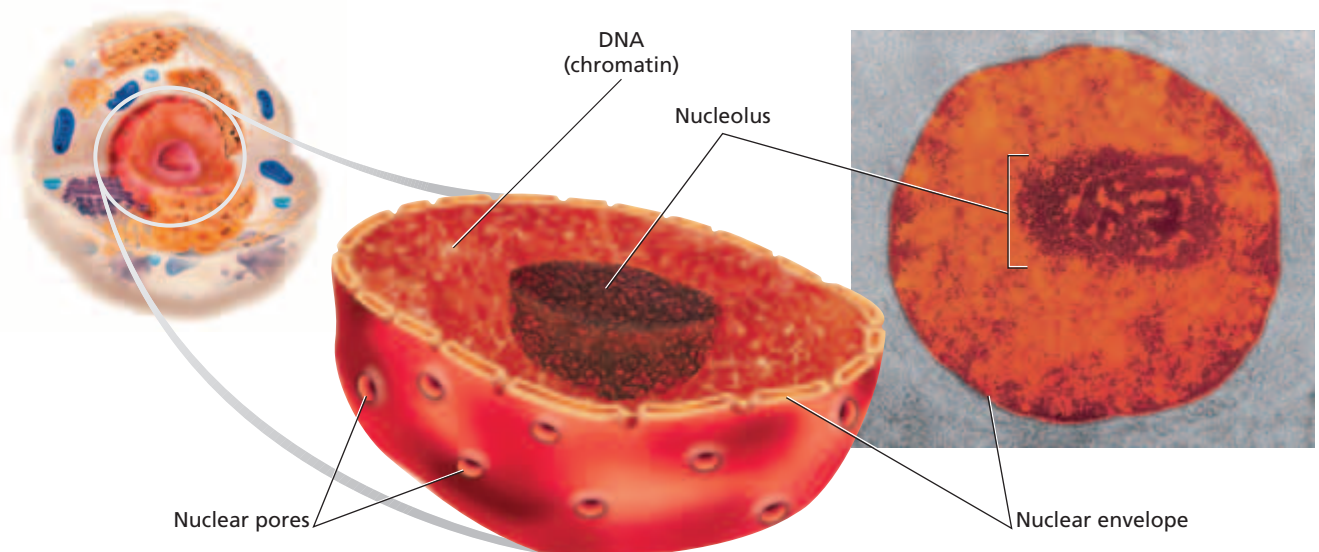
The nucleus is surrounded by a double membrane called the **nuclear envelope**. The nuclear envelope is made up of two phospholipid bilayers. Covering the surface of the nuclear envelope are tiny, protein-lined holes, which are called *nuclear pores*. The nuclear pores provide passageways for RNA and other materials to enter and leave the nucleus.

Nucleolus

Most nuclei contain at least one denser area, called the **nucleolus** (noo-KLEE-uh-luhs). The nucleolus (plural, *nucleoli*) is the site where DNA is concentrated when it is in the process of making ribosomal RNA. **Ribosomes** (RIE-buh-SOHMZ) are organelles made of protein and RNA that direct protein synthesis in the cytoplasm.

FIGURE 4-12

The nucleus of a cell is surrounded by a double membrane called the nuclear envelope. The nucleus stores the cell's DNA.



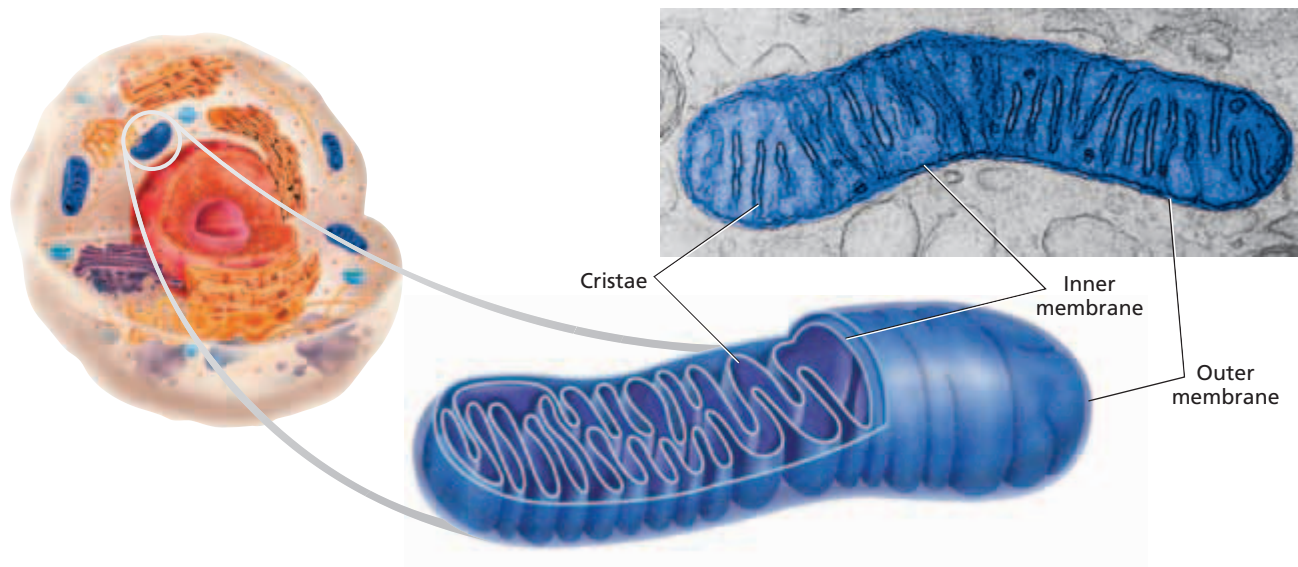


FIGURE 4-13

Mitochondria convert organic molecules into energy for the cell. Mitochondria have an inner membrane and an outer membrane. The folds of the inner membrane, called *cristae*, are the site of energy conversion.

MITOCHONDRIA

Mitochondria (MIET-oh-KAHN-dree-uh) (singular, *mitochondrion*) are tiny organelles that transfer energy from organic molecules to adenosine triphosphate (ATP). ATP ultimately powers most of the cell's chemical reactions. Highly active cells, such as muscle cells, can have hundreds of mitochondria. Cells that are not very active, such as fat-storage cells, have few mitochondria.

Like a nucleus, a mitochondrion has an inner and an outer phospholipid membrane, as shown in Figure 4-13. The outer membrane separates the mitochondrion from the cytosol. The inner membrane has many folds, called *cristae* (KRIS-tee). Cristae contain proteins that carry out energy-harvesting chemical reactions.

Mitochondrial DNA

Mitochondria have their own DNA and can reproduce only by the division of preexisting mitochondria. Scientists think that mitochondria originated from prokaryotic cells that were incorporated into ancient eukaryotic cells. This symbiotic relationship provided the prokaryotic invaders with a protected place to live and provided the eukaryotic cell with an increased supply of ATP.



FIGURE 4-14

Ribosomes are the organelles responsible for building protein. Ribosomes have a large and small subunit, each made of protein and ribosomal RNA. Some ribosomes are free in the cell. Others are attached to the rough endoplasmic reticulum.

RIBOSOMES

Ribosomes are small, roughly spherical organelles that are responsible for building protein. Ribosomes do not have a membrane. They are made of protein and RNA molecules. Ribosome assembly begins in the nucleolus and is completed in the cytoplasm. One large and one small subunit come together to make a functioning ribosome, shown in Figure 4-14. Some ribosomes are free within the cytosol. Others are attached to the rough endoplasmic reticulum.

ENDOPLASMIC RETICULUM

The **endoplasmic reticulum** (EN-doh-PLAZ-mik ri-TIK-yuh-luhm), abbreviated ER, is a system of membranous tubes and sacs, called *cisternae* (sis-TUHR-nee). The ER functions primarily as an intracellular highway, a path along which molecules move from one part of the cell to another. The amount of ER inside a cell fluctuates, depending on the cell's activity. There are two types of ER: rough and smooth. The two types of ER are thought to be continuous.

Rough Endoplasmic Reticulum

The rough endoplasmic reticulum is a system of interconnected, flattened sacs covered with ribosomes, as shown in Figure 4-15. The rough ER produces phospholipids and proteins. Certain types of proteins are made on the rough ER's ribosomes. These proteins are later exported from the cell or inserted into one of the cell's own membranes. For example, ribosomes on the rough ER make digestive enzymes, which accumulate inside the endoplasmic reticulum. Little sacs or vesicles then pinch off from the ends of the rough ER and store the digestive enzymes until they are released from the cell. Rough ER is most abundant in cells that produce large amounts of protein for export, such as cells in digestive glands and antibody-producing cells.

Smooth Endoplasmic Reticulum

The smooth ER lacks ribosomes and thus has a smooth appearance. Most cells contain very little smooth ER. Smooth ER builds lipids such as cholesterol. In the ovaries and testes, smooth ER produces the steroid hormones estrogen and testosterone. In skeletal and heart muscle cells, smooth ER releases calcium, which stimulates contraction. Smooth ER is also abundant in liver and kidney cells, where it helps detoxify drugs and poisons. Long-term abuse of alcohol and other drugs causes these cells to produce more smooth ER. Increased amounts of smooth ER in liver cells is one of the factors that can lead to drug tolerance. As Figure 4-15 shows, rough ER and smooth ER form an interconnected network.

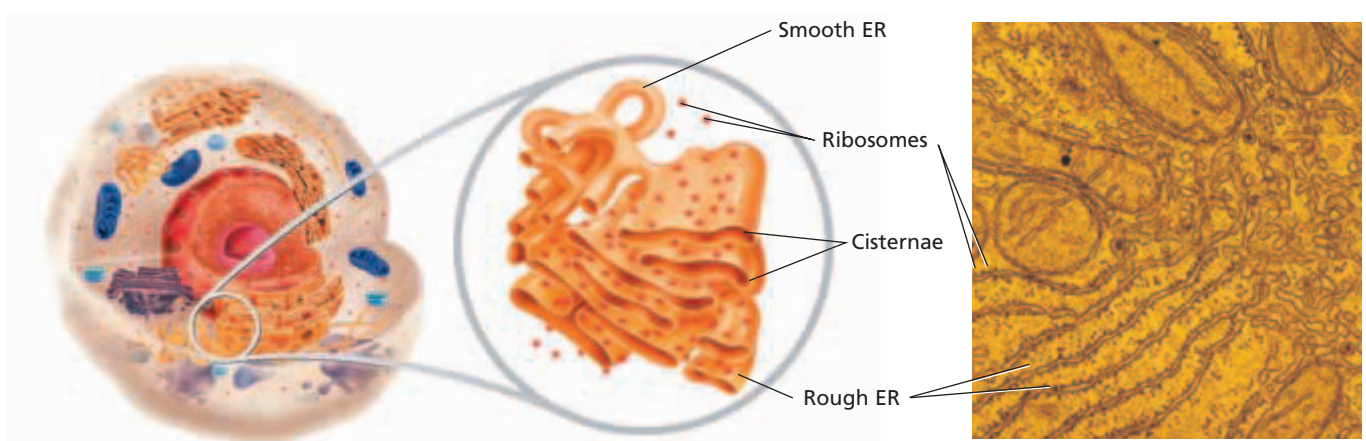
Word Roots and Origins

reticulum

from the Latin *rete*, meaning "net";
reticulum means "little net"

FIGURE 4-15

The endoplasmic reticulum (ER) serves as a site of synthesis for proteins, lipids, and other materials. The dark lines in the photo represent the membranes of the ER, and the narrow lighter areas between the dark lines show the channels and spaces (cisternae) inside the ER.



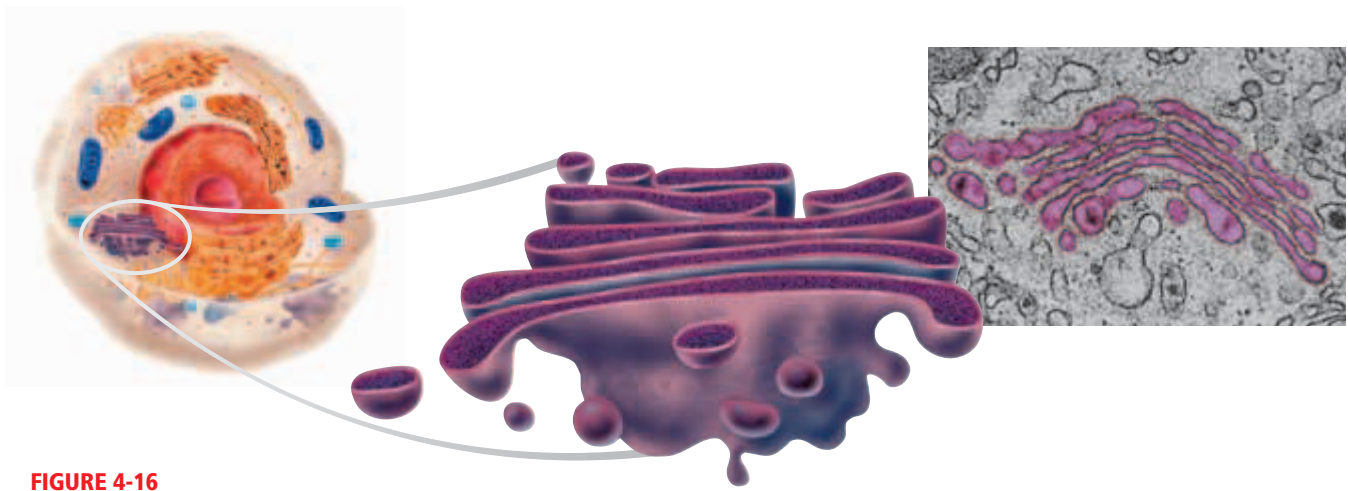


FIGURE 4-16

The Golgi apparatus modifies many cellular products and prepares them for export.

GOLGI APPARATUS

The **Golgi apparatus**, shown in Figure 4-16, is another system of flattened, membranous sacs. The sacs nearest the nucleus receive vesicles from the ER containing newly made proteins or lipids. Vesicles travel from one part of the Golgi apparatus to the next and transport substances as they go. The stacked membranes modify the vesicle contents as they move along. The proteins get “address labels” that direct them to various other parts of the cell. During this modification, the Golgi apparatus can add carbohydrate labels to proteins or alter new lipids in various ways.

VESICLES

Cells contain several types of vesicles, which perform various roles. Vesicles are small, spherically shaped sacs that are surrounded by a single membrane and that are classified by their contents. Vesicles often migrate to and merge with the plasma membrane. As they do, they release their contents to the outside of the cell.

Lysosomes

Lysosomes (LIE-suh-SOHMZ) are vesicles that bud from the Golgi apparatus and that contain digestive enzymes. These enzymes can break down large molecules, such as proteins, nucleic acids, carbohydrates, and phospholipids. In the liver, lysosomes break down glycogen in order to release glucose into the bloodstream. Certain white blood cells use lysosomes to break down bacteria. Within a cell, lysosomes digest worn-out organelles in a process called *autophagy* (aw-TAHF-uh-jee).

Lysosomes are also responsible for breaking down cells when it is time for the cells to die. The digestion of damaged or extra cells by the enzymes of their own lysosomes is called *autolysis* (aw-TAHL-uh-sis). Lysosomes play a very important role in maintaining an organism’s health by destroying cells that are no longer functioning properly.

Peroxisomes

Peroxisomes are similar to lysosomes but contain different enzymes and are not produced by the Golgi apparatus. Peroxisomes are abundant in liver and kidney cells, where they neutralize free radicals (oxygen ions that can damage cells) and detoxify alcohol and other drugs. Peroxisomes are named for the hydrogen peroxide, H_2O_2 , they produce when breaking down alcohol and killing bacteria. Peroxisomes also break down fatty acids, which the mitochondria can then use as an energy source.

Other Vesicles

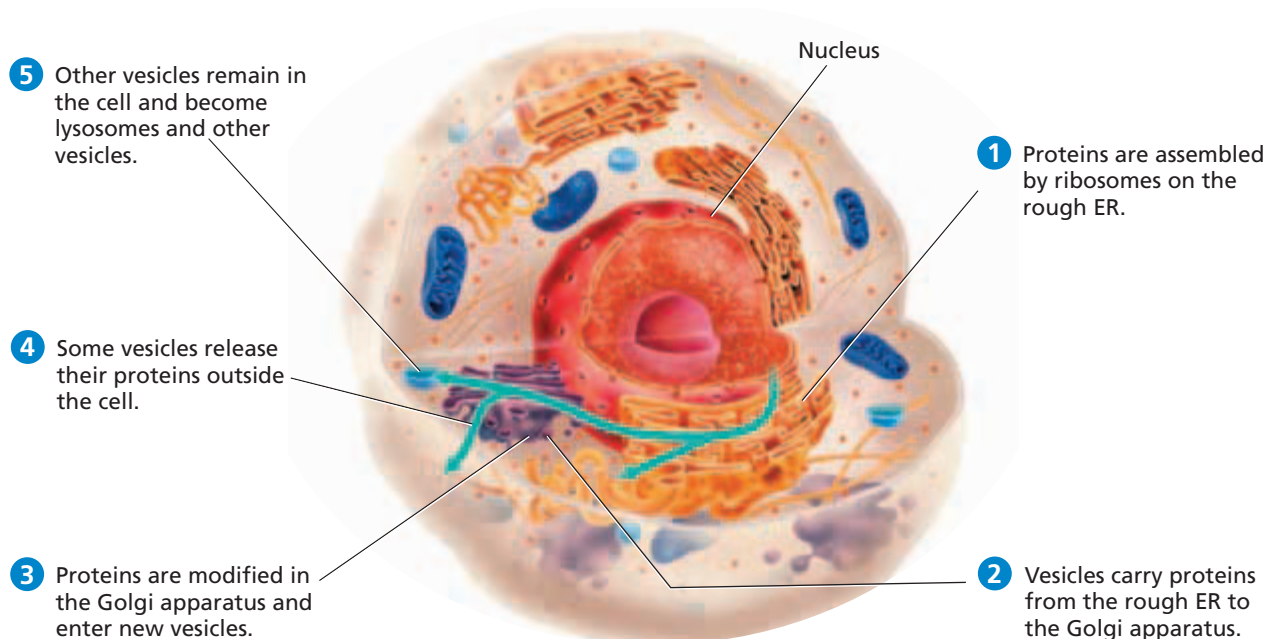
Specialized peroxisomes, called *glyoxysomes*, can be found in the seeds of some plants. They break down stored fats to provide energy for the developing plant embryo. Some cells engulf material by surrounding it with plasma membrane. The resulting pocket buds off to become a vesicle inside the cell. This vesicle is called an *endosome*. Lysosomes fuse with endosomes and digest the engulfed material. Food vacuoles are vesicles that store nutrients for a cell. Contractile vacuoles are vesicles that can contract and dispose of excess water inside a cell.

Protein Synthesis

One of the major functions of a cell is the production of protein. The path some proteins take from synthesis to export can be seen in Figure 4-17. In step ①, proteins are assembled by ribosomes on the rough ER. Then, in step ②, vesicles transport proteins to the Golgi apparatus. In step ③, the Golgi modifies proteins and packages them in new vesicles. In step ④, vesicles release proteins that have destinations outside the cell. In step ⑤, vesicles containing enzymes remain inside the cell as lysosomes, peroxisomes, endosomes, or other types of vesicles.

FIGURE 4-17

The rough ER, Golgi apparatus, and vesicles work together to transport proteins to their destinations inside and outside the cell.



CYTOSKELETON

The **cytoskeleton** is a network of thin tubes and filaments that crisscrosses the cytosol. The tubes and filaments give shape to the cell from the inside in the same way that tent poles support the shape of a tent. The cytoskeleton also acts as a system of internal tracks, shown in Figure 4-18, on which items move around inside the cell. The cytoskeleton's functions are based on several structural elements. Three of these are microtubules, microfilaments, and intermediate filaments, shown and described in Table 4-2.

Microtubules

Microtubules are hollow tubes made of a protein called *tubulin*. Each tubulin molecule consists of two slightly different subunits. Microtubules radiate outward from a central point called the *centrosome* near the nucleus. Microtubules hold organelles in place, maintain a cell's shape, and act as tracks that guide organelles and molecules as they move within the cell.

Microfilaments

Finer than microtubules, **microfilaments** are long threads of the beadlike protein actin and are linked end to end and wrapped around each other like two strands of a rope. Microfilaments contribute to cell movement, including the crawling of white blood cells and the contraction of muscle cells.

Intermediate Filaments

Intermediate filaments are rods that anchor the nucleus and some other organelles to their places in the cell. They maintain the internal shape of the nucleus. Hair-follicle cells produce large quantities of intermediate filament proteins. These proteins make up most of the hair shaft.

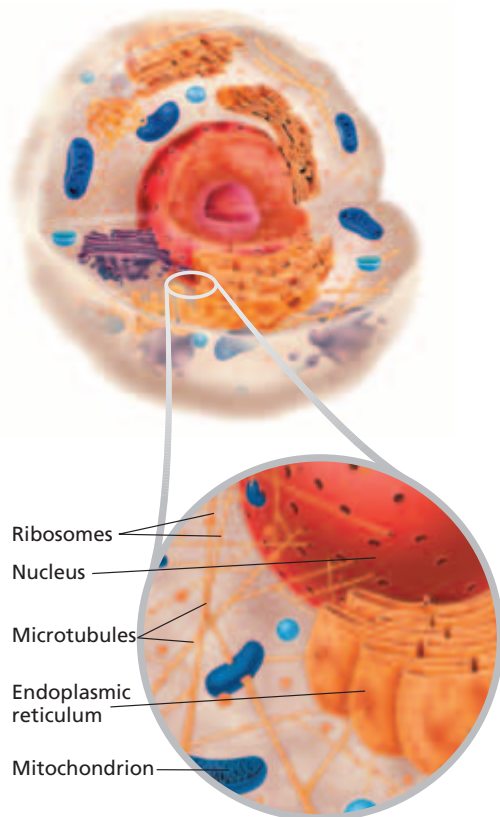
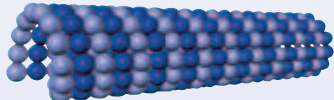


FIGURE 4-18

Microtubules provide a path for organelles and molecules as they move throughout the cell.

TABLE 4-2 The Structure of the Cytoskeleton

Property	Microtubules	Microfilaments	Intermediate filaments
Structure	hollow tubes made of coiled protein	two strands of intertwined protein	protein fibers coiled into cables
Protein subunits	tubulin, with two subunits: α and β tubulin	actin	one of several types of fibrous proteins
Main function	maintenance of cell shape; cell motility (in cilia and flagella); chromosome movement; organelle movement	maintenance and changing of cell shape; muscle contraction; movement of cytoplasm; cell motility; cell division	maintenance of cell shape; anchor nucleus and other organelles; maintenance of shape of nucleus
Shape			

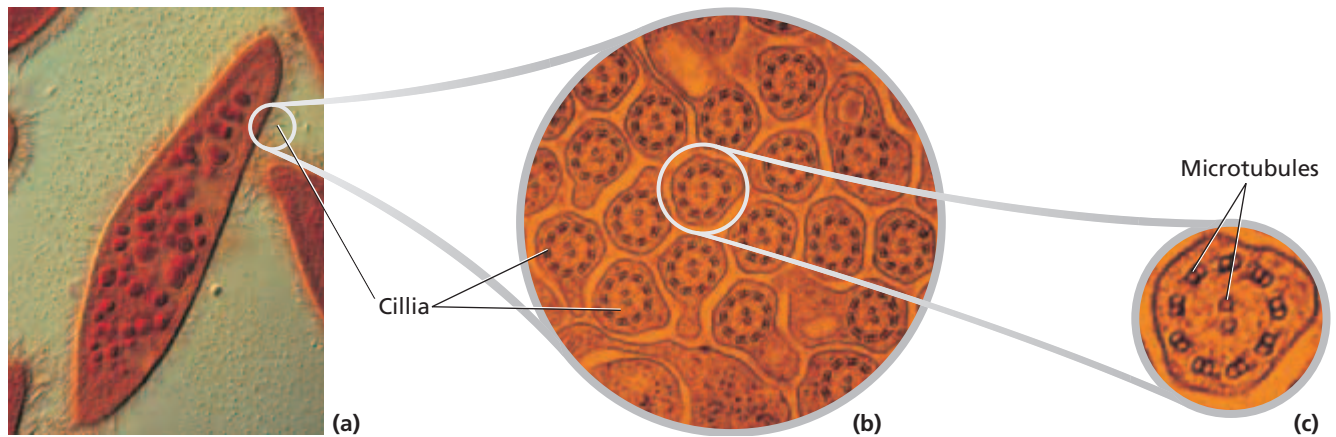


FIGURE 4-19

A SEM of a paramecium shows cilia on the surface of the cell (a). A TEM of a cross section of those cilia (b) reveals the internal structure of the cilia. The characteristic 9+2 configuration of microtubules can be clearly seen (c).

Cilia and Flagella

Cilia (SIL-ee-uh) and **flagella** (fluh-JEL-uh) are hairlike structures that extend from the surface of the cell, where they assist in movement. Cilia are short and are present in large numbers on certain cells, whereas flagella are longer and are far less numerous on the cells where they occur. Cilia and flagella have a membrane on their outer surface and an internal structure of nine pairs of microtubules around two central tubules, as Figure 4-19 shows.

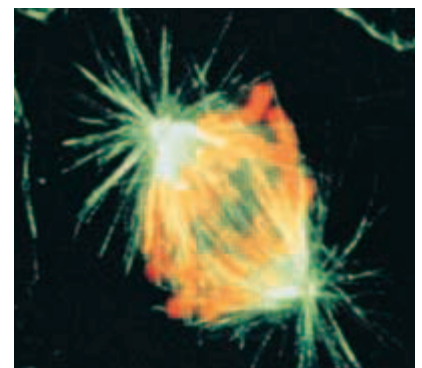
Cilia on cells in the inner ear vibrate and help detect sound. Cilia cover the surfaces of many protists and “row” the protists through water like thousands of oars. On other protists, cilia sweep water and food particles into a mouthlike opening. Many kinds of protists use flagella to propel themselves, as do human sperm cells.

Centrioles

Centrioles consist of two short cylinders of microtubules at right angles to each other and are situated in the cytoplasm near the nuclear envelope. Centrioles occur in animal cells, where they organize the microtubules of the cytoskeleton during cell division, as shown in Figure 4-20. Plant cells lack centrioles. Basal bodies have the same structure that centrioles do. Basal bodies are found at the base of cilia and flagella and appear to organize the development of cilia and flagella.

FIGURE 4-20

During cell division, centrioles organize microtubules that pull the chromosomes (orange) apart. The centrioles are at the center of rays of microtubules, which have been stained green with a fluorescent dye.



SECTION 3 REVIEW

1. Explain how the fluid mosaic model describes the plasma membrane.
2. List three cellular functions that occur in the nucleus.
3. Describe the organelles that are found in a eukaryotic cell.
4. Identify two characteristics that make mitochondria different from other organelles.
5. Contrast three types of cytoskeletal fibers.

CRITICAL THINKING

6. **Relating Concepts** If a cell has a high energy requirement, would you expect the cell to have many mitochondria or few mitochondria? Why?
7. **Analyzing Information** How do scientists think that mitochondria originated? Why?
8. **Analyzing Statements** It is not completely accurate to say that organelles are floating freely in the cytosol. Why not?

Science in Action

How Do Cells Secrete Proteins?

The invention of electron microscopes allowed biologists to see the detail of the tiny structures inside cells. But it was clever experimentation by George Palade that revealed how those tiny structures help a cell survive.



George Palade

HYPOTHESIS: Membranous Organelles Secrete Proteins

Six years after he graduated from medical school in Romania, George Palade began conducting research at the Rockefeller Institute for Medical Research in New York City. He studied the network of membranous organelles in cells of the guinea pig pancreas. This network included the rough endoplasmic reticulum (ER), the smooth ER, the Golgi apparatus, lysosomes, and secretory granules.

Palade knew about the structure of these organelles. He also knew that ribosomes on the rough ER were associated with making proteins. But what Palade and other scientists did not know was how the proteins were secreted from cells once they were made on the ribosomes. Palade suspected that the membranous organelles played a role.

METHODS: Track Proteins

Part of Palade's genius was inventing a way to make thin slices of tissue from a guinea pig's pancreas and keep the cells of the tissue alive. He also developed a way to track newly made proteins, a method called the *pulse-chase technique*. With this technique, Palade added "labeled" amino acids (made with radioactive atoms) to the pancreas cells for a fixed amount of time. This was the "pulse." The cells used the labeled amino acids and their own "unlabeled" amino acids (without radioactive atoms) to make proteins. Palade would then "chase" out any labeled amino acid that the cells had not used to build proteins by adding an excess of unlabeled amino acid.

RESULTS: Black Dots Move as Time Passes

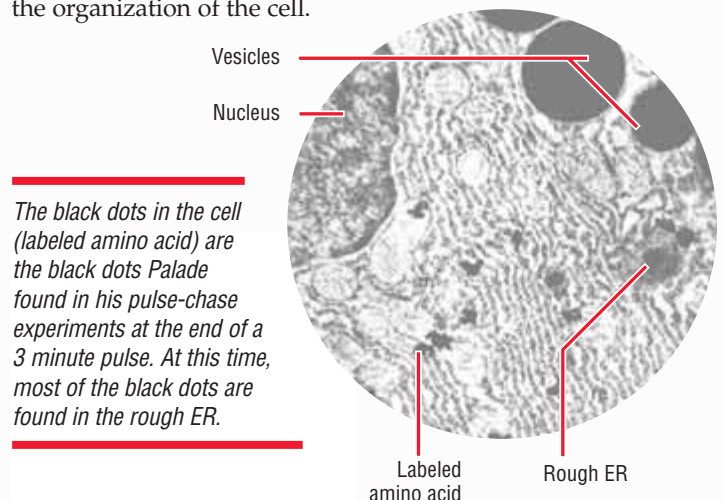
At first, the black dots that represented labeled amino acids were found in the rough ER. Photos taken at later time periods showed the black dots in vesicles close to smooth ER, then in smooth ER, then in the Golgi apparatus, and finally in vesicles close to the edge of the cell.

CONCLUSION: Secreted Proteins Follow a Specific Path

Palade concluded that secreted proteins move from the rough ER to the smooth ER in vesicles that are pinched off from the rough ER membrane. The proteins then move from the smooth ER to the Golgi apparatus (again, in vesicles). From the Golgi apparatus, the proteins move in vesicles to the edge of the cell. Finally, the vesicles fuse with the plasma membrane.

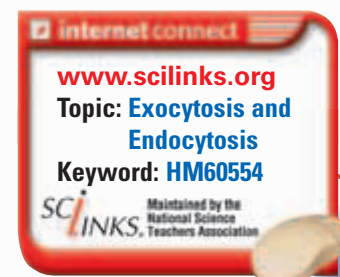
Further Experiments and a Nobel Prize

Palade published the work in 1964. In 1974, George Palade and two other researchers, Albert Claude and Christian De Duve, were awarded the Nobel Prize in medicine for their discoveries about the organization of the cell.



REVIEW

1. What did the pulse-chase experiments allow Palade to observe?
2. Summarize the results of Palade's pulse-chase experiments.
3. **Critical Thinking** Was it important to use living tissue for the experiments?



UNIQUE FEATURES OF PLANT CELLS

Plant cells have three kinds of structures that are not found in animal cells and that are extremely important to plant survival: plastids, central vacuoles, and cell walls.

PLANT CELLS

Most of the organelles and other parts of the cell just described are common to all eukaryotic cells. However, plant cells have three additional kinds of structures that are extremely important to plant function: cell walls, large central vacuoles, and plastids.

To understand why plant cells have structures not found in animal cells, consider how a plant's lifestyle differs from an animal's. Plants make their own carbon-containing molecules directly from carbon taken in from the environment. Plant cells take carbon dioxide gas from the air, and in a process called *photosynthesis*, they convert carbon dioxide and water into sugars. The organelles and structures in plant cells are shown in Figure 4-21.

OBJECTIVES

- **List** three structures that are present in plant cells but not in animal cells.
- **Compare** the plasma membrane, the primary cell wall, and the secondary cell wall.
- **Explain** the role of the central vacuole.
- **Describe** the roles of plastids in the life of a plant.
- **Identify** features that distinguish prokaryotes, eukaryotes, plant cells, and animal cells.

VOCABULARY

cell wall
central vacuole
plastid
chloroplast
thylakoid
chlorophyll

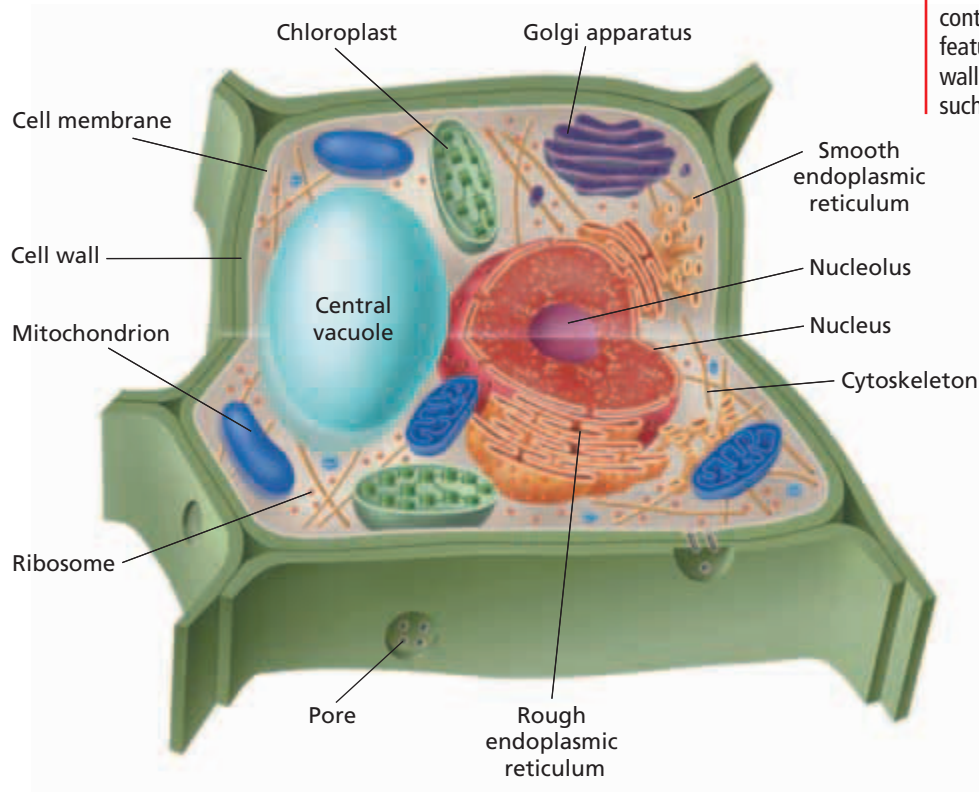


FIGURE 4-21

In addition to containing almost all of the types of organelles that animal cells contain, plant cells contain three unique features. Those features are the cell wall, the central vacuole, and plastids, such as chloroplasts.

CELL WALL

The **cell wall** is a rigid layer that lies outside the cell's plasma membrane. Plant cell walls contain a carbohydrate called *cellulose*. Cellulose is embedded in a matrix of proteins and other carbohydrates that form a stiff box around each cell. Pores in the cell wall allow water, ions, and some molecules to enter and exit the cell.

Primary and Secondary Cell Walls

The main component of the cell wall, cellulose, is made directly on the surface of the plasma membrane by enzymes that travel along the membrane. These enzymes are guided by microtubules inside the plasma membrane. Growth of the primary cell wall occurs in one direction, based on the orientation of the microtubules. Other components of the cell wall are made in the ER. These materials move in vesicles to the Golgi and then to the cell surface.

Some plants also produce a secondary cell wall. When the cell stops growing, it secretes the secondary cell wall between the plasma membrane and the primary cell wall. The secondary cell wall is very strong but can no longer expand. The wood in desks and tabletops is made of billions of secondary cell walls. The cells inside the walls have died and disintegrated.

CENTRAL VACUOLE

Plant cells may contain a reservoir that stores large amounts of water. The **central vacuole** is a large, fluid-filled organelle that stores not only water but also enzymes, metabolic wastes, and other materials. The central vacuole, shown in Figure 4-22, forms as other smaller vacuoles fuse together. Central vacuoles can make up 90 percent of the plant cell's volume and can push all of the other organelles into a thin layer against the plasma membrane. When water is plentiful, it fills a plant's vacuoles. The cells expand

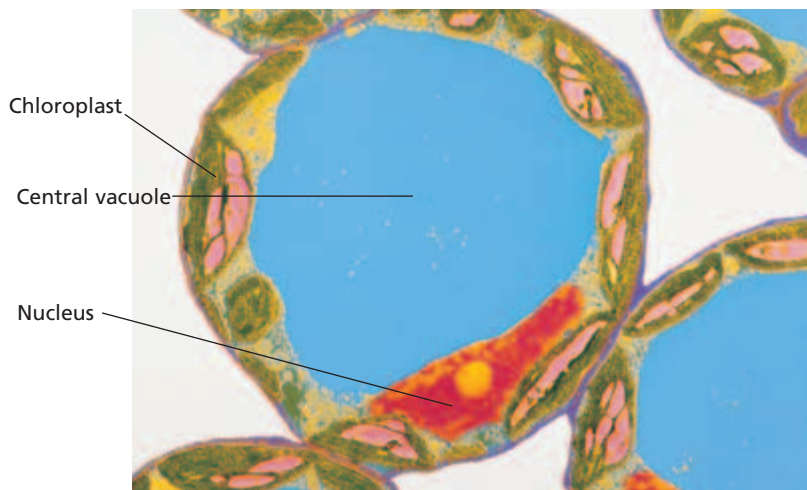
and the plant stands upright. In a dry period, the vacuoles lose water, the cells shrink, and the plant wilts.

Other Vacuoles

Some vacuoles store toxic materials. The vacuoles of acacia trees, for example, store poisons that provide a defense against plant-eating animals. Tobacco plant cells store the toxin nicotine in a storage vacuole. Other vacuoles store plant pigments, such as the colorful pigments found in rose petals.

FIGURE 4-22

The central vacuole occupies up to 90 percent of the volume of some plant cells. The central vacuole stores water and helps keep plant tissue firm.



PLASTIDS

Plastids are another unique feature of plant cells. **Plastids** are organelles that, like mitochondria, are surrounded by a double membrane and contain their own DNA. There are several types of plastids, including chloroplasts, chromoplasts, and leucoplasts.

Chloroplasts

Chloroplasts use light energy to make carbohydrates from carbon dioxide and water. As Figure 4-23 shows, each chloroplast contains a system of flattened, membranous sacs called **thylakoids**. Thylakoids contain the green pigment **chlorophyll**, the main molecule that absorbs light and captures light energy for the cell. Chloroplasts can be found not only in plant cells but also in a wide variety of eukaryotic algae, such as seaweed.

Chloroplast DNA is very similar to the DNA of certain photosynthetic bacteria. Plant cell chloroplasts can arise only by the division of preexisting chloroplasts. These facts may suggest that chloroplasts are descendants of ancient prokaryotic cells. Like mitochondria, chloroplasts are also thought to be the descendants of ancient prokaryotic cells that were incorporated into plant cells through a process called *endosymbiosis*.

Chromoplasts

Chromoplasts are plastids that contain colorful pigments and that may or may not take part in photosynthesis. Carrot root cells, for example, contain chromoplasts filled with the orange pigment carotene. Chromoplasts in flower petal cells contain red, purple, yellow, or white pigments.

Other Plastids

Several other types of plastids share the general features of chloroplasts but differ in content. For example, amyloplasts store starch. Chloroplasts, chromoplasts, and amyloplasts arise from a common precursor, called a *proplastid*.

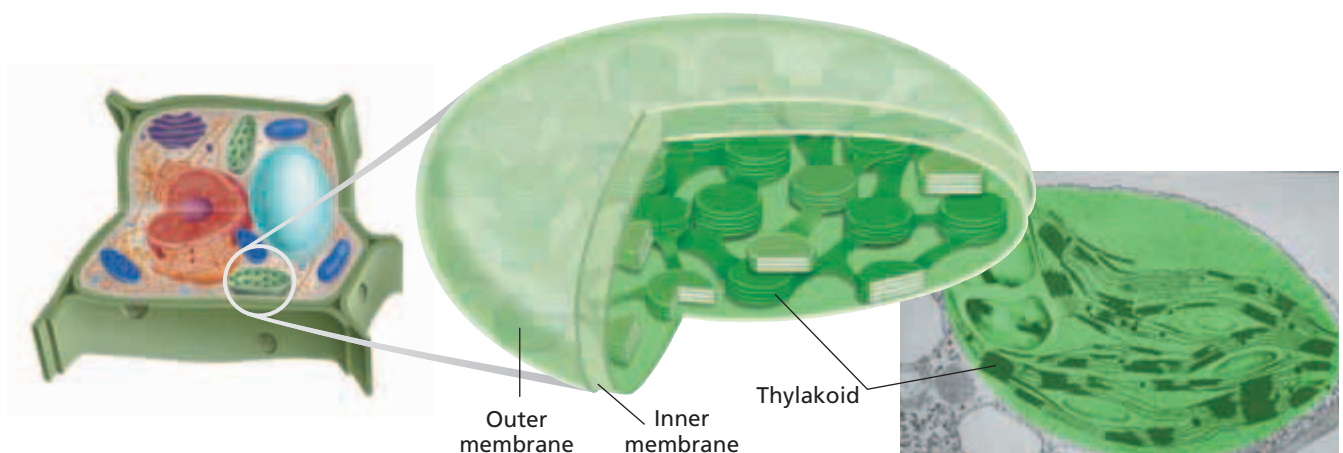
Word Roots and Origins

chloroplast

from the Greek *chloros*, meaning "pale green," and *plastos*, meaning "formed"

FIGURE 4-23

A chloroplast captures energy from sunlight and uses that energy to convert carbon dioxide and water into sugar and other carbohydrates.



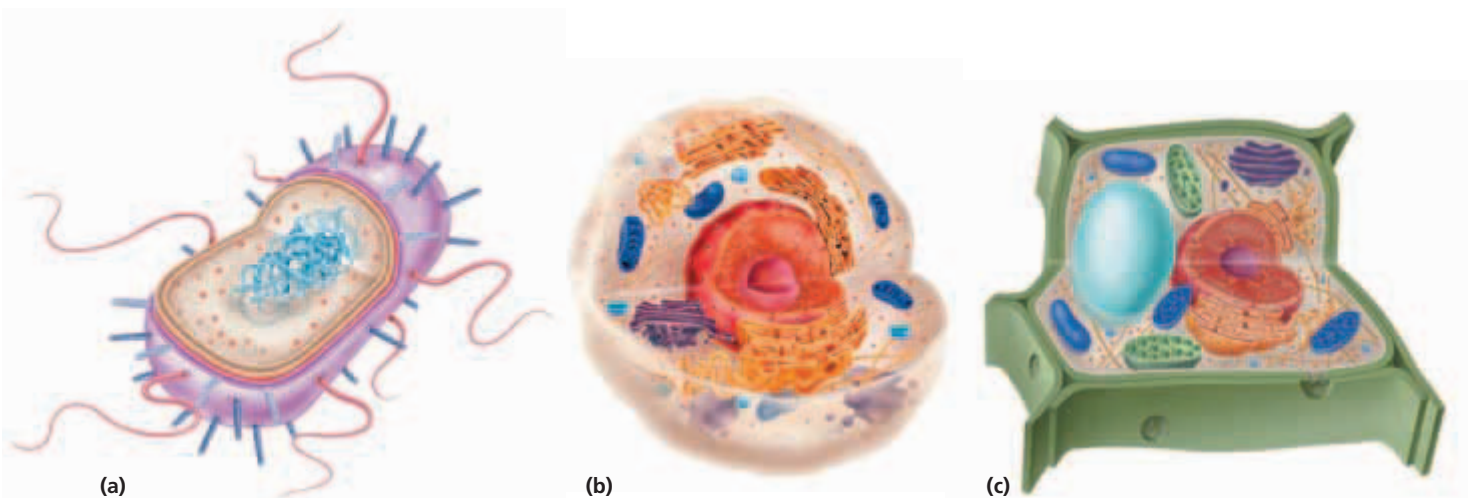


FIGURE 4-24

Prokaryotes (a) can be distinguished from eukaryotes (b and c) in that prokaryotes lack a nucleus and membrane-bound organelles. Plant cells (c) have the same organelles that animal cells do and have a cell wall, a central vacuole, and plastids.

COMPARING CELLS

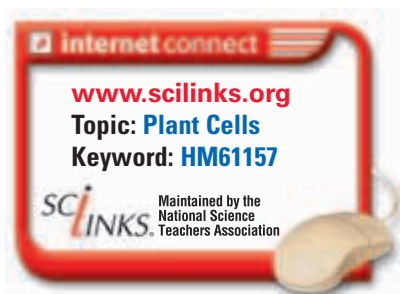
All cells share common features, such as a cell membrane, cytoplasm, ribosomes, and genetic material. But there is a high level of diversity among cells, as shown in Figure 4-24. There are significant differences between prokaryotes and eukaryotes. In addition, plant cells have features that are not found in animal cells.

Prokaryotes Versus Eukaryotes

Prokaryotes differ from eukaryotes in that prokaryotes lack a nucleus and membrane-bound organelles. Prokaryotes have a region, called a *nucleoid*, in which their genetic material is concentrated. However, prokaryotes lack an internal membrane system.

Plant Cells Versus Animal Cells

Three unique features distinguish plant cells from animal cells. One is the production of a cell wall by plant cells. Plant cells contain a large central vacuole. Third, plant cells contain a variety of plastids, which are not found in animal cells. Cell walls, central vacuoles, and plastids are unique features that are important to plant function.



SECTION 4 REVIEW

1. Identify three unique features of plant cells.
2. List the differences between the plasma membrane, the primary cell wall, and the secondary cell wall.
3. Identify three functions of plastids.
4. Name three things that may be stored in vacuoles.
5. Describe the features that distinguish prokaryotes from eukaryotes and plant cells from animal cells.

CRITICAL THINKING

6. **Evaluating Viewpoints** One student says vacuoles keep plants from wilting. Another says cell walls do this. Who is right? Explain.
7. **Making Comparisons** If you discovered a new cell, what characteristics would you use to determine which kind of cell it is? Explain.
8. **Analyzing Information** Tobacco plant cells contain a toxic chemical. Why don't tobacco plant cells poison themselves? Explain.

CHAPTER HIGHLIGHTS

SECTION 1

The History of Cell Biology

- All living things are made up of one or more cells. Robert Hooke discovered cells. Anton van Leeuwenhoek was the first to observe living cells.
- The cell theory states all living organisms are made of one or more cells, cells are the basic units of structure and function, and cells come only from pre-existing cells.
- All living things are made of organized parts, obtain energy from their surroundings, perform chemical reactions, change with time, respond to their environment, and reproduce.

Vocabulary

cell (p. 69)

cell theory (p. 70)

SECTION 2

Introduction to Cells

- A cell's shape reflects its function.
- Cell size is limited by a cell's surface area-to-volume ratio.
- The three basic parts of a cell are the plasma membrane, the cytoplasm, and the nucleus.
- Prokaryotes are organisms that lack a nucleus or membrane-bound organelles.
- In multicellular eukaryotes, cells organize into tissues, organs, organ systems, and finally organisms.

Vocabulary

plasma membrane (p. 74)

cytoplasm (p. 74)

cytosol (p. 74)

nucleus (p. 74)

prokaryote (p. 75)

eukaryote (p. 75)

organelle (p. 75)

tissue (p. 76)

organ (p. 76)

organ system (p. 76)

SECTION 3

Cell Organelles and Features

- Cell membranes are made of two phospholipid layers and proteins.
- The nucleus directs the cell's activities and stores DNA.
- Mitochondria harvest energy from organic compounds and transfer it to ATP.
- Ribosomes are either free or attached to the rough ER and play a role in protein synthesis.
- The rough ER prepares proteins for export or insertion into the cell membrane. The smooth ER builds lipids and participates in detoxification of toxins.
- The Golgi processes and packages proteins.
- Vesicles are classified by their contents.
- The cytoskeleton is made of protein fibers that help cells move and maintain their shape.

Vocabulary

phospholipid bilayer (p. 77)

chromosome (p. 79)

nuclear envelope (p. 79)

nucleolus (p. 79)

ribosome (p. 79)

mitochondrion (p. 80)

endoplasmic reticulum (p. 81)

Golgi apparatus (p. 82)

lysosome (p. 82)

cytoskeleton (p. 84)

microtubule (p. 84)

microfilament (p. 84)

cilium (p. 85)

flagellum (p. 85)

centriole (p. 85)

SECTION 4

Unique Features of Plant Cells

- Plant cells have cell walls, central vacuoles, and plastids.
- In plant cells, a rigid cell wall covers the cell membrane and provides support and protection.
- Large central vacuoles store water, enzymes, and waste products and provide support for plant tissue.
- Plastids store starch and pigments. The chloroplast converts light energy into chemical energy by photosynthesis.
- Prokaryotes, animal cells, and plant cells can be distinguished from each other by their unique features.

Vocabulary

cell wall (p. 88)

central vacuole (p. 88)

plastid (p. 89)

chloroplast (p. 89)

thylakoid (p. 89)

chlorophyll (p. 89)

CHAPTER REVIEW

USING VOCABULARY

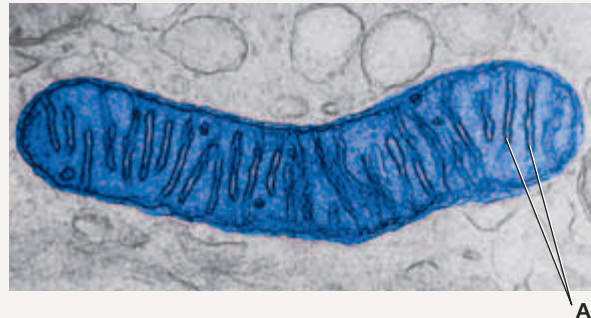
- For each pair of terms, explain how the meanings of the terms differ.
 - nucleolus* and *nucleus*
 - cell wall* and *cell membrane*
 - ribosomes* and *endoplasmic reticulum*
 - chromatin* and *chromosomes*
 - mitochondria* and *chloroplast*
- Explain the relationship between cilia and flagella.
- Use the following terms in the same sentence: *rough ER*, *smooth ER*, *Golgi apparatus*, *vesicle*, and *ribosome*.
- Word Roots and Origins** The word root *eu* means “true,” *pro* means “before,” and *kary* means “nucleus.” Using this information, explain what the terms *eukaryote* and *prokaryote* suggest about their evolution.

UNDERSTANDING KEY CONCEPTS

- Name** the scientist that first observed nonliving cells.
- Discuss** the roles of Schleiden, Schwann, and Virchow in the development of the cell theory.
- Analyze** the three parts of the cell theory.
- Identify** the characteristics shared by all living organisms.
- State** the relationship between cell shape and cell function.
- Identify** the factors that limit the growth of cells.
- Draw** the three major parts of a eukaryotic cell.
- Compare** the structure of a prokaryotic cell with that of a eukaryotic cell.
- Sequence** the relationship between organs, cells, organ systems, and tissues.
- Explain** why a cell membrane is called a *fluid mosaic*.
- Describe** the parts of a nucleus.
- Propose** why muscle cells have more mitochondria than other kinds of eukaryotic cells have.
- Describe** the role of ribosomes in cells.
- Compare** the functions of the Golgi apparatus with those of the ER.
- Discuss** the structure and function of vesicles.
- Describe** the structures that make up the cytoskeleton.
- Summarize** the differences between plant cells and animal cells.
- Propose** how the cell wall helps give support to a plant cell.
- Compare** primary and secondary cell walls.
- Predict** what would happen to a plant with a genetic defect that produced no central vacuole.
- Compare** mitochondria and chloroplasts.
- CONCEPT MAPPING** Use the following  terms to create a concept map that compares animal cells with plant cells: *cell membrane*, *cell wall*, *central vacuole*, *chloroplasts*, and *mitochondria*.

CRITICAL THINKING

- Interpreting Graphics** Answer the following questions based on the figure below.
 - Identify the structures labeled A in the micrograph.
 - Explain the significance of the shape of these structures.



- Applying Concepts** Using your knowledge of the relationship between surface area and volume, explain why small pieces of a food cook faster than larger pieces of the same food.
- Making Comparisons** The coils of a radiator provide a large surface area from which heat is radiated into a room. Which cell organelles have a structure similar to that of a radiator? How is their structure related to their function?
- Evaluating Differences** Explain why colonial organisms are not considered multicellular. What features make colonial organisms different from multicellular organisms?
- Applying Information** Write a report summarizing the roles of different types of cell-membrane proteins in the preservation of body organs donated for transplant.

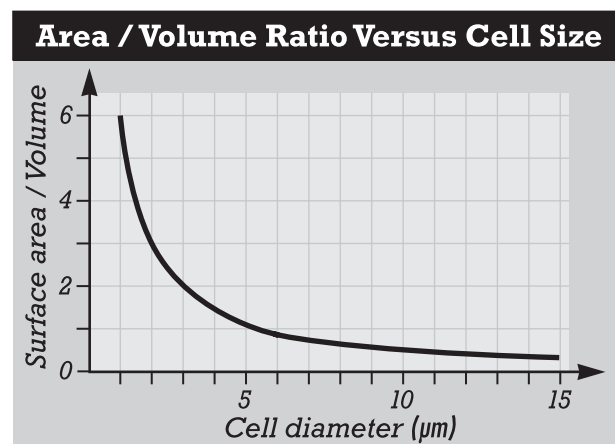


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- The eukaryotic nucleus houses all of the following except the
 - RNA
 - DNA
 - nucleolus
 - endoplasmic reticulum
- Which structure contributes to support and movement within a cell?
 - crista
 - cell wall
 - ribosome
 - microfilament
- Which of the following statements about RNA is true?
 - RNA is found only in proteins.
 - RNA is found only in the nucleus.
 - RNA is found only in the cytoplasm.
 - RNA is found in the nucleus and cytoplasm.

INTERPRETING GRAPHICS: The graph below shows the relationship between cell size and surface area-to-volume ratio. Use the graph below to answer the questions that follow.

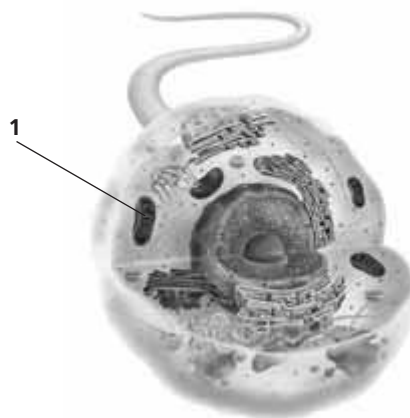


- By what percentage does the surface area-to-volume ratio change when a cell grows from 1 μm to 2 μm in diameter?
 - 10 percent
 - 20 percent
 - 50 percent
 - 90 percent
- What is the maximum diameter that this cell could attain before the surface area-to-volume ratio would fall below 1?
 - 2 μm
 - 5 μm
 - 10 μm
 - 15 μm

DIRECTIONS: Complete the following analogy.

6. Mitochondria : energy release :: ribosome :
- cell support
 - protein synthesis
 - cellular digestion
 - cellular transport

INTERPRETING GRAPHICS: The figure below shows a diagram of a cell. Use the figure to answer the question that follows.



7. What is the function of the structure labeled 1?
- to make ATP
 - to make proteins
 - to make carbohydrates
 - to move proteins through the cell

SHORT RESPONSE

A cell's shape is generally related to its function.

Skin cells are flat and platelike. Nerve cells have long extensions. Explain the relationship between the shape of skin and nerve cells and their function in the body.

EXTENDED RESPONSE

Despite the diversity among cells, eukaryotic cells share many common features.

Part A Describe the structure and function of the organelles found in an animal cell.

Part B Summarize the differences that distinguish animal cells from bacteria and plant cells.

Test TIP

For short-response and essay questions that ask you to compare and contrast, be sure to include both the shared characteristics (compare) and unique characteristics (contrast) for each item.

Comparing Animal and Plant Cells

OBJECTIVES

- Examine the similarities and differences between the structure of cells in animals and the structure of cells in plants.

PROCESS SKILLS

- hypothesizing
- classifying
- observing

MATERIALS

- lab apron
- safety goggles
- compound light microscope
- forceps
- microscope slides and coverslips
- dropper bottle of Lugol's iodine solution
- prepared slides of human epithelial cells
- sprigs of *Elodea*
- prepared slides of three unknowns

Background

1. In this investigation, you will use a compound light microscope to observe cells from animals and plants. First, you will view a prepared slide of human epithelial cells taken from the skin lining the mouth. Then, you will make your own slide of a leaf from *Elodea*, a pond weed shown in the photograph on the next page.



2. Based on your observations of human epithelial cells and *Elodea* leaf cells, you will be asked to classify three slides of unknown cells as either animal or plant cells.
3. Before you examine any cells, list the structural characteristics that distinguish animal cells from plant cells.

PART A Animal Cells

1.  **CAUTION** Handle glass microscope slides carefully. Dispose of broken glass separately in a container designated by your teacher.
2.  **CAUTION** Do not use electrical equipment with wet hands or near water.
3. Examine a prepared slide of epithelial cells under low, power. Locate cells that are separate from each other, and place them in the center of the field of view. Examine the cells under high power. Adjust the diaphragm to reduce the light intensity and achieve greater clarity.
4. In your lab report, make a drawing of two or three cells as they appear under high power. Identify and label the cell membrane, the cytoplasm, the nuclear envelope, and the nucleus of one of the cells in your drawing.

PART B Plant Cells

5. Carefully tear off a small leaf near the top of an *Elodea* sprig. Using forceps, place the whole leaf in a drop of water on a slide. Place a coverslip on top of the leaf.
6. Observe the leaf under low power. The outermost part of the cell is the cell wall. The many small, green organelles in the cells are chloroplasts.
7. Locate a cell that you can see clearly, and move the slide so that the cell is in the center of the field of view. Examine this cell under high power, and use the fine adjustment to bring the cell into focus.
8. Find an *Elodea* cell that is large enough to allow you to see the cell wall and the chloroplasts clearly. In your lab report, make a drawing of this cell. Label the cell wall and at least one chloroplast in your drawing.



9. The chloroplasts may be moving in some of the cells. If you observe no movement, warm the slide in your hand or shine a bright lamp on it for a minute or two. Then, reexamine the slide under high power, and look for the movement of the cell's contents. This movement is called *cytoplasmic streaming*.
10. Because the cell membrane is pressed against the cell wall, you may not see it. Also, the abundance of chloroplasts may hide other organelles in the cells. You can make the cell membrane, vacuole, nucleus, and nucleolus more visible by making a stained wet-mount slide of *Elodea*.
11.    Put on a lab apron and safety goggles. Prepare a wet-mount slide of *Elodea* as you did in step 4, but substitute Lugol's iodine solution for the water. Allow the iodine solution to diffuse throughout the leaf.

12. Observe the stained cells under low and high power. Make a drawing of a stained *Elodea* cell in your lab report. Label the central vacuole, nucleus, nucleolus, chloroplasts, cell wall, and cell membrane if they are visible.

PART C Identifying Unknown Cells

13. Make a data table like the one below to record your observations of the unknown specimens.
14. Obtain prepared slides of three unknown specimens from your teacher.
15. Observe each specimen under low and high power. In your data table, record the code number assigned to each unknown, each specimen's classification as plant or animal, and your reasons for classifying each specimen.
16.   Clean up your materials, and wash your hands before leaving the lab.

Analysis and Conclusions

1. According to your observations in this investigation, list several ways that plant and animal cells are structurally similar and several ways that they are different.
2. What do you think might be the function of cytoplasmic streaming in a plant cell? Lugol's iodine solution causes cytoplasmic streaming to stop. Why do you think this happens?
3. Which organelles that you read about in this chapter did you not see in this investigation? Why do you think you were unable to see these organelles in your slides?

Further Inquiry

Use library resources to locate electron micrographs of cell structures that you were unable to see with the compound light microscope.

CLASSIFICATION OF UNKNOWN SPECIMENS

Unknown (code number)	Classification (plant or animal)	Reasons for classification

HOMEOSTASIS AND CELL TRANSPORT

The process of phagocytosis is just one of the many ways cells transport materials in and out of the cell. An example of phagocytosis is shown here as a macrophage engulfs a human tumor cell. (SEM 3,520×)



SECTION 1 *Passive Transport*

SECTION 2 *Active Transport*



Unit 1—Cell Transport
and Homeostasis
Topics 3–6

PASSIVE TRANSPORT

*Cell membranes help organisms maintain homeostasis by controlling what substances may enter or leave cells. Some substances can cross the cell membrane without any input of energy by the cell in a process known as **passive transport**.*

DIFFUSION

The simplest type of passive transport is diffusion. **Diffusion** is the movement of molecules from an area of higher concentration to an area of lower concentration. This difference in the concentration of molecules across a distance is called a **concentration gradient**. Consider what happens when you add a sugar cube to a beaker of water. As shown in Figure 5-1, the sugar cube sinks to the bottom of the beaker. This sinking makes the concentration of sugar molecules greater at the bottom of the beaker than at the top. As the cube dissolves, the sugar molecules begin to diffuse slowly through the water, moving towards the lower concentration at the top.

Diffusion is driven entirely by the molecules' kinetic energy. Molecules are in constant motion because they have kinetic energy. Molecules move randomly, traveling in a straight line until they hit an object, such as another molecule. When they hit something, they bounce off and move in a new direction, traveling in another straight line. If no object blocks their movement, they continue on their path. Thus, molecules tend to move from areas where they are more concentrated to areas where they are less concentrated, or "down" their concentration gradient.

In the absence of other influences, diffusion will eventually cause the molecules to be in **equilibrium**—the concentration of molecules will be the same throughout the space the molecules occupy. Returning to the example in Figure 5-1, if the beaker of water is left undisturbed, at some point the concentration of sugar molecules will be the same throughout the beaker. The sugar concentration will then be at equilibrium.

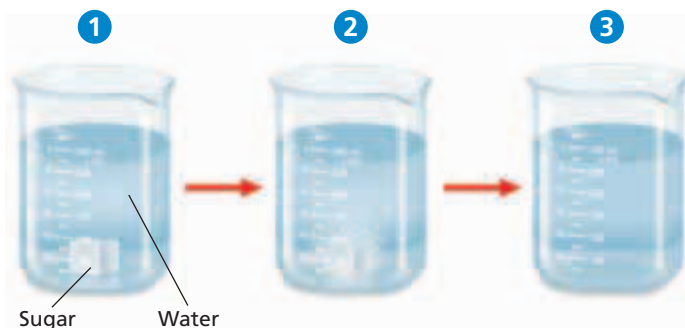


FIGURE 5-1

Sugar molecules, initially in a high concentration at the bottom of a beaker, ①, will move about randomly through diffusion, ②, and eventually reach equilibrium, ③. At equilibrium the sugar concentration will be the same throughout the beaker. Diffusion occurs naturally because of the kinetic energy the molecules possess.

OBJECTIVES

- **Explain** how an equilibrium is established as a result of diffusion.
- **Distinguish** between diffusion and osmosis.
- **Explain** how substances cross the cell membrane through facilitated diffusion.
- **Explain** how ion channels assist the diffusion of ions across the cell membrane.

VOCABULARY

passive transport
diffusion
concentration gradient
equilibrium
osmosis
hypotonic
hypertonic
isotonic
contractile vacuole
turgor pressure
plasmolysis
cytolysis
facilitated diffusion
carrier protein
ion channel



Quick Lab

Observing Diffusion

Materials disposable gloves, lab apron, safety goggles, 600 mL beaker, 25 cm dialysis tubing, funnel, 15 mL starch solution (10 percent), 20 drops Lugol's solution, 300 mL water, 100 mL graduated cylinder, 20 cm piece of string (2)

Procedure



1. Put on your disposable gloves, lab apron, and safety goggles.
2. Pour 300 mL of water in the 600 mL beaker.
3. Add 20 drops of Lugol's solution to the water.
4. Open the dialysis tubing, and tie one end tightly with a piece of string.
5. Using the funnel, pour 15 mL of 10 percent starch solution into the dialysis tubing.
6. Tie the other end of the dialysis tubing tightly with the second piece of string, forming a sealed bag around the starch solution.
7. Place the bag into the solution in the beaker, and observe the setup for a color change.

Analysis What happened to the color in the bag? What happened to the color of the water around the bag? Explain your observations.

It is important to understand that even at equilibrium the random movement of molecules continues. But because there is an equal concentration of molecules everywhere, molecules are just as likely to move in one direction as in any other. The random movements of many molecules in many directions balance one another, and equilibrium is maintained.

Diffusion Across Membranes

Cell membranes allow some molecules to pass through, but not others. If a molecule can pass through a cell membrane, it will diffuse from an area of higher concentration on one side of the membrane to an area of lower concentration on the other side. Diffusion across a membrane is also called *simple diffusion*, and only allows certain molecules to pass through the membrane.

The simple diffusion of a molecule across a cell membrane depends on the size and type of molecule and on the chemical nature of the membrane. A membrane can be made, in part, of a phospholipid bilayer, and certain proteins can form pores in the membrane. Molecules that can dissolve in lipids may pass directly through the membrane by diffusion. For example, because of their nonpolar nature, both carbon dioxide and oxygen dissolve in lipids. Molecules that are very small but not soluble in lipids may diffuse across the membrane by moving through the pores in the membrane.

OSMOSIS

A solution is composed of a solute dissolved in a solvent. In the sugar water described in Figure 5-1, the solute was sugar and the solvent was water, and the solute molecules diffused through the solvent. It is also possible for solvent molecules to diffuse. In the case of cells, the solutes are organic and inorganic compounds, and the solvent is water. The process by which water molecules diffuse across a cell membrane from an area of higher concentration to an area of lower concentration is called **osmosis** (ahs-MOH-sis). Because water is moving from a higher to lower concentration, osmosis does not require cells to expend energy. Therefore, osmosis is the passive transport of water.

Direction of Osmosis

The net direction of osmosis depends on the relative concentration of solutes on the two sides of the membrane. Examine Table 5-1. When the concentration of solute molecules outside the cell is *lower* than the concentration in the cytosol, the solution outside is **hypotonic** to the cytosol. In this situation, water diffuses *into* the cell until equilibrium is established. When the concentration of solute molecules outside the cell is *higher* than the concentration in the cytosol, the solution outside is **hypertonic** to the cytosol. In this situation, water diffuses *out of* the cell until equilibrium is established.

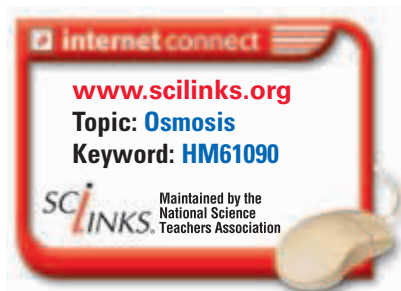
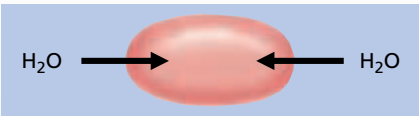
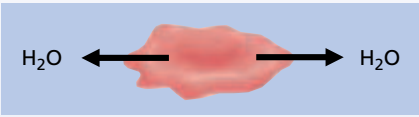
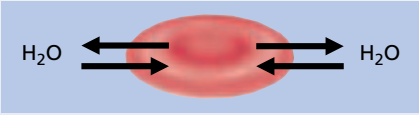


TABLE 5-1 Direction of Osmosis

Condition	Net movement of water
External solution is hypotonic to cytosol	into the cell 
External solution is hypertonic to cytosol	out of the cell 
External solution is isotonic to cytosol	none 

When the concentrations of solutes outside and inside the cell are equal, the outside solution is said to be **isotonic** to the cytosol. Under these conditions, water diffuses into and out of the cell at equal rates, so there is no net movement of water.

Notice that the prefixes *hypo-*, *hyper-*, and *iso-* refer to the relative solute concentrations of two solutions. Thus, if the solution outside the cell is *hypotonic* to the cytosol, then the cytosol must be *hypertonic* to that solution. Conversely, if the solution outside is *hypertonic* to the cytosol, then the cytosol must be *hypotonic* to the solution. Water tends to diffuse from hypotonic solutions to hypertonic solutions.

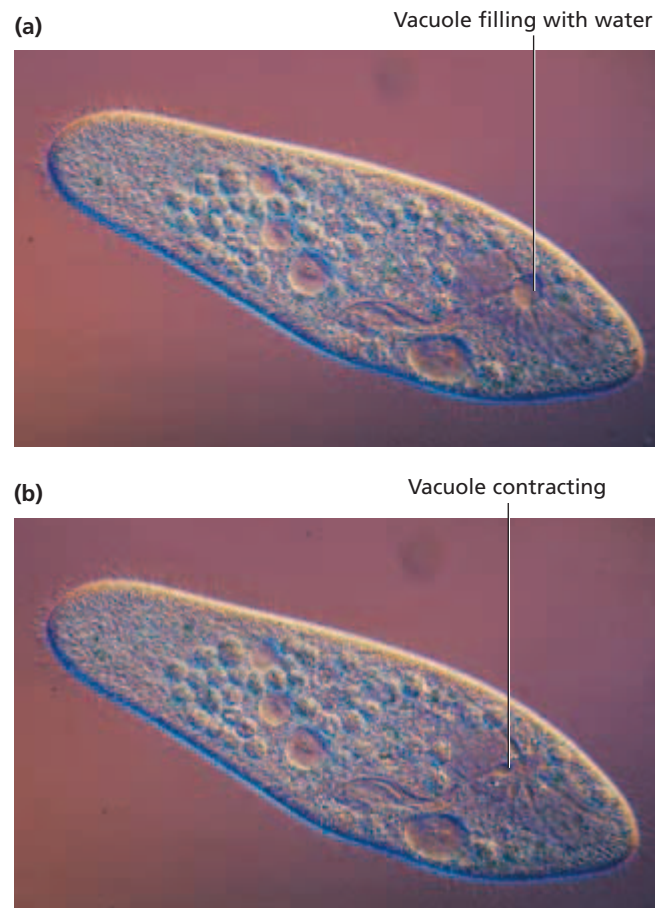
How Cells Deal with Osmosis

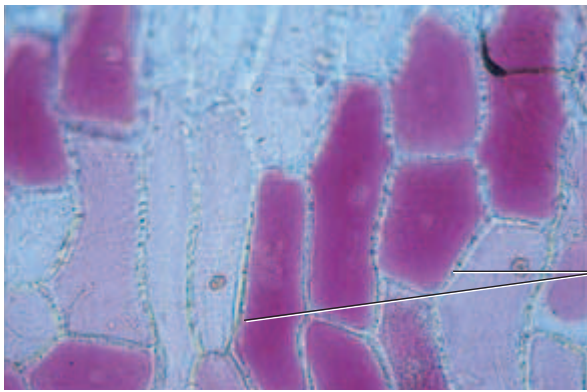
Cells that are exposed to an isotonic external environment usually have no difficulty keeping the movement of water across the cell membrane in balance. This is the case with the cells of vertebrate animals on land and of most other organisms living in the sea. In contrast, many cells function in a hypotonic environment. Such is the case for unicellular freshwater organisms. Water constantly diffuses into these organisms. Because they require a relatively lower concentration of water in the cytosol to function normally, unicellular organisms must rid themselves of the excess water that enters by osmosis.

Some of them, such as the paramecia shown in Figure 5-2, do this with **contractile vacuoles** (kon-TRAK-til VAK-yōō-ōl), which are organelles that remove water. Contractile vacuoles collect the excess water and then contract, pumping the water out of the cell. Unlike diffusion and osmosis, this pumping action is not a form of passive transport because it requires the cell to expend energy.

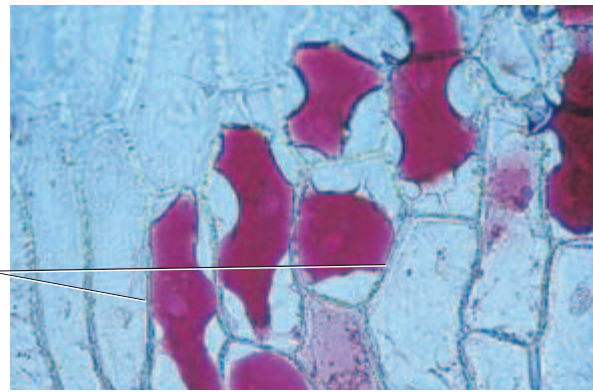
FIGURE 5-2

The paramecia shown below live in fresh water, which is hypotonic to their cytosol. (a) Contractile vacuoles collect excess water that moves by osmosis into the cytosol. (b) The vacuoles then contract, returning the water to the outside of the cell. (LM 315×)





(a) HYPOTONIC



(b) HYPERTONIC

FIGURE 5-3

These two photographs show cells in the skin of a red onion. (a) In a hypotonic environment, the cells are pressed against the cell walls. (b) In a hypertonic environment, the cells contract and pull away from the cell walls.

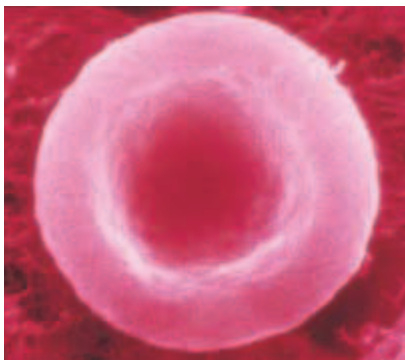
Word Roots and Origins

cytolysis

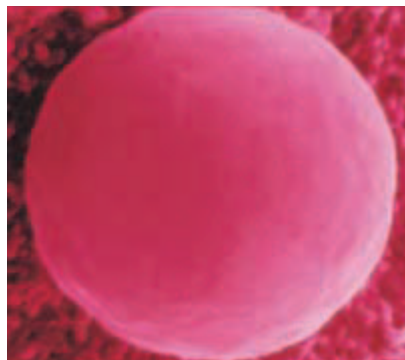
from the Greek *kytos*, meaning "hollow vessel," and *lysis*, meaning "loosening"

FIGURE 5-4

(a) In an environment that is isotonic to the cytosol, a human red blood cell keeps its normal shape—round and dimpled. (b) In a hypotonic environment, the cell gains water and swells. (c) In a hypertonic environment, the cell loses water and becomes shriveled.



(a) ISOTONIC



(b) HYPOTONIC



(c) HYPERTONIC

Other cells, including many of those in multicellular organisms, respond to hypotonic environments by pumping solutes out of the cytosol. This lowers the solute concentration in the cytosol, bringing it closer to the solute concentration in the environment. As a result, water molecules are less likely to diffuse into the cell.

Most plant cells, like animal cells, live in a hypotonic environment. In fact, the cells that make up plant roots may be surrounded by water. This water moves into plant cells by osmosis. These cells swell as they fill with water until the cell membrane is pressed against the inside of the cell wall, as Figure 5-3a shows. The cell wall is strong enough to resist the pressure exerted by the water inside the expanding cell. The pressure that water molecules exert against the cell wall is called **turgor pressure** (TER-GOR PRESH-er).

In a hypertonic environment, water leaves the cells through osmosis. As shown in Figure 5-3b, the cells shrink away from the cell walls, and turgor pressure is lost. This condition is called **plasmolysis** (plaz-MAHL-uh-sis), and is the reason that plants wilt if they don't receive enough water.

Some cells cannot compensate for changes in the solute concentration of their environment. Red blood cells in humans, for instance, lack contractile vacuoles, solute pumps, and cell walls. As shown in Figure 5-4, these cells lose their normal shape when they are exposed to an environment that is not isotonic to their cytosol. In a hypertonic environment, water leaves the cells, making them shrink and shrivel. In a hypotonic environment, water diffuses into the cells, causing them to swell and eventually burst. The bursting of cells is called **cytolysis** (sie-TAHL-uh-sis).

FACILITATED DIFFUSION

Another type of passive transport is called **facilitated diffusion**. This process is used for molecules that cannot readily diffuse through cell membranes, even when there is a concentration gradient across the membrane. Such molecules may not be soluble in lipids, or they may be too large to pass through the pores in the membrane. In facilitated diffusion, the movement of these kinds of molecules across the cell membrane is assisted by specific proteins in the membrane. These proteins are known as **carrier proteins**.

The carrier proteins that serve in facilitated diffusion transport molecules from an area of higher concentration on one side of the membrane to an area of lower concentration on the other side. Because the molecules are moving from a higher to lower concentration, which does not require any additional energy, facilitated diffusion is passive transport.

Figure 5-5 shows a model of how facilitated diffusion is thought to work. According to the model, a molecule binds to a specific carrier protein that transports it. As soon as the molecule binds to the carrier protein, the carrier protein then changes shape. This altered shape may shield the molecule from the hydrophobic interior of the lipid bilayer. Once shielded, the molecule can be transported through the cell membrane. On the other side of the membrane, the molecule is released from the carrier protein, and the protein returns to its original shape. The carrier protein is now free to bind to another molecule.

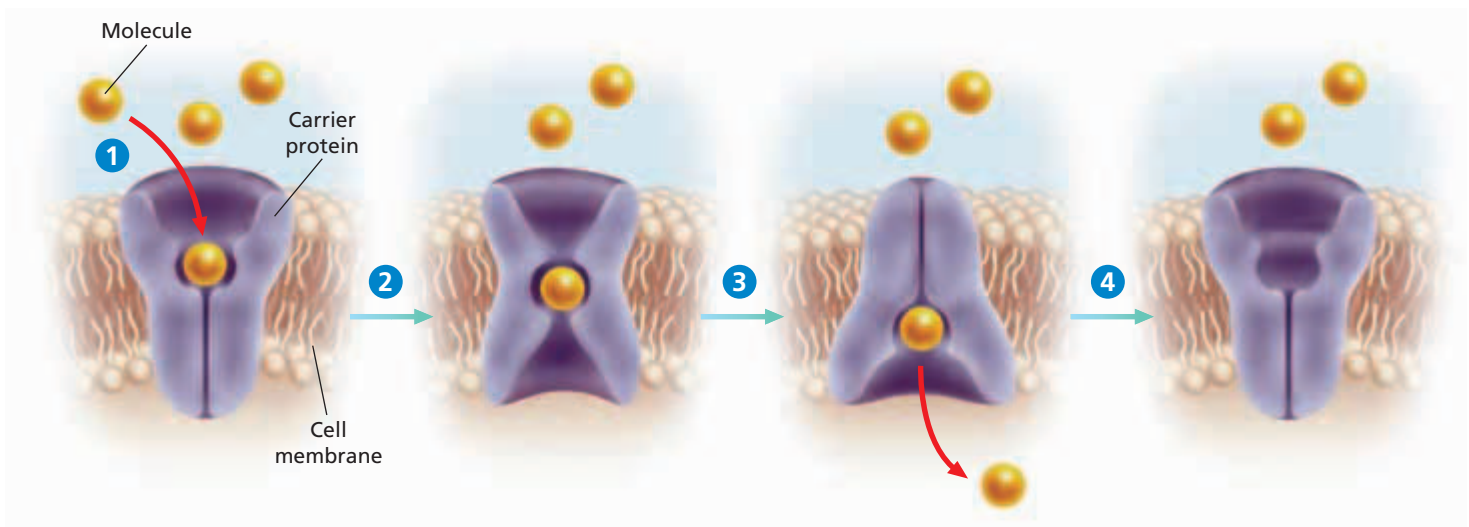
A good example of facilitated diffusion is the transport of glucose. Many cells depend on glucose for much of their energy needs. But glucose molecules are too large to diffuse easily across cell membranes. When the level of glucose within a cell is lower than the level of glucose outside the cell, carrier proteins transport glucose into the cell.

FIGURE 5-5

Facilitated diffusion occurs in four steps.

- 1 A molecule, such as glucose, binds to a carrier protein on one side of the cell membrane.
- 2 The carrier protein changes shape, shielding the molecule from the interior of the membrane.
- 3 The molecule is released on the other side of the membrane.
- 4 The carrier protein then returns to its original shape.

FACILITATED DIFFUSION



Eco Connection

Purifying Water with Membranes

The tendency for water molecules to diffuse across membranes can be used to extract pure water from a mixture of water and solutes. If a dilute solution is separated from a more concentrated solution by a selectively permeable membrane, osmosis will occur, as water molecules diffuse from the dilute solution to the concentrated solution. However, if enough external pressure is applied to the concentrated solution, the opposite will happen: water molecules will diffuse from the concentrated solution to the dilute solution. This process, called *reverse osmosis*, effectively moves most of the water to one side of the membrane and leaves most of the solutes on the other side.

Reverse osmosis was initially developed for desalination plants, which produce fresh water from sea water. It is now also used to purify polluted water from a variety of sources, including manufacturing facilities and sanitary landfills. After the polluted water from these sources is purified through reverse osmosis, it is clean enough to be returned safely to the environment.

The transport of glucose illustrates two important properties of facilitated diffusion. First, facilitated diffusion can help substances move either into or out of a cell, depending on the concentration gradient. Thus, when the level of glucose is higher inside a cell than it is outside the cell, facilitated diffusion speeds the diffusion of glucose out of the cell. Second, the carrier proteins involved in facilitated diffusion are each specific for one type of molecule. For example, the carrier protein that helps with the diffusion of glucose and other simple sugars does not assist with the diffusion of amino acids.

DIFFUSION THROUGH ION CHANNELS

Another type of transport involves membrane proteins known as **ion channels**. When ion channels transport ions from higher to lower concentrations they are a form of passive transport. Ion channels transport ions such as sodium (Na^+), potassium (K^+), calcium (Ca^{2+}), and chloride (Cl^-). These ions are important for a variety of cell functions. Because they are not soluble in lipids, however, ions cannot diffuse across the phospholipid bilayer without assistance. Ion channels provide small passageways across the cell membrane through which ions can diffuse. Each type of ion channel is usually specific for one type of ion. For example, sodium channels will allow Na^+ ions to go through, but will not allow Ca^{2+} ions or Cl^- to enter the cell.

Some ion channels are always open. Others have “gates” that open to allow ions to pass or close to stop their passage. The gates may open or close in response to three kinds of stimuli: stretching of the cell membrane, electrical signals, or chemicals in the cytosol or external environment. These stimuli therefore control the ability of specific ions to cross the cell membrane.

SECTION 1 REVIEW

1. Toward what condition does diffusion eventually lead, in the absence of other influences?
2. How is osmosis related to diffusion?
3. If the concentration of solute molecules outside a cell is lower than the concentration in the cytosol, is the external solution hypotonic, hypertonic, or isotonic to the cytosol?
4. What role do carrier proteins play in facilitated diffusion?
5. How is facilitated diffusion similar to diffusion through ion channels?

CRITICAL THINKING

6. **Applying Information** Sea water has a higher concentration of solutes than do human body cells. Why might drinking large amounts of sea water be dangerous for humans?
7. **Relating Concepts** What would happen to a grape placed in a bowl with highly concentrated sugar water?
8. **Applying Information** Using what you know about osmosis, explain what would happen to a jellyfish placed in a freshwater lake.

ACTIVE TRANSPORT

*In many cases, cells must move materials from an area of lower concentration to an area of higher concentration, or “up” their concentration gradient. Such movement of materials is known as **active transport**. Unlike passive transport, active transport requires a cell to expend energy.*

CELL MEMBRANE PUMPS

Ion channels and carrier proteins not only assist in passive transport but also help with some types of active transport. The carrier proteins that serve in active transport are often called *cell membrane “pumps”* because they move substances from lower to higher concentrations.

Carrier proteins involved in facilitated diffusion and those involved in active transport are very similar. In both, the molecule first binds to a specific kind of carrier protein on one side of the cell membrane. Once it is bound to the molecule, the protein changes shape, shielding the molecule from the hydrophobic interior of the phospholipid bilayer. The protein then transports the molecule through the membrane and releases it on the other side. However, cell membrane pumps require energy. Most often the energy needed for active transport is supplied directly or indirectly by ATP.

Sodium-Potassium Pump

One example of active transport in animal cells involves a carrier protein known as the **sodium-potassium pump**. As its name suggests, this protein transports Na^+ ions and K^+ ions up their concentration gradients. To function normally, some animal cells must have a higher concentration of Na^+ ions outside the cell and a higher concentration of K^+ ions inside the cell. The sodium-potassium pump maintains these concentration differences.

Follow the steps in Figure 5-6 to see how the sodium-potassium pump operates. First, three Na^+ ions bind to the carrier protein on the cytosol side of the membrane, as shown in step ①. At the same time, the carrier protein removes a phosphate group from a molecule of ATP. As you can see in step ②, the phosphate group from the ATP molecule binds to the carrier protein. Step ③ shows how the removal of the phosphate group from ATP supplies the energy needed to change the shape of the carrier protein. With its new shape, the protein carries the three Na^+ ions through the membrane and then forces the Na^+ ions outside the cell where the Na^+ concentration must remain high.

OBJECTIVES

- **Distinguish** between passive transport and active transport.
- **Explain** how the sodium-potassium pump operates.
- **Compare** endocytosis and exocytosis.

VOCABULARY

active transport
sodium-potassium pump
endocytosis
vesicle
pinocytosis
phagocytosis
phagocyte
exocytosis



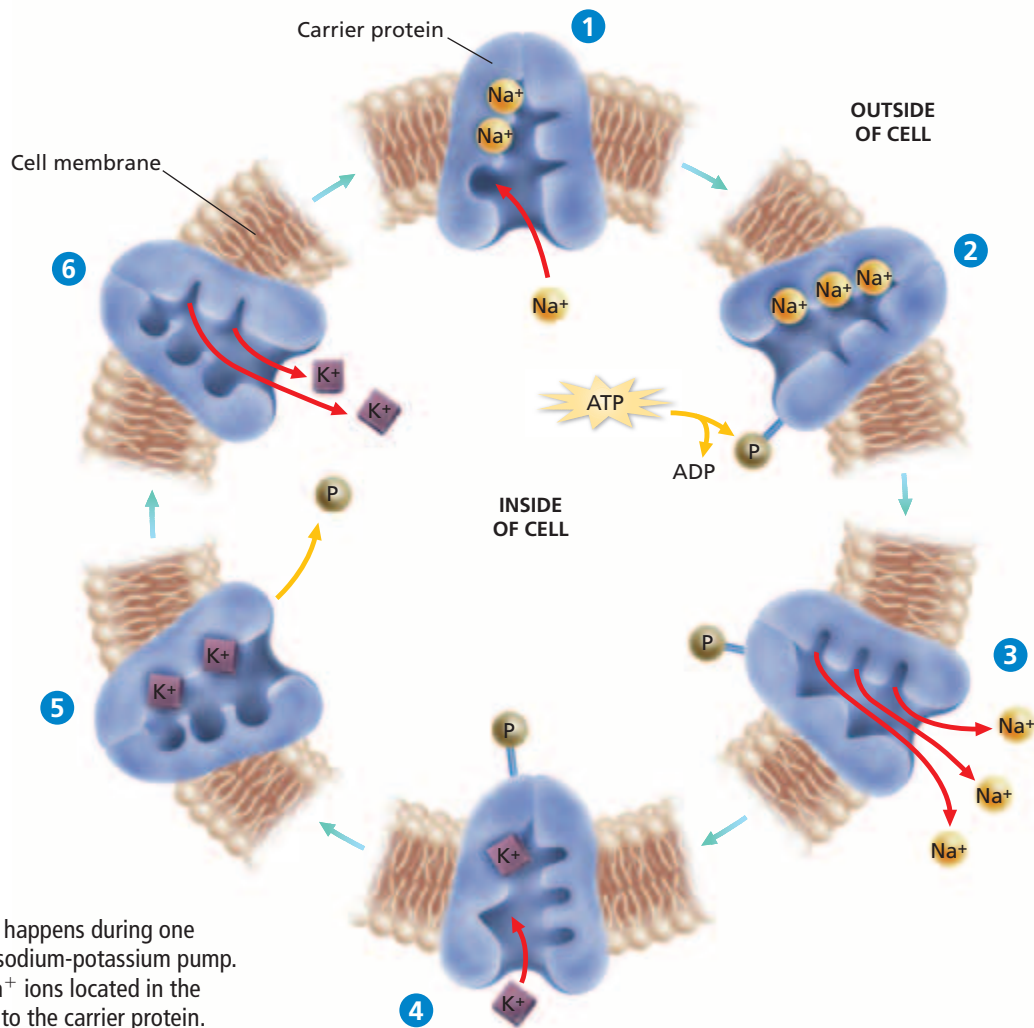


FIGURE 5-6

Follow what happens during one cycle of the sodium-potassium pump.

- 1 Three Na^+ ions located in the cytosol bind to the carrier protein.
- 2 A phosphate group, represented by the letter P in the diagram, is removed from ATP and bound to the carrier protein.
- 3 The binding of the phosphate group changes the shape of the carrier protein, allowing the three Na^+ ions to be released into the cell's environment.
- 4 Two K^+ ions located outside the cell bind to the carrier protein.
- 5 The phosphate group is released, restoring the original shape of the carrier protein.
- 6 The two K^+ ions are released into the cytosol, and the cycle is ready to repeat.

At this point, the carrier protein has the shape it needs to bind two K^+ ions outside the cell, as step 4 shows. When the K^+ ions bind, the phosphate group is released, as indicated in step 5, and the carrier protein restores its original shape. As shown in step 6 this time, the change in shape causes the carrier protein to release the two K^+ ions inside the cell. At this point the carrier protein is ready to begin the process again. Thus, a complete cycle of the sodium-potassium pump transports three Na^+ ions out of the cell and two K^+ ions into the cell. At top speed, the sodium-potassium pump can transport about 450 Na^+ ions and 300 K^+ ions per second.

The exchange of three Na^+ ions for two K^+ ions creates an electrical gradient across the cell membrane. That is, the outside of the membrane becomes positively charged relative to the inside of the membrane, which becomes relatively negative. In this way, the two sides of the cell membrane are like the positive and negative terminals of a battery. This difference in charge is important for the conduction of electrical impulses along nerve cells. The sodium-potassium pump is only one example of a cell membrane pump. Other pumps work in similar ways to transport important metabolic materials across cell membranes.

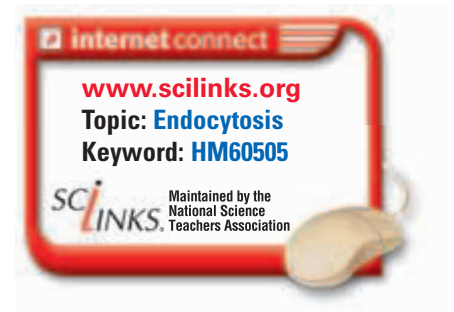
MOVEMENT IN VESICLES

Some substances, such as macromolecules and nutrients, are too large to pass through the cell membrane by the transport processes you have studied so far. Cells employ two other transport mechanisms—endocytosis and exocytosis—to move such substances into or out of cells. Endocytosis and exocytosis are also used to transport large quantities of small molecules into or out of cells at a single time. Both endocytosis and exocytosis require cells to expend energy. Therefore, they are types of active transport.

Endocytosis

Endocytosis (EN-doh-sie-TOH-sis) is the process by which cells ingest external fluid, macromolecules, and large particles, including other cells. As you can see in Figure 5-7, these external materials are enclosed by a portion of the cell's membrane, which folds into itself and forms a pouch. The pouch then pinches off from the cell membrane and becomes a membrane-bound organelle called a **vesicle**. Some of the vesicles fuse with lysosomes, and their contents are digested by lysosomal enzymes. Other vesicles that form during endocytosis fuse with other membrane-bound organelles.

Two main types of endocytosis are based on the kind of material that is taken into the cell: **pinocytosis** (PIEN-oh-sie-TOH-sis) involves the transport of solutes or fluids, and **phagocytosis** (FAG-oh-sie-TOH-sis) is the movement of large particles or whole cells. Many unicellular organisms feed by phagocytosis. In addition, certain cells in animals use phagocytosis to ingest bacteria and viruses that invade the body. These cells, known as **phagocytes**, allow lysosomes to fuse with the vesicles that contain the ingested bacteria and viruses. Lysosomal enzymes then destroy the bacteria and viruses before they can harm the animal.



Word Roots and Origins

vesicle

from the Latin *vesicula*,
meaning "bladder" or "sac"

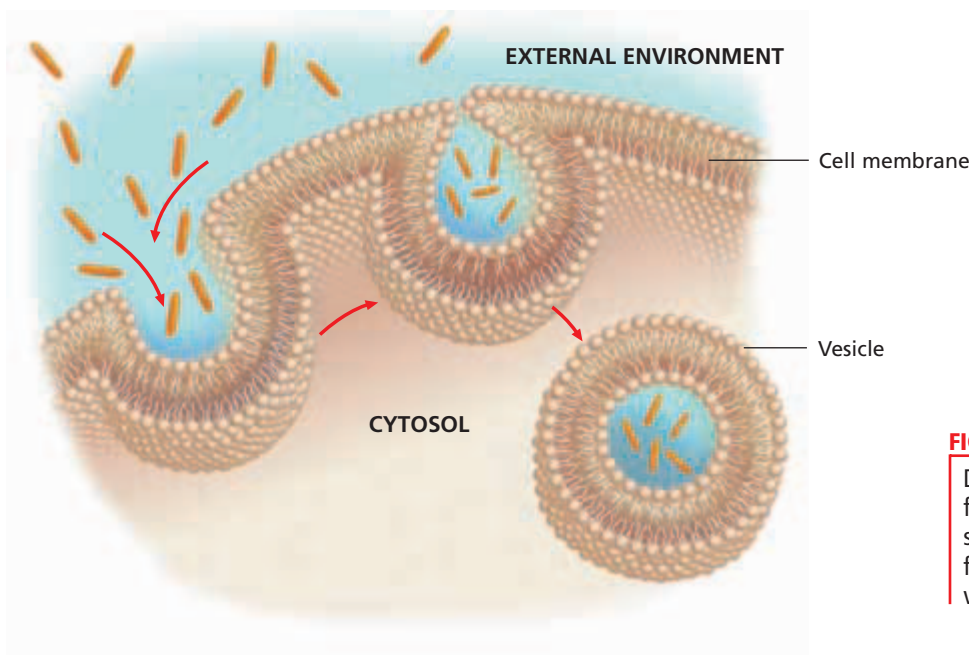


FIGURE 5-7

During endocytosis, the cell membrane folds around food or liquid and forms a small pouch. The pouch then pinches off from the cell membrane to become a vesicle.

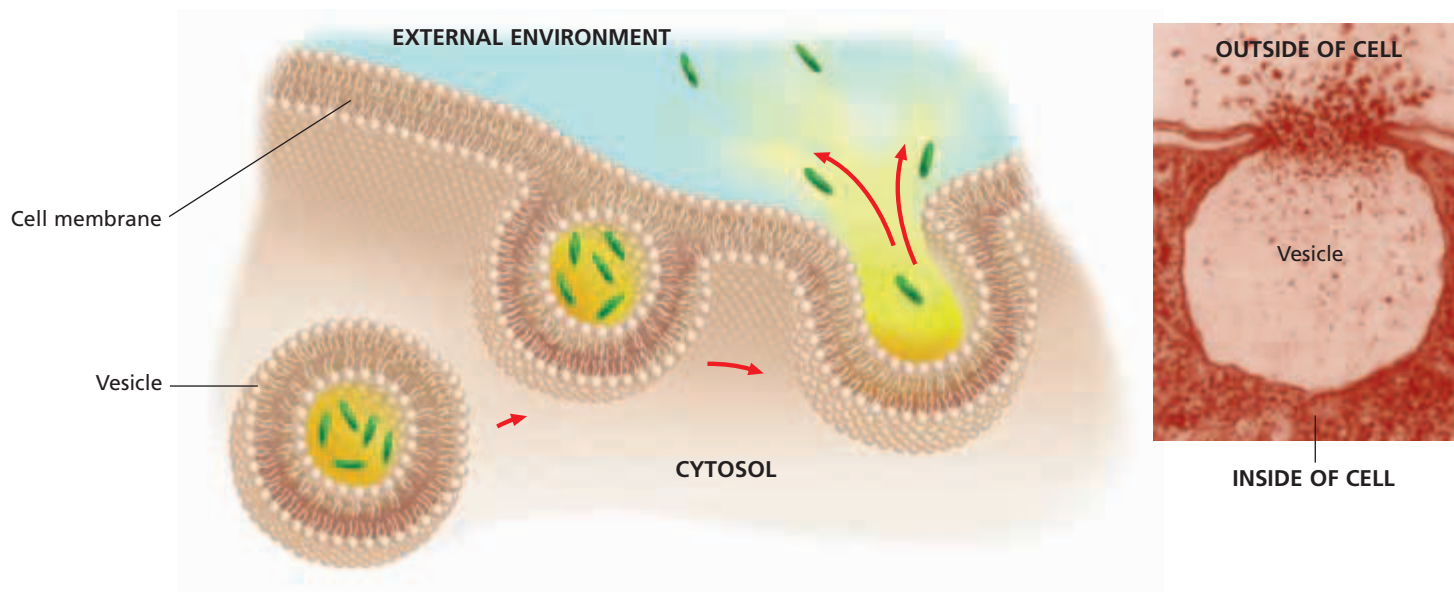


FIGURE 5-8

During exocytosis, a vesicle moves to the cell membrane, fuses with it, and then releases its contents to the outside of the cell.

Exocytosis

Exocytosis (EK-soh-sie-TOH-sis) is the process by which a substance is released from the cell through a vesicle that transports the substance to the cell surface and then fuses with the membrane to let the substance out of the cell. This process, illustrated in Figure 5-8, is basically the reverse of endocytosis. During exocytosis, vesicles release their contents into the cell's external environment. Figure 5-8 also shows a photo of a vesicle during exocytosis.

Cells may use exocytosis to release large molecules such as proteins, waste products, or toxins that would damage the cell if they were released within the cytosol. Recall that proteins are made on ribosomes and packaged into vesicles by the Golgi apparatus. The vesicles then move to the cell membrane and fuse with it, delivering the proteins outside the cell. Cells in the nervous and endocrine systems also use exocytosis to release small molecules that control the activities of other cells.

SECTION 2 REVIEW

1. Explain the difference between passive transport and active transport.
2. What functions do carrier proteins perform in active transport?
3. What provides the energy that drives the sodium-potassium pump?
4. Explain the difference between pinocytosis and phagocytosis.
5. Describe the steps involved in exocytosis.
6. How do endocytosis and exocytosis differ? How can that difference be seen?

CRITICAL THINKING

7. **Analyzing Information** During intense exercise, potassium tends to accumulate in the fluid surrounding muscle cells. What membrane protein helps muscle cells counteract this tendency? Explain your answer.
8. **Evaluating Differences** How does the sodium-potassium pump differ from facilitated diffusion?
9. **Relating Concepts** The vesicles formed during pinocytosis are much smaller than those formed during phagocytosis. Explain.

CHAPTER HIGHLIGHTS

SECTION 1

Passive Transport

- Passive transport involves the movement of molecules across the cell membrane without an input of energy by the cell.
- Diffusion is the movement of molecules from an area of higher concentration to an area of lower concentration, driven by the molecules' kinetic energy until equilibrium is reached.
- Molecules can diffuse across a cell membrane by dissolving in the phospholipid bilayer or by passing through pores in the membrane.
- Osmosis is the diffusion of water across a membrane. The net direction of osmosis is determined by the relative solute concentrations on the two sides of the membrane.
- When the solute concentration outside the cell is lower than that in the cytosol, the solution outside is hypotonic to the cytosol, and water will diffuse into the cell.
- When the solute concentration outside the cell is higher than that in the cytosol, the solution outside is hypertonic to the cytosol, and water will diffuse out of the cell.
- When the solute concentrations outside and inside the cell are equal, the solution outside is isotonic, and there will be no net movement of water.
- To remain alive, cells must compensate for the water that enters the cell in hypotonic environments and leaves the cell in hypertonic environments.
- In facilitated diffusion, a molecule binds to a carrier protein on one side of the cell membrane. The carrier protein then changes its shape and transports the molecule down its concentration gradient to the other side of the membrane.
- Ion channels are proteins, or groups of proteins, that provide small passageways across the cell membrane through which specific ions can diffuse.

Vocabulary

passive transport (p. 97)

diffusion (p. 97)

concentration gradient (p. 97)

equilibrium (p. 97)

osmosis (p. 98)

hypotonic (p. 98)

hypertonic (p. 98)

isotonic (p. 99)

contractile vacuole (p. 99)

turgor pressure (p. 100)

plasmolysis (p. 100)

cytolysis (p. 100)

facilitated diffusion (p. 101)

carrier protein (p. 101)

ion channel (p. 102)

SECTION 2

Active Transport

- Active transport moves molecules across the cell membrane from an area of lower concentration to an area of higher concentration. Unlike passive transport, active transport requires cells to expend energy.
- Some types of active transport are performed by carrier proteins called cell membrane pumps.
- One example of a cell membrane pump is the sodium-potassium pump. It moves three Na^+ ions into the cell's external environment for every two K^+ ions it moves into the cytosol. ATP supplies the energy that drives the pump.
- Endocytosis and exocytosis are active transport mechanisms in which large substances enter or leave cells inside vesicles.
- In endocytosis, the cell membrane folds around something in the external environment and forms a pouch. The pouch then pinches off and becomes a vesicle in the cytoplasm.
- Endocytosis includes pinocytosis, in which the vesicle contains solutes or fluids, and phagocytosis, in which the vesicle contains large particles or cells.
- In exocytosis, vesicles made by the cell fuse with the cell membrane, releasing their contents into the external environment.

Vocabulary

active transport (p. 103)

sodium-potassium

pump (p. 103)

endocytosis (p. 105)

vesicle (p. 105)

pinocytosis (p. 105)

phagocytosis (p. 105)

phagocyte (p. 105)

exocytosis (p. 106)

CHAPTER REVIEW

USING VOCABULARY

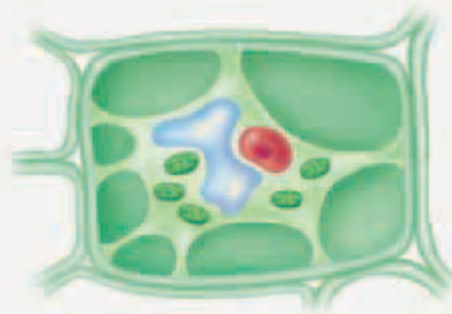
- For each pair of terms, explain how the meanings of the terms differ.
 - diffusion* and *facilitated diffusion*
 - hypotonic* and *hypertonic*
 - plasmolysis* and *cytolysis*
 - pinocytosis* and *phagocytosis*
- Explain the relationship between plasmolysis and turgor pressure in plant cells.
- Use the following terms in the same sentence: *carrier protein*, *concentration gradient*, *facilitated diffusion*, and *passive transport*.
- Word Roots and Origins** The prefix *pino-* means “to drink,” *phago-* means “to eat,” and *cyto-* means “cell.” Using this information, explain why the words *pinocytosis* and *phagocytosis* are good names for the processes they describe.

UNDERSTANDING KEY CONCEPTS

- Explain** why diffusion eventually results in equilibrium.
- Identify** whether all molecules diffuse through all cell membranes. Explain your answer.
- Distinguish** between diffusion and osmosis.
- Describe** what it means to say that two solutions are isotonic.
- Define** the term *contractile vacuole*. What is a contractile vacuole’s function?
- Determine** how hypotonic, hypertonic and turgor pressure are interrelated.
- Summarize** the factors that determine the direction of net movement of water across a cell membrane.
- Determine** how the phospholipid bilayer of a membrane forms a barrier to molecules.
- Explain** how substances cross a cell membrane through facilitated diffusion.
- Describe** how ion channels assist in the diffusion of ions through a cell membrane.
- Distinguish** between passive transport and active transport.
- Explain** how a cell that consumes glucose can speed up its intake of glucose from the environment.
- Identify** how ATP is involved in maintaining the sodium and potassium gradients across a cell membrane.
- Distinguish** between the processes of endocytosis and exocytosis.
- CONCEPT MAPPING** Use the following terms to create a concept map that shows how cells maintain homeostasis: *active transport*, *carrier protein*, *cell transport*, *concentration gradient*, *diffusion*, *endocytosis*, *exocytosis*, *facilitated diffusion*, *homeostasis*, *osmosis*, *passive transport*, and *sodium-potassium pump*.

CRITICAL THINKING

- Making Comparisons** There is a higher concentration of air molecules inside an inflated balloon than there is outside the balloon. Because of their constant random motion, the molecules inside press against the balloon and keep it taut. How is the pressure exerted by these air molecules similar to turgor pressure? How is it different?
- Comparing Concepts** Sometimes, water seeps through the concrete wall of a basement after a heavy rain, and the homeowner must remove the water with a sump pump. How can this situation be compared to the action of a unicellular non-photosynthetic organism that lives in a pond?
- Predicting Results** If a cell were exposed to a poison that blocked the cell’s ability to manufacture ATP, what effect would that have on the cell membrane’s transport processes?
- Interpreting Graphics** The drawing below shows a plant cell after the solute concentration of its environment has been changed. How would you describe the new external environment?



- INTERACTIVE BIOLOGY TUTOR** **Analyzing Information** Write a report summarizing the roles of osmosis and diffusion in the preservation of body organs donated for transplants. Why must organs be preserved in special solutions prior to a transplant? Find out what kinds of substances these solutions contain.

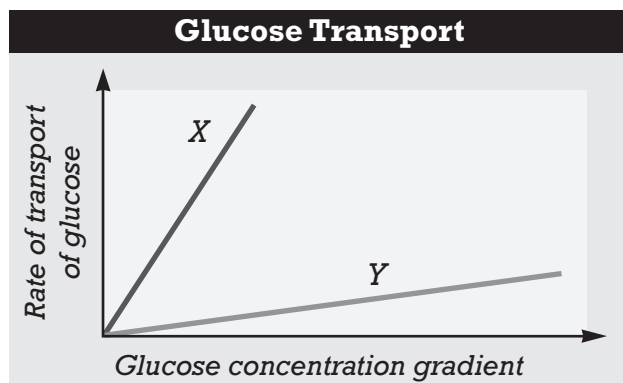


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- During diffusion, molecules tend to move in what direction?
 - the molecules involved in diffusion never move
 - in a direction that doesn't depend on the concentration gradient
 - from an area of lower concentration to an area of higher concentration
 - from an area of higher concentration to an area of lower concentration
- Ion channels aid the movement of which substances?
 - ions across a cell membrane
 - water across a cell membrane
 - molecules up a concentration gradient
 - carrier proteins within the lipid bilayer
- The sodium-potassium pump transports which of the following?
 - both Na^+ and K^+ into the cell
 - both Na^+ and K^+ out of the cell
 - Na^+ into the cell and K^+ out of the cell
 - Na^+ out of the cell and K^+ into the cell
- Which process do some animal cells use to engulf, digest, and destroy invading bacteria?
 - exocytosis
 - pinocytosis
 - phagocytosis
 - All of the above

INTERPRETING GRAPHICS: The graph below shows the rate of glucose transport across a cell membrane versus the concentration gradient. Use the graph that follows to answer the question.

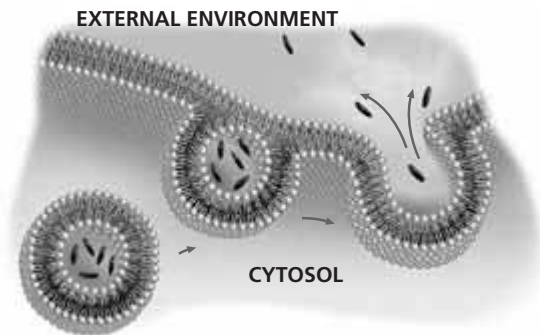


- Which line represents the diffusion of glucose through the lipid bilayer?
 - line X
 - line Y
 - both lines X and Y
 - neither line X nor Y

DIRECTIONS: Complete the following analogy.

6. passive transport : osmosis :: active transport :
- cytolysis
 - diffusion
 - ion channel
 - endocytosis

INTERPRETING GRAPHICS: The diagram below shows one form of cellular transport. Use the diagram to answer the question that follows.



- What form of cellular transport is being illustrated in the above diagram?
 - osmosis
 - exocytosis
 - facilitated diffusion
 - a cell membrane pump

SHORT RESPONSE

When a cell takes in substances through endocytosis, the outside of the cell membrane becomes the inside of the vesicle.

What might this suggest about the structure of the cell membrane?

EXTENDED RESPONSE

Some cells have carrier proteins that transport sugar molecules and hydrogen ions into the cytosol at the same time. These proteins move sugar up their gradient as hydrogen ions move down their gradient.

Part A How would the transport of sugar into these cells affect the pH of the cells' external environment?

Part B What would happen to the transport of sugar if hydrogen ions were removed from the external environment?

Test TIP

Pay close attention to questions that contain words such as *not*, *only*, *rather*, and *some*, because these words can often help you eliminate answer choices.

Analyzing the Effect of Cell Size on Diffusion

OBJECTIVES

- **Relate** a cell's size to its surface area-to-volume ratio.
- **Predict** how the surface area-to-volume ratio of a cell will affect the diffusion of substances into the cell.

PROCESS SKILLS

- making observations
- using scientific methods
- designing an experiment
- collecting data
- organizing data
- graphing
- analyzing data

MATERIALS

- safety goggles
- lab apron
- disposable gloves
- block of phenolphthalein agar (3×6 cm)
- plastic knife
- metric ruler
- 250 mL beaker
- 150 mL vinegar
- plastic spoon
- paper towel

Background

1. Substances enter and leave a cell in several ways including by diffusion.
2. Describe the process of diffusion.
3. Define the terms *surface area* and *volume*.
4. How efficiently a cell can exchange substances depends on the surface area-to-volume ratio. Surface area-to-volume ratio can be calculated by dividing surface area by volume of the cell (surface area $\div$ volume of cell).

PART A Designing an Experiment

1. Using the materials listed for this lab, design an experiment to test how the surface area-to-volume ratio affects the diffusion of a substance into a cell.

You Choose

As you design your experiment, decide the following:

- a. what question you will explore
- b. what hypothesis you will test
- c. how many "cells" (agar cubes) you will have and what sizes they will be
- d. how long to leave the "cells" in the vinegar
- e. how to determine how far the vinegar diffused into a "cell"
- f. how to prevent contamination of agar cubes as you handle them
- g. what data to record in your data table



2. Write a procedure for your experiment. Make a list of all the safety precautions you will take. Have your teacher approve your procedure and safety precautions before you begin the experiment.

PART B Conducting Your Experiment

3.     **CAUTION** Always wear safety goggles and a lab apron to protect your eyes and clothing. Put on safety goggles, a lab apron, and disposable gloves. Do not touch or taste any chemicals. Carry out the experiment you designed. Record your observations in your data table.
4.  **CAUTION** Know the location of the emergency shower and eyewash station and how to use them. If you get a chemical on your skin or clothing, wash it off at the sink while calling to the teacher. Notify the teacher of a spill. Spills should be cleaned up promptly, according to your teacher's directions.
5.  **CAUTION** Glassware is fragile. Notify the teacher of broken glass or cuts. Do not clean up broken glass or spills with broken glass unless the teacher tells you to do so.
6.  Clean up your work area and all lab equipment. Return lab equipment to its proper place. Dispose of solutions, broken glass, and agar in the designated waste containers. Do not pour chemicals down the drain or put lab materials in the trash unless your teacher tells you to do so. Wash your hands thoroughly before you leave the lab and after you finish all work.

Analysis and Conclusions

1. Describe any changes in the appearance of the cubes.
2. Make a graph using your group's data. Plot "Diffusion distance (mm)" on the vertical axis. Use graph paper to plot "Surface area-to-volume ratio" on the horizontal axis.
3. Using the graph you made in item 2, make a statement about the relationship between the surface area-to-volume ratio and the distance a substance diffuses.
4. Make a graph using your group's data. Use graph paper to plot "Rate of diffusion (mm/min)" (distance vinegar moved time) on the vertical axis. Plot "Surface area-to-volume ratio" on the horizontal axis.
5. Using the graph you made in item 4, make a statement about the relationship between the surface area-to-volume ratio and the rate of diffusion of a substance.
6. In what ways do your agar models simplify or fail to simulate the features of real cells?
7. Calculate the surface area and volume of a cube with a side length of 5 cm. Calculate the surface area and volume of a cube with a side length of 10 cm. Determine the surface area-to-volume ratio of each of these cubes. Which cube has the greater surface area-to-volume ratio?
8. How does the size of a cell affect the diffusion of substances into the cell?

Further Inquiry

Write a new question about cell size and diffusion that also explores hypotonic and hypertonic solutions that could be explored with another investigation.

PHOTOSYNTHESIS

Alfalfa, corn, and soybeans are growing on this farmland in Pennsylvania. Through photosynthesis, these plants obtain energy from the sun and store it in organic compounds. Humans and other living organisms depend on organic compounds—and therefore on photosynthesis—to obtain the energy necessary for living.

SECTION 1 *The Light Reactions*

SECTION 2 *The Calvin Cycle*



Unit 2—Photosynthesis
Topics 1–6

THE LIGHT REACTIONS

All organisms use energy to carry out the functions of life. Where does this energy come from? Directly or indirectly, almost all of the energy in living systems comes from the sun. Energy from the sun enters living systems when plants, algae, some unicellular protists, and some prokaryotes absorb sunlight and use it to make organic compounds.

OBTAINING ENERGY

Organisms can be classified according to how they get energy. Organisms that use energy from sunlight or from chemical bonds in inorganic substances to make organic compounds are called **autotrophs** (AWT-oh-TROHFS). Most autotrophs use the process of **photosynthesis** to convert light energy from the sun into chemical energy in the form of organic compounds, primarily carbohydrates.

The tree in Figure 6-1 is an autotroph because it converts light energy from the sun into organic compounds. The tree uses these compounds for energy. The caterpillar also depends on the tree for energy, as it cannot manufacture organic compounds itself. Animals and other organisms that must get energy from food instead of directly from sunlight or inorganic substances are called **heterotrophs** (HEHT-uh-oh-TROHFS). The bird is also a heterotroph. The food that fuels the bird originates with an autotroph (the tree), but it passes indirectly to the bird through the caterpillar. In similar ways, almost all organisms ultimately depend on autotrophs to obtain the energy necessary to carry out the processes of life.

Photosynthesis involves a complex series of chemical reactions in which the product of one reaction is consumed in the next reaction. A series of chemical reactions linked in this way is referred to as a *biochemical pathway*.

OBJECTIVES

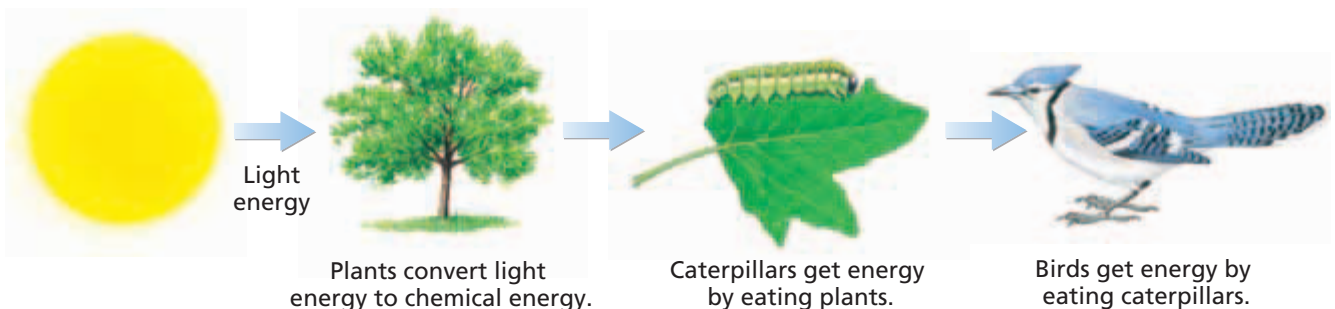
- **Explain** why almost all organisms depend on photosynthesis.
- **Describe** the role of chlorophylls and other pigments in photosynthesis.
- **Summarize** the main events of the light reactions.
- **Explain** how ATP is made during the light reactions.

VOCABULARY

autotroph
photosynthesis
heterotroph
light reactions
chloroplast
thylakoid
granum
stroma
pigment
chlorophyll
carotenoid
photosystem
primary electron acceptor
electron transport chain
chemiosmosis

FIGURE 6-1

The tree depends on the sun for energy. Like all heterotrophs, the caterpillar and bird depend on an autotroph (the tree and its leaves) for energy.



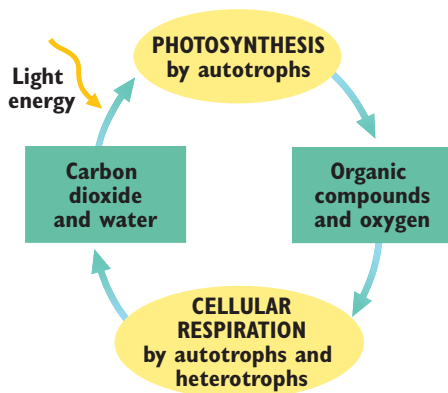


FIGURE 6-2

Many autotrophs produce organic compounds and oxygen through photosynthesis. Both autotrophs and heterotrophs produce carbon dioxide through cellular respiration.

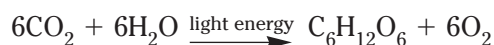
OVERVIEW OF PHOTOSYNTHESIS

Figure 6-2 shows how autotrophs use photosynthesis to produce organic compounds from carbon dioxide (CO_2) and water. The oxygen (O_2) and some of the organic compounds produced are then used by cells in a process called *cellular respiration*. During cellular respiration, CO_2 and water are produced. Thus, the products of photosynthesis are reactants in cellular respiration. Conversely, the products of cellular respiration are reactants in photosynthesis.

Photosynthesis can be divided into two stages:

- 1. Light Reactions** Light energy (absorbed from the sun) is converted to chemical energy, which is temporarily stored in ATP and the energy carrier molecule *NADPH*.
- 2. Calvin Cycle** Organic compounds are formed using CO_2 and the chemical energy stored in ATP and *NADPH*.

Photosynthesis can be summarized by the following equation:



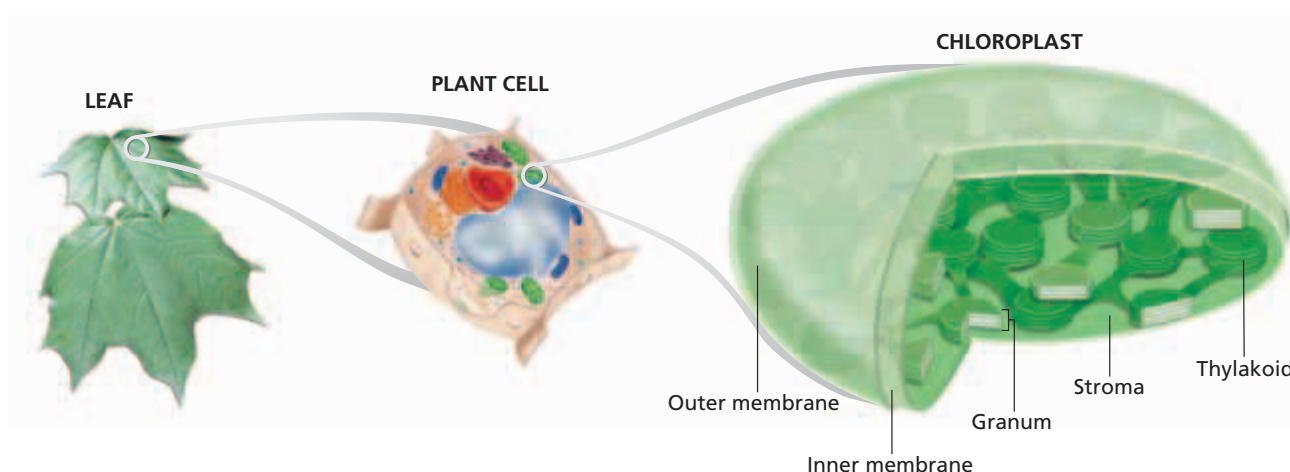
This equation, however, does not explain how photosynthesis occurs. It is helpful to examine the two stages separately in order to better understand the overall process of photosynthesis.

CAPTURING LIGHT ENERGY

The first stage of photosynthesis includes the **light reactions**, so named because they require light to happen. The light reactions begin with the absorption of light in **chloroplasts**, organelles found in the cells of plants and algae. Most chloroplasts are similar in structure. As shown in Figure 6-3, each chloroplast is surrounded by a pair of membranes. Inside the inner membrane is another system of membranes called **thylakoids** (THIE-luh-koydz) that are arranged as flattened sacs. The thylakoids are connected and layered to form stacks called **grana** (GRAY-nuh) (singular, **granum**). Surrounding the grana is a solution called the **stroma** (STROH-muh).

FIGURE 6-3

Photosynthesis in eukaryotes occurs inside the chloroplast. The light reactions of photosynthesis take place in the thylakoids, which are stacked to form grana.



Light and Pigments

To understand how chloroplasts absorb light in photosynthesis, it is important to understand some of the properties of light. Light from the sun appears white, but it is actually made of a variety of colors. As shown in Figure 6-4, white light can be separated into its component colors by passing the light through a prism. The resulting array of colors, ranging from red at one end to violet at the other, is called the *visible spectrum*.

When white light strikes an object, its component colors can be reflected, transmitted, or absorbed by the object. Many objects contain **pigments**, compounds that absorb light. Most pigments absorb certain colors more strongly than others. By absorbing certain colors, a pigment subtracts those colors from the visible spectrum. Therefore, the light that is reflected or transmitted by the pigment no longer appears white. For example, the lenses in green-tinted sunglasses contain a pigment that reflects and transmits green light and absorbs the other colors. As a result, the lenses look green.

Chloroplast Pigments

Located in the membrane of the thylakoids are several pigments, the most important of which are called **chlorophylls** (KLAWR-uh-FILZ). There are several different types of chlorophylls. The two most common types are known as chlorophyll *a* and chlorophyll *b*. As Figure 6-5 shows, chlorophyll *a* absorbs less blue light but more red light than chlorophyll *b* absorbs. Neither chlorophyll *a* nor chlorophyll *b* absorbs much green light. Instead, they allow green light to be reflected or transmitted. For this reason, leaves and other plant structures that contain large amounts of chlorophyll look green.

Only chlorophyll *a* is directly involved in the light reactions of photosynthesis. Chlorophyll *b* assists chlorophyll *a* in capturing light energy, and therefore chlorophyll *b* is called an accessory pigment. Other compounds found in the thylakoid membrane, including the yellow, orange, and brown **carotenoids** (kuh-RAHT'n-OYDZ), also function as accessory pigments. Looking again at Figure 6-5, notice that the pattern of light absorption of one of the carotenoids differs from the pattern of either type of chlorophyll. By absorbing colors that chlorophyll *a* cannot absorb, the accessory pigments enable plants to capture more of the energy in light.

In the leaves of a plant, the chlorophylls are generally much more abundant and therefore mask the colors of the other pigments. But in the nonphotosynthetic parts of a plant, such as fruits and flowers, the colors of the other pigments may be quite visible. During the fall, many plants lose their chlorophylls, and their leaves take on the rich hues of the carotenoids.

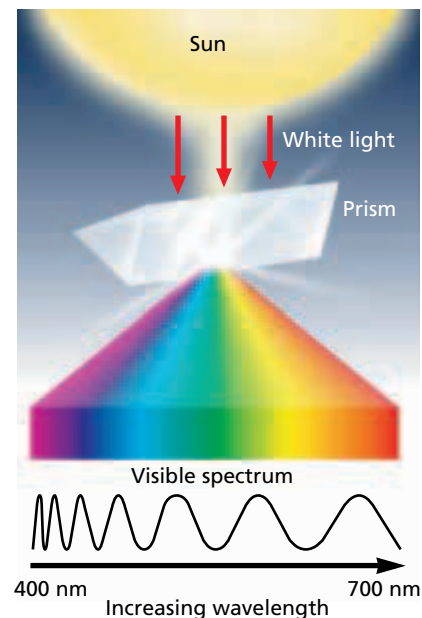


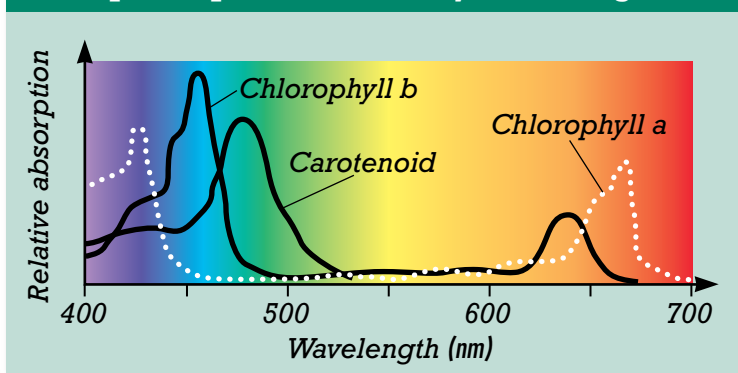
FIGURE 6-4

White light contains a variety of colors called the *visible spectrum*. Each color has a different wavelength, measured in nanometers.

FIGURE 6-5

The three curves on this graph show how three pigments involved in photosynthesis differ in the colors of light they absorb. Where a curve has a peak, much of the light at that wavelength is absorbed. Where a curve has a trough, much of the light at that wavelength is reflected or transmitted.

Absorption Spectra of Photosynthetic Pigments



Word Roots and Origins

thylakoid

from the Greek *thylakos*,
meaning "pocket"

CONVERTING LIGHT ENERGY TO CHEMICAL ENERGY

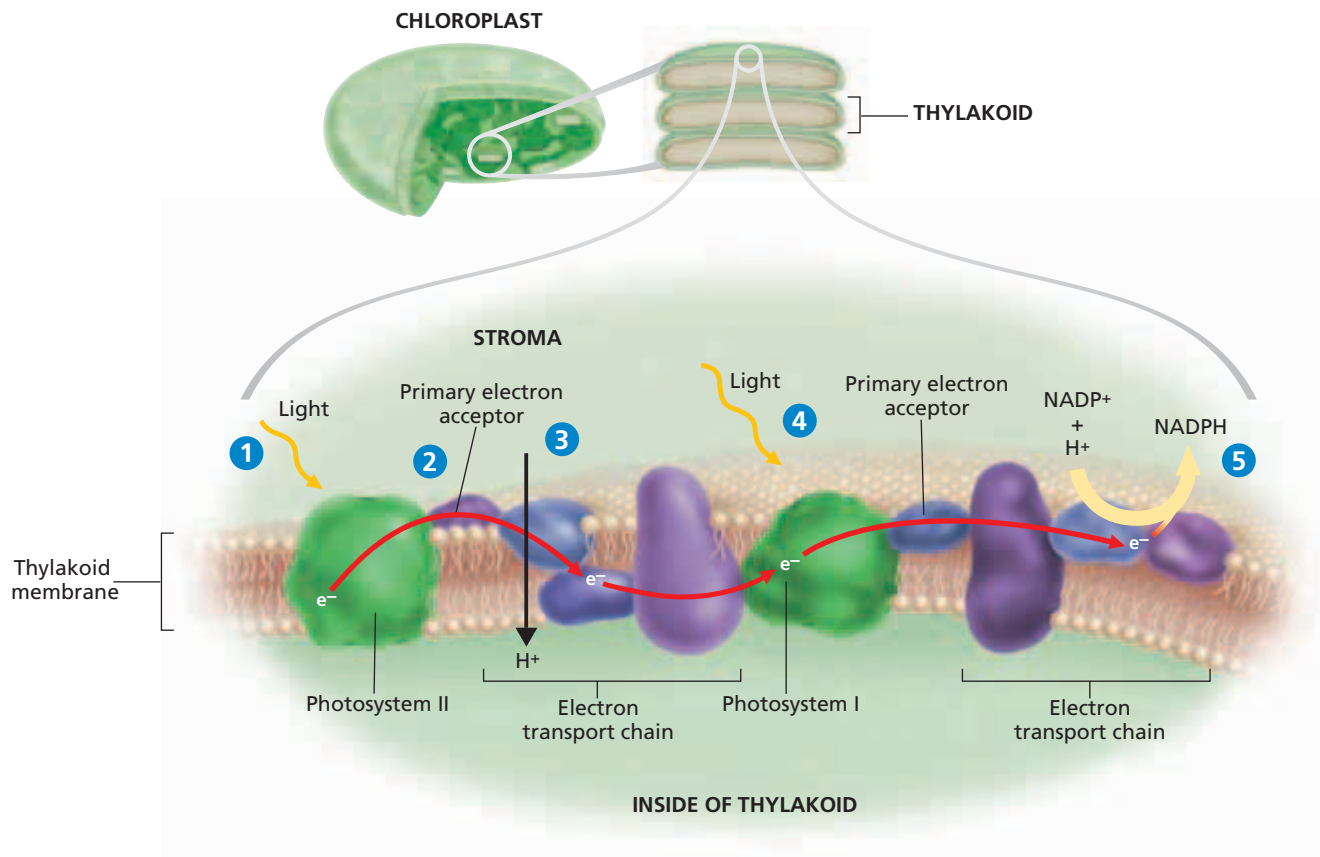
Once the pigments in the chloroplast have captured light energy, the light energy must then be converted to chemical energy. The chemical energy is temporarily stored in ATP and NADPH. During these reactions, O_2 is given off. The chlorophylls and carotenoids are grouped in clusters of a few hundred pigment molecules in the thylakoid membrane. Each cluster of pigment molecules and the proteins that the pigment molecules are embedded in are referred to collectively as a **photosystem**. Two types of photosystems are known: *photosystem I* and *photosystem II*. They contain similar kinds of pigments, but they have different roles in the light reactions.

The light reactions begin when accessory pigment molecules in both photosystems absorb light. By absorbing light, those molecules acquire some of the energy carried by the light. In each photosystem, the acquired energy is passed quickly to the other pigment molecules until it reaches a specific pair of chlorophyll *a* molecules. Chlorophyll *a* can also absorb light. The events that occur next can be divided into five steps, as shown in Figure 6-6.

In step 1, light energy forces electrons to enter a higher energy level in the two chlorophyll *a* molecules of photosystem II. These energized electrons are said to be "excited."

FIGURE 6-6

The light reactions, which take place in the thylakoid membrane, use energy from sunlight to produce ATP, NADPH, and oxygen.



The excited electrons have enough energy to leave the chlorophyll *a* molecules. Because they have lost electrons, the chlorophyll *a* molecules have undergone an oxidation reaction. Remember that each oxidation reaction must be accompanied by a reduction reaction. So, some substance must accept the electrons that the chlorophyll *a* molecules have lost, as shown in step ②. The acceptor of the electrons lost from chlorophyll *a* is a molecule in the thylakoid membrane called the **primary electron acceptor**.

In step ③, the primary electron acceptor donates the electrons to the first of a series of molecules located in the thylakoid membrane. These molecules are called an **electron transport chain** because they transfer electrons from one molecule to the next. As the electrons pass from molecule to molecule in the chain, they lose most of the energy that they acquired when they were excited. The energy they lose is used to move protons (H^+) into the thylakoid.

In step ④, light is absorbed by photosystem I. This happens at the same time that light is absorbed by photosystem II. Electrons move from a pair of chlorophyll *a* molecules in photosystem I to another primary electron acceptor. The electrons lost by these chlorophyll *a* molecules are replaced by the electrons that have passed through the electron transport chain from photosystem II.

The primary electron acceptor of photosystem I donates electrons to a different electron transport chain. This chain brings the electrons to the side of the thylakoid membrane that faces the stroma. There the electrons combine with a proton and $NADP^+$, an organic molecule that accepts electrons during oxidation/reduction reactions. As you can see in step ⑤, this reaction causes $NADP^+$ to be reduced to NADPH.

Replacing Electrons in Light Reactions

In step ④, electrons from chlorophyll molecules in photosystem II replace electrons that leave chlorophyll molecules in photosystem I. If this replacement did not occur, both electron transport chains would stop, and photosynthesis would not happen. The replacement electrons for photosystem II are provided by water molecules. As Figure 6-7 shows, an enzyme inside the thylakoid splits water molecules into protons, electrons, and oxygen. The following equation summarizes the reaction:



For every two molecules of water that are split, four electrons become available to replace those lost by the chlorophyll molecules in photosystem II. The protons that are produced are left inside the thylakoid, and the oxygen diffuses out of the chloroplast and can then leave the plant. Thus, oxygen is not needed for photosynthesis to occur but is essential for cellular respiration in most organisms, including plants themselves.



FIGURE 6-7

The splitting of water inside the thylakoid releases electrons, which replace the electrons that leave photosystem II.

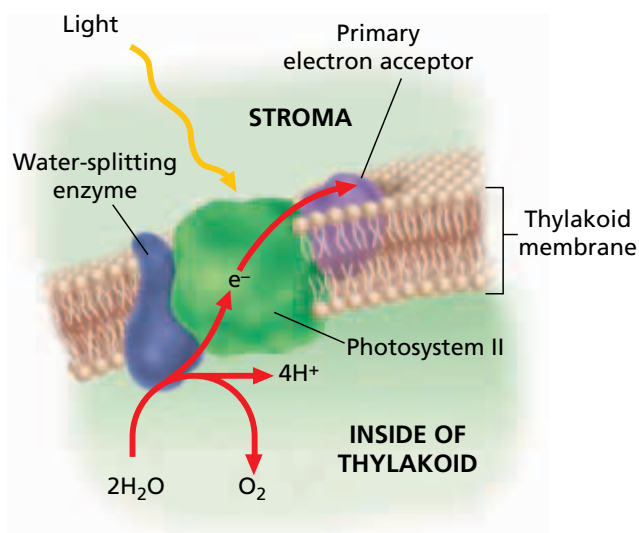
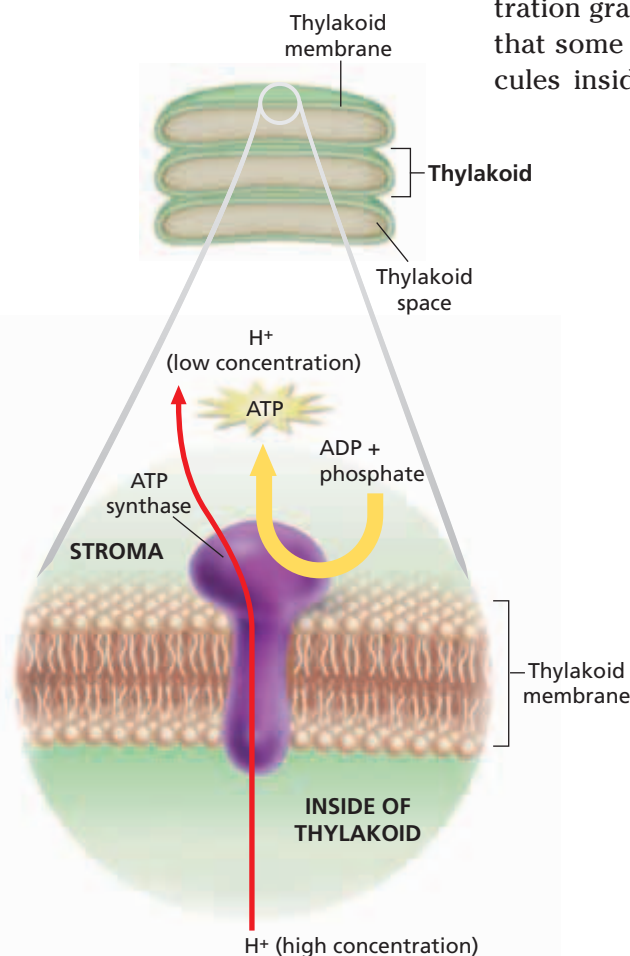


FIGURE 6-8

During chemiosmosis, the movement of protons into the stroma of the chloroplast releases energy, which is used to produce ATP.



Making ATP in Light Reactions

ATP is the main energy currency of cells. An important part of the light reactions is the synthesis of ATP through a process called **chemiosmosis** (KEM-i-ahs-MOH-sis). Chemiosmosis relies on a concentration gradient of protons across the thylakoid membrane. Recall that some protons are produced from the splitting of water molecules inside the thylakoid. Other protons are pumped from the stroma to the interior of the thylakoid. The energy required to pump these protons is supplied by the excited electrons as they pass along the electron transport chain of photosystem II. Both of these mechanisms act to build up a concentration gradient of protons. That is, the concentration of protons is higher inside the thylakoid than in the stroma.

The concentration gradient of protons represents potential energy. That energy is harnessed by an enzyme called *ATP synthase*, which is located in the thylakoid membrane, as Figure 6-8 shows. ATP synthase makes ATP by adding a phosphate group to adenosine diphosphate, or ADP. The energy driving this reaction is provided by the movement of protons from inside the thylakoid to the stroma through the ATP synthase complex. Thus, ATP synthase converts the potential energy of the proton concentration gradient into chemical energy stored in ATP. As you learned earlier, some of the protons in the stroma are used to make NADPH from NADP^+ at the end of the electron transport chain of photosystem I. Together, NADPH and ATP provide energy for the second set of reactions in photosynthesis, which are described in the next section.

SECTION 1 REVIEW

1. Explain why both autotrophs and heterotrophs depend on photosynthesis to obtain the energy they need for life processes.
2. Describe the role of chlorophylls in the biochemical pathways of photosynthesis.
3. List the three substances that are produced when water molecules are broken down during the light reactions.
4. Explain why the splitting of water is important to the continuation of the light reactions.
5. Name the product of the process known as *chemiosmosis*.

CRITICAL THINKING

6. **Making Comparisons** Thinking about the main role of pigments in photosynthesis, explain how the pigments in colored objects such as clothes differ from plant pigments.
7. **Analyzing Information** The molecule that precedes the electron transport chains of both photosystem I and photosystem II is an electron acceptor. What is the original molecule that is the electron donor for both of these systems?
8. **Predicting Results** Explain how the light reactions would be affected if there were no concentration gradient of protons across the thylakoid membrane.

Science in Action

How Can Scientists Replicate Photosynthesis in the Lab?

Life on Earth is powered by the sun's energy. The chloroplasts found in plants and some other organisms can efficiently capture and convert solar energy to sugars through a process known as *photosynthesis*. Now, scientists have developed an artificial photosystem that may one day allow for solar-powered food or fuel production in the lab.



Dr. Darius Kuciauskas

HYPOTHESIS: Synthetic Compounds Can Imitate a Photosynthetic System

In nature, a photosystem contains pigments and proteins embedded in a linear sequence in the membrane of a chloroplast. This arrangement allows for effective capture and orderly transfer of electrons from one photosystem to another and eventually to a final electron acceptor for powering synthesis of sugar.

Dr. Darius Kuciauskas (koo-si-AHS-kuhs) of Virginia Commonwealth University and his colleagues thought they could produce an artificial photosystem that carried out the initial steps of photosynthesis. Kuciauskas and his team predicted that when they shined light on four pigment molecules with energy-conducting chemical bonds, the pigment molecules would capture and channel light energy through the “wirelike” bonds. The electrons traveling through the bonds would then pass to, and electrically charge, an acceptor molecule that was placed at the center of the pigment complex.



Kuciauskas and his colleagues use laser systems such as this one to study light absorption in molecules that might one day be used in an artificial photosystem.

METHODS: Design and Build a Pigment-Protein Light-Harvesting Complex

Kuciauskas and his team set out to create the photosystem. The team started with purple pigment molecules called *porphyrins*. These molecules absorb light energy and are related to naturally occurring chlorophyll. The scientists joined four of the porphyrins to each other. To the center-most porphyrin, they attached a ball-shaped cluster of 60 carbon atoms called C_{60} , or *fullerene*.

RESULTS: The Artificial Photosystem Captured and Transferred Electrons

The four porphyrin pigments captured light energy, and the bonds passed electrons to the C_{60} acceptor molecule. The acceptor held the charge for only one microsecond—about one millionth of a second.

CONCLUSION: Artificial Photosystems Are Possible

The Kuciauskas team concluded that their complex does act as an artificial photosystem, at least for a fleeting instant. Kuciauskas and his colleagues have continued their work, aiming to modify the complex further so it can hold its energy longer. Ultimately, they hope their photosystem can power chemical reactions for making important organic molecules that serve as food or fuel.

REVIEW

1. What are the essential parts of a photosystem?
2. Describe the method by which the Kuciauskas team built an artificial photosystem.
3. **Critical Thinking** Review the methods that Kuciauskas and his team used to test their hypothesis. Determine one variable the scientists might change to modify their results.

Internet connect

www.scilinks.org

Topic: **Plant Photosynthetic Pigments**

Keyword: **HM61164**

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SECTION 2

OBJECTIVES

- **Summarize** the main events of the Calvin cycle.
- **Describe** what happens to the compounds that are made in the Calvin cycle.
- **Distinguish** between C_3 , C_4 , and CAM plants.
- **Summarize** how the light reactions and the Calvin cycle work together to create the continuous cycle of photosynthesis.
- **Explain** how environmental factors influence photosynthesis.

VOCABULARY

Calvin cycle
carbon fixation
stomata
 C_4 pathway
CAM pathway

THE CALVIN CYCLE

In the second set of reactions in photosynthesis, plants use the energy that was stored in ATP and NADPH during the light reactions to produce organic compounds in the form of sugars. These organic compounds are then consumed by autotrophs and heterotrophs alike for energy. The most common way that plants produce organic compounds is called the Calvin cycle.

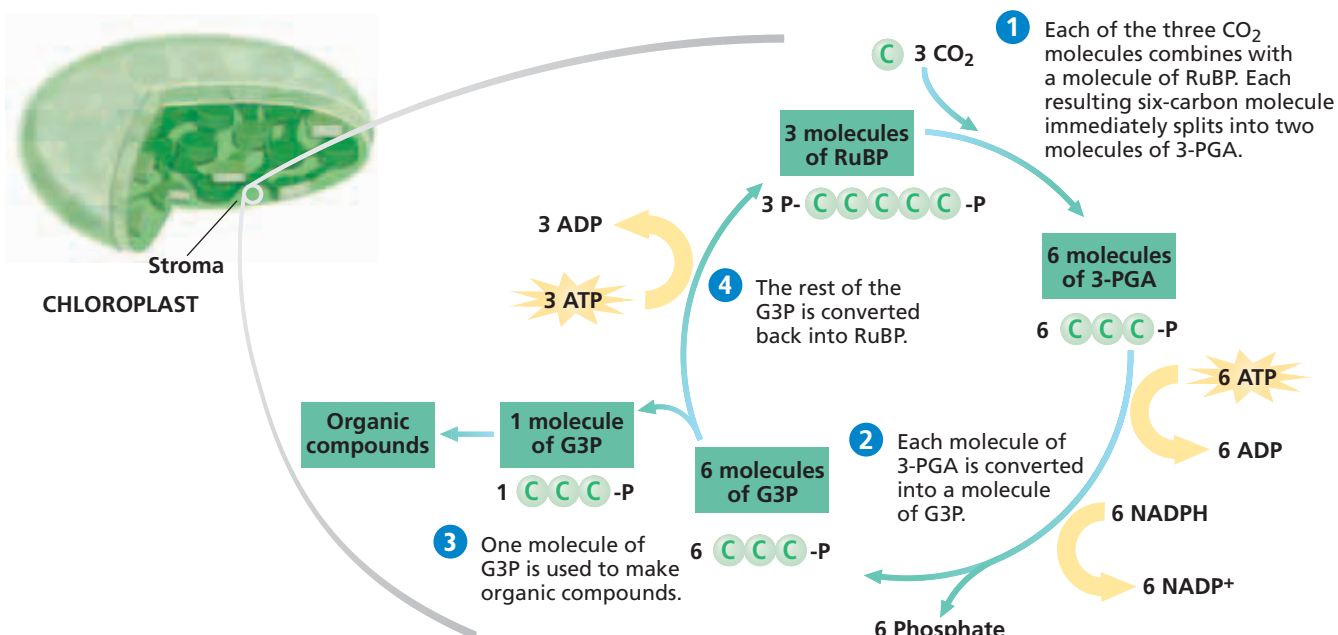
CARBON FIXATION

The **Calvin cycle** is a series of enzyme-assisted chemical reactions that produces a three-carbon sugar. In the Calvin cycle, carbon atoms from CO_2 in the atmosphere are bonded, or “fixed,” into organic compounds. This incorporation of CO_2 into organic compounds is called **carbon fixation**. A total of three CO_2 molecules must enter the Calvin cycle to produce each three-carbon sugar that will be used to make the organic compounds. The Calvin cycle occurs within the stroma of the chloroplast. Figure 6-9 shows the events that occur when three CO_2 molecules enter the Calvin cycle.

In step 1, CO_2 diffuses into the stroma from the surrounding cytosol. An enzyme combines each CO_2 molecule with a five-carbon molecule called *ribulose biphosphate* (RuBP). The six-carbon molecules that result are very unstable, and they each immediately split into two three-carbon molecules. These three-carbon molecules are called *3-phosphoglycerate* (3-PGA).

FIGURE 6-9

The Calvin cycle, in which carbon is fixed into organic compounds, takes place in the stroma of the chloroplast.



In step ②, each molecule of 3-PGA is converted into another three-carbon molecule, glyceraldehyde 3-phosphate (G3P), in a two-part process. First, each 3-PGA molecule receives a phosphate group from a molecule of ATP. The resulting compound then receives a proton (H^+) from NADPH and releases a phosphate group, producing G3P. The ADP, $NADP^+$, and phosphate that are also produced can be used again in the light reactions to make more ATP and NADPH.

In step ③, one of the G3P molecules leaves the Calvin cycle and is used to make organic compounds (carbohydrates) in which energy is stored for later use.

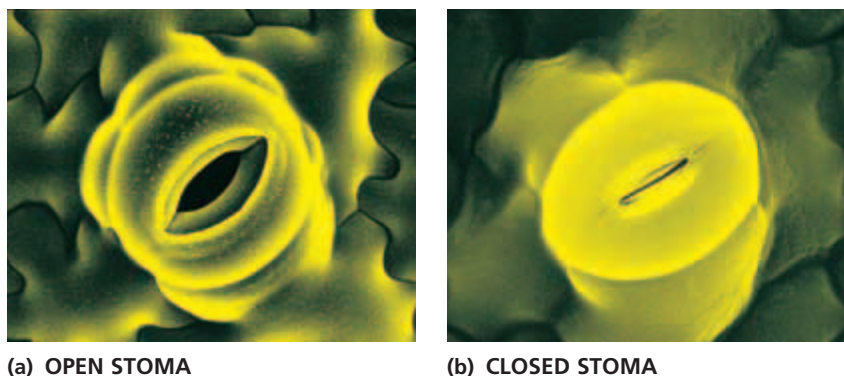
In step ④, the remaining G3P molecules are converted back into RuBP through the addition of phosphate groups from ATP molecules. The resulting RuBP molecules then enter the Calvin cycle again.

The Calvin cycle (named for Melvin Calvin, the American biochemist who worked out the chemical reactions in the cycle) is the most common pathway for carbon fixation. Plant species that fix carbon exclusively through the Calvin cycle are known as C_3 plants because of the three-carbon compound that is initially formed in this process.

ALTERNATIVE PATHWAYS

Many plant species that evolved in hot, dry climates fix carbon through alternative pathways. Under hot and dry conditions, plants can rapidly lose water to the air through small pores called **stomata** (STOH-muh-tuh). Stomata (singular, *stoma*), shown in Figure 6-10, are usually located on the undersurface of the leaves. Plants can reduce water loss by partially closing their stomata when the air is hot and dry.

Stomata are the major passageways through which CO_2 enters and O_2 leaves a plant. When a plant's stomata are partly closed, the level of CO_2 in the plant falls as CO_2 is consumed in the Calvin cycle. At the same time, the level of O_2 in the plant rises as the light reactions generate O_2 . Both a low CO_2 level and a high O_2 level inhibit carbon fixation by the Calvin cycle. Alternative pathways for carbon fixation help plants deal with this problem.



Word Roots and Origins

stomata

from the Greek *stoma*, meaning "mouth"

FIGURE 6-10

These photos show stomata in the leaf of a tobacco plant, *Nicotiana tabacum*. (a) When a stoma is open, water, carbon dioxide, and other gases can pass through it to enter or leave a plant (814 $\times$). (b) When a stoma is closed, passage through it is greatly restricted (878 $\times$).

The C₄ Pathway

One alternative pathway enables certain plants to fix CO₂ into four-carbon compounds. This pathway is thus called the **C₄ pathway**, and plants that use it are known as *C₄ plants*. During the hottest part of the day, C₄ plants have their stomata partially closed. However, certain cells in C₄ plants have an enzyme that can fix CO₂ into four-carbon compounds even when the CO₂ level is low and the O₂ level is high. These compounds are then transported to other cells, where CO₂ is released and enters the Calvin cycle. C₄ plants include corn, sugar cane, and crab grass. Such plants lose only about half as much water as C₃ plants when producing the same amount of carbohydrates. Many plants that use the C₄ pathway evolved in tropical climates.

The CAM Pathway

Cactuses, pineapples, and certain other plants have a different adaptation to hot, dry climates. Such plants fix carbon through a pathway called the **CAM pathway**. CAM is an abbreviation for *crassulacean acid metabolism*, because this water-conserving pathway was first discovered in plants of the family Crassulaceae, such as the jade plant. Plants that use the CAM pathway open their stomata at night and close them during the day—just the opposite of what other plants do. At night, CAM plants take in CO₂ and fix it into a variety of organic compounds. During the day, CO₂ is released from these compounds and enters the Calvin cycle. Because CAM plants have their stomata open at night, when the temperature is lower, they grow fairly slowly. However, they lose less water than either C₃ or C₄ plants.

Eco Connection

Photosynthesis and the Global Greenhouse

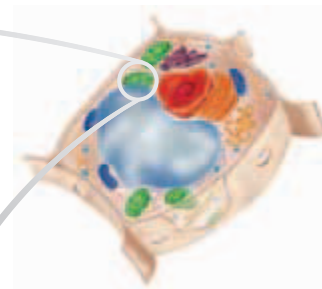
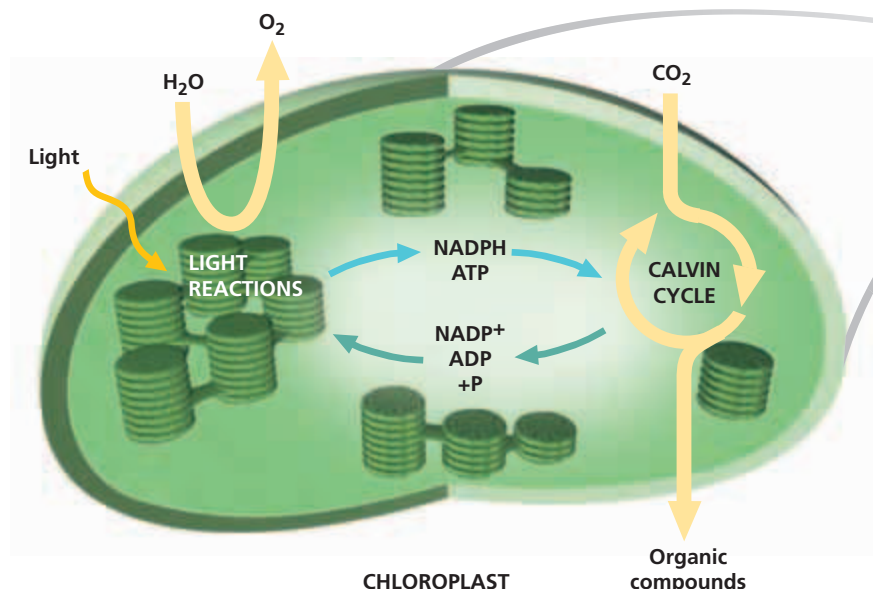
With the beginning of the Industrial Revolution around 1850, the atmospheric concentration of CO₂ started to increase. This increase has resulted largely from the burning of fossil fuels, which releases CO₂ as a byproduct. You might expect plants to benefit from the buildup of CO₂ in the atmosphere. In fact, the rise in CO₂ levels may harm photosynthetic organisms more than it helps them. CO₂ and other gases in the atmosphere retain some of the Earth's heat, causing the Earth to become warmer. This warming could reduce the amount of worldwide precipitation, creating deserts that would be inhospitable to most plants.

A SUMMARY OF PHOTOSYNTHESIS

Photosynthesis happens in two stages, both of which occur inside the chloroplasts of plant cells and algae.

1. The light reactions—Energy is absorbed from sunlight and converted into chemical energy, which is temporarily stored in ATP and NADPH.
2. The Calvin cycle—Carbon dioxide and the chemical energy stored in ATP and NADPH are used to form organic compounds.

Photosynthesis itself is an ongoing cycle: the products of the light reactions are used in the Calvin cycle, and some of the products of the Calvin cycle are used in the light reactions. The other products of the Calvin cycle are used to produce a variety of organic compounds, such as amino acids, lipids, and carbohydrates. Many plants produce more carbohydrates than they need. These extra carbohydrates can be stored as starch in the chloroplasts and in structures such as roots and fruits. These stored carbohydrates provide the chemical energy that both autotrophs and heterotrophs depend on.



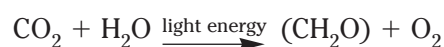
PLANT CELL

FIGURE 6-11

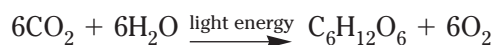
Photosynthesis is an ongoing cycle.

Figure 6-11 above illustrates how the light reactions in the thylakoid and the Calvin cycle in the stroma actually work together as one continuous cycle—photosynthesis. Amazingly, this complicated process can occur in every one of the thousands of chloroplasts present in a plant if the conditions are right.

Recall that water is split during the light reactions, yielding electrons, protons, and oxygen as a byproduct. Thus, the simplest overall equation for photosynthesis, including both the light reactions and the Calvin cycle, can be written as follows:



In the equation above, (CH_2O) represents the general formula for a carbohydrate. It is often replaced in this equation by the carbohydrate glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, giving the following equation:



However, glucose is not actually a direct product of photosynthesis. Glucose is often included in the equation mainly to emphasize the relationship between photosynthesis and cellular respiration, in which glucose plays a key role. Just as the light reactions and the Calvin cycle together create an ongoing cycle, photosynthesis and cellular respiration together also create an ongoing cycle.

The light reactions are sometimes referred to as *light-dependent reactions*, because the energy from light is required for the reactions to occur. The Calvin cycle is sometimes referred to as the *light-independent reactions* or the *dark reactions*, because the Calvin cycle does not require light directly. But the Calvin cycle usually proceeds during daytime, when the light reactions are producing the materials that the Calvin cycle uses to fix carbon into organic compounds.



Quick Lab

Analyzing Photosynthesis

Materials disposable gloves, lab apron, safety goggles, 250 mL Erlenmeyer flasks (3), bromothymol blue, 5 cm sprigs of elodea (2), water, drinking straw, plastic wrap, 100 mL graduated cylinder

Procedure



1. Put on your disposable gloves, lab apron, and safety goggles.
2. Label the flasks "1," "2," and "3." Add 200 mL of water and 20 drops of bromothymol blue to each flask.
3. Put the drinking straw in flask 1, and blow into the blue solution until the solution turns yellow. Repeat this step with flask 2.
4. Put one elodea sprig in flask 1. Do nothing to flask 2. Put the other elodea sprig in flask 3.
5. Cover all flasks with plastic wrap. Place the flasks in a well-lighted location, and leave them overnight. Record your observations.

Analysis Describe your results. Explain what caused one of the solutions to change color. Why did the other solutions not change color? Which flask is the control in this experiment?

FACTORS THAT AFFECT PHOTOSYNTHESIS

Photosynthesis is affected by the environment. Three factors with the most influence are light intensity, CO_2 level, and temperature.

Light Intensity

As shown in Figure 6-12a, the rate of photosynthesis increases as light intensity increases. Higher light intensity excites more electrons in both photosystems. Thus, the light reactions occur more rapidly. However, at some point all of the available electrons are excited, and the maximum rate of photosynthesis is reached. The rate stays level regardless of further increases in light intensity.

FIGURE 6-12

Environmental factors affect the rate of photosynthesis in plants. (a) As light intensity increases, the rate of photosynthesis increases and then levels off at a maximum. (b) As temperature increases, the rate of photosynthesis increases to a maximum and then decreases with further rises in temperature.

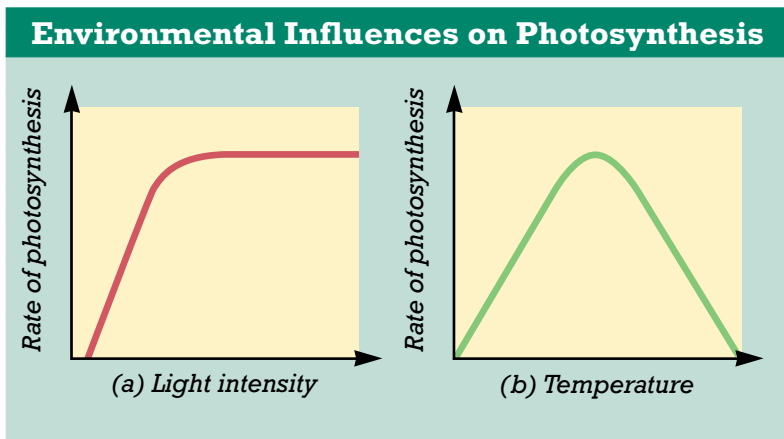
Carbon Dioxide Levels

Increasing levels of CO_2 also stimulate photosynthesis until the rate of photosynthesis levels off. Thus, a graph of the rate of photosynthesis versus CO_2 concentration would resemble Figure 6-12a.

Temperature

Increasing temperature accelerates the chemical reactions involved in photosynthesis. As a result, the rate of photosynthesis increases

as temperature increases, over a certain range. This effect is illustrated by the left half of the curve in Figure 6-12b. The rate peaks at a certain temperature, at which many of the enzymes that catalyze the reactions become ineffective. Also, the stomata begin to close, limiting water loss and CO_2 entry into the leaves. These conditions cause the rate of photosynthesis to decrease when the temperature is further increased, as shown by the right half of the curve in Figure 6-12b.



SECTION 2 REVIEW

1. Name the part of the chloroplast where the Calvin cycle takes place.
2. Describe what can happen to the three-carbon molecules made in the Calvin cycle.
3. Distinguish between C_3 , C_4 , and CAM plants.
4. Explain why the light reactions and the Calvin cycle are dependent on each other.
5. Explain why increased light intensity might not result in an increased rate of photosynthesis.

CRITICAL THINKING

6. **Predicting Results** What would happen to photosynthesis if all of the three-carbon sugars produced in the Calvin cycle were used to make organic compounds?
7. **Inferring Relationships** Explain how a global temperature increase could affect plants.
8. **Evaluating Differences** Explain how the world would be different if C_4 plants and CAM plants had not evolved.

CHAPTER HIGHLIGHTS

SECTION 1

The Light Reactions

- Photosynthesis converts light energy into chemical energy through series of reactions known as biochemical pathways. Almost all life depends on photosynthesis.
- Autotrophs use photosynthesis to make organic compounds from carbon dioxide and water. Heterotrophs cannot make their own organic compounds from inorganic compounds and therefore depend on autotrophs.
- White light from the sun is composed of an array of colors called the visible spectrum. Pigments absorb certain colors of light and reflect or transmit the other colors.
- The light reactions of photosynthesis begin with the absorption of light by chlorophyll *a* and accessory pigments in the thylakoids.
- Excited electrons that leave chlorophyll *a* travel along two electron transport chains, resulting in the production of NADPH. The electrons are replaced when water is split into electrons, protons, and oxygen in the thylakoid. Oxygen is released as a byproduct of photosynthesis.
- As electrons travel along the electron transport chains, protons move into the thylakoid and build up a concentration gradient. The movement of protons down this gradient of protons and through ATP synthase results in the synthesis of ATP through chemiosmosis.

Vocabulary

autotroph (p. 113)
photosynthesis (p. 113)
heterotroph (p. 113)
light reactions (p. 114)
chloroplast (p. 114)

thylakoid (p. 114)
granum (p. 114)
stroma (p. 114)
pigment (p. 115)
chlorophyll (p. 115)

carotenoid (p. 115)
photosystem (p. 116)
primary electron
acceptor (p. 117)

electron transport
chain (p. 117)
chemiosmosis (p. 118)

SECTION 2

The Calvin Cycle

- The ATP and NADPH produced in the light reactions drive the second stage of photosynthesis, the Calvin cycle. In the Calvin cycle, CO₂ is incorporated into organic compounds, a process called carbon fixation.
- The Calvin cycle produces a compound called G3P. Most G3P molecules are converted into RuBP to keep the Calvin cycle operating. However, some G3P molecules are used to make other organic compounds, including amino acids, lipids, and carbohydrates.
- Plants that fix carbon using only the Calvin cycle are known as C₃ plants. Some plants that evolved in hot, dry climates fix carbon through alternative pathways—the C₄ and CAM pathways. These plants carry out carbon fixation and the Calvin cycle either in different cells or at different times.
- Photosynthesis occurs in two stages. In the light reactions, energy is absorbed from sunlight and converted into chemical energy; in the Calvin cycle, carbon dioxide and chemical energy are used to form organic compounds.
- The rate of photosynthesis increases and then reaches a plateau as light intensity or CO₂ concentration increases. Below a certain temperature, the rate of photosynthesis increases as temperature increases. Above that temperature, the rate of photosynthesis decreases as temperature increases.

Vocabulary

Calvin cycle (p. 120)
carbon fixation (p. 120)

stomata (p. 121)
C₄ pathway (p. 122)

CAM pathway (p. 122)

CHAPTER REVIEW

USING VOCABULARY

- For each pair of terms, explain the relationship between the terms.
 - electron transport chain* and *primary electron acceptor*
 - photosystem* and *pigment*
 - thylakoid* and *stroma*
 - carbon fixation* and *C₄ pathway*
- Use the following terms in the same sentence: *photosynthesis*, *light reactions*, *pigment*, and *photosystem*.
- Choose the term that does not belong in the following group, and explain why it does not belong: *electron transport chain*, *chemiosmosis*, *Calvin cycle*, and *photosystem II*.
- Word Roots and Origins** The prefix *chloro-* is derived from the Greek word that means “green.” Using this information, explain why the chlorophylls are well named.

UNDERSTANDING KEY CONCEPTS

- Distinguish** between autotrophs and heterotrophs.
- Identify** the primary source of energy for humans.
- Relate** the types of pigments involved in photosynthesis and their roles.
- Compare** the different roles of photosystem I and photosystem II in photosynthesis.
- Describe** what happens to the extra energy in excited electrons as they pass along an electron transport chain in a chloroplast.
- Explain** how oxygen is generated in photosynthesis.
- Summarize** the events of chemiosmosis.
- State** the three major steps of the Calvin cycle.
- Explain** what happens to the 3-carbon compounds that do not leave the Calvin cycle to be made into organic compounds.
- Propose** what might happen to photosynthesis if ATP were not produced in the light reactions.
- Relate** the rate of photosynthesis to carbon dioxide levels.
- Unit 2–Photosynthesis**



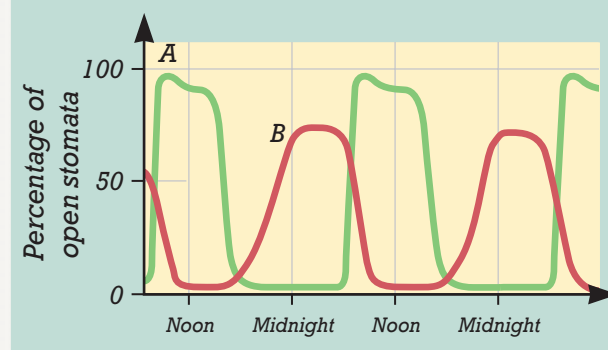
Many plants have stomata that take in CO₂ at night and release it during the day. Write a report about these types of plants, and summarize why this adaptation is an advantage for plants living in a hot, dry climate.

- CONCEPT MAPPING** Create a concept map that shows the steps of photosynthesis. Include the following terms in your concept map: *light reactions*, *Calvin cycle*, *photosystem I*, *photosystem II*, *carbon fixation*, *accessory pigment*, *chlorophyll*, *electron transport chain*, *ATP synthase*, *chemiosmosis*, and *photosynthesis*.

CRITICAL THINKING

- Predicting Results** When the CO₂ concentration in the cells of a C₃ plant is low compared with the O₂ concentration, an enzyme combines RuBP with O₂ rather than with CO₂. What effect would this enzymatic change have on photosynthesis? Under what environmental conditions would it be most likely to occur?
- Evaluating Information** All of the major components of the light reactions, including the pigment molecules clustered in photosystems I and II, are located in the thylakoid membrane. What is the advantage of having these components confined to the same membrane rather than dissolved in the stroma or the cytosol?
- Inferring Relationships** When the sun's rays are blocked by a thick forest, clouds, or smoke from a large fire, what effect do you think there will be on photosynthesis? How might it affect the levels of atmospheric carbon dioxide and oxygen? What experiments could scientists conduct in the laboratory to test your predictions?
- Interpreting Graphics** The graph below shows how the percentage of stomata that are open varies over time for two different plants. One curve represents the stomata of a geranium, and the other curve represents the stomata of a pineapple. Which curve corresponds to the pineapple stomata? Explain your reasoning.

Daily Cycle of Stomatal Opening



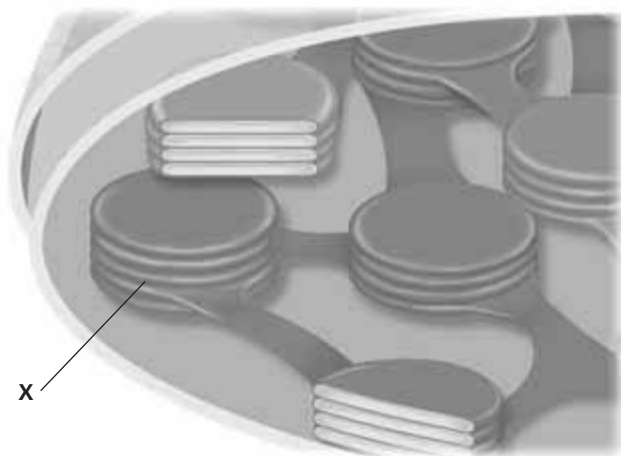


Standardized Test Preparation

DIRECTIONS: Choose the letter of the answer choice that best answers the question.

- Which of the following is a reactant in the Calvin cycle?
 - O_2
 - CO_2
 - H_2O
 - $C_6H_{12}O_6$
- Which of the following statements is correct?
 - Accessory pigments are not involved in photosynthesis.
 - Accessory pigments add color to plants but do not absorb light energy.
 - Accessory pigments absorb colors of light that chlorophyll *a* cannot absorb.
 - Accessory pigments receive electrons from the electron transport chain of photosystem I.
- Oxygen is produced at what point during photosynthesis?
 - when CO_2 is fixed
 - when water is split
 - when ATP is converted into ADP
 - when 3-PGA is converted into G3P

INTERPRETING GRAPHICS: The diagram below shows a portion of a chloroplast. Use the diagram to answer the question that follows.



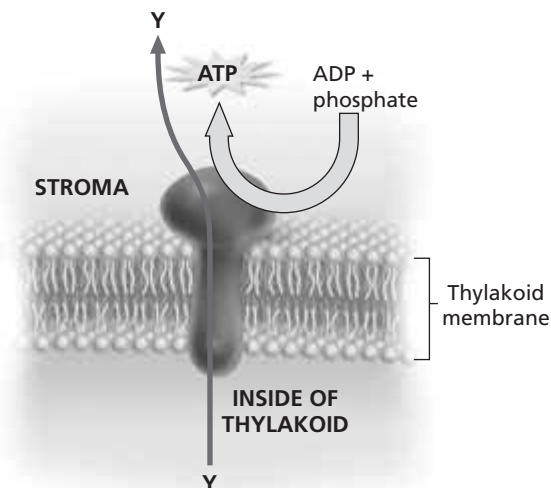
- Which of the following correctly identifies the structure marked *X* and the activities that take place there?
 - stroma—Calvin cycle
 - stroma—light reactions
 - thylakoid—Calvin cycle
 - thylakoid—light reactions

DIRECTIONS: Complete the following analogy.

5. light reactions : ATP :: Calvin cycle :

- H^+
- O_2
- G3P
- H_2O

INTERPRETING GRAPHICS: The diagram below shows a step in the process of chemiosmosis. Use the diagram to answer the question that follows.



- What is the substance identified as *Y* in the image?
 - H^+
 - NAD^+
 - NADPH
 - ADP synthase

SHORT RESPONSE

Chloroplasts are organelles with areas that conduct different specialized activities.

Where in the chloroplast do the light reactions and the Calvin cycle occur?

EXTENDED RESPONSE

The reactions of photosynthesis make up a biochemical pathway.

Part A What are the reactants and products for both the light reactions and the Calvin cycle?

Part B Explain how the biochemical pathway of photosynthesis recycles many of its own reactants, and identify the recycled reactants.

Test TIP

For a question involving a biochemical pathway, write out all of the reactants and products of each step *before* answering the question.

Measuring the Rate of Photosynthesis

OBJECTIVES

- Measure the amount of oxygen produced by an elodea sprig.
- Determine the rate of photosynthesis for elodea.

PROCESS SKILLS

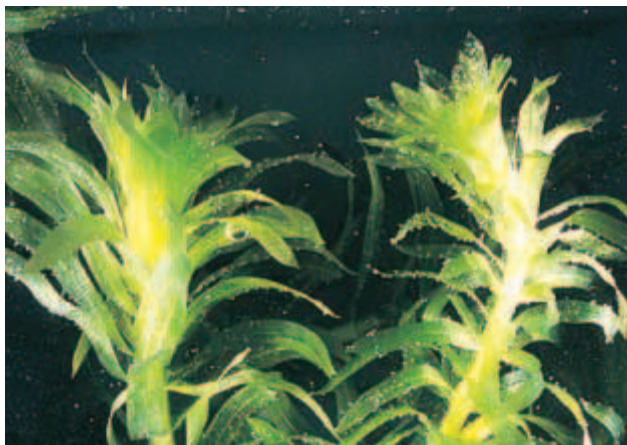
- making observations
- measuring
- collecting data
- graphing

MATERIALS

- 500 mL of 5% baking-soda-and-water solution
- 600 mL beaker
- 20 cm long elodea sprigs (2–3)
- glass funnel
- test tube
- metric ruler

Background

1. Summarize the main steps of photosynthesis.
2. State the source of the oxygen produced during photosynthesis.
3. Identify factors that can affect the rate of photosynthesis.



Procedure

PART A Setting Up the Experiment



CAUTION Always wear safety goggles and a lab

apron to protect your eyes and clothing. Glassware is fragile. Notify the teacher of broken glass or cuts. Do not clean up broken glass or spills with broken glass unless the teacher tells you to do so. Wear disposable polyethylene gloves when handling any plant. Do not eat any part of a plant or plant seed used in the lab. Wash hands thoroughly after handling any part of a plant.

1. Add 450 mL of baking-soda-and-water solution to a beaker.
2. Put two or three sprigs of elodea in the beaker. The baking soda will provide the elodea with the carbon dioxide it needs for photosynthesis.
3. Place the wide end of the funnel over the elodea. The end of the funnel with the small opening should be pointing up. The elodea and the funnel should be completely under the solution.
4. Fill a test tube with the remaining baking-soda-and-water solution. Place your thumb over the end of the test tube. Turn the test tube upside down, taking care that no air enters. Hold the opening of the test tube under the solution, and place the test tube over the small end of the funnel. Try not to let any solution leak out of the test tube as you do this.
5. Place the beaker setup in a well-lighted area near a lamp or in direct sunlight.

PART B Collecting Data

6. Create a data table like the one below.
7. Record that there was 0 mm gas in the test tube on day 0. (If you were unable to place the test tube without getting air in the tube, measure the height of the column of air in the test tube in millimeters. Record this value for day 0.) In this lab, change in gas volume is indicated by a linear measurement expressed in millimeters.
8. For days 1 through 5, measure the amount of gas in the test tube. Record the measurements in your data table under the heading "Total amount of gas present (mm)."
9. Calculate the amount of gas produced each day by subtracting the amount of gas present on the previous day from the amount of gas present today. Record these amounts under the heading "Amount of gas produced per day (mm)."

Analysis and Conclusions

1. Using the data from your table, prepare a graph.
2. Using information from your graph, describe what happened to the amount of gas in the test tube.
3. How much gas was produced in the test tube after day 5?
4. Write the equation for photosynthesis. Explain each part of the equation. For example, which ingredients are necessary for photosynthesis to take place? Which substances are produced by photosynthesis? Which gas is produced that we need in order to live?



5. What may happen to the oxygen level if an animal, such as a snail, were put in the beaker with the elodea sprig while the elodea sprig was making oxygen?

Further Inquiry

Design an experiment for predicting the effects of temperature on the amount of oxygen produced or rate of photosynthesis by elodea.

AMOUNT OF GAS PRESENT IN THE TEST TUBE

Days of exposure to light	Total amount of gas present (mm)	Amount of gas produced per day (mm)
0		
1		
2		
3		
4		
5		

CELLULAR RESPIRATION



Like other heterotrophs, the giant panda, *Ailuropoda melanoleuca*, obtains organic compounds by consuming other organisms. Biochemical pathways within the panda's cells transfer energy from those compounds to ATP.

SECTION 1 *Glycolysis and Fermentation*

SECTION 2 *Aerobic Respiration*



Unit 3—Cellular Respiration
Topics 1–6



For project ideas from *Scientific American*, visit go.hrw.com and type in the keyword **HM6SAA**.

GLYCOLYSIS AND FERMENTATION

Most foods contain usable energy, stored in complex organic compounds such as proteins, carbohydrates, and fats. All cells break down organic compounds into simpler molecules, a process that releases energy to power cellular activities.

HARVESTING CHEMICAL ENERGY

Cellular respiration is the complex process in which cells make adenosine triphosphate (ATP) by breaking down organic compounds. Recall that autotrophs, such as plants, use photosynthesis to convert light energy from the sun into chemical energy, which is stored in organic compounds. Both autotrophs and heterotrophs undergo cellular respiration to break these organic compounds into simpler molecules and thus release energy. Some of the energy is used to make ATP. The energy in ATP is then used by cells to do work.

Overview of Cellular Respiration

Figure 7-1 shows that autotrophs and heterotrophs use cellular respiration to make carbon dioxide (CO_2) and water from organic compounds and oxygen (O_2). ATP is also produced during cellular respiration. Autotrophs then use the CO_2 and water to produce O_2 and organic compounds. Thus, the products of cellular respiration are reactants in photosynthesis. Conversely, the products of photosynthesis are reactants in cellular respiration. Cellular respiration can be divided into two stages:

- 1. Glycolysis** Organic compounds are converted into three-carbon molecules of **pyruvic** (pie-ROO-vik) **acid**, producing a small amount of ATP and **NADH** (an electron carrier molecule). Glycolysis is an **anaerobic** (AN-uhr-oh-bik) process because it does not require the presence of oxygen.
- 2. Aerobic Respiration** If oxygen is present in the cell's environment, pyruvic acid is broken down and NADH is used to make a large amount of ATP through the process known as **aerobic** (uhr-OH-bik) **respiration** (covered later).

Pyruvic acid can enter other pathways if there is no oxygen present in the cell's environment. The combination of glycolysis and these anaerobic pathways is called *fermentation*.

OBJECTIVES

- **Identify** the two major steps of cellular respiration.
- **Describe** the major events in glycolysis.
- **Compare** lactic acid fermentation with alcoholic fermentation.
- **Calculate** the efficiency of glycolysis.

VOCABULARY

cellular respiration
pyruvic acid
NADH
anaerobic
aerobic respiration
glycolysis
NAD⁺
fermentation
lactic acid fermentation
alcoholic fermentation
kilocalorie

FIGURE 7-1

Both autotrophs and heterotrophs produce carbon dioxide and water through cellular respiration. Many autotrophs produce organic compounds and oxygen through photosynthesis.

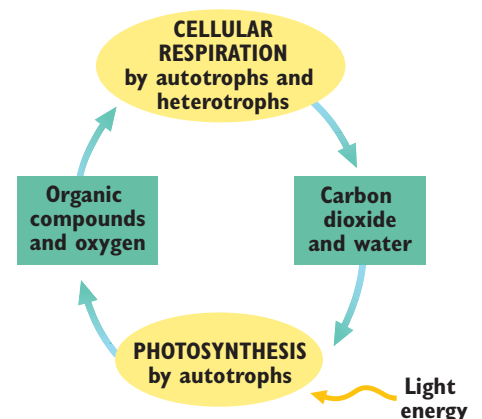
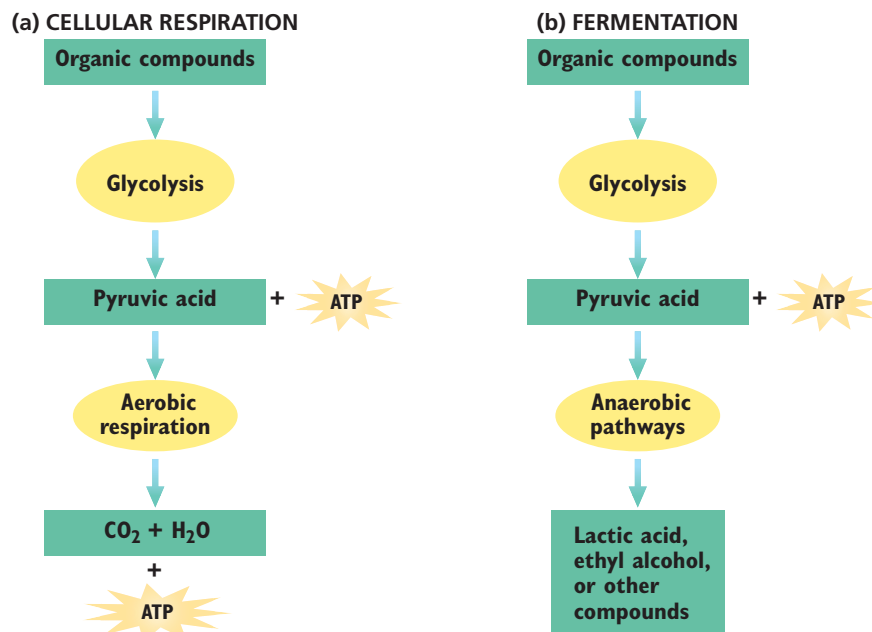




FIGURE 7-2

Organisms use cellular respiration to harness energy from organic compounds in food. (a) Glycolysis, the first stage of cellular respiration, produces a small amount of ATP. Most of the ATP produced in cellular respiration results from aerobic respiration, which is the second stage of cellular respiration. (b) In some cells, glycolysis may result in fermentation if oxygen is not present.



Many of the reactions in cellular respiration are redox reactions. Recall that in a *redox reaction*, one reactant is oxidized (loses electrons) while another is reduced (gains electrons). Although many kinds of organic compounds can be oxidized in cellular respiration, it is customary to focus on the simple sugar called *glucose* ($C_6H_{12}O_6$). The following equation summarizes cellular respiration:



This equation, however, does not explain how cellular respiration occurs. It is useful to examine each of the two stages, summarized in Figure 7-2a. (Figure 7-2b illustrates the differences between cellular respiration and fermentation.) The first stage of cellular respiration is glycolysis.

GLYCOLYSIS

Glycolysis is a biochemical pathway in which one six-carbon molecule of glucose is oxidized to produce two three-carbon molecules of pyruvic acid. Like other biochemical pathways, glycolysis is a series of chemical reactions catalyzed by specific enzymes. All of the reactions of glycolysis take place in the cytosol and occur in four main steps, as illustrated in Figure 7-3 on the next page.

In step **1**, two phosphate groups are attached to one molecule of glucose, forming a new six-carbon compound that has two phosphate groups. The phosphate groups are supplied by two molecules of ATP, which are converted into two molecules of ADP in the process.

In step **2**, the six-carbon compound formed in step **1** is split into two three-carbon molecules of glyceraldehyde 3-phosphate (G3P). Recall that G3P is also produced by the Calvin cycle in photosynthesis.

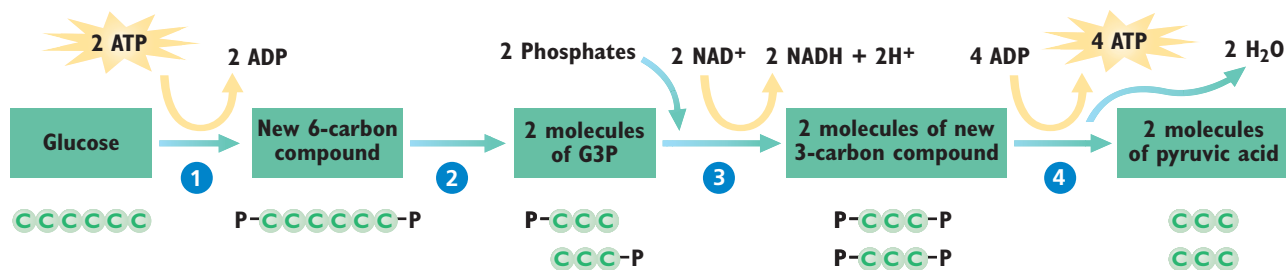


FIGURE 7-3

Glycolysis takes place in the cytosol of cells and involves four main steps. A net yield of two ATP molecules is produced for every molecule of glucose that undergoes glycolysis.

In step ③, the two G3P molecules are oxidized, and each receives a phosphate group. The product of this step is two molecules of a new three-carbon compound. As shown in Figure 7-3, the oxidation of G3P is accompanied by the reduction of two molecules of nicotinamide adenine dinucleotide (NAD^+) to NADH. NAD^+ is similar to NADP^+ , a compound involved in the light reactions of photosynthesis. Like NADP^+ , NAD^+ is an organic molecule that accepts electrons during redox reactions.

In step ④, the phosphate groups added in step ① and step ③ are removed from the three-carbon compounds formed in step ③. This reaction produces two molecules of pyruvic acid. Each phosphate group is combined with a molecule of ADP to make a molecule of ATP. Because a total of four phosphate groups were added in step ① and step ③, four molecules of ATP are produced.

Notice that two ATP molecules were used in step ①, but four were produced in step ④. Therefore, glycolysis has a net yield of two ATP molecules for every molecule of glucose that is converted into pyruvic acid. What happens to the pyruvic acid depends on the type of cell and on whether oxygen is present.

FERMENTATION

When oxygen is present, cellular respiration continues as pyruvic acid enters the pathways of aerobic respiration. (Aerobic respiration is covered in detail in the next section.) In anaerobic conditions (when oxygen is absent), however, some cells can convert pyruvic acid into other compounds through additional biochemical pathways that occur in the cytosol. The combination of glycolysis and these additional pathways, which regenerate NAD^+ , is known as **fermentation**. The additional fermentation pathways do not produce ATP. However, if there were not a cellular process that recycled NAD^+ from NADH, glycolysis would quickly use up all the NAD^+ in the cell. Glycolysis would then stop. ATP production through glycolysis would therefore also stop. The fermentation pathways thus allow for the continued production of ATP.

There are many fermentation pathways, and they differ in terms of the enzymes that are used and the compounds that are made from pyruvic acid. Two common fermentation pathways result in the production of lactic acid and ethyl alcohol.

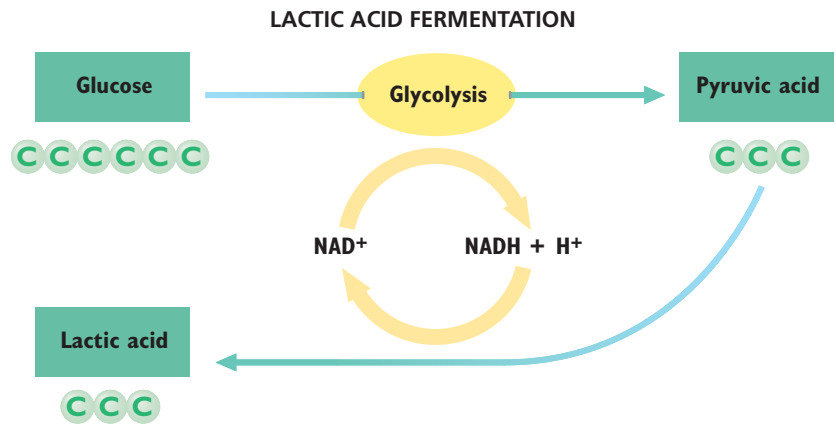
Word Roots and Origins

fermentation

from the Latin *fermentum*, meaning “leaven” or anything that causes baked goods to rise, such as yeast

FIGURE 7-4

Some cells engage in lactic acid fermentation when oxygen is absent. In this process, pyruvic acid is reduced to lactic acid and NADH is oxidized to NAD^+ .



Lactic Acid Fermentation

In **lactic acid fermentation**, an enzyme converts pyruvic acid made during glycolysis into another three-carbon compound, called lactic acid. As Figure 7-4 shows, lactic acid fermentation involves the transfer of one hydrogen atom from NADH and the addition of one free proton (H^+) to pyruvic acid. In the process, NADH is oxidized to form NAD^+ . The resulting NAD^+ is used in glycolysis, where it is again reduced to NADH. Thus, the regeneration of NAD^+ in lactic acid fermentation helps to keep glycolysis operating.

Lactic acid fermentation by microorganisms plays an essential role in the manufacture of many dairy products, as illustrated in Figure 7-5. Milk will ferment naturally if not refrigerated properly or consumed in a timely manner. Such fermentation of milk is considered “spoiling.” But ever since scientists discovered the microorganisms that cause this process, fermentation has been used in a controlled manner to produce cheese, buttermilk, yogurt, sour cream, and other cultured dairy products. Only harmless, active microorganisms are used in the fermentation of dairy products.

Lactic acid fermentation also occurs in your muscle cells during very strenuous exercise, such as sprinting. During this kind of exercise, muscle cells use up oxygen more rapidly than it can be delivered to them. As oxygen becomes depleted, the muscle cells begin to switch from cellular respiration to lactic acid fermentation. Lactic acid accumulates in the muscle cells, making the cells’ cytosol more acidic. The increased acidity may reduce the capacity of the cells to contract, resulting in muscle fatigue, pain, and even cramps. Eventually, the lactic acid diffuses into the blood and is transported to the liver, where it can be converted back into pyruvic acid.

FIGURE 7-5

In cheese making, fungi or bacteria are added to large vats of milk. The microorganisms carry out lactic acid fermentation, converting some of the sugar in the milk to lactic acid.



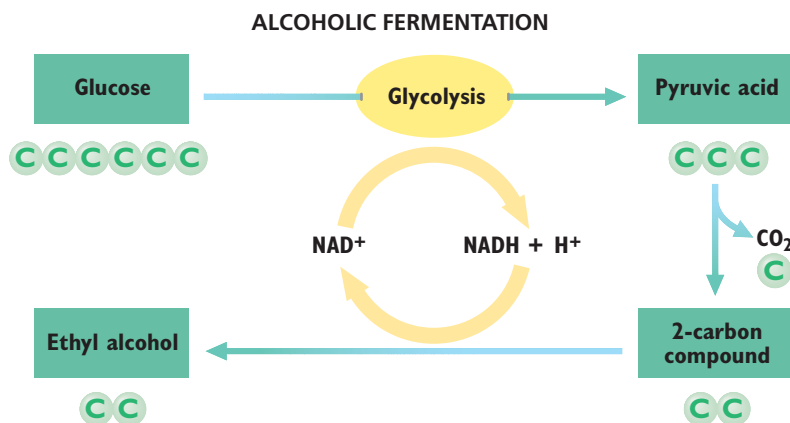


FIGURE 7-6

Some cells engage in alcoholic fermentation, converting pyruvic acid into ethyl alcohol. Again, NADH is oxidized to NAD⁺.

Alcoholic Fermentation

Some plant cells and unicellular organisms, such as yeast, use a process called **alcoholic fermentation** to convert pyruvic acid into ethyl alcohol. After glycolysis, this pathway requires two steps, which are shown in Figure 7-6. In the first step, a CO₂ molecule is removed from pyruvic acid, leaving a two-carbon compound. In the second step, two hydrogen atoms are added to the two-carbon compound to form ethyl alcohol. As in lactic acid fermentation, these hydrogen atoms come from NADH and H⁺, regenerating NAD⁺ for use in glycolysis.

Alcoholic fermentation by yeast cells such as those in Figure 7-7 is the basis of the wine and beer industry. Yeasts are a type of fungi. These microorganisms cannot produce their own food. But supplied with food sources that contain sugar (such as fruits and grains), yeast cells will perform the reactions of fermentation, releasing ethyl alcohol and carbon dioxide in the process. The ethyl alcohol is the ‘alcohol’ in alcoholic beverages. To make table wines, the CO₂ that is generated in the first step of fermentation is allowed to escape. To make sparkling wines, such as champagne, CO₂ is retained within the mixture, “carbonating” the beverage.

Bread making also depends on alcoholic fermentation performed by yeast cells. In this case, the CO₂ that is produced by fermentation makes the bread rise by forming bubbles inside the dough, and the ethyl alcohol evaporates during baking.

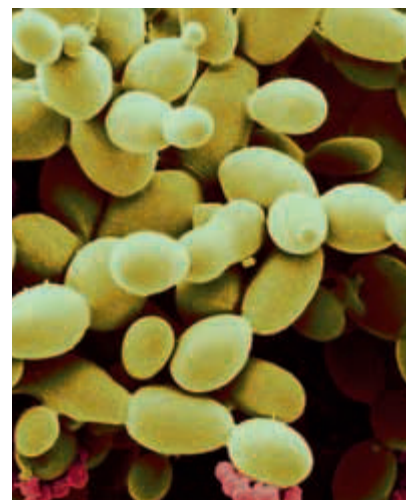


FIGURE 7-7

The yeast *Saccharomyces cerevisiae* is used in alcohol production and bread making.

EFFICIENCY OF GLYCOLYSIS

How efficient is glycolysis in obtaining energy from glucose and using it to make ATP from ADP? To answer this question, one must compare the amount of energy available in glucose with the amount of energy contained in the ATP that is produced by glycolysis. In such comparisons, energy is often measured in units of **kilocalories** (kcal). One kilocalorie equals 1,000 calories (cal).

Word Roots and Origins

kilocalorie

from the Greek *chilioi*, meaning “thousand,” and the Latin *calor*, meaning “heat”



Scientists have calculated that the complete oxidation of a standard amount of glucose releases 686 kcal. The production of a standard amount of ATP from ADP absorbs a minimum of about 7 kcal, depending on the conditions inside the cell. Recall that two ATP molecules are produced from every glucose molecule that is broken down by glycolysis.

$$\begin{aligned}\text{Efficiency of glycolysis} &= \frac{\text{Energy required to make ATP}}{\text{Energy released by oxidation of glucose}} \\ &= \frac{2 \times 7 \text{ kcal}}{686 \text{ kcal}} \times 100\% = 2\%\end{aligned}$$

You can see that the two ATP molecules produced during glycolysis receive only a small percentage of the energy that could be released by the complete oxidation of each molecule of glucose. Much of the energy originally contained in glucose is still held in pyruvic acid. Even if pyruvic acid is converted into lactic acid or ethyl alcohol, no additional ATP is synthesized. It's clear that glycolysis alone or as part of fermentation is not very efficient in transferring energy from glucose to ATP.

Organisms probably evolved to use glycolysis very early in the history of life on Earth. The first organisms were bacteria, and they produced all of their ATP through glycolysis. It took more than a billion years for the first photosynthetic organisms to appear. The oxygen they released as a byproduct of photosynthesis may have stimulated the evolution of organisms that make most of their ATP through aerobic respiration.

By themselves, the anaerobic pathways provide enough energy for many present-day organisms. However, most of these organisms are unicellular, and those that are multicellular are very small. All of them have limited energy requirements. Larger organisms have much greater energy requirements that cannot be satisfied by glycolysis alone. These larger organisms meet their energy requirements with the more efficient pathways of aerobic respiration.

SECTION 1 REVIEW

1. Explain the role of organic compounds in cellular respiration.
2. For each six-carbon molecule that begins glycolysis, identify how many molecules of ATP are used and how many molecules of ATP are produced.
3. Distinguish between the products of the two types of fermentation discussed in this section.
4. Calculate the efficiency of glycolysis if 12 kcal of energy are required to transfer energy from glucose to ATP.

CRITICAL THINKING

5. **Applying Information** A large amount of ATP in a cell inhibits the enzymes that drive the first steps of glycolysis. How will this inhibition of enzymes eventually affect the amount of ATP in the cell?
6. **Predicting Results** How might the efficiency of glycolysis change if this process occurred in only one step? Explain your answer.
7. **Relating Concepts** In what kind of environment would you expect to find organisms that carry out fermentation?

AEROBIC RESPIRATION

In most cells, glycolysis does not result in fermentation. Instead, when oxygen is available, pyruvic acid undergoes aerobic respiration, the pathway of cellular respiration that requires oxygen. Aerobic respiration produces nearly 20 times as much ATP as is produced by glycolysis alone.

OVERVIEW OF AEROBIC RESPIRATION

Aerobic respiration has two major stages: the Krebs cycle and the electron transport chain, which is associated with chemiosmosis (using the energy released as protons move across a membrane to make ATP). In the Krebs cycle, the oxidation of glucose that began with glycolysis is completed. As glucose is oxidized, NAD^+ is reduced to NADH. In the electron transport chain, NADH is used to make ATP. Although the Krebs cycle also produces a small amount of ATP, most of the ATP produced during aerobic respiration is made through the activities of the electron transport chain and chemiosmosis. The reactions of the Krebs cycle, the electron transport chain, and chemiosmosis occur only if oxygen is present in the cell.

In prokaryotes, the reactions of the Krebs cycle and the electron transport chain take place in the cytosol of the cell. In eukaryotic cells, however, these reactions take place inside mitochondria rather than in the cytosol. The pyruvic acid that is produced in glycolysis diffuses across the double membrane of a mitochondrion and enters the mitochondrial matrix. The **mitochondrial matrix** is the space inside the inner membrane of a mitochondrion. Figure 7-8 illustrates the relationships between these mitochondrial parts. The mitochondrial matrix contains the enzymes needed to catalyze the reactions of the Krebs cycle.

OBJECTIVES

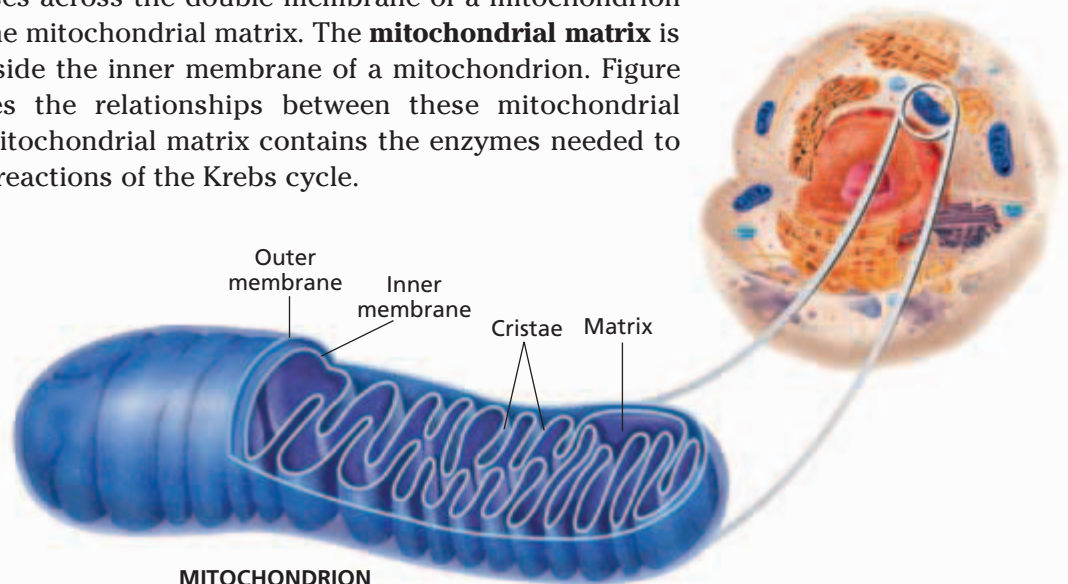
- **Relate** aerobic respiration to the structure of a mitochondrion.
- **Summarize** the events of the Krebs cycle.
- **Summarize** the events of the electron transport chain and chemiosmosis.
- **Calculate** the efficiency of aerobic respiration.
- **Contrast** the roles of glycolysis and aerobic respiration in cellular respiration.

VOCABULARY

mitochondrial matrix
acetyl CoA
Krebs cycle
oxaloacetic acid
citric acid
FAD

FIGURE 7-8

In eukaryotic cells, the reactions of aerobic respiration occur inside mitochondria. The Krebs cycle takes place in the mitochondrial matrix, and the electron transport chain is located in the inner membrane.



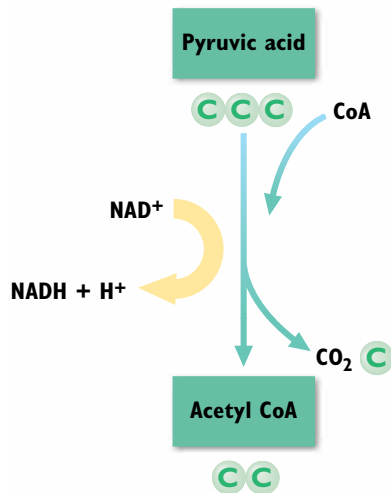


FIGURE 7-9

Glycolysis yields two molecules of pyruvic acid. In aerobic respiration, each molecule of pyruvic acid reacts with coenzyme A (CoA) to form a molecule of acetyl CoA. Notice that CO₂, NADH, and H⁺ are also produced in this reaction.

When pyruvic acid enters the mitochondrial matrix, it reacts with a molecule called coenzyme A to form acetyl (uh-SEET-uhl) coenzyme A, abbreviated **acetyl CoA** (uh-SEET-uhl KOH-AY). This reaction is illustrated in Figure 7-9. The acetyl part of acetyl CoA contains two carbon atoms, but as you learned earlier, pyruvic acid is a three-carbon compound. The carbon atom that is lost in the conversion of pyruvic acid to acetyl CoA is released in a molecule of CO₂. This reaction reduces a molecule of NAD⁺ to NADH.

THE KREBS CYCLE

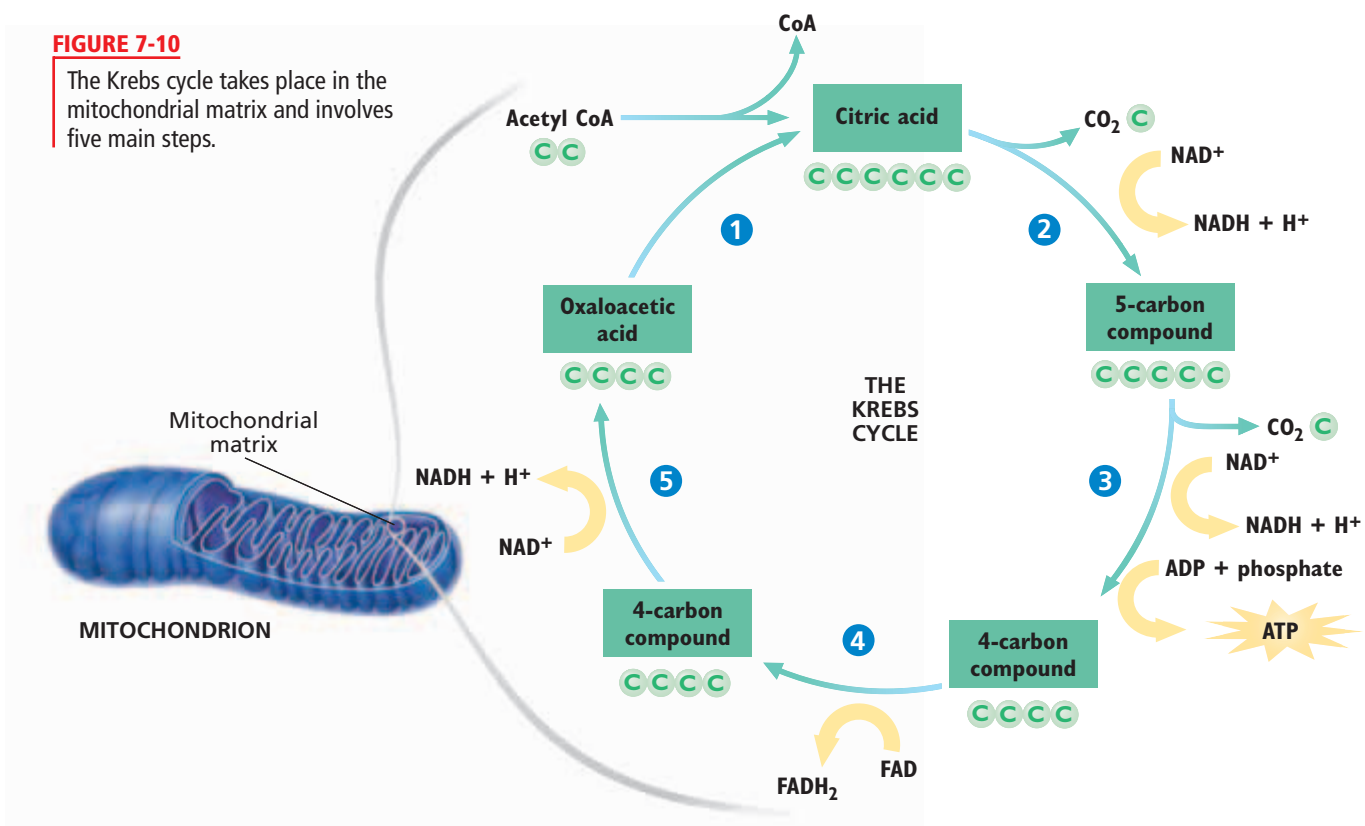
The **Krebs cycle** is a biochemical pathway that breaks down acetyl CoA, producing CO₂, hydrogen atoms, and ATP. The reactions that make up the cycle were identified by Hans Krebs (1900–1981), a German biochemist. The Krebs cycle has five main steps. In eukaryotic cells, all five steps occur in the mitochondrial matrix. Examine Figure 7-10 as you read about the steps in the Krebs cycle.

In step **1**, a two-carbon molecule of acetyl CoA combines with a four-carbon compound, **oxaloacetic acid** (AHKS-uh-loh-uh-SEET-ik) **acid**, to produce a six-carbon compound, **citric acid** (SI-trik) **acid**. Notice that this reaction regenerates coenzyme A.

In step **2**, citric acid releases a CO₂ molecule and a hydrogen atom to form a five-carbon compound. By losing a hydrogen atom with its electron, citric acid is oxidized. The electron in the hydrogen atom is transferred to NAD⁺, reducing it to NADH.

FIGURE 7-10

The Krebs cycle takes place in the mitochondrial matrix and involves five main steps.



In step ③, the five-carbon compound formed in step ② also releases a CO_2 molecule and a hydrogen atom, forming a four-carbon compound. Again, NAD^+ is reduced to NADH. Notice that in this step a molecule of ATP is also synthesized from ADP.

In step ④, the four-carbon compound formed in step ③ releases a hydrogen atom to form another four-carbon compound. This time, the hydrogen atom is used to reduce FAD to FADH_2 . FAD, or flavin adenine dinucleotide, is a molecule very similar to NAD^+ . Like NAD^+ , FAD accepts electrons during redox reactions.

In step ⑤, the four-carbon compound formed in step ④ releases a hydrogen atom to regenerate oxaloacetic acid, which keeps the Krebs cycle operating. The electron in the hydrogen atom reduces NAD^+ to NADH.

Recall that in glycolysis one glucose molecule produces two pyruvic acid molecules, which can then form two molecules of acetyl CoA. Thus, one glucose molecule is completely broken down in two turns of the Krebs cycle. These two turns produce four CO_2 molecules, two ATP molecules, and hydrogen atoms that are used to make six NADH and two FADH_2 molecules. The CO_2 diffuses out of the cells and is given off as waste. The ATP can be used for energy. But note that each glucose molecule yields only two molecules of ATP through the Krebs cycle—the same number as in glycolysis.

The bulk of the energy released by the oxidation of glucose still has not been transferred to ATP. Glycolysis of one glucose molecule produces two NADH molecules, and the conversion of the two resulting molecules of pyruvic acid to acetyl CoA produces two more. Adding the six NADH molecules from the Krebs cycle gives a total of 10 NADH molecules for every glucose molecule that is oxidized. These 10 NADH molecules and the two FADH_2 molecules from the Krebs cycle drive the next stage of aerobic respiration. That is where most of the energy transfer from glucose to ATP actually occurs.

ELECTRON TRANSPORT CHAIN AND CHEMIOSMOSIS

The electron transport chain, linked with chemiosmosis, constitutes the second stage of aerobic respiration. Recall that the *electron transport chain* is a series of molecules in a membrane that transfer electrons from one molecule to another. In eukaryotic cells, the electron transport chain and the enzyme ATP synthase are embedded in the inner membrane of the mitochondrion in folds called *cristae*. In prokaryotes, the electron transport chain is in the cell membrane. ATP is produced by the electron transport chain when NADH and FADH_2 release hydrogen atoms, regenerating NAD^+ and FAD. To understand how ATP is produced, you must follow what happens to the electrons and protons that make up these hydrogen atoms.



Quick Lab

Comparing CO_2 Production

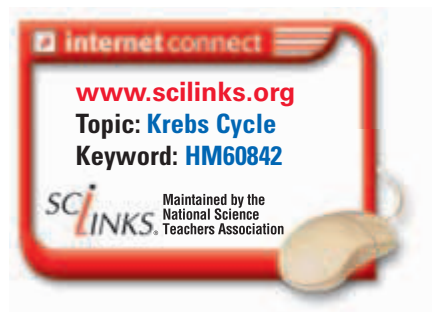
Materials disposable gloves, lab apron, safety goggles, 250 mL flask, 100 mL graduated cylinder, phenolphthalein solution, pipet, drinking straw, water, clock, sodium hydroxide solution

Procedure



1. Put on your disposable gloves, lab apron, and safety goggles.
2. Add 50 mL of water and four drops of phenolphthalein to the flask.
3. Use the straw to gently blow into the solution for 1 minute. Add the sodium hydroxide one drop at a time, and gently swirl the flask. Record the number of drops you use.
4. When the liquid turns pink, stop adding drops.
5. Empty and rinse your flask as your teacher directs, and repeat step 2. Walk vigorously for 2 minutes, and repeat steps 3 and 4.

Analysis Which trial produced the most carbon dioxide? Which trial used the most energy?



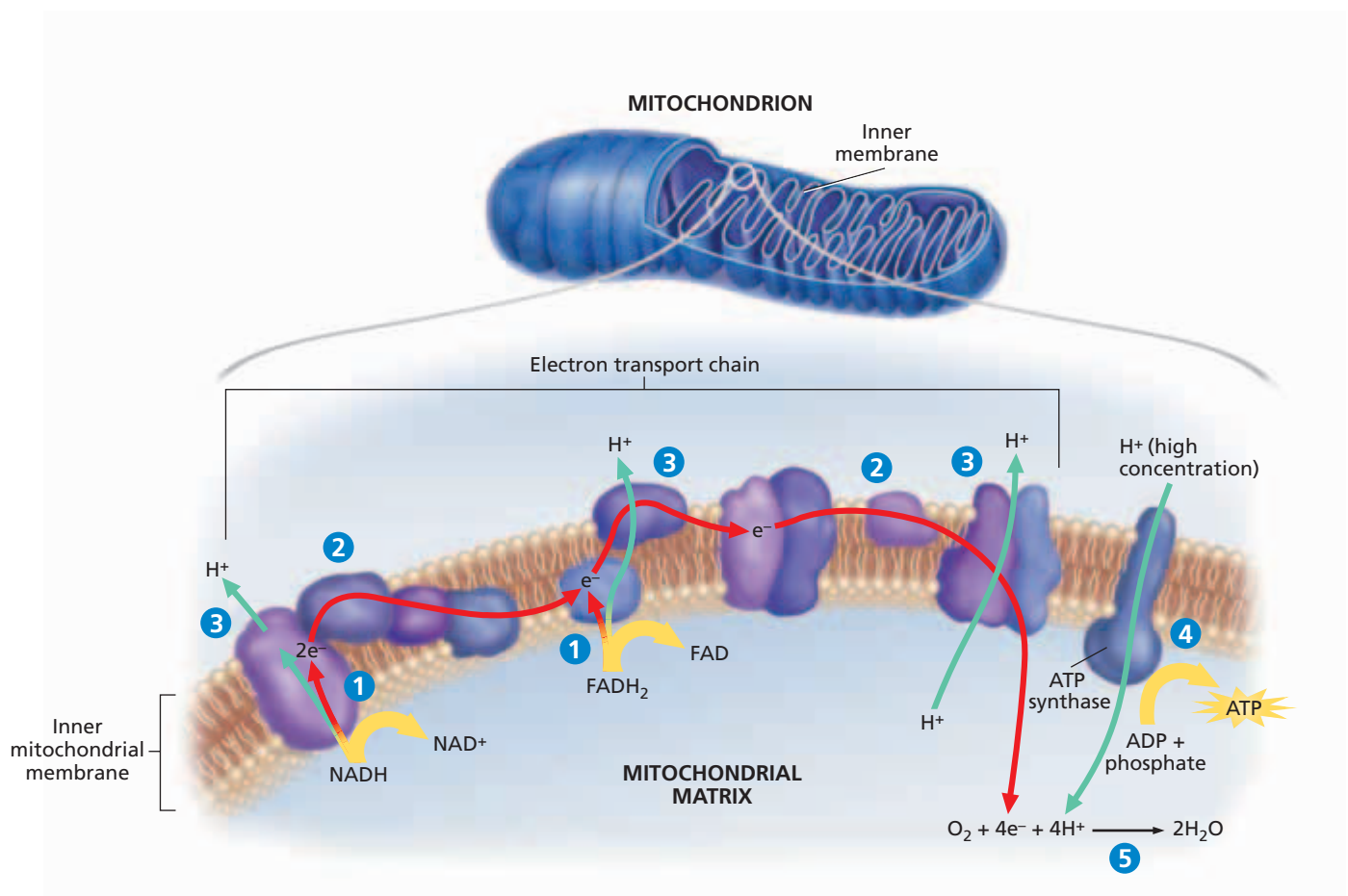


FIGURE 7-11

Electron transport and chemiosmosis take place along the inner mitochondrial membrane and involve five steps.

The electrons in the hydrogen atoms from NADH and FADH₂ are at a high energy level. In the electron transport chain, these electrons are passed along a series of molecules embedded in the inner mitochondrial membrane, as shown in Figure 7-11. In step 1, NADH and FADH₂ give up electrons to the electron transport chain. NADH donates electrons at the beginning, and FADH₂ donates them farther down the chain. These molecules also give up protons (hydrogen ions, H⁺). In step 2, the electrons are passed down the chain. As they move from molecule to molecule, they lose energy. In step 3, the energy lost from the electrons is used to pump protons from the matrix, building a high concentration of protons between the inner and outer membranes. Thus, a concentration gradient of protons is created across the inner membrane. An electrical gradient is also created, as the protons carry a positive charge.

In step 4, the concentration and electrical gradients of protons drive the synthesis of ATP by chemiosmosis, the same process that generates ATP in photosynthesis. ATP synthase molecules are embedded in the inner membrane, near the electron transport chain molecules. As protons move through ATP synthase and down their concentration and electrical gradients, ATP is made from ADP and phosphate. In step 5, oxygen is the final acceptor of electrons that have passed down the chain. Oxygen also accepts protons that were part of the hydrogen atoms supplied by NADH and FADH₂. The protons, electrons, and oxygen all combine to form water, as shown by the equation in step 5.

MITOCHONDRIA: Many Roles in Disease

Every cell contains very small organelles that are known as *mitochondria*. Mitochondria generate almost all of the ATP that fuels the activity in living organisms. Scientists have known for years that certain diseases are directly caused by mitochondrial dysfunction. However, new research shows that mitochondria may play roles in the symptoms of aging and may contribute to the development of Alzheimer's disease and cancer.

Mitochondrial Diseases

Mitochondria are very unusual organelles, because they have their own DNA. Mutations in mitochondrial DNA are responsible for several rare but serious disorders. Examples include *Leigh's syndrome*, a potentially deadly childhood disease that causes loss of motor and verbal skills, and *Pearson's syndrome*, which causes childhood bone marrow dysfunction and pancreatic failure.

Mitochondria in Aging

Mitochondria may play a role in causing some problems associated with aging. Chemical reactions of the Krebs cycle and electron transport chain sometimes release stray electrons that "leak out" of mitochondria into the cell. These electrons can combine with oxygen to form free radicals. Free radicals are especially reactive atoms or groups of atoms with one or more unpaired electrons. Free radicals quickly react with other molecules, such as DNA and protein; these reactions may disrupt cell activity. Biologists think that many characteristics of human aging, from wrinkles to mental decline, may be brought on partly by the damage caused by free radicals.

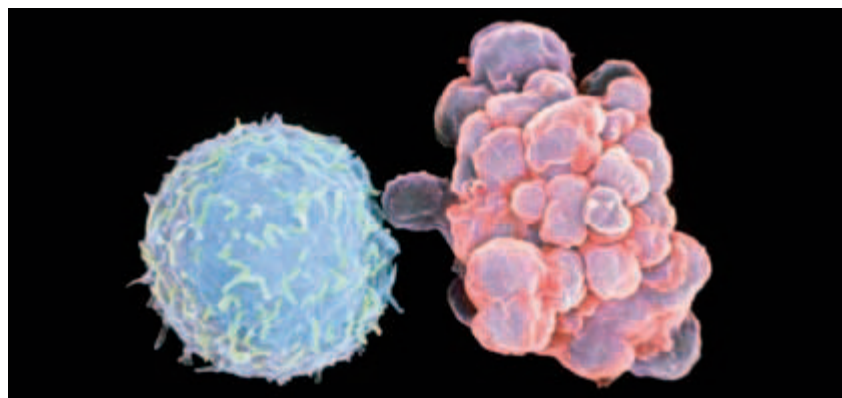
Mitochondria in Other Diseases

Recent research also shows that mitochondria may be important in diseases related to *apoptosis*, or programmed cell death. Scientists have shown that

signals from the mitochondria are instrumental in starting and/or continuing the apoptosis process. Yet sometimes, mitochondria mistakenly push or fail to push the "self-destruct button" in cells. In cases of stroke and Alzheimer's disease, for example, mitochondria may cause too many cells to die, which may lead to mental lapses and other symptoms. In the case of cancer, mitochondria may fail to initiate apoptosis. This failure could allow tumor cells to grow and invade healthy tissues.

Promise of New Treatments

Researchers are now investigating mitochondria as targets for drug treatments to prevent or treat a variety of conditions. Conversely, researchers are also studying how certain conditions impair mitochondrial function. One day, scientists may use knowledge about mitochondria to help ease the symptoms of aging and to cure or prevent many diseases.



Mitochondria may play a role in programmed cell death, or apoptosis. A white blood cell undergoing apoptosis (right) looks very different from a normal white blood cell (left). (SEM 2,600×)

REVIEW

1. How do mitochondria contribute to free radical formation?
2. How could research on mitochondria be helpful to society?
3. **Critical Thinking** Evaluate the following statement: Mitochondria—we can't live with them; we can't live without them.



The Importance of Oxygen

ATP can be synthesized by chemiosmosis only if electrons continue to move from molecule to molecule in the electron transport chain. The last molecule in the electron transport chain must pass electrons on to a final electron acceptor. Otherwise, the electron transport chain would come to a halt. Consider what would happen if cars kept entering a dead-end, one-way street. At some point, no more cars could enter the street. Similarly, if the last molecule could not “unload” the electrons it accepts, then no more electrons could enter the electron transport chain and ATP synthesis would stop. By accepting electrons from the last molecule in the electron transport chain, oxygen allows additional electrons to pass along the chain. As a result, ATP can continue to be made through chemiosmosis.

EFFICIENCY OF CELLULAR RESPIRATION

How many ATP molecules are made in cellular respiration? Refer to Figure 7-12 as you calculate the total. Recall that glycolysis and the Krebs cycle each produce two ATP molecules directly for every glucose molecule that is oxidized. Furthermore, each NADH molecule that supplies the electron transport chain can generate three ATP molecules, and each FADH_2 molecule can generate two ATP molecules. Thus, the 10 NADH and two FADH_2 molecules made through glycolysis, conversion of pyruvic acid to acetyl CoA, and the Krebs cycle can produce up to 34 ATP molecules by the electron transport chain and chemiosmosis. Adding the four ATP molecules from glycolysis and the Krebs cycle gives a maximum yield of 38 ATP molecules per molecule of glucose.

FIGURE 7-12

Follow each pathway to see how one glucose molecule can generate up to 38 ATP molecules in cellular respiration when oxygen is present.

